

**1997–2023 IMO Hong Kong Team
Selection Test
Solutions**

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Solutions

Year 1997

Test 1

1. By the QM-AM inequality, we have

$$x + y + z \leq \sqrt{3(x^2 + y^2 + z^2)}.$$

This implies

$$\begin{aligned} \frac{xyz(x + y + z + \sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)(xy + yz + zx)} &\leq \frac{(\sqrt{3} + 1)xyz\sqrt{x^2 + y^2 + z^2}}{(x^2 + y^2 + z^2)(xy + yz + zx)} \\ &= \frac{(\sqrt{3} + 1)xyz}{\sqrt{x^2 + y^2 + z^2} \cdot (xy + yz + zx)}. \end{aligned}$$

Next, by the AM-GM inequality, we have

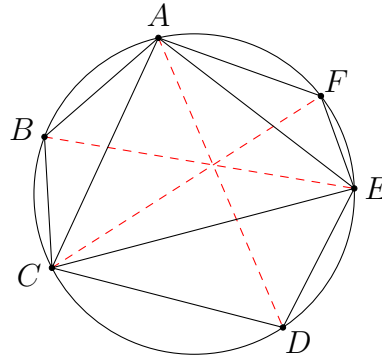
$$x^2 + y^2 + z^2 \geq 3\sqrt[3]{x^2y^2z^2}, \quad xy + yz + zx \geq 3\sqrt[3]{x^2y^2z^2}.$$

It follows that

$$\frac{xyz(x + y + z + \sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)(xy + yz + zx)} \leq \frac{(\sqrt{3} + 1)xyz}{\sqrt{3}\sqrt[3]{xyz} \cdot 3\sqrt[3]{x^2y^2z^2}} = \frac{3 + \sqrt{3}}{9}.$$

2. By Ceva's theorem, AD, BE, CF are concurrent if and only if

$$\frac{\sin \angle CAD}{\sin \angle EAD} \times \frac{\sin \angle AEB}{\sin \angle CEB} \times \frac{\sin \angle ECF}{\sin \angle ACF} = 1.$$



By the extended sine law, we have

$$\frac{\sin \angle CAD}{\sin \angle EAD} = \frac{CD}{DE}, \quad \frac{\sin \angle AEB}{\sin \angle CEB} = \frac{AB}{BC}, \quad \frac{\sin \angle ECF}{\sin \angle ACF} = \frac{EF}{FA}.$$

Thus, AD, BE, CF are concurrent if and only if

$$\frac{CD}{DE} \times \frac{AB}{BC} \times \frac{EF}{FA} = 1.$$

This is exactly $AB \cdot CD \cdot EF = BC \cdot DE \cdot FA$.

3. All integers $n = 3^k$ with $k \in \mathbb{Z}^+$ satisfy $n \mid 2^n + 1$. Indeed, since $3 \mid 2 + 1$, by the lifting the exponent lemma, we have

$$v_3(2^{3^k} + 1^{3^k}) = v_3(2 + 1) + v_3(3^k) = 1 + k.$$

This implies $3^{k+1} \mid 2^n + 1$, and hence $n \mid 2^n + 1$.

The only prime number n satisfying $n \mid 2^n + 1$ is $n = 3$. Indeed, let $n = p$ be a prime. By the Fermat little theorem, we have

$$2^p + 1 \equiv 2 + 1 = 3 \pmod{p}.$$

This is congruent to 0 modulo p if and only if $p \mid 3$, i.e. $p = 3$.

4. The only possibilities are $(x_1, x_2) = (c, c)$ for some nonzero integer c .

Rewrite the recurrence relation as

$$\frac{2}{x_{n-1}} = \frac{1}{x_n} + \frac{1}{x_{n-2}}.$$

Let $y_n = \frac{1}{x_n}$ for all n . This implies

$$y_{n-1} - y_{n-2} = y_n - y_{n-1}.$$

Therefore, $\{y_n\}$ is an arithmetic sequence. Let $y_n = a + nd$ for some constants a and d . Then we have

$$x_n = \frac{1}{y_n} = \frac{1}{a + nd}.$$

If $d \neq 0$, then we have $|a + nd| \rightarrow \infty$ when $n \rightarrow \infty$, and hence $x_n \rightarrow 0$. However, $x_n \neq 0$. This shows x_n cannot be an integer for large n . So it suffices to consider $d = 0$. In that case $\{x_n\}$ is a constant sequence. In order that there are infinitely many integer terms, the constant must be an integer. So $x_1 = x_2 = c$ for some integer $c \neq 0$. Note that a constant sequence satisfies the recurrence relation since

$$c = \frac{c^2}{2c - c}.$$

5. By the pigeonhole principle, two of the numbers

$$1, 11, 111, \dots$$

leave the same remainder when divided by X . Their difference, which is of the form $11\cdots 100\cdots 0$, is divisible by X . Therefore, we can transform X to $11\cdots 100\cdots 0$, and then to $11\cdots 1$ by deleting the zeros.

If the current number is 1, we are done. Otherwise, we have

$$\underbrace{11\cdots 1}_{k \text{ times}} \times 82 = 9 \underbrace{11\cdots 1}_{(k-2) \text{ times}} 02.$$

So we can transform $11\cdots 1$ to $911\cdots 102$, and then to $911\cdots 12$. Next, note that

$$\underbrace{911\cdots 12}_m \times 9 = 82 \underbrace{00\cdots 08}_m.$$

So we can transform the number to $8200\cdots 08$, and then to 828. Afterwards, we apply the operations as follows.

$$828 \xrightarrow{\times 25} 20700 \longrightarrow 27 \xrightarrow{\times 4} 108 \longrightarrow 18 \xrightarrow{\times 5} 90 \longrightarrow 9$$

This completes the proof.

Test 2

1. (i) This is an alternative form of Legendre's formula. Here is a simple proof of the result. Let $n = \overline{a_s a_{s-1} \cdots a_0}_{(2)}$ be the binary representation of n . Then we have

$$\begin{aligned} f(n!) &= \sum_{j=1}^s \left\lfloor \frac{n}{2^j} \right\rfloor = \sum_{j=1}^s \overline{a_s a_{s-1} \cdots a_j}_{(2)} = \sum_{j=1}^s \sum_{k=j}^s a_k \cdot 2^{k-j} = \sum_{k=1}^s \sum_{j=1}^k a_k \cdot 2^{k-j} \\ &= \sum_{k=1}^s a_k (2^k - 1) = \sum_{k=1}^s a_k \cdot 2^k - \sum_{k=1}^s a_k = n - g(n). \end{aligned}$$

- (ii) By part (a), we have

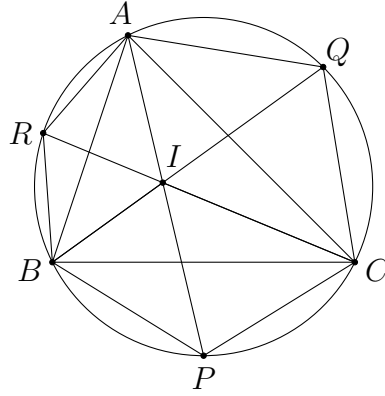
$$\begin{aligned} f\left(\frac{(2n)!}{n!n!}\right) &= f((2n)!) - 2f(n!) \\ &= [2n - g(2n)] - 2[n - g(n)] \\ &= 2g(n) - g(2n) \\ &= 2g(n) - g(n) \\ &= g(n). \end{aligned}$$

Thus, 4 divides $\binom{2n}{n}$ if and only if $g(n) \geq 2$. This holds if and only if $n = 100\cdots 0_{(2)} = 2^k$ for some nonnegative integer k .

2. Let I be the incentre of $\triangle ABC$. It is well-known that $PB = PI = PC$, etc. By the triangle inequality, we have

$$\begin{aligned} BI + CI &> BC, \\ CI + AI &> CA, \\ AI + BI &> AB, \\ 2IP &= BP + CP > BC, \\ 2IQ &= CQ + AQ > CA, \\ 2IR &= AR + BR > AB. \end{aligned}$$

Adding these inequalities and dividing both sides by 2, we obtain the desired inequality.



3. f can be any function such that $f(2^k m) = f(2^k)$ for any $k \geq 0$ and odd m , where $f(0), f(1), f(2), f(2^2), \dots$ are arbitrary integers.

We label the equation as follows.

$$f(x) + f(y) = f(x + 2xy) + f(y - 2xy) \quad (1)$$

Putting $x = 1$ and $y = n$ in (1), we obtain

$$f(1) + f(n) = f(2n + 1) + f(-n). \quad (2)$$

Putting $x = n$ and $y = -1$ in (1), we obtain

$$f(n) + f(-1) = f(-n) + f(2n - 1). \quad (3)$$

Since $f(1) = f(-1)$, by comparing (2) and (3), we immediately obtain

$$f(2n + 1) = f(2n - 1)$$

for any $n \in \mathbb{Z}$. It follows that $f(m) = f(1)$ for any odd m . Using (2), we obtain

$$f(n) = f(-n)$$

for any $n \in \mathbb{Z}$.

Next, we put $x = -2k - 1$ and $y = n$ in (1). Since both x and $x + 2xy = x(1 + 2y)$ are odd, we obtain

$$f(n) = f((4k + 3)n).$$

Replacing k by $-(k + 1)$, and using $f(m) = f(-m)$, we get

$$f(n) = f((-4k - 1)n) = f((4k + 1)n).$$

Combining these, we know that $f(n) = f(mn)$ for any odd m . Therefore, by writing $n = 2^k m$ where m is odd, we find that

$$f(2^k m) = f(2^k).$$

It remains to check that all such functions satisfy all the conditions. Firstly, $f(-1) = f(1)$ is true. Secondly, Note that $v_2(x + 2xy) = v_2(x(1 + 2y)) = v_2(x)$ and similarly $v_2(y - 2xy) = v_2(y)$. Therefore, we must have

$$f(x + 2xy) = f(x), \quad f(y - 2xy) = f(y).$$

This proves (1), and we are done.

4. If the current number is $10a + b$ where $0 \leq b \leq 9$, then the next number is $a + 4b$. Note that

$$10(a + 4b) = (10a + b) + 39b.$$

Therefore, $13 \mid a + 4b$ if and only if $13 \mid 10a + b$. Since $13 \mid 1001$, all the numbers in the sequence are divisible by 13. The only possible prime number in this sequence is 13.

Now, since $1001 \rightarrow 104 \rightarrow 26 \rightarrow 26 \rightarrow \dots$, 13 will never appear after 1001. However, since $13 \rightarrow 13 \rightarrow \dots$, it cannot appear before 1001 as well. Thus, there is no prime in the sequence.

Test 3

1. The answer is any power of 2.

Suppose $n = a + (a + 1) + \dots + (a + k)$ for some positive integers a and k . Then we have

$$2n = (2a + k)(k + 1).$$

Let $n = 2^r m$ where m is odd. Note that $2a + k$ and $k + 1$ have different parities. Therefore, 2^{r+1} must divide one of $2a + k$ and $k + 1$, and hence the other number divides m . Since $2a + k, k + 1 > 1$, we must have $m \geq 3$. This implies $n \neq 2^r$ for any $r \geq 0$.

On the other hand, for $m \geq 3$, we can find a solution. Indeed, if $2^{r+1} > m$, then we set

$$(a, k) = \left(2^r - \frac{m-1}{2}, m-1 \right)$$

so that $a, k \in \mathbb{Z}^+$ and $(2a+k)(k+1) = 2^{r+1}m = 2n$. Similarly, if $2^{r+1} < m$, then we set

$$(a, k) = \left(\frac{m+1}{2} - 2^r, 2^{r+1} - 1 \right)$$

so that $a, k \in \mathbb{Z}^+$ and $(2a+k)(k+1) = 2^{r+1}m = 2n$.

2. Let $c_j = x - b_j > 0$ for each j . Note that

$$f(y) = (y - b_1)(y - b_2) \cdots (y - b_n).$$

Therefore, we have

$$\begin{aligned} f(x+1) &= (x+1 - b_1)(x+1 - b_2) \cdots (x+1 - b_n) \\ &= (1 + c_1)(1 + c_2) \cdots (1 + c_n). \end{aligned}$$

Now, by the AM-GM inequality, we have

$$\frac{1}{c_1} + \frac{1}{c_2} + \cdots + \frac{1}{c_n} \geq n \sqrt[n]{\frac{1}{c_1 c_2 \cdots c_n}}.$$

Thus, it suffices to prove

$$\prod_{j=1}^n \sqrt[n]{\frac{(1+c_j)^n}{c_j}} \geq 2n. \tag{1}$$

Indeed, since $n \geq 2$, we get

$$\frac{(1+c_j)^n}{c_j} \geq \frac{1}{c_j} + n + \frac{n(n-1)}{2} c_j \geq n + 2 \sqrt{\frac{1}{c_j} \cdot \frac{n(n-1)}{2} c_j} \geq 2n$$

by the AM-GM inequality and a simple inequality $2\sqrt{\frac{n(n-1)}{2}} \geq n$ (which is equivalent to $n \geq 2$). This implies (1). Equality holds when $n = 2$ and $c_j = 1$, i.e. $x-1 = b_1 = b_2 = \cdots = b_n$.

3. This is a simple corollary of Brokard's theorem. Alternatively, we provide an elementary proof as follows.

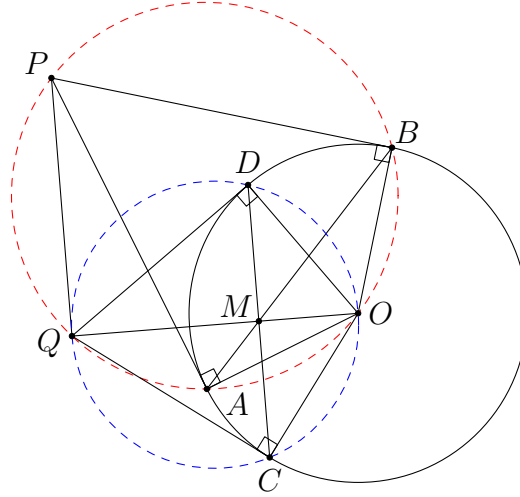
Note that O, M, Q are collinear since all of them lie on the perpendicular bisector of CD . By the property of tangents, we know that Q, C, O, D are concyclic. This yields

$$MQ \times MO = MC \times MD = MA \times MB.$$

Thus, Q, A, O, B are concyclic, and hence P, Q, A, O, B are concyclic. Therefore, we obtain

$$\angle OQP = \angle OAP = 90^\circ.$$

This means $OQ \perp PQ$.



4. No. We consider the numbers modulo 3.

- If $3 \mid a, b$, then $3 \mid a + b$ and $3 \mid ab$. The number of multiples of 3 remains unchanged.
- If $3 \mid a$ and $3 \nmid b$, then $3 \nmid a + b$ and $3 \mid ab$. The number of multiples of 3 remains unchanged.
- If $a \equiv b \equiv \pm 1 \pmod{3}$, then $3 \nmid a + b$ and $3 \nmid ab$. The number of multiples of 3 remains unchanged.
- If $a \equiv 1 \pmod{3}$ and $b \equiv 2 \pmod{3}$, then $3 \mid a + b$ and $ab \equiv 2 \pmod{3}$. The number of multiples of 3 is increased by 1, and there is a number which is congruent to 2 modulo 3.

Now, note that $3 \mid 21, 27, 180, 540$ and $64 \equiv 1 \pmod{3}$. If the 5 numbers can appear on the board at the same time, there must be an instant when the number of multiples of 3 changes from 3 to 4. This must be the last case in the above list, hence the 5 numbers are now congruent to $0, 0, 0, 0, 2$ modulo 3 respectively. Afterwards, no matter which 2 numbers we choose, the new numbers are still congruent to $0, 0, 0, 0, 2$ modulo 3 respectively. Therefore, it is impossible to generate $21, 27, 64, 180, 540$. This is a contradiction.

Year 1998

Test 1

1. Let $\ell = \text{lcm}(3, 5, \dots, 2n+1)$, and let $k = v_3(\ell)$. Note that $k \geq 1$ since $n \geq 1$. Since $2 \cdot 3^k$ is even, and $3 \cdot 3^k > 2n+1$ (or otherwise $3^{k+1} \mid \ell$), only the term 3^k among $3, 5, \dots, 2n+1$ is divisible by 3^k . Therefore, $\frac{\ell}{3}S = \frac{\ell}{3^{k+1}} + m$ for some integer m . But then $\frac{\ell}{3^{k+1}}$ is not an integer. Thus, S cannot be an integer.

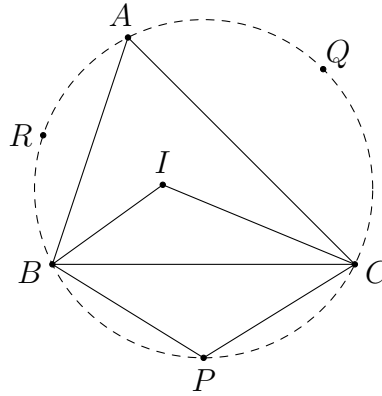
2. By the Fermat little theorem, we have $10^{k(p-1)} \equiv 1 \pmod{p}$ for any $k \in \mathbb{N}$. This implies

$$p \mid 10^{k(p-1)} - 1 = \underbrace{99\dots 9}_{k(p-1) \text{ times}}.$$

3. Since $\angle BIC = 90^\circ + \frac{A}{2}$, the circumcentre P of $\triangle BIC$ must lie on the different side of BC as A and I . As

$$\angle BPC = 360^\circ - 2\angle BIC = 360 - 2\left(90^\circ + \frac{A}{2}\right) = 180^\circ - A,$$

the point P lies on (ABC) . By symmetry, Q and R also lie on (ABC) as desired.



4. This is a special case of Jacobi's theorem. Alternatively, by Ceva's theorem, it suffices to prove

$$\frac{\sin \angle ACX}{\sin \angle BCX} \times \frac{\sin \angle BAY}{\sin \angle CAZ} \times \frac{\sin \angle CBZ}{\sin \angle ABZ} = 1.$$

Applying the sine law to $\triangle ACX$ and $\triangle BCX$, we obtain

$$\sin \angle ACX = AX \times \frac{\sin \angle CAX}{CX}, \quad \sin \angle BCX = BX \times \frac{\sin \angle CBX}{CX}.$$

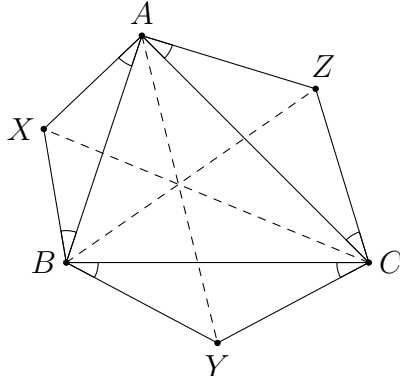
Combining these, since $AX = BX$, we get

$$\frac{\sin \angle ACX}{\sin \angle BCX} = \frac{\sin \angle CAX}{\sin \angle CBX}.$$

Similarly, we find that

$$\frac{\sin \angle ACX}{\sin \angle BCX} \times \frac{\sin \angle BAY}{\sin \angle CAZ} \times \frac{\sin \angle CBZ}{\sin \angle ABZ} = \frac{\sin \angle CAX}{\sin \angle CBX} \times \frac{\sin \angle ABY}{\sin \angle ACY} \times \frac{\sin \angle BCZ}{\sin \angle BAZ}.$$

Note that $\angle CAX = \angle BAZ$, $\angle ABY = \angle CBX$ and $\angle BCZ = \angle ACY$. Therefore, this is equal to 1, and we are done.



5. By the extended sine law, we have

$$a_j = 2R_j \sin A_j, \quad b_j = 2R_j \sin B_j, \quad c_j = 2R_j \sin C_j.$$

Replacing all side lengths on the left-hand side, we need to prove

$$\sin^m A_1 \sin^n A_2 + \sin^m B_1 \sin^n B_2 + \sin^m C_1 \sin^n C_2 \leq \frac{9}{4}.$$

Since $\sin \theta \leq 1$ for any θ , it suffices to prove this inequality for $m = n = 1$. This means we need to prove

$$a_1 a_2 + b_1 b_2 + c_1 c_2 \leq 9R_1 R_2.$$

Recall that

$$9R_j^2 - (a_j^2 + b_j^2 + c_j^2) = O_j H_j^2 \geq 0,$$

where O_j and H_j are the circumcentre and orthocentre of $\triangle A_j B_j C_j$ respectively. By the Cauchy-Schwarz inequality, we obtain

$$(a_1 a_2 + b_1 b_2 + c_1 c_2)^2 \leq (a_1^2 + b_1^2 + c_1^2)(a_2^2 + b_2^2 + c_2^2) \leq (9R_1^2)(9R_2^2) = 81R_1^2 R_2^2.$$

This implies $a_1 a_2 + b_1 b_2 + c_1 c_2 \leq 9R_1 R_2$. The proof is complete.

6. (a) Note that the numbers are

$$\begin{aligned} a_k + (-1)^{k+1}(k+1) &= \binom{p-2}{k} + (-1)^{k+1}(k+1) \\ &= \frac{(p-2)(p-3)\cdots(p-1-k)}{k!} + (-1)^{k+1}(k+1) \end{aligned}$$

where $1 \leq k \leq p-2$. Since

$$(p-2)(p-3)\cdots(p-1-k) \equiv (-2)(-3)\cdots(-1-k) = (-1)^k(k+1)! \pmod{p}$$

and $p \nmid k!$, we have

$$a_k + (-1)^{k+1}(k+1) \equiv (-1)^k(k+1) + (-1)^{k+1}(k+1) = 0 \pmod{p}.$$

(b) The answer is 1728.

For each choice of x_1, x_2, x_3 satisfying the inequalities, we have

$$x_4 = 100 - x_1 - x_2 - x_3 \geq 100 - 10 - 11 - 12 > 50.$$

Therefore, it suffices to count the number of triples (x_1, x_2, x_3) satisfying the inequalities. The answer is

$$(10 - (-1) + 1)(11 - 0 + 1)(12 - 1 + 1) = 12^3 = 1728.$$

Test 2

1. The answer is 5873.

Let $a_1 = x$ and $a_2 = y$. Then the sequence is

$$x, y, y - x, -x, -y, x - y, x, y, \dots$$

Note that the sequence repeats every 6 terms. Also, the sum of every 6 consecutive terms is 0. Thus, the sum of the first $1997 = 6 \times 332 + 5$ terms is $x + y + (y - x) + (-x) + (-y) = y - x$, and the sum of the first $1879 = 6 \times 313 + 1$ terms is x . Given that $y - x = 1879$ and $x = 1997$, we have

$$y = (y - x) + x = 1879 + 1997 = 3876.$$

Therefore, the sum of the first $2000 = 6 \times 333 + 2$ terms is

$$x + y = 1997 + 3876 = 5873.$$

2. Let $x = \sqrt{a-1}$, $y = \sqrt{b-1}$ and $z = \sqrt{c-1}$. Then we have $x, y, z \geq 0$. We need to prove

$$x + y + z \leq \sqrt{(z^2 + 1)[(x^2 + 1)(y^2 + 1) + 1]}.$$

By the Cauchy-Schwarz inequality, we have

$$\begin{aligned} (z^2 + 1)[(x^2 + 1)(y^2 + 1) + 1] &= (1 + z^2)[(x^2 + 1)(y^2 + 1) + 1] \\ &\geq \left(\sqrt{(x^2 + 1)(y^2 + 1)} + z \right)^2. \end{aligned}$$

Also, we have

$$(x^2 + 1)(y^2 + 1) = (x^2 + 1)(1 + y^2) \geq (x + y)^2.$$

Combining these, we obtain

$$\sqrt{(z^2 + 1)[(x^2 + 1)(y^2 + 1) + 1]} \geq \sqrt{(x^2 + 1)(y^2 + 1)} + z \geq x + y + z.$$

Equality holds when $x = \frac{1}{y}$ and $z = \frac{1}{\sqrt{(x^2 + 1)(y^2 + 1)}}$. This means

$$a = 1 + t, \quad b = 1 + \frac{1}{t}, \quad c = 1 + \frac{t}{(t+1)^2}$$

for some $t > 0$.

3. (IMO Shortlist 1988) We have $f(1997) = 1997$.

Label the equation as follows.

$$f(f(m) + f(n)) = m + n \tag{1}$$

If $f(a) = f(b)$ for some $a, b \in \mathbb{Z}^+$, then by putting $m = a$ and $m = b$ in (1), we obtain

$$a + n = f(f(a) + f(n)) = f(f(b) + f(n)) = b + n.$$

This implies $a = b$. So f is injective. Now, using (1), we have

$$f(f(m+1) + f(1)) = (m+1) + 1 = m+2 = f(f(m) + f(2)).$$

Using the injectivity, this becomes

$$f(m+1) + f(1) = f(m) + f(2).$$

Let $d = f(2) - f(1)$. By induction, we can easily prove that

$$f(m) = f(1) + (m-1)d$$

for any $m \in \mathbb{Z}^+$. In other words, $f(m) = am + b$ for some integers a, b . Now, we have

$$f(f(m) + f(n)) = a(am + b + an + b) + b = a^2m + a^2n + (2ab + b).$$

This is equal to $m + n$ for any $m, n \in \mathbb{Z}^+$ if and only if $a^2 = 1$ and $2ab + b = 0$. Clearly, the only solution with $a \geq 0$ is $(a, b) = (1, 0)$. This means $f(m) = m$ for any $m \in \mathbb{Z}^+$. In particular, we have $f(1997) = 1997$.

4. If $n = p^k$ for some odd prime p and positive integer k , then we have

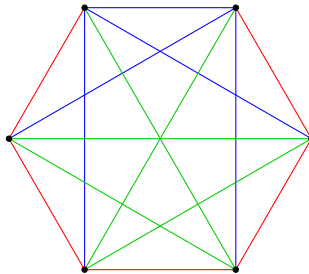
$$\frac{\sigma(n)}{n} = \frac{1 + p + \cdots + p^k}{p^k} = 1 + \frac{1}{p} + \cdots + \frac{1}{p^k} < \frac{1}{1 - \frac{1}{p}} = \frac{p}{p-1} < 2.$$

If $n = p^k q^\ell$ for some odd primes p, q and positive integers k, ℓ , then we have

$$\frac{\sigma(n)}{n} = \frac{1 + p + \cdots + p^k}{p^k} \cdot \frac{1 + q + \cdots + q^\ell}{q^\ell} < \frac{p}{p-1} \cdot \frac{q}{q-1} \leq \frac{3}{2} \cdot \frac{5}{4} < 2.$$

This shows $\sigma(n) < 2n$ if n has at most 2 distinct prime factors. Therefore, in order that n is a perfect number, it must have at least 3 distinct prime factors.

5. (a) We can colour the edges as shown. Since the subgraph formed by edges of the same colour connects all 6 vertices but there are only 5 edges, it is a tree and hence has no closed circuit.



- (b) There are $\binom{n}{2} \geq \frac{n(n-1)}{2}$ edges in total. By the pigeonhole principle, there are $\frac{n(n-1)}{2k}$ edges which are of the same colour. Consider the subgraph formed by these edges. The number of vertices is at most n . Since

$$\frac{n(n-1)}{2k} \geq \frac{n(2k+1-1)}{2k} = n,$$

the number of edges of this subgraph is at least the number of vertices. Thus, there must be a closed circuit. This gives a monochromatic closed circuit in G .

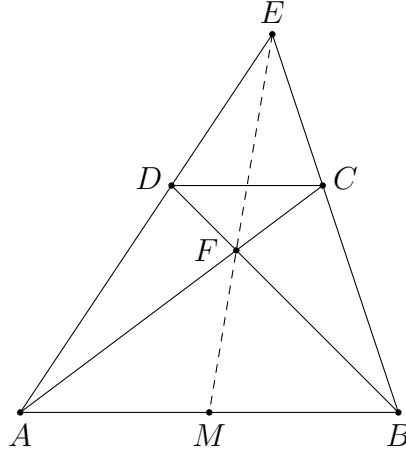
6. See the solution of 2004 Test 1 Problem 4.

Year 1999

Test 1

1. (i) Let AD meet BC at E , AC meet BD at F , and let M be the midpoint of AB . Since $AB > CD$, the points D and C must lie on the segments AE and BE respectively.

Since $\triangle EDC \sim \triangle EAB$, we have $\frac{EA}{EB} = \frac{DA}{CB} > 1$. This implies $EA > EB$, and hence $\angle CBA > \angle DAB$.



- (ii) By Steiner's theorem, E, F, M are collinear. Since $EA > EB$, both E and F lie on the same side as B to the perpendicular bisector of AB . This implies $FA > FB$. Now, as $\triangle FAB \sim \triangle FCD$, we have $\frac{AC}{BD} = \frac{FA}{FB} > 1$. This means $AC > BD$.

2. (a) Taking logarithm, it is the same as proving

$$a_1 \ln a_1 + a_2 \ln a_2 + \cdots + a_n \ln a_n \geq \frac{a_1 + a_2 + \cdots + a_n}{n} \ln(a_1 a_2 \cdots a_n).$$

This means

$$\sum_{k=1}^n a_k \ln a_k \geq \frac{1}{n} \left(\sum_{k=1}^n a_k \right) \left(\sum_{k=1}^n \ln a_k \right).$$

WLOG assume $a_1 \geq a_2 \geq \cdots \geq a_n$. Then we have $\ln a_1 \geq \ln a_2 \geq \cdots \geq \ln a_n$. The above inequality holds by Chebyshev's inequality.

- (b) Firstly, we have

$$\begin{aligned} xy + yz + zx - 3xyz &= (xy + yz + zx)(x + y + z) - 3xyz \\ &= x^2y + x^2z + y^2x + y^2z + z^2x + z^2y \geq 0. \end{aligned}$$

Equality holds when $(x, y, z) = (1, 0, 0)$ or its permutation.

Secondly, we have

$$\begin{aligned}
& \frac{1}{4} - (xy + yz + zx - 3xyz) \\
&= \frac{1}{4}(x + y + z)^3 - (xy + yz + zx)(x + y + z) + 3xyz \\
&= \frac{1}{4}((x^3 + y^3 + z^3) - (x^2y + x^2z + y^2x + y^2z + z^2x + z^2y) + 6xyz).
\end{aligned}$$

By Schur's inequality, we have

$$x^3 + y^3 + z^3 + 3xyz \geq x^2y + x^2z + y^2x + y^2z + z^2x + z^2y.$$

Together with $3xyz \geq 0$, we see the above inequality holds. Equality holds when $(x, y, z) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$ or its permutation.

3. (IMO 1979 Problem 1 modified) We have

$$\begin{aligned}
& 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1334} + \frac{1}{1335} \\
&= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{1334} + \frac{1}{1335}\right) - 2\left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{1334}\right) \\
&= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{1334} + \frac{1}{1335}\right) - \left(1 + \frac{1}{2} + \cdots + \frac{1}{667}\right) \\
&= \frac{1}{668} + \frac{1}{669} + \cdots + \frac{1}{1335} \\
&= \left(\frac{1}{668} + \frac{1}{1335}\right) + \left(\frac{1}{669} + \frac{1}{1334}\right) + \cdots + \left(\frac{1}{1001} + \frac{1}{1002}\right) \\
&= 2003 \cdot \left(\frac{1}{668 \cdot 1335} + \frac{1}{669 \cdot 1334} + \cdots + \frac{1}{1001 \cdot 1002}\right).
\end{aligned}$$

One may check that 2003 is a prime. Thus, this factor cannot be cancelled out by the denominator. Thus, p must be divisible by 2003.

4. The solutions are $f(x) = 0$, $f(x) = 1$, $f(x) = \omega$ where ω is an $(m-1)$ st root of unity, and $f(x) = x^m$.

If $f(x) = c$ is a constant polynomial, then the relation holds if and only if $c = c^m$. Clearly, the solutions are $c = 0, 1$ and all the $(m-1)$ st roots of unity.

If f is a non-constant polynomial, then $f(x)$ attains infinitely many values. Thus, there are infinitely many y such that $f(y) = y^m$. Since f is a polynomial, this implies $f(x) = x^m$ for any x .

It is easy to check that all these are solutions.

5. Suppose the members are A, B, C, D, E, F . By the pigeonhole principle, either A knows at least $\left\lceil \frac{5}{2} \right\rceil = 3$ other members, or A does not know at least 3 other members. WLOG assume it is the former case. Suppose A knows B, C, D . If any two of B, C, D know each other, say B and C , then we are done since all of A, B, C know each other. Otherwise, any two of B, C, D do not know each other, which is also our goal. This completes the proof.

6. (IMO Shortlist 1998 C3 simplified)

- (a) Consider the same problem but we require the numbers at the end to be in ascending order (i.e. descending order is not allowed). Let m_k be the least number of moves needed to guarantee that this can be done when there are k numbers. By symmetry, this number is the same if we require the numbers at the end to be in descending order. Clearly, we have $n_k \leq m_k$.

We claim that $m_{k+1} \leq m_k + 1$ for $k \leq 4$. Given a permutation of $1, 2, \dots, k+1$, one of the numbers 1 and $k+1$ must be located at the first two or the last two positions since there are at most 5 numbers in total. We can use at most 1 move to put this number to the first or the last position. For the remaining numbers, we need at most m_k moves to sort them in the correct order, so that all numbers are in monotonic order. This proves the inequality.

It follows that $n_4 \leq m_4 \leq m_3 + 1 \leq m_2 + 2 = 3$ (obviously, $m_2 = 1$). Next, we show that 2413 cannot be changed to 1234 or 4321 in 2 steps. This can be verified by checking all possibilities as follows. (Note that swapping the same block twice is useless.)

$$\underline{2}413 \longrightarrow 4213 \longrightarrow 4123, 4231, 1243$$

$$2\underline{4}13 \longrightarrow 2143 \longrightarrow 1243, 2134$$

$$24\underline{1}3 \longrightarrow 2431 \longrightarrow 4231, 2341, 2134$$

Therefore, we must have $n_4 = 3$.

- (b) We have $n_5 = 4$.

For the upper bound, we use the inequality established in part (a) to obtain

$$n_5 \leq m_5 \leq m_4 + 1 \leq 4.$$

We now show that 31524 cannot be changed to 12345 or 54321 in 3 steps.

Again, we list out all possibilities.

0 moves	1 move	2 moves	3 moves
<u>3</u> 1524	1 <u>3</u> 524	15324	51324, 15234, 15342, 12354
	13 <u>5</u> 24	13254	31254, 12354, 13245
	135 <u>2</u> 4	13542	31542, 15342, 13452, 53142, 13245
	<u>1</u> 3524	53124	35124, 51324, 53214, 53142, 53421
<u>3</u> 1 <u>5</u> 24	<u>3</u> 5124	53124	51324, 53214, 53142, 13524, 53421
	<u>3</u> 51 <u>2</u> 4	35214	53214, 32514, 35241, 31254
	351 <u>2</u> 4	35142	53142, 31542, 35412
	351 <u>2</u> 4	35421	53421, 34521, 35241, 35412, 32451, 31245
<u>3</u> 15 <u>2</u> 4	<u>3</u> 1254	13254	12354, 13524, 13245
	<u>3</u> 12 <u>5</u> 4	32154	23154, 32514, 32145, 12354
	312 <u>5</u> 4	31245	13245, 32145, 31425, 34215, 31542, 35421
	<u>3</u> 12 <u>5</u> 4	35214	53214, 32514, 35124, 35241
<u>3</u> 15 <u>2</u> 4	<u>3</u> 1542	13542	15342, 13452, 13524, 53142, 13245
	<u>3</u> 1542	35142	53142, 35412, 35124
	315 <u>4</u> 2	31452	13452, 34152, 31425, 35412
	<u>3</u> 15 <u>4</u> 2	31245	13245, 32145, 31425, 31254, 34215, 35421

7. (a) We start from an arbitrary vertex A_1 , and find one of its neighbour A_2 . Since $\deg A_2 = 2$, A_2 has another neighbour A_3 . Repeating the process, we must eventually find a vertex A_k whose another neighbour is A_1 since there are finitely many vertices and all of $A_2, A_3, \dots, A_{k-1}$ already have 2 neighbours. Now, all vertices already have degree 2. Therefore, this must be the whole graph or otherwise G cannot be connected. Thus, $A_1 A_2 \dots A_k$ is a polygon.
- (b) It suffices to show that G has a $(\Delta_G + 1)$ -colouring. Indeed, we can just assign an arbitrary colour to the vertices one by one such that each vertex has a different colour as all its neighbours which are already coloured. This is possible since there are $\Delta_G + 1$ colours but each vertex has degree at most Δ_G .
- (c) G can be a cycle graph with an odd number of vertices and any complete graph.

Firstly, a cycle graph with an odd number of vertices satisfies $\chi_G = 3$ and $\Delta_G = 2$. Indeed, $\Delta_G = 2$ holds by definition. If at most 2 colours are used, there is a unique way to colour the vertices $A_1, A_2, \dots$ of the polygon $A_1 A_2 \dots A_k$ such that A_{j+1} has a different colour as A_j . But since k is odd, A_k has the same colour as A_1 , contradiction. Therefore, at least 3 colours are needed. Since $\chi_G \leq \Delta_G + 1 = 3$, we must have $\chi_G = 3 = \Delta_G + 1$.

Secondly, a complete graph with k vertices satisfies $\chi_G = k$ and $\Delta_G = k - 1$.

Both are obvious since any pair of vertices is joined by an edge (hence all vertices have different colours).

- (d) (Brook's theorem) We prove the result by induction on Δ_G , and for each Δ_G , we induct on the number k of vertices.

The base case $\Delta_G = 2$ is trivial since the graph is just a path or a cycle. Suppose $\Delta_G \geq 3$. Now, the base case is $k = \Delta_G + 1$. As G is not a complete graph, there is a vertex A which has degree at most $\Delta_G - 1$. We can colour all other vertices in different colours (Δ_G colours are used). Then since $\deg A \leq \Delta_G - 1$, we can always assign a colour to A which is different from the colours of all its neighbours.

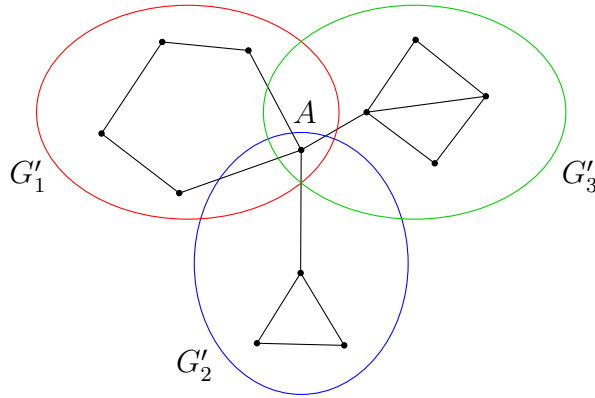
Now, consider the inductive step, where $k \geq \Delta_G + 2$.

Case 1. There exists a vertex A such that $G - \{A\}$ is disconnected.

Suppose the connected components of $G - \{A\}$ are $G_1, G_2, \dots, G_m$. For each G_j , consider the connected subgraph G'_j induced by G containing all the vertices in G_j together with A . Since $m \geq 2$, G'_j has fewer vertices. Also, the maximum degree of G'_j cannot exceed Δ_G . Thus, we can find a Δ_G -colouring of G'_j by the inductive hypothesis unless G'_j is a complete graph or an odd cycle. But even in these cases, we have

$$\Delta_{G'_j} = \deg_{G'_j} A \leq \deg_G A - 1 \leq \Delta_G - 1.$$

This shows we can still find a colouring using at most $\Delta_G - 1 + 1 = \Delta_G$ colours.

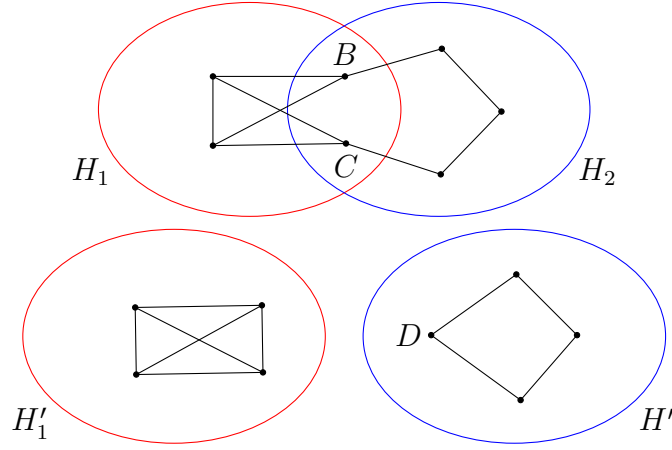


Now, we have found a Δ_G -colouring for each G'_j . WLOG assume A has colour 1 in all such colourings. Then we can combine the colourings used for all G'_j to give a colouring for G . The condition is satisfied, since any two of $G_1, G_2, \dots, G_m$ are disconnected.

Case 2. $G - \{A\}$ is connected for all vertex A , and there exist two non-adjacent vertices B and C such that $G - \{B, C\}$ is disconnected.

Suppose the connected components of $G - \{B, C\}$ are $G_1, G_2, \dots, G_m$. Note that B must be adjacent to some vertex in each G_j , or otherwise we can remove C to get at least two connected components (one is G_j), contradicting the assumption. The same holds for C . For each G_j , let H_j be the induced subgraph of G containing B, C and all vertices in G_j .

Now, the degrees of B and C in H_j are at most $\Delta_G - 1$ since $m \geq 2$. Consider the graph H'_j obtained from H_j by including the edge $B - C$. Then we still have $\Delta_{H'_j} \leq \Delta_G$. So there is a Δ_G -colouring for H'_j unless it is an exceptional case. If such a colouring exists for all H'_j , WLOG we may assume B has colour 1 and C has colour 2 in all colourings (note that their colours are different since $B - C$ is an edge). Then we can combine all the colourings to give a Δ_G -colouring for G .



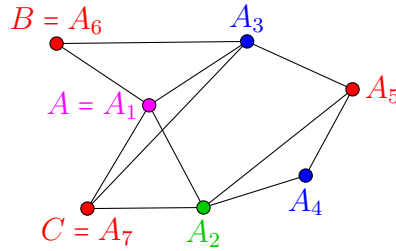
Note that the same argument works as long as there is a Δ_G colouring for each H'_j . In particular, the argument works if some H'_j is an odd cycle (since $\Delta_G \geq 3$) or a complete graph with at most Δ_G vertices. So the only case left is that H'_1 , for example, is a complete graph with $\Delta_G + 1$ vertices. Since the degrees of B and C in G are at most Δ_G , each of B and C can only be adjacent to one more vertex not in G_1 . Therefore, we have $m = 2$, and each of B and C is adjacent to only one vertex in G_2 .

Now, we can identify B and C in H_2 as the same point D to form a new connected graph H' . Since $\deg D \leq 2$ and the degrees of all other vertices are unchanged, we have $\Delta_{H'} \leq \Delta_G$ and H' has fewer vertices than G . Therefore, there is a Δ_G -colouring of H' . This also holds even if H' is an odd cycle or a complete graph (which has at most 3 vertices as $\deg D \leq 2$). This yields a Δ_G -colouring of H_2 such that B and C have the same colour (as D). Then we can colour the remaining $\Delta_G - 1$ vertices in G_1 using colours different from that of B and C . This gives a Δ_G -colouring of G .

Case 3. $G - \{B, C\}$ is connected for all non-adjacent vertices B and C .

Let A be a vertex with $\deg A = \Delta_G$. If all neighbours of A are adjacent, then there is already a $(\Delta_G + 1)$ -clique. This is impossible since the graph is connected and not complete, but we cannot add more edges to this clique (or Δ_G is increased). Suppose B and C are some non-adjacent neighbours of A .

Starting from $A_1 = A$, we construct a sequence of vertices as follows. Suppose we have constructed $A_1, A_2, \dots, A_n$. Since $G - \{B, C\}$ is connected, we can always find some $A_{n+1} \neq B, C$ such that A_{n+1} is adjacent to one of $A_1, A_2, \dots, A_n$ unless all vertices are used. We repeat the same process until we have obtained $A_1, A_2, \dots, A_{k-2}$. Then we define $A_{k-1} = B$ and $A_k = C$.



We colour the vertices from A_k back to A_1 as follows. We can assign colour 1 to A_k and A_{k-1} since they are non-adjacent. For each A_j we can always assign a colour to it which is different from its neighbours in $A_k, A_{k-1}, \dots, A_{j+1}$ since $\deg A_j \leq \Delta_G$ and one of the neighbours of A_j is $A_1, A_2, \dots, A_{j-1}$ by construction. We can do this for $A_{k-2}, A_{k-3}, \dots, A_2$. Lastly, for A_1 , it has Δ_G neighbours, where 2 of them (A_{k-1} and A_k) have the same colour. So we can still find a new colour for A_1 . This yields a Δ_G -colouring for G .

The proof is complete by combining all these cases and using induction.

- (e) An example is a complete graph with k vertices. By part (c), we have $\chi(G) = k$. Also, $\overline{G}$ has no edges, and so $\chi(\overline{G}) = 1$. This gives

$$\chi(G) + \chi(\overline{G}) = k + 1$$

as desired.

Year 2000

Test 1

1. The remainder is 1.

Note that $1999 \equiv -16 = -2^4 \pmod{31}$. Therefore, we have

$$1999^{2000} \equiv (2^4)^{2000} = (2^5)^{1600} = 32^{1600} \equiv 1 \pmod{31}.$$

Thus, the remainder is 1.

2. We have

$$\begin{aligned} & (a^3 + b^3 + c^3) \left(\frac{1 + ab^2}{c^3} + \frac{1 + bc^2}{a^3} + \frac{1 + ca^2}{b^3} \right) \\ &= \sum_{\text{sym}} \frac{a^3}{c^3} + \sum_{\text{cyc}} \frac{a^4 b^2}{c^3} + \sum_{\text{cyc}} \frac{ab^5}{c^3} + \sum_{\text{cyc}} ab^2 + 3. \end{aligned}$$

Applying the AM-GM inequality to each sum, and using $abc = 1$, this is bounded below by

$$6 + 3abc + 3abc + 3abc + 3 = 18.$$

Equality holds when $a = b = c = 1$.

3. No. In a move, if we choose 100 cups where k of them are facing downward and $100 - k$ of them are facing upward, then the number of cups facing downward is changed by $(100 - k) - k = 100 - 2k$. Since there is an even number of cups facing downward initially, there is always an even number of cups facing downward. Thus, it is impossible to make all 1999 cups face downward.

It is possible to make all cups face downward if there are 1998 cups. If we turn over cups $1, 2, \dots, 100$ and then turn over cups $2, 3, \dots, 101$, we see that only cups 1 and 101 are turned over. This shows we can turn over any 2 cups after 2 moves. Since $2 \mid 1998$, we can repeat the same process to turn over all cups.

4. Using the concyclic points, we have

$$\angle CAQ = \angle CBQ = \angle DBP = \angle DEP$$

and

$$\angle QCA = \angle QBA = \angle PBE = \angle PDE.$$

This implies $\triangle QAC \sim \triangle PED$. Let

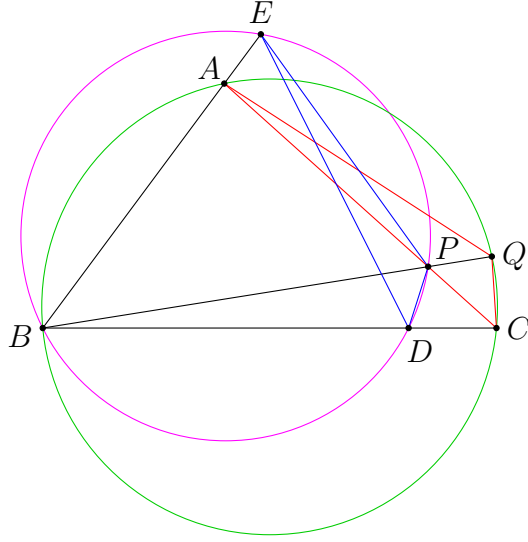
$$\frac{QA}{PE} = \frac{QC}{PD} = \frac{AC}{ED} = k.$$

Applying Ptolemy's theorem to the cyclic quadrilateral $EBDP$, we obtain

$$BD \times PE + BE \times PD = BP \times DE.$$

It follows that

$$\begin{aligned} AQ + CQ &= kPE + kPD \\ &= \frac{k}{AC}(BD \times PE + BE \times PD) \\ &= \frac{k}{AC}(BP \times DE) \\ &= BP. \end{aligned}$$



5. We prove that any simple graph with $2n$ vertices and without triangles has at most n^2 edges. We prove this by induction. When $n = 1$, clearly there is at most $n^2 = 1$ edge between 2 vertices. Assume there are at most k^2 edges when there are $2k$ vertices. Consider a simple graph with $2(k + 1)$ vertices and without triangles.

We are done if there is no edge. So we may assume there is an edge $A - B$. For any other vertex C , at most one of the pair $\{A, C\}$ and $\{B, C\}$ is joined by an edge since there is no triangle. Thus, there are at most $2k$ edges containing exactly one of A and B . Among the remaining $2k$ vertices, there are at most k^2 edges by the inductive hypothesis. Therefore, the number of edges is at most

$$1 + 2k + k^2 = (k + 1)^2.$$

This proves the inductive step, and so we are done.

6. The only possible primes are 2, 5, 257.

If $n = pm$ for some odd prime p , then we have

$$n^n + 1 = (n^m)^p + 1 = (n^m + 1)(n^{m(p-1)} + n^{m(p-2)} + \dots + 1).$$

Clearly, both factors are larger than 1, and hence $n^n + 1$ is not a prime. Therefore, n must be a power of 2.

Note that

$$16^{16} + 1 > 2^{64} = 2^4 \times (2^{10})^6 > 10 \times (10^3)^6 = 10^{19}.$$

So it remains to check $n = 1, 2, 4, 8$. Indeed, we have

$$\begin{aligned} 1^1 + 1 &= 2, \\ 2^2 + 1 &= 5, \\ 4^4 + 1 &= 257, \\ 8^8 + 1 &= (2^8)^3 + 1 = (2^8 + 1)(2^{16} - 2^8 + 1). \end{aligned}$$

Among these, only 2, 5, 257 are primes.

Test 2

1. Note that $46410 = 2 \times 3 \times 5 \times 7 \times 13 \times 17$. It suffices to check

$$m = n^{30} - n^{14} - n^{18} + n^2$$

is divisible by each of $p = 2, 3, 5, 7, 13, 17$.

If $p \mid n$, then we must have $p \mid m$. If $p \nmid n$, then we have

$$n^{p-1} \equiv 1 \pmod{p}$$

by the Fermat little theorem. Since

$$m = n^2(n^{12} - 1)(n^{16} - 1),$$

we have $p \mid m$ if $p - 1 \mid 12$ or $p - 1 \mid 16$. This clearly holds for $p = 2, 3, 5, 7, 13, 17$. So we are done.

2. (a) By the Euler-Fermat theorem, since $\varphi(10) = 4$ and $4n + k \equiv k \pmod{4}$, we have

$$a^{4n+k} \equiv a^k \pmod{10}$$

for any positive integers a, n, k . This means a^{4n+k} and a^k have the same rightmost digit.

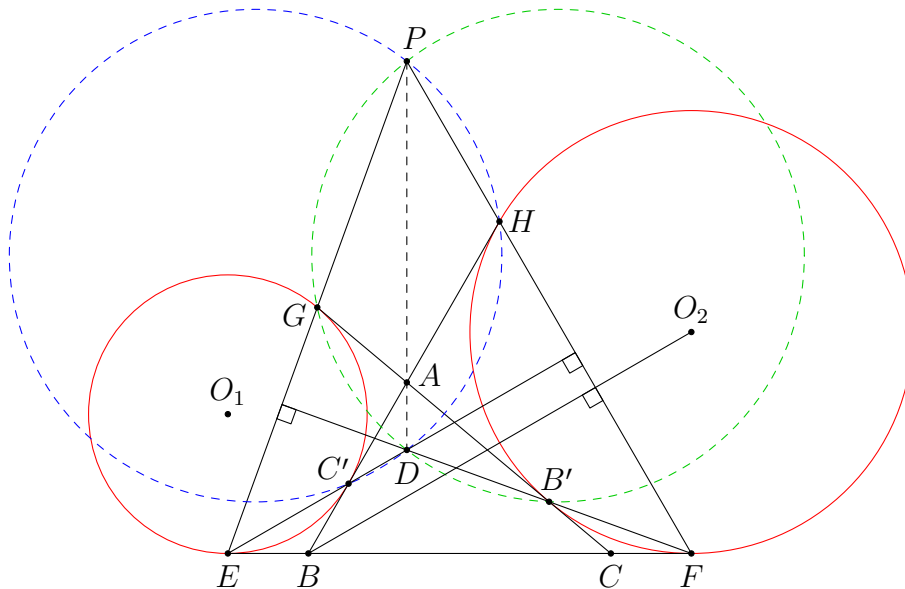
(b) The rightmost digit is 0.

By part (a), since $1999 \equiv 3 \pmod{4}$, we have

$$2^{1999} + 7^{1999} + 9^{1999} \equiv 2^3 + 7^3 + 9^3 \equiv 8 + 3 + 9 \equiv 0 \pmod{10}.$$

This shows the rightmost digit of the given number is 0.

3. Let EC' meet FB' at D . Note that EC' is parallel to the internal angle bisector of $\angle CBA$, which is BO_2 . Therefore, $ED \perp PF$. Similarly, $FD \perp PE$. This implies D is the orthocentre of $\triangle PEF$, and hence $PD \perp EF$. It suffices to show P, A, D are collinear, since this would imply $PA \perp BC$.



Consider (PGB') and $(PC'H)$. Since $AG \times AB' = AC' \times AH$, the point A lies on the radical axis of these circles. Thus, PA is the radical axis. It remains to show that D also lies on this radical axis. Indeed, we shall prove that D lies on both circles. We have

$$\angle GB'D = \angle CB'F = \angle B'FC = \angle GPD.$$

The last equality holds since D is the orthocentre of $\triangle PEF$. This implies D lies on (PGB') . The other assertion can be proved similarly. So we are done.

4. The answer is $\frac{1}{2}(\sqrt{al} + \sqrt{bm} + \sqrt{cn} + \sqrt{dr})^2$.

Indeed, by the Cauchy-Schwarz inequality, we have

$$(ax + by + cz + dw) \left(\frac{\ell}{x} + \frac{m}{y} + \frac{n}{z} + \frac{r}{w} \right) \geq (\sqrt{a\ell} + \sqrt{bm} + \sqrt{cn} + \sqrt{dr})^2.$$

Using the condition $ax + by + cz + dw = 2$, we get

$$F \geq \frac{1}{2}(\sqrt{a\ell} + \sqrt{bm} + \sqrt{cn} + \sqrt{dr})^2.$$

Equality holds when $x : y : z : w = \sqrt{\frac{\ell}{a}} : \sqrt{\frac{m}{b}} : \sqrt{\frac{n}{c}} : \sqrt{\frac{r}{d}}$. Clearly, there exists a unique choice for such a tuple (x, y, z, w) satisfying $ax + by + cz + dw = 2$. Therefore, the minimum of F is

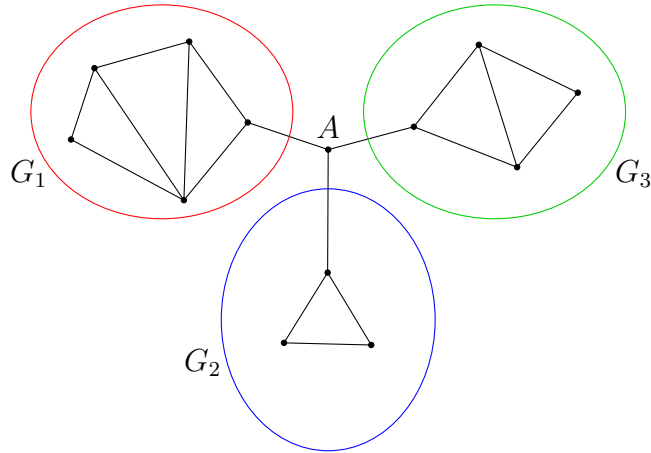
$$\frac{1}{2}(\sqrt{a\ell} + \sqrt{bm} + \sqrt{cn} + \sqrt{dr})^2.$$

5. (a) (1999 Test 1 Problem 7(b)) See the solution above.

(b) (1999 Test 1 Problem 7(d) simplified) We give an alternative proof for this simpler version. We prove that there is a Δ_G -colouring by induction on the number n of vertices.

The base case is $n = 3$, and the graph is $A - B - C$. Clearly, we can assign colour 1 to A and C , and colour 2 to B . This gives $\chi_G \leq 2 = \Delta_G$.

Assume the assertion holds when there are fewer than n vertices. Consider a simple connected graph G with n vertices which is not regular. Since it is not regular, there is a vertex A having degree at most $\Delta_G - 1$. Let $G_1, G_2, \dots, G_m$ be the connected components of $G - \{A\}$. We claim that there is a Δ_G -colouring for each G_j .



Suppose on the contrary that G_1 has no Δ_G -colouring. By the inductive hypothesis, G_1 must be regular. If $\Delta_{G_1} \leq \Delta_G - 1$, then by part (a), we have

$$\chi_{G_1} \leq \Delta_{G_1} + 1 \leq \Delta_G.$$

This shows a Δ_G -colouring exists. Therefore, we may assume $\Delta_{G_1} = \Delta_G$. But then this implies all vertices in G_1 has degree Δ_G . This is impossible since one of these vertices is adjacent to A , and hence its degree in G is at least $\Delta_G + 1$. This proves our claim.

Now, we choose any Δ_G -colouring for each G_j . Lastly, we can assign a colour to A which is different from the colours of all its neighbours since $\deg A \leq \Delta_G - 1$. This completes the proof by induction.

Year 2001

Test 1

1. (IMO 1985 Problem 5) Note that KN and AC are not parallel, or otherwise $M = B$. Let D be the radical centre of $(AKNC)$, $(AMBC)$ and $(MBKN)$. This means D is the concurrent point of AC, KN, MB . WLOG assume it is the configuration as shown. Since

$$\angle KMD = \angle KNB = \angle KAC,$$

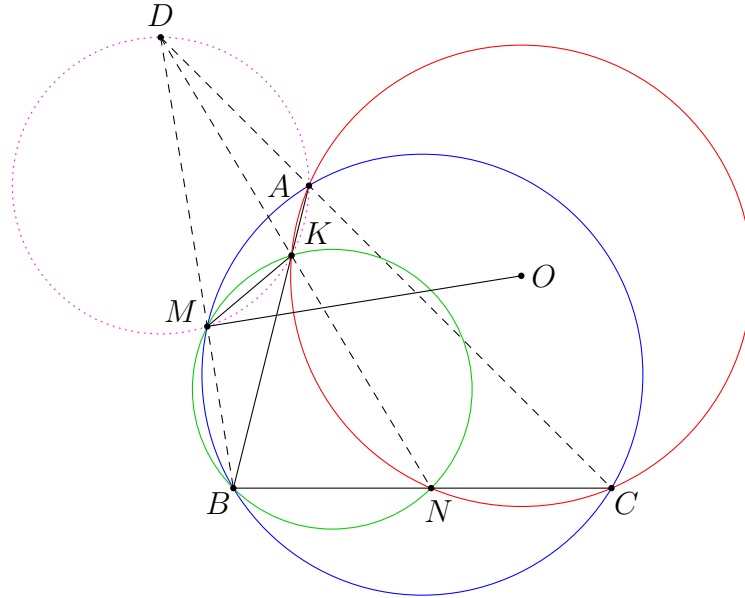
the points D, M, K, A are concyclic. Let R be the radius of $(AKNC)$. By the power of a point theorem, we have

$$\begin{aligned} DO^2 - R^2 &= DA \times DC = DM \times DB, \\ BO^2 - R^2 &= BK \times BA = BM \times BD. \end{aligned}$$

Subtracting these, we obtain

$$\begin{aligned} DO^2 - BO^2 &= DM \times DB - BM \times BD \\ &= (DM - BM)(DM + BM) \\ &= DM^2 - BM^2. \end{aligned}$$

This implies $OM \perp DB$, and hence $\angle OMB = 90^\circ$.



2. n can be any positive integer of the form $4k - 2$ where $k \in \mathbb{Z}^+$.

When $n = 4k - 2$, we have

$$n^8 - n^2 = n^2(n^6 - 1) = 4(2k - 1)^2(n^6 - 1).$$

Since both $2k-1$ and n^6-1 are odd, this number is not divisible by 8, and hence not divisible by 72.

We now claim that for any other n , $n^8 - n^2$ is divisible by both 8 and 9. If n is even, n must be a multiple of 4. In this case $n^8 - n^2$ is divisible by n^2 , and hence divisible by 8. If n is odd, it is well-known that $8 \mid n^2 - 1$. Thus,

$$n^8 - n^2 = n^2(n^2 - 1)(n^4 + n^2 + 1)$$

is divisible by 8.

Next, if $3 \mid n$, then $n^8 - n^2$ is divisible by n^2 , and hence divisible by 9. If $3 \nmid n$, then $3 \mid n^2 - 1$. By the lifting the exponent lemma, we have

$$3^2 \mid (n^2)^3 - 1^3 = n^6 - 1,$$

and hence $9 \mid n^2(n^6 - 1) = n^8 - n^2$. Therefore, $9 \mid n^8 - n^2$ in any case.

Combining these, we know that $72 \mid n^8 - n^2$ if $n \not\equiv 2 \pmod{4}$. So we are done.

3. (Ukraine TST 1999 Problem 6) No. Suppose on the contrary that $f(x)$ is reducible. Then so is

$$4^{2^n} f(x) = (4x^2 + 4x)^{2^n} + 4^{2^n} = ((2x + 1)^2 - 1)^{2^n} + 4^{2^n}.$$

Let $y = 2x + 1$. Then we have $x = \frac{y-1}{2}$, and hence

$$g(y) = (y^2 - 1)^{2^n} + 4^{2^n} \tag{1}$$

is reducible over $\mathbb{Q}$. By Gauss's lemma, g is reducible over $\mathbb{Z}$.

We claim that g can be factorized into even polynomials over $\mathbb{Z}$. Consider a non-constant irreducible factor h of g . We are done if h is an even polynomial (since then $\frac{g}{h}$ must be even). Assume h is not even. As $h(y) \mid g(y)$, we have $h(-y) \mid g(-y) = g(y)$. Since h is irreducible and is not even, $h(y)$ and $h(-y)$ must be relatively prime. It follows that

$$g(y) = h(y)h(-y)p(y)$$

for some $p \in \mathbb{Z}[y]$. Note that $h(y)h(-y)$ is an even polynomial. So we are done unless p is a constant polynomial. Let $p(y) = c$, and let a be the leading coefficient of h . Then the leading coefficient of g is a^2c . In view of (1), we must have $c = 1$ and $a = \pm 1$. Let b be the constant term of h . Then the constant term of g is $b^2c = b^2$. Comparing with (1), we obtain $4^{2^n} + 1 = b^2$. This implies $4^{2^n} = (4^{2^{n-1}})^2$ and b^2 are consecutive perfect squares, which is impossible as $4^{2^n} > 1$. Therefore, we have proven that g can be factorized into even polynomials over $\mathbb{Z}$.

As a result, by letting $z = y^2$, we find that $(z - 1)^{2^n} + 4^{2^n}$ is also reducible. Let $z = 4t + 1$. This shows $4^{2^n}(t^{2^n} + 1)$ is reducible, and so are $t^{2^n} + 1$ and $(s + 1)^{2^n} + 1$. However, note that

$$(s + 1)^{2^n} + 1 = s^{2^n} + \sum_{r=1}^{2^n-1} \binom{2^n}{r} s^r + 2.$$

The leading coefficient is odd, while all other coefficients are even. Also, the constant term is not a multiple of 4. By Eisenstein's criterion, we know that $(s + 1)^{2^n} + 1$ is irreducible. This is a contradiction. Therefore, $f(x)$ is irreducible.

4. By the AM-GM inequality, we have

$$(a + b + 4c)^2 \geq \left(2\sqrt{(a + b)(4c)}\right)^2 = 16(a + b)c$$

and

$$\frac{100abc}{a + b + c} \leq \frac{100c}{a + b + c} \left(\frac{a + b}{2}\right)^2 = \frac{25(a + b)^2 c}{a + b + c}.$$

Therefore, it suffices to prove

$$d^2 + 16cd \geq \frac{25cd^2}{c + d}$$

where $d = a + b$. Indeed,

$$\begin{aligned} & d^2 + 16cd \geq \frac{25cd^2}{c + d} \\ \Leftrightarrow & d(16c + d)(c + d) \geq 25cd^2 \\ \Leftrightarrow & d(16c^2 - 8cd + d^2) \geq 0 \\ \Leftrightarrow & d(4c - d)^2 \geq 0. \end{aligned}$$

This is obviously true. Equality holds when $a = b = 2c$.

5. The largest possible value of $|M|$ is 53.

Note that there are $\frac{120}{2} = 60$ multiples of 2 and $\frac{120}{3} = 40$ multiples of 3 in S . Therefore, we need to remove at least 53 multiples of 2 and 33 multiples of 3 to ensure that no 8 numbers remaining have greatest common divisor divisible by 2 or 3. Among these removed numbers, at most $\frac{120}{6} = 20$ of them are repeated, which happens when they are multiples of 6. Also, we have to remove at least one number from $\{5, 25, 35, 55, 65, 85, 95, 115\}$ since all of them are multiples of 5 and none of them is a multiple of 2 or 3. Thus, at least

$$53 + 33 - 20 + 1 = 67$$

numbers should be removed. So we have $|M| \leq 120 - 67 = 53$.

We now give an example of M having 53 elements such that no 8 numbers have greatest common divisor larger than 1. Suppose M contains

$$2, 4, 8, 16, 32, 64, 118, 3, 9, 27, 81, 87, 93, 111$$

and all other numbers in S which are not multiples of 2 or 3 except 35. By the inclusion-exclusion principle, there are

$$7 + 7 + 120 - \frac{120}{2} - \frac{120}{3} + \frac{120}{6} - 1 = 53$$

numbers in total. Since $17 \times 8 > 120$, it suffices to check we cannot find 8 numbers which are simultaneously multiples of 2, 3, 5, 7, 11 or 13. This can be seen by listing out the multiples of these primes in M as follows.

$$2 : 2, 4, 8, 16, 32, 64, 118$$

$$3 : 3, 9, 27, 81, 87, 93, 111$$

$$5 : 5, 25, 55, 65, 85, 95, 115$$

$$7 : 7, 49, 77, 91, 119$$

$$11 : 11, 55, 77$$

$$13 : 13, 65, 91$$

Year 2002

Test 1

1. The answer is any integer of the form $105k + 59$ where $k \in \mathbb{Z}$.

Since $n \equiv 2 \equiv -1 \pmod{3}$ and $n \equiv -1 \pmod{5}$, we have

$$n \equiv -1 \pmod{15}.$$

Testing $n = -1, 14, 29, 44, 59$, we see that $n = 59$ satisfies $n \equiv 3 \pmod{7}$. Therefore, $n = 59$ is one solution. By the Chinese remainder theorem, since 3, 5, 7 are pairwise relatively prime and $3 \times 5 \times 7 = 105$, we have

$$n \equiv 59 \pmod{105}.$$

2. (a) The answer is $a_n = 2 \cdot 3^n - 2^n$.

The characteristic equation is $\lambda^2 - 5\lambda + 6 = 0$. The roots are $\lambda = 3, 2$. Therefore, we have

$$a_n = A \cdot 3^n + B \cdot 2^n$$

for some constants A and B . Putting $n = 0$ and $n = 1$, we need to solve

$$\begin{cases} A + B = 1, \\ 3A + 2B = 4. \end{cases}$$

We easily deduce $A = (3A + 2B) - 2(A + B) = 2$ and $B = 1 - A = -1$. Thus, we have

$$a_n = 2 \cdot 3^n - 2^n.$$

- (b) The answer is $b_n = (n + 6)2^{n-1}$.

The characteristic equation is $\lambda^2 - 4\lambda + 4 = 0$, with a double root $\lambda = 2$. Therefore, we have

$$b_n = (Cn + D)2^n$$

for some constants C and D . Putting $n = 0$ and $n = 1$, we need to solve

$$\begin{cases} D = 3, \\ 2(C + D) = 7. \end{cases}$$

Clearly, $C = \frac{7}{2} - D = \frac{1}{2}$. Thus, we have

$$b_n = (n + 6)2^{n-1}.$$

3. (Nesbitt's inequality) We have

$$\begin{aligned}
& \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2} \\
\Leftrightarrow & \frac{a+b+c}{b+c} + \frac{a+b+c}{c+a} + \frac{a+b+c}{a+b} \geq \frac{9}{2} \\
\Leftrightarrow & (a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) \geq \frac{9}{2} \\
\Leftrightarrow & [(b+c) + (c+a) + (a+b)] \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) \geq 9.
\end{aligned}$$

This is true by the Cauchy-Schwarz inequality. Equality holds when $a = b = c$.

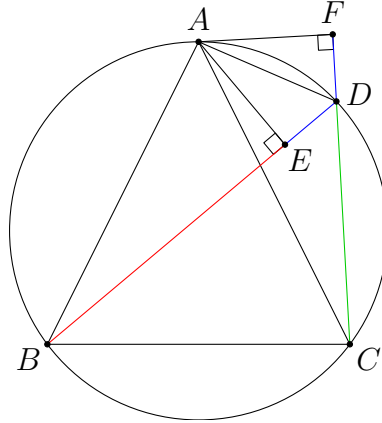
4. (Archimedes' broken-chord theorem) Let F be the foot of perpendicular from A to CD . Note that

$$\angle ADE = \angle ADB = \angle ACB = \angle ABC = \angle ADF.$$

Also, we have $\angle AED = \angle AFD$ and $AD = AD$. This implies $\triangle ADE \cong \triangle ADF$, and hence $AE = AF$ and $DE = DF$.

Next, observe that $\angle EBA = \angle FCA$ and $\angle AEB = \angle AFC$. Together with $AE = AF$, we find that $\triangle AEB \cong \triangle AFC$. Therefore, we have

$$BE = CF = CD + DF = CD + ED.$$



5. Taking modulo 4, we obtain

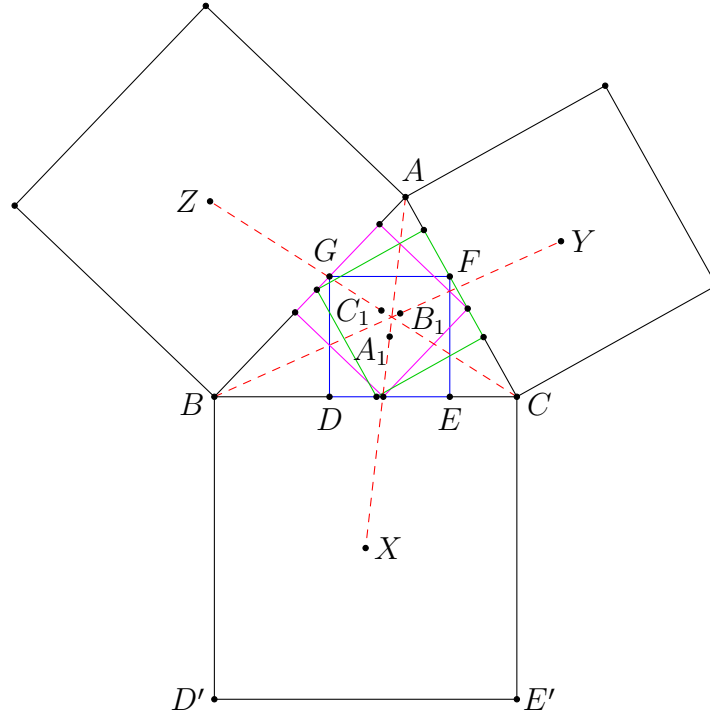
$$(-a+b)(-b-c)(c+d)(-d-a) \equiv 1 \pmod{4}.$$

This implies all factors on the left are odd, and an even number of them is congruent to 3 modulo 4 and the others are congruent to 1 modulo 4. It follows that their sum is a multiple of 4, i.e.

$$4 \mid (-a+b) + (-b-c) + (c+d) + (-d-a) = -2a.$$

This shows a is even.

6. (IMO Shortlist 2001 G1) Let $DEFG$ be the square with A_1 as centre such that D, E lie on BC , F lies on CA , and G lies on AB . By considering a homothety with centre A , we can map $DEFG$ to a square $D'E'CB$ since $GF \parallel BC$. Let X be the centre of $D'E'CB$. Due to the homothety, A, A_1, X are collinear. Similarly, let Y and Z be the centres of the squares constructed outside $\triangle ABC$ having CA and AB as a side respectively. It suffices to prove AX, BY, CZ are concurrent. This follows from Jacobi's theorem since $\angle ZAB = \angle YAC = 45^\circ$, $\angle XBC = \angle ZBA = 45^\circ$ and $\angle YCA = \angle XCB = 45^\circ$.



7. Consider the 2002 intervals $I_k = \left(\frac{k}{2002}, \frac{k+1}{2002} \right)$ where $k = -1001, -1000, \dots, 1000$. Since y is irrational, the numbers $y - y', 2y - (2y)', \dots, 2001y - (2001y)'$ are irrational numbers between $-\frac{1}{2}$ and $\frac{1}{2}$. Thus, each of them belongs to one of the I_k 's.

We claim that one of the numbers $my - (my)'$ (with $1 \leq m \leq 2001$) belongs to I_{-1} or I_0 . Suppose on the contrary that all these numbers belong to the other 2000 intervals. By the pigeonhole principle, two of them, say $my - (my)'$ and $ny - (ny)'$ with $m > n$, belong to the same interval I_k . Then we have

$$(m-n)y - [(my)' - (ny)'] = [my - (my)'] - [ny - (ny)'] < \frac{k+1}{2002} - \frac{k}{2002} = \frac{1}{2002}.$$

Therefore, $(my)' - (ny)'$ is the closest integer to $(m-n)y$, and we know that $(m-n)y - [(m-n)y]'$ belongs to I_{-1} or I_0 . This is a contradiction. Therefore,

we can find some $my - (my)'$ belonging to I_{-1} or I_0 . For this m , we have

$$\langle my \rangle = |my - (my)'| < \frac{1}{2002} < \frac{1}{2001}.$$

This completes the proof.

Year 2003

Test 1

1. Using the AM-GM inequality, we have

$$\begin{aligned}(a+b+c)^2 &= (a^2+bc) + (b^2+ca) + (c^2+ab) + \frac{ab+ca}{2} + \frac{bc+ab}{2} + \frac{ca+bc}{2} \\ &\geq 2a\sqrt{bc} + 2b\sqrt{ca} + 2c\sqrt{ab} + a\sqrt{bc} + b\sqrt{ca} + c\sqrt{ab} \\ &= 3(a\sqrt{bc} + b\sqrt{ca} + c\sqrt{ab}).\end{aligned}$$

Equality holds when $a = b = c$.

2. Using the parallel lines, we have $\triangle ADE \sim \triangle ABM$ and $\triangle CMO \sim \triangle CED$. This implies

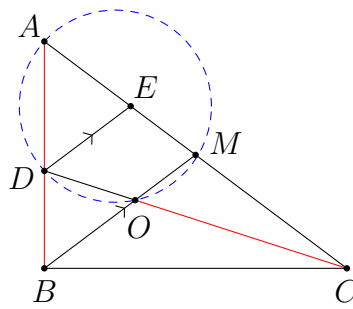
$$\frac{AB}{BD} = \frac{AM}{ME} = \frac{CM}{ME} = \frac{CO}{OD}.$$

As $AB = CO$, we obtain $BD = OD$.

Now,

$$\begin{aligned}&ADOM \text{ is cyclic} \\ \Leftrightarrow &\angle BAM = \angle DOB \\ \Leftrightarrow &\angle BAM = \angle DBO \\ \Leftrightarrow &\angle BAM = \angle ABM \\ \Leftrightarrow &MB = MA.\end{aligned}$$

Since $MA = MC$, this is equivalent to $\angle ABC = 90^\circ$, i.e. $AB \perp BC$ as desired.



3. Consider an arbitrary club C_1 . Since it shares a common member with each of the other $n^2 - n + 1$ clubs, by the pigeonhole principle, there is a member in C_1 who is also a member of at least

$$\left\lceil \frac{n^2 - n + 1}{n} \right\rceil = n$$

clubs. Suppose X is a member of the clubs $C_1, C_2, \dots, C_{n+1}$. For any other club C_k , since it shares a common member with each of $C_1, C_2, \dots, C_{n+1}$, by the pigeonhole principle, there is a member in C_k who is also a member of at least

$$\left\lceil \frac{n+1}{n} \right\rceil = 2$$

clubs among $C_1, C_2, \dots, C_{n+1}$. As the only common member in $C_1, C_2, \dots, C_{n+1}$ is X , this member must also be X . This shows X is a member of C_k for all k .

4. (USAMO 1982 Problem 4 modified) We can take $k = 20$. Then for any $n \in \mathbb{Z}^+$, we have

$$20 \cdot 22^n + 1 \equiv (-1)(1) + 1 = 0 \pmod{21},$$

and hence $20 \cdot 22^n + 1$ is a composite number.

5. (Superabundant numbers/IMO Shortlist 1983) When $n = m!$, we have

$$\frac{\sigma(m!)}{m!} \geq \frac{1}{m!} \left(m! + \frac{m!}{2} + \frac{m!}{3} + \dots + \frac{m!}{m} \right) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m}.$$

Since the harmonic series diverges, the sequence $\left\{ \frac{\sigma(n)}{n} \right\}$ is unbounded. Therefore, after finding some redundant numbers $n_1, n_2, \dots, n_k$, we can always find the next number n_{k+1} such that $\frac{\sigma(n_{k+1})}{n_{k+1}} > \frac{\sigma(n_k)}{n_k}$. This implies n_{k+1} is redundant, and so there are infinitely many redundant numbers.

6. At least 86 rows are needed.

Suppose there are 86 (type A) countries with 23 representatives each, 10 countries with 2 representatives each, and 4 countries with 1 representative each. Then there are $86 + 10 + 4 = 100$ countries and $86 \times 23 + 10 \times 2 + 4 \times 1 = 2002$ representatives. No two type A countries may sit in the same row since $23 \times 2 > 45$. Thus, at least 86 rows are needed.

We now show that it is always sufficient to use 86 rows. We arrange the countries in decreasing order of the number of representatives. We first put all representatives from country 1 in the first row. Next, if all representatives from country 2 can be seated in the same row as country 1, we put them in that row, or otherwise they are put in the next row. Similarly, if all representatives from country $k+1$ can be seated in the same row as country k , we put them in that row, or otherwise they are put in the next row. We do the same thing until all representatives are seated or all 86 rows are used up.

It suffices to consider the case when all 86 rows are used up but some representatives still do not have a sit. We claim that each row consists of at least 23

representatives. If not, there are at least 23 empty seats in some row j , which means the next country to be put has at least 24 representatives. But then there should be at least 24 representatives in row j as well according to the order of the countries. This is a contradiction. It follows that the number of representatives remaining is at most

$$2002 - 23 \times 86 = 24.$$

Among these representatives, suppose c of them come from the same country, where c is maximized. First of all, we have $c \leq 22$ since otherwise at least 99 countries have at least 23 representatives (except possibly the last country), which yields the contradiction that the total number of representatives is at least

$$23 \times 99 = 2277 > 2002.$$

Now, we partition the remaining representatives into at most two groups by putting those c representatives in one group. We claim that all groups have at most 22 representatives. If not, since $c \leq 22$, it must be the case that $c = 1$ and there are 23 more representatives. By the choice of c , all these 24 representatives come from different countries. But this is impossible since there are at most $100 - 86 = 14$ countries left.

Therefore, it remains to show that there are at least 2 rows with 22 empty seats. Indeed, if this does not hold, then the total number of representatives is at least

$$24 \times 85 + 23 = 2063 > 2002.$$

This is a contradiction, and so we are done.

7. The minimum value is $3n^2 - 3n$.

We have

$$\begin{aligned} \sum_{k=1}^{n-1} kx_k(2n-k) &= 2n(2n-2) - \sum_{k=1}^{n-1} k^2x_k \\ &= 2n(2n-2) - \sum_{k=1}^{n-1} x_k - \sum_{k=1}^{n-1} (k-1)(k+1)x_k \\ &\geq 2n(2n-2) - n - \sum_{k=1}^{n-1} (k-1)nx_k \\ &= 2n(2n-2) - n - n \sum_{k=1}^{n-1} kx_k + n \sum_{k=1}^{n-1} x_k \\ &= 2n(2n-2) - n - n(2n-2) + n^2 \\ &= 3n^2 - 3n. \end{aligned}$$

Equality holds when $x_1 = n-1$, $x_2 = x_3 = \dots = x_{n-2} = 0$ and $x_{n-1} = 1$. Therefore, the minimum value is $3n^2 - 3n$.

Year 2004

Test 1

1. (IMO 1972 Problem 1 modified) There are $2^{10} - 1 = 1023$ proper subsets of S , each containing at most 9 elements. The sum of all elements in such a proper subset ranges from 0 to

$$117 + 116 + \cdots + 109 = \frac{(117 + 109) \times 9}{2} = 1017.$$

There are only 1018 possibilities. By the pigeonhole principle, there are two proper subsets having the same sum. Removing their common elements if necessary, we obtain two disjoint subsets having the same sum.

2. By the Cauchy-Schwarz inequality, we have

$$(bc + ca + ab) \left(\frac{a^4}{bc} + \frac{b^4}{ca} + \frac{c^4}{ab} \right) \geq (a^2 + b^2 + c^2)^2.$$

Note that $a^2 + b^2 + c^2 \geq ab + bc + ca$ since this is the same as

$$\frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2) \geq 0.$$

Therefore, we get

$$\frac{a^4}{bc} + \frac{b^4}{ca} + \frac{c^4}{ab} \geq \frac{(a^2 + b^2 + c^2)^2}{bc + ca + ab} \geq a^2 + b^2 + c^2.$$

Equality holds when $a = b = c$.

3. (a) Suppose there exist $a, b \in \mathbb{Z}^+$ such that $a + b = 2003$ and $2003 \mid ab$. Since 2003 is a prime, we must have $2003 \mid a$ or $2003 \mid b$. WLOG assume $2003 \mid a$. Then we have $a + b \geq 2003 + 1 > 2003$, which is a contradiction. Therefore, 2003 is not good.
- (b) No. Suppose there exist $a, b \in \mathbb{Z}^+$ such that $a + b = 2002$ and $2002 \mid ab$. Note that $2002 = 2 \times 7 \times 11 \times 13$. For each prime $p \mid 2002$, we have $p \mid ab$, and so $p \mid a$ or $p \mid b$. In either case, since $p \mid a + b$, we must have $p \mid a$ and $p \mid b$. As this holds for $p = 2, 7, 11, 13$, we must have $2002 \mid a, b$. This yields $a + b \geq 2002 + 2002 > 2002$, which is a contradiction. Therefore, 2002 is not good.
- (c) All positive integers of the form $4p_1p_2 \cdots p_s$ or $8p_1p_2 \cdots p_s$ where $p_1, p_2, \dots, p_s$ are distinct odd primes (possibly $s = 0$) are good but not very good.

In view of the proof of part (b), all squarefree positive integers are not good. For any non-squarefree positive integer n , we can write $n = p^2m$ for some

prime p and positive integer m . Since we can take $a = pm$ and $b = (p-1)pm$ such that

$$a + b = p^2m = n$$

and

$$ab = (p-1)p^2m^2 = (p-1)mn,$$

we know that n is good by definition. Also, note that a and b are distinct unless $p = 2$. This shows n is very good if n is divisible by the square of an odd prime.

Thus, it remains to consider $n = 2^k m$ for some $k \geq 2$ and odd m which is squarefree. If $k \geq 4$, we can take $a = 2^{k-2}m$ and $b = 3 \cdot 2^{k-2}m$ such that

$$a + b = 2^k m = n$$

and

$$ab = 3 \cdot 2^{2k-4}m^2 = 3 \cdot 2^{k-4}mn.$$

Since $a \neq b$, this shows n is very good.

Lastly, if $k = 2, 3$, we show that n is not very good. Indeed, suppose $a + b = n$ and $n \mid ab$. As in part (b), we deduce $m \mid a$ and $m \mid b$. Let $a = mc$ and $b = md$. Then we need $c + d = 2^k$ and $2^k \mid cd$. It is routine to check that there is no solution except $c = d$ when $k = 2, 3$ (note that there are only a few pairs of (c, d) satisfying $c + d = 2^k$). Therefore, n is not very good.

4. (1998 Test 2 Problem 6 generalized) Let the perpendicular lines drawn from M and N to BD and AC meet at X . Let XP meet AD at Y . Let MX meet BD at E , and let NX meet AC at F . Note that E, X, F, P are concyclic since $\angle PEX + \angle PFX = 180^\circ$.

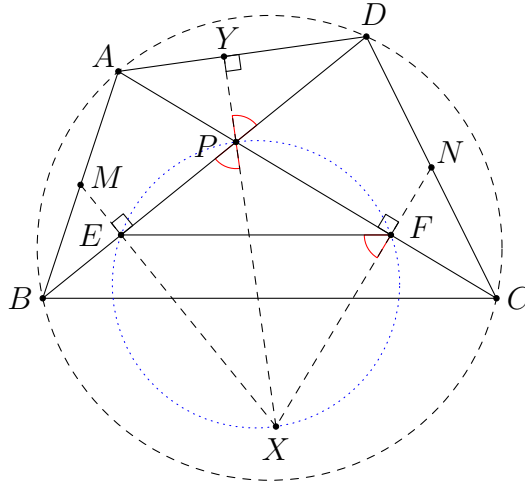
Suppose A, B, C, D are concyclic. Then we have $\triangle PAB \sim \triangle PDC$. Note that the pairs M, N and E, F are corresponding points under this similarity. Therefore, we have

$$\frac{PE}{EB} = \frac{PF}{FC},$$

and hence $EF \parallel BC$. It follows that

$$\begin{aligned} \angle YDP + \angle DPY &= \angle ADB + \angle EPX \\ &= \angle ACB + \angle EFX \\ &= \angle PFE + \angle EFX \\ &= 90^\circ. \end{aligned}$$

This proves $XY \perp AD$, and so the perpendicular line drawn from P to AD also passes through X .



Conversely, suppose $XY \perp AD$. We now prove that A, B, C, D are concyclic. Indeed, suppose the point C moves on the ray AP beyond P , where A, B, D, P are fixed. We want to find the position of C such that the midpoint N of CD lies on the perpendicular line drawn from X to AP . If there are two such positions, named C_1 and C_2 , then the line joining the midpoints of C_1D and C_2D , which is XF , is parallel to C_1C_2 by the midpoint theorem. This is impossible. Thus, there is at most one such position. Therefore, C must be the second intersection point of (ABD) and AP by false position. This proves A, B, C, D are concyclic.

5. We consider the following mapping between the set S of all positive integers formed by n 1's and n 2's and the set T of all positive integers with n digits formed by 1, 2, 3, 4 such that the numbers of 1's and 2's are equal.

For each $m \in S$, we can pair up every two consecutive digits of m from left to right. This gives n pairs in total. Then we replace the pairs 11, 22, 12, 21 by 1, 2, 3, 4 respectively. If there are a, b, c, d pairs of 11, 22, 12, 21 respectively, then there are $2a + c + d$ 1's and $2b + c + d$ 2's in m . Thus, we must have $a = b$. This shows the numbers of 1's and 2's in the image are the same, and so the image belongs to T .

Clearly, the mapping is reversible. For each $m \in T$, We just need to replace the digits 1, 2, 3, 4 in m by 11, 22, 12, 21 respectively. If there are a, a, c, d copies of 1, 2, 3, 4 respectively, then there are $2a + c + d$ 1's and $2a + c + d$ 2's in the image, and so the image belongs to S . This shows the mapping is a bijection, and hence

$$f(n) = |S| = |T| = g(n).$$

6. This is an immediate consequence of Lucas' theorem. We give an elementary proof as follows.

Let $n = ap + r$ where $0 \leq r \leq p - 1$. Then we have

$$\begin{aligned} \binom{n}{p} &= \frac{n(n-1)\cdots(n-p+1)}{p!} \\ &= \frac{(ap+r)(ap+r-1)\cdots(ap+1)(a)(ap-1)(ap-2)\cdots(ap+r-p+1)}{(p-1)!}. \end{aligned}$$

Note that the numerator is congruent to

$$r(r-1)\cdots(1)(a)(p-1)(p-2)\cdots(r+1) = a \cdot (p-1)!$$

modulo p . Therefore, we find that

$$\binom{n}{p} \equiv a = \left\lfloor \frac{n}{p} \right\rfloor \pmod{p}.$$

This proves the desired result.

Year 2005

Test 1

1. There is no such function.

Putting $n = m = 2$ in (ii), we obtain

$$f(4) = \frac{f(2)f(2)}{f(2)} = f(2).$$

Then by condition (iii), we have

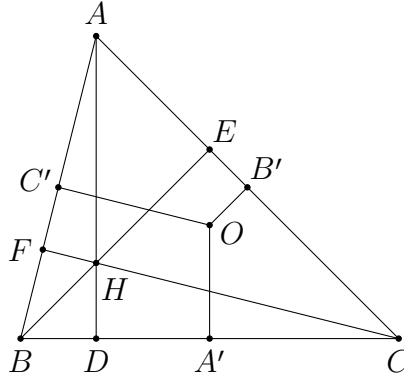
$$2 = f^{2004}(2) = f^{2003}(f(2)) = f^{2003}(f(4)) = f^{2004}(4) = 4.$$

This is a contradiction. So there is no such function.

2. It is well-known that $OA' = \frac{1}{2}AH$, etc. Therefore, we need to prove

$$AH \times HD = BH \times HE = CH \times HF.$$

Indeed, $AH \times HD = BH \times HE$ is true since A, B, D, E are concyclic. By symmetry, the other equality also holds.



3. (IMO Shortlist 1987 generalized) By the Cauchy-Schwarz inequality, we have

$$((q+r) + (r+p) + (p+q)) \left(\frac{1}{q+r} + \frac{1}{r+p} + \frac{1}{p+q} \right) \geq (1+1+1)^2 = 9.$$

Dividing both sides by $2(p+q+r)$, we obtain

$$\frac{1}{q+r} + \frac{1}{r+p} + \frac{1}{p+q} \geq \frac{9}{2(p+q+r)}.$$

Next, WLOG assume $p \geq q \geq r$. Then we have $\frac{1}{q+r} \geq \frac{1}{r+p} \geq \frac{1}{p+q}$ and $p^m \geq q^m \geq r^m$. By Chebyshev's inequality, we obtain

$$\frac{p^m}{q+r} + \frac{q^m}{r+p} + \frac{r^m}{p+q} \geq \frac{p^m + q^m + r^m}{3} \cdot \left(\frac{1}{q+r} + \frac{1}{r+p} + \frac{1}{p+q} \right).$$

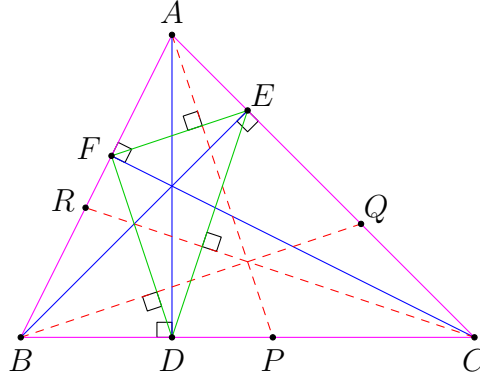
By the power mean inequality, since $m > 1$, we have

$$\frac{p^m + q^m + r^m}{3} \geq \left(\frac{p + q + r}{3} \right)^m.$$

Combining these and the result of the first part, we obtain

$$\frac{p^m}{q+r} + \frac{q^m}{r+p} + \frac{r^m}{p+q} \geq \left(\frac{p+q+r}{3} \right)^m \cdot \frac{9}{2(p+q+r)} = \frac{(p+q+r)^{m-1}}{2 \cdot 3^{m-2}}.$$

4. By a corollary of Carnot's theorem, the perpendicular lines drawn from A, B, C to EF, FD, DE are concurrent if and only if the perpendicular lines drawn from D, E, F to BC, CA, AB are concurrent. The latter is clearly true since the perpendicular lines are just AD, BE, CF , which are concurrent at the orthocentre of $\triangle ABC$. This proves AP, BQ, CR are concurrent.



5. The answer is 0 if $n = 1$, and is $\sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n - k - 1)!$ if $n \geq 2$, or any equivalent expression.

The result is obvious for $n = 1$. Consider $n \geq 2$. For each $k = 0, 1, \dots, n$, let N_k be the number of ways to arrange the $2n$ people such that A_i, B_i are next to each other for $i = 1, 2, \dots, k$ (and probably for some more pairs).

For each k , we need to arrange k pairs and the remaining $2n - 2k$ people. There are $(k + 2n - 2k - 1)! = (2n - k - 1)!$ ways to do so. Within each pair, there are 2 ways to arrange the two people. Therefore, we have

$$N_k = 2^k (2n - k - 1)!.$$

Now, by the inclusion-exclusion principle, since there are $\binom{n}{k}$ ways to choose k pairs, the number of ways to arrange the $2n$ people such that no pair A_i, B_i sits next to each other is

$$\sum_{k=0}^n (-1)^k \binom{n}{k} N_k = \sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n - k - 1)!.$$

6. The answer is 1003.

Note that

$$r = \frac{2004 + \sqrt{2004^2 + 4}}{2}$$

is an irrational number greater than 1. Since $a_n \in \mathbb{Z}^+$, we have $ra_n \notin \mathbb{Q}$. Thus, we obtain

$$a_{n+1} < ra_n < a_{n+1} + 1$$

for any $n \geq 0$. Equivalently, we have

$$a_n - \frac{1}{r} < \frac{a_{n+1}}{r} < a_n.$$

As $r > 1$, this yields $a_n - 1 < \frac{a_{n+1}}{r} < a_n$, and hence

$$\left\lfloor \frac{a_{n+1}}{r} \right\rfloor = a_n - 1.$$

Next, observe that $r = 2004 + \frac{1}{r}$. It follows that

$$a_{n+1} = [ra_n] = \left[2004a_n + \frac{a_n}{r} \right] = 2004a_n + a_{n-1} - 1$$

since $2004a_n \in \mathbb{Z}$ and $\left\lfloor \frac{a_n}{r} \right\rfloor = a_{n-1} - 1$ from above. Thus, we obtain

$$a_{n+1} \equiv a_{n-1} - 1 \pmod{2004}$$

for any $n \in \mathbb{Z}^+$. Then we can easily obtain

$$a_{2004} \equiv a_{2002} - 1 \equiv a_{2000} - 2 \equiv \cdots \equiv a_0 - 1002 \equiv 1003 \pmod{2004}.$$

Test 2

1. Firstly, since

$$\angle AED + \angle AFD = 90^\circ + 90^\circ = 180^\circ,$$

the points A, E, D, F are concyclic.

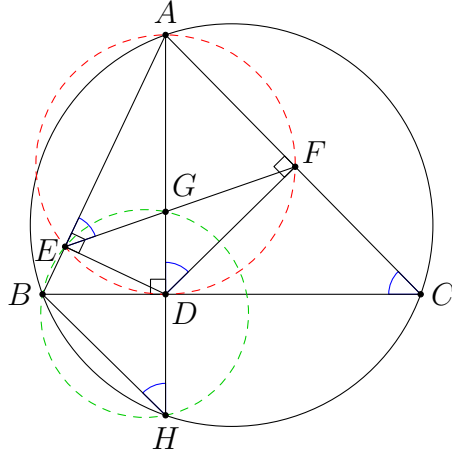
Secondly, since

$$\angle AHB = \angle ACB = 90^\circ - \angle DAF = \angle FDA = \angle FEA,$$

the points E, B, H, G are concyclic.

Thirdly, AD is tangent to (BDE) since $\angle ADB = \angle DEB = 90^\circ$. It follows from the power of points that

$$AD^2 = AE \times AB = AG \times AH.$$



2. Construct a point E on the extension of AC such that $CD = CE$. Using the condition, we find that

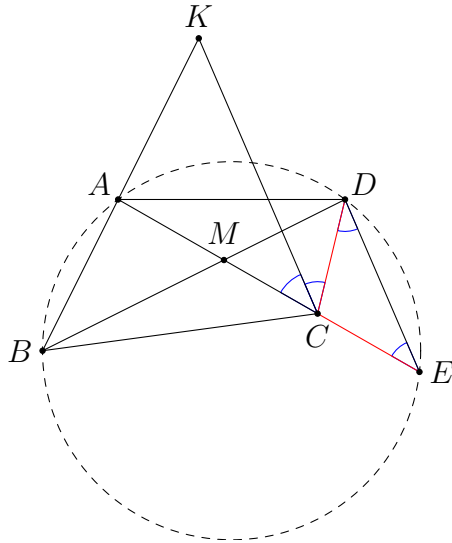
$$\begin{aligned} MB \cdot MD &= MA \cdot MC + MA \cdot CD \\ &= MA(MC + CE) \\ &= MA \cdot ME. \end{aligned}$$

This implies A, B, E, D are concyclic.

Now, since $\angle CDE = \angle CED = \frac{1}{2} \angle DCA = \angle KCA$, we obtain

$$\angle BKC = \angle BAC - \angle KCA = \angle BDE - \angle CDE = \angle BDC$$

as desired.



3. The smallest possible value of x is $\frac{3\sqrt{7}}{4}$.

Let a, b, c be the side lengths of a triangle, and let

$$d = \frac{a+b-c}{2}, \quad e = \frac{b+c-a}{2}, \quad f = \frac{c+a-b}{2}.$$

Note that $d, e, f > 0$, and either all of them are integers or each of them is an odd integer divided by 2. The triangle is uniquely determined by d, e, f . By Heron's formula, we have

$$x = \sqrt{def(d+e+f)}.$$

We list out all possibilities for which $x^2 \leq \frac{63}{16}$. WLOG assume $d \geq e \geq f$.

- If d, e, f are integers, then we must have $d = e = f = 1$, since otherwise

$$x^2 \geq (2)(1)(1)(4) = 8 > \frac{63}{16}.$$

When $d = e = f = 1$, we have $x^2 = 3$.

- If $2d, 2e, 2f$ are odd integers, then we must have $f = \frac{1}{2}$, since otherwise

$$x^2 \geq \left(\frac{3}{2}\right)\left(\frac{3}{2}\right)\left(\frac{3}{2}\right)\left(\frac{9}{2}\right) = \frac{243}{16} > \frac{63}{16}.$$

- If we further have $e = \frac{1}{2}$, let $d = \frac{2k-1}{2}$ for some $k \in \mathbb{Z}^+$. Then we get

$$x^2 = \frac{2k-1}{8} \cdot \frac{2k+1}{2} = \frac{4k^2-1}{16}.$$

Thus, x^2 can be $\frac{3}{16}, \frac{15}{16}, \frac{35}{16}, \frac{63}{16}, \dots$

- If $e \geq \frac{3}{2}$, then we must have $d = e = \frac{3}{2}$, since otherwise

$$x^2 \geq \left(\frac{5}{2}\right)\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\left(\frac{9}{2}\right) = \frac{135}{16} > \frac{63}{16}.$$

When $d = e = \frac{3}{2}$, we have $x^2 = \frac{63}{16}$.

Now, notice that among all possible values of x^2 not exceeding $\frac{63}{16}$, only $\frac{63}{16}$ has appeared twice. Thus, this is the smallest value of x^2 , and the smallest value of x is $\frac{3\sqrt{7}}{4}$. Indeed, when $(a, b, c) = (2, 2, 3), (1, 4, 4)$, we have $x = \frac{3\sqrt{7}}{4}$.

4. The ratio can be $\frac{5}{3}$ or $\frac{5}{4}$.

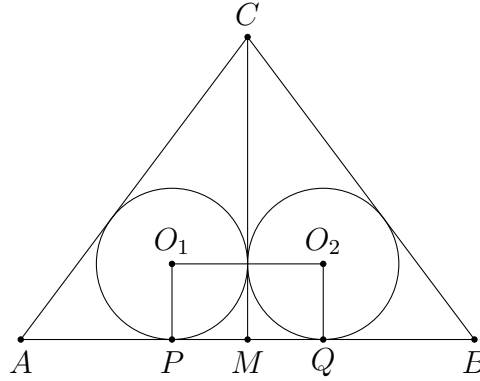
Let $a = BC$, $b = CA$, $c = AB$, $d = a + b$, $s = \frac{a + b + c}{2}$, $x = MA$, $y = MB$ and $z = MC$. Also, let r be the inradius of $\triangle ABC$, and let r' be the inradius of $\triangle AMC$.

Firstly, by considering $\triangle AMC$, we obtain $r' = \frac{2[AMC]}{b + x + z}$. Similarly, by considering $\triangle BMC$, we obtain $r' = \frac{2[BMC]}{a + y + z}$. Equating these, and using $\frac{[AMC]}{[BMC]} = \frac{x}{y}$, we obtain

$$\frac{b + x + z}{a + y + z} = \frac{x}{y}.$$

This implies $\frac{b + z}{a + z} = \frac{x}{y}$. Since $x + y = c$, we obtain

$$x = \frac{(b + z)c}{a + b + 2z} \quad \text{and} \quad y = \frac{(a + z)c}{a + b + 2z}. \quad (1)$$



Secondly, we have

$$O_1P = AP \tan \frac{A}{2} = \frac{b + x - z}{2} \cdot \frac{r}{s - a} = \frac{(b + x - z)r}{b + c - a}$$

and similarly $O_2Q = \frac{(a + y - z)r}{a + c - b}$. Since these are equal, we have

$$\frac{b + x - z}{b + c - a} = \frac{a + y - z}{a + c - b}.$$

Using (1), this becomes

$$\frac{(a + b + 2z)(b - z) + (b + z)c}{b + c - a} = \frac{(a + b + 2z)(a - z) + (a + z)c}{a + c - b}.$$

Clearing denominators and simplifying, we get

$$4(a-b)z^2 = (a-b)(a+b+c)(a+b-c).$$

This implies $a = b$ or $z = \sqrt{s(s-c)}$. If $a = b$, then we have $x = y = \frac{c}{2}$ by (1), and hence

$$z = \sqrt{a^2 - x^2} = \sqrt{\left(a + \frac{c}{2}\right)\left(a - \frac{c}{2}\right)} = \sqrt{s(s-c)}.$$

Therefore,

$$z = \sqrt{s(s-c)} \tag{2}$$

holds in any case.

Thirdly, we use the given condition. Note that PQO_2O_1 is a rectangle, and so

$$[PQO_2O_1] = PQ \cdot r' = \left(\frac{x+z-b}{2} + \frac{y+z-a}{2}\right)r' = \frac{(c+2z-d)r'}{2}.$$

Next, we have

$$[ABC] = [AMC] + [BMC] = \frac{b+x+z}{2} \cdot r' + \frac{a+y+z}{2} \cdot r' = \frac{(c+2z+d)r'}{2}.$$

Since $6[PQO_2O_1] = [ABC]$, we get $6(c+2z-d) = c+2z+d$. This implies

$$z = \frac{7d-5c}{10}. \tag{3}$$

Comparing (2) and (3), we solve

$$\left(\frac{7d-5c}{10}\right)^2 = s(s-c) = \frac{(d+c)(d-c)}{4}.$$

This is the same as $(3d-5c)(4d-5c) = 0$. Therefore, we have

$$\frac{d}{c} = \frac{5}{3}, \frac{5}{4}.$$

Both values are possible. For example, when $(a, b, c) = (5, 5, 6), (5, 5, 8)$, both $\triangle AMC$ and $\triangle BMC$ are right-angled triangles with side lengths 3, 4, 5. It is easy to find $r' = 1$ and verify $6[PQO_2O_1] = 12 = [ABC]$.

Year 2006

Test 1

1. The solutions are $(x, y) = (0, 0), (4, 5), (5, 4)$.

Let $a = x + y$ and $b = x - y$. The equation can be rewritten as

$$28(x + y) = 3((x + y)^2 + 3(x - y)^2),$$

which means $28a = 3(a^2 + 3b^2)$. Clearly, $3 \mid a$. Let $a = 3c$. The equation becomes

$$28c = 9c^2 + 3b^2.$$

We further find that $3 \mid c$. Let $c = 3d$. The equation becomes

$$28d = 27d^2 + b^2.$$

Now, as $d(28 - 27d) = b^2 \geq 0$, we must have $0 \leq d \leq \frac{28}{27}$. The only integer solutions are $d = 0, 1$.

When $d = 0$, $b = 0$. One finds that $a = 0$, and hence $x = y = 0$.

When $d = 1$, $b = \pm 1$. One finds that $a = 9$, and hence $(x, y) = (4, 5), (5, 4)$.

All these are solutions since the steps are reversible.

2. (Ackermann function/IMO 1981 Problem 6 modified)

- (a) The answer is $f(3, 2005) = 2^{2008} - 3$.

We first label the equations as follows:

$$f(0, y) = y + 1, \tag{1}$$

$$f(x + 1, 0) = f(x, 1), \tag{2}$$

$$f(x + 1, y + 1) = f(x, f(x + 1, y)). \tag{3}$$

By putting $y = 1$ in (1), we get $f(0, 1) = 2$. By putting $x = 0$ in (2), we get $f(1, 0) = f(0, 1) = 2$.

Using (3) and (1), we find that

$$f(1, y) = f(0, f(1, y - 1)) = f(1, y - 1) + 1.$$

By induction, one easily obtains

$$f(1, y) = f(1, 0) + y = y + 2.$$

Similarly, we have

$$f(2, y) = f(1, f(2, y - 1)) = f(2, y - 1) + 2.$$

By induction, we obtain

$$f(2, y) = f(2, 0) + 2y = f(1, 1) + 2y = 2y + 3.$$

Next, we have

$$f(3, y) = f(2, f(3, y - 1)) = 2f(3, y - 1) + 3.$$

Adding

$$\begin{aligned} f(3, y) &= 2f(3, y - 1) + 3, \\ 2f(3, y - 1) &= 2^2 f(3, y - 2) + 2 \cdot 3, \\ 2^2 f(3, y - 2) &= 2^3 f(3, y - 3) + 2^2 \cdot 3, \\ &\dots, \\ 2^{y-1} f(3, 1) &= 2^y f(3, 0) + 2^{y-1} \cdot 3, \end{aligned}$$

we obtain

$$\begin{aligned} f(3, y) &= 2^y f(3, 0) + 3(1 + 2 + 2^2 + \dots + 2^{y-1}) \\ &= 2^y f(2, 1) + 3(2^y - 1) = 2^{y+3} - 3. \end{aligned}$$

In particular, we have $f(3, 2005) = 2^{2008} - 3$.

- (b) The answer is $f(4, 2005) = g(2008) - 3$, where $g(n)$ is defined by $g(1) = 2$ and $g(n + 1) = 2^{g(n)}$ for any $n \in \mathbb{Z}^+$.

We prove by induction that $f(4, y) = g(y + 3) - 3$. Firstly, by (2), we have

$$f(4, 0) = f(3, 1) = 2^4 - 3 = 2^{2^2} - 3 = g(3) - 3.$$

This proves the base case.

Assume $f(4, y) = g(y + 3) - 3$ for some $y \in \mathbb{Z}^+$. Using (3), we find that

$$f(4, y + 1) = f(3, f(4, y)) = 2^{f(4, y) + 3} - 3 = 2^{g(y + 3)} - 3 = g(y + 4) - 3.$$

This proves the inductive step.

Therefore, we have $f(4, 2005) = g(2008) - 3$.

3. $\angle B$ can be 22.5° or 67.5° .

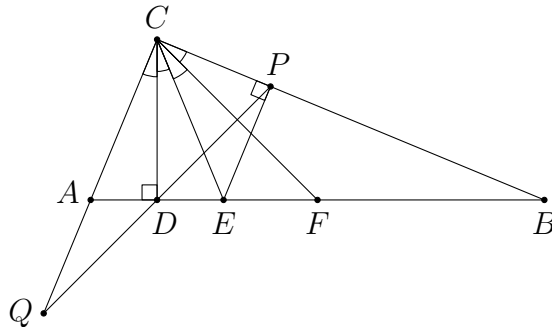
WLOG assume $AC < BC$. Let D be the foot of altitude from C , let E be the foot of internal angle bisector from C , and let F be the midpoint of AB . Let P be the projection of E on BC , and let DP meet AC at Q .

Firstly, since $\angle ACD = \angle ECD$ and $CD \perp AE$, we have $\triangle ACD \cong \triangle ECD$. Secondly, note that C, D, E, P are concyclic since

$$\angle CDE + \angle CPE = 90^\circ + 90^\circ = 180^\circ.$$

It follows that

$$\angle CPQ = \angle CPD = \angle CED = \angle CAD = \angle CAB.$$



Together with the common angle at C , we know that $\triangle CPQ \sim \triangle CAB$ (and D lies between P and Q). As $\angle DCQ = \angle FCB$, the points D and F are corresponding points under this similarity. Thus, D is the midpoint of PQ . Now, since D is also the midpoint of AE , the quadrilateral $AQEP$ is a parallelogram. This implies $CQ \parallel PE$, and hence $\angle ACB = 90^\circ$. Then we have

$$\angle ABC = \angle FCB = \frac{90^\circ}{4} = 22.5^\circ.$$

If $AC > BC$, then the roles of A and B are swapped, and hence $\angle ABC = 67.5^\circ$.

4. (a) The smallest possible value of Q is $\frac{1}{3^{2005}}$.

By the power mean inequality, we have

$$S_n \geq 3 \left(\frac{S_1}{3} \right)^n = \frac{1}{3^{n-1}}$$

for any positive integer n . Therefore, we have

$$Q = S_3 S_{2004} \geq \frac{1}{3^2} \cdot \frac{1}{3^{2003}} = \frac{1}{3^{2005}}.$$

Equality holds when $x = y = z = \frac{1}{3}$. Therefore, the minimum value of Q is $\frac{1}{3^{2005}}$.

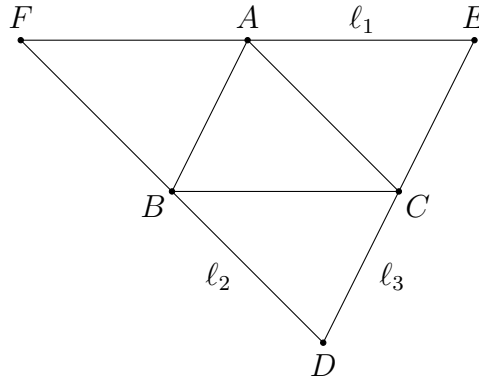
- (b) P is larger than Q .

Indeed, we have

$$\begin{aligned} & P \geq Q \\ \Leftrightarrow & (x^2 + y^2 + z^2)(x^{2005} + y^{2005} + z^{2005}) \geq (x^3 + y^3 + z^3)(x^{2004} + y^{2004} + z^{2004}) \\ \Leftrightarrow & \sum_{\text{svm}} x^{2005} y^2 \geq \sum_{\text{svm}} x^{2004} y^3. \end{aligned}$$

This is true by Muirhead's theorem. Equality holds when $x = y = z$. Since it is given that x, y, z are pairwise distinct, the inequality is strict, and hence $P > Q$.

5. Let A, B, C be points in S such that $[ABC]$ is maximized. Let ℓ_1 be the line passing through A which is parallel to BC . If there exists a point $X \in S$ which lies on different side of ℓ_1 as B , then the distance from X to BC is greater than that from A to BC . This yields the contradiction $[XBC] > [ABC]$. Therefore, X must lie on the same side of ℓ_1 as B .



Similarly, let ℓ_2 be the line passing through B which is parallel to CA , and let ℓ_3 be the line passing through C which is parallel to AB . Then any point in S must lie inside the triangle $\triangle DEF$ bounded by ℓ_1, ℓ_2, ℓ_3 . Note that $\triangle ABC$ is the median triangle of $\triangle DEF$. So $[DEF] = 4[ABC] < 4$. Thus, we can simply take $\triangle DEF$.

6. We construct the numbers recursively. Firstly, we start from the following 2^2 integers:

$$77, 78, 87, 88,$$

each of which has 2^1 digits, and at most 2^0 digit of each pair is the same.

Suppose we have constructed $k = 2^{n+1}$ integers $a_1, a_2, \dots, a_k$ with digits 7, 8, each of which has 2^n digits, and at most 2^{n-1} digits of each pair are the same. We construct $2k = 2^{n+2}$ integers as follows:

$$\overline{a_1 a_1}, \overline{a_2 a_2}, \dots, \overline{a_k a_k}, \overline{a_1 b_1}, \overline{a_2 b_2}, \dots, \overline{a_k b_k},$$

where b_j means the integer having the same number of digits as a_j such that each digit is different from that of a_j (and is still 7 or 8). For example, when $n = 1$, we construct

$$7777, 7878, 8787, 8888, 7788, 7887, 8778, 8877.$$

Note that each of the integers constructed has $2 \cdot 2^n = 2^{n+1}$ digits. Also, one can easily prove by induction that $b_1, b_2, \dots, b_k$ is a permutation of $a_1, a_2, \dots, a_k$.

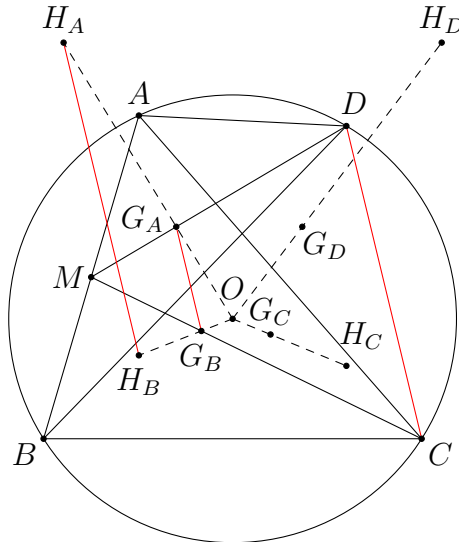
- For $\overline{a_i a_i}$ and $\overline{a_j a_j}$ (with $i \neq j$), at most $2 \cdot 2^{n-1} = 2^n$ of their digits are the same by the inductive hypothesis. Similarly, the same holds for the pair $\overline{a_i a_i}$ and $\overline{a_j b_j}$ unless $a_i = b_j$, and for the pair $\overline{a_i b_i}$ and $\overline{a_j b_j}$.
- For $\overline{a_i a_i}$ and $\overline{a_j b_j}$ with $a_i = b_j$, the first 2^n digits are different, while the last 2^n digits are the same.
- For $\overline{a_i a_i}$ and $\overline{a_i b_i}$, the first 2^n digits are the same, while the last 2^n digits are different.

Thus, in any case, each pair of integers shares at most 2^n same digits. Repeating the same procedure, we eventually obtain 2^{2006} integers satisfying all the conditions.

Test 2

1. Let O be the centre of $(ABCD)$. Let H_A, H_B, H_C, H_D and G_A, G_B, G_C, G_D be the orthocentres and centroids of $\triangle DAB$, $\triangle ABC$, $\triangle BCD$ and $\triangle CDA$ respectively. Let M be the midpoint of AB . Recall that O lies on $H_A G_A$, etc., with $OG_A : G_A H_A = 1 : 2$. Also, we have $DG_A : G_A M = 2 : 1 = CG_B : G_B M$. From these, we know that $H_A H_B \parallel G_A G_B \parallel DC$, and

$$H_A H_B = 3G_A G_B = CD.$$



By symmetry, we have $H_B H_C = DA$, $H_C H_D = AB$, $H_D H_A = BC$, $H_A H_C = CA$ and $H_B H_D = DB$. Thus, $H_A H_B H_C H_D$ is congruent to $CDAB$. It follows

that $H_A H_B H_C H_D$ is concyclic. Now, note that $G_A G_B G_C G_D$ is the image of $H_A H_B H_C H_D$ under the homothety with centre O and ratio $\frac{1}{3}$. Therefore, $G_A G_B G_C G_D$ is cyclic.

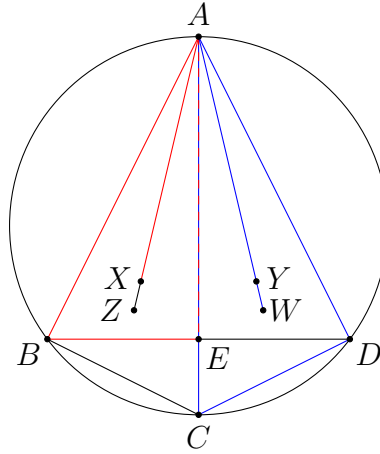
2. Since $\angle BAE = \angle CAD$ (in view of $BC = CD$) and $\angle EBA = \angle DCA$, we have $\triangle ABE \sim \triangle ACD$. Since X and W are incentres of $\triangle ABE$ and $\triangle ACD$ respectively, they are corresponding points under this similarity. It follows that

$$\frac{AX}{AW} = \frac{AB}{AC}.$$

Similarly, we have $\frac{AY}{AZ} = \frac{AD}{AC}$. Now,

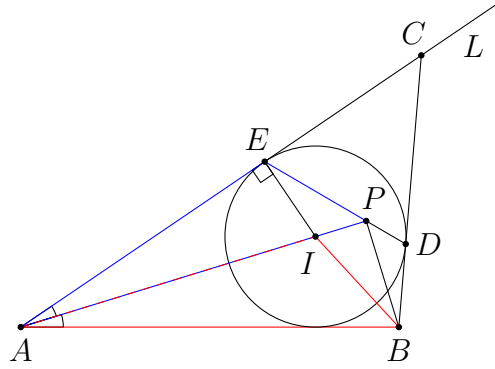
$$\begin{aligned} & W, X, Y, Z \text{ are concyclic} \\ \Leftrightarrow & AX \times AZ = AY \times AW \\ \Leftrightarrow & \frac{AX}{AW} = \frac{AY}{AZ} \\ \Leftrightarrow & \frac{AB}{AC} = \frac{AD}{AC} \\ \Leftrightarrow & AB = AD. \end{aligned}$$

This completes the proof.



3. Let I be the incentre of $\triangle ABC$, and let P be the intersection point of AI and DE . We claim that $\angle APB = 90^\circ$. Once this is proved, since the line AP is fixed, the point P is independent of C , and so P is the desired fixed point.

Note that $\angle BAI = \angle PAE$ and $\angle AIB = 90^\circ + \frac{C}{2} = \angle AEP$. These imply $\triangle IAB \sim \triangle EAP$. By spiral similarity, we have $\triangle PAB \sim \triangle EAI$, and hence $\angle APB = \angle AEI = 90^\circ$ as desired.

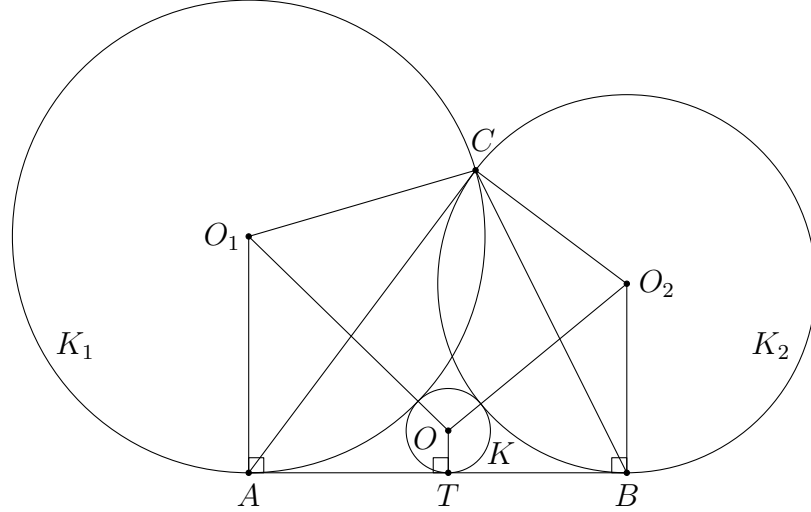


4. (a) The answer is $r = \frac{abc^2}{4R(a+b)^2}$ or any equivalent expression.

Let O, O_1, O_2 be the centres of K, K_1, K_2 respectively. Let R_1 and R_2 be the radii of K_1 and K_2 respectively. We have

$$R_1 = AO_1 = \frac{AC}{2 \cos \angle CAO_1} = \frac{b}{2 \sin A} = \frac{bR}{a}$$

by the extended sine law. By symmetry, we have $R_2 = \frac{aR}{b}$.



Now, let T be the tangential point of K with AB . By Pythagorus' theorem, we have

$$AT = \sqrt{OO_1^2 - (AO_1 - TO)^2} = \sqrt{(R_1 + r)^2 - (R_1 - r)^2} = 2\sqrt{R_1 r}.$$

Similarly, we have $BT = 2\sqrt{R_2 r}$. This yields

$$c = AB = 2\sqrt{R_1 r} + 2\sqrt{R_2 r},$$

and hence

$$r = \frac{c^2}{4(\sqrt{R_1} + \sqrt{R_2})^2} = \frac{c^2}{4(R_1 + R_2 + 2\sqrt{R_1 R_2})} = \frac{abc^2}{4R(a+b)^2}.$$

(b) We have $\angle C = 90^\circ$.

Note that $\frac{ab}{(a+b)^2} \leq \frac{1}{4}$ (since this is the same as $(a-b)^2 \geq 0$) and $c \leq 2R$.

By part (a), we have

$$r = \frac{abc^2}{4R(a+b)^2} \leq \frac{4R^2}{16R} = \frac{R}{4}.$$

Therefore, equality should hold. This means $a = b$ and $c = 2R$. In order that c is the diameter of the circumcircle, we must have $\angle C = 90^\circ$.

Year 2007

Test 1

1. The maximum value is $-2\sqrt{2}$.

By the Cauchy-Schwarz inequality, we have

$$16s^2 = (9 + 4 + 1 + 2)(p^2 + q^2 + r^2 + 4) \geq (3p + 2q + r + 2\sqrt{2})^2.$$

This implies $4|s| \geq 3p + 2q + r + 2\sqrt{2}$, and hence

$$3p + 2q + r - 4|s| \leq -2\sqrt{2}.$$

Equality holds when $3 : 2 : 1 : \sqrt{2} = p : q : r : 2$, i.e.

$$(p, q, r, s) = (3\sqrt{2}, 2\sqrt{2}, \sqrt{2}, \pm 4\sqrt{2}).$$

2. The minimum value is $3 - 2\sqrt{2}$.

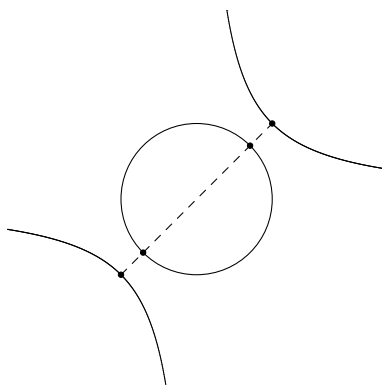
Condition (i) can be rewritten as

$$(\sin A - |\sin B|)(\cos B - |\cos A|) = 0.$$

If $\sin A = |\sin B|$, then we have $\sin^2 A = \sin^2 B = 1 - \cos^2 B$.

If $\cos B = |\cos A|$, then we have $\cos^2 B = \cos^2 A = 1 - \sin^2 A$.

Therefore, in any case, $(\sin A, \cos B)$ is a point on the unit circle on the coordinate plane. Since the steps are reversible, it can be any point on the unit circle.



Next, $(\tan C, \cot C)$ can be any point on the hyperbola $xy = 1$. Therefore, we are asked to find the minimum distance between a point on the unit circle and a point on the hyperbola. In view of the geometry, since the line $y = x$ is an axis of symmetry of both geometric objects, the minimum distance is attained

between points on the line $y = x$. By symmetry, it suffices to consider $x > 0$. Clearly, $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ is a solution to $x^2 + y^2 = 1$ and $y = x$, while $(1, 1)$ is a solution to $xy = 1$ and $y = x$. Therefore, the minimum distance is

$$\left(1 - \frac{1}{\sqrt{2}}\right)^2 + \left(1 - \frac{1}{\sqrt{2}}\right)^2 = 3 - 2\sqrt{2},$$

and we are done.

3. By the cosine formula, we have

$$\begin{aligned} & \frac{\cos A}{a^3} + \frac{\cos B}{b^3} + \frac{\cos C}{c^3} \geq \frac{3}{2abc} \\ \Leftrightarrow & \frac{\frac{b^2 + c^2 - a^2}{2a^3bc} + \frac{c^2 + a^2 - b^2}{2ab^3c} + \frac{a^2 + b^2 - c^2}{2abc^3}}{\frac{b^2 + c^2 - a^2}{a^2} + \frac{c^2 + a^2 - b^2}{b^2} + \frac{a^2 + b^2 - c^2}{c^2}} \geq \frac{3}{2abc} \\ \Leftrightarrow & \frac{b^2}{a^2} + \frac{c^2}{a^2} + \frac{c^2}{b^2} + \frac{a^2}{b^2} + \frac{a^2}{c^2} + \frac{b^2}{c^2} \geq 6. \end{aligned}$$

This is true by the AM-GM inequality. Equality holds when $\triangle ABC$ is equilateral.

4. (IMO 1970 Problem 6 simplified)

(a) We first show that there must be a non-acute triangle among any 4 points. Consider the convex hull of A, B, C, D .

- If the convex hull is a quadrilateral $ABCD$, then since

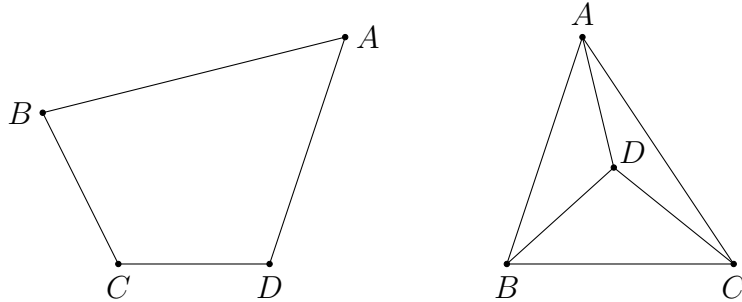
$$\angle ABC + \angle BCD + \angle CDA + \angle DAB = 360^\circ,$$

one of these angles is at least 90° . This gives rise to a non-acute triangle.

- If the convex hull is a triangle, say $\triangle ABC$, then since

$$\angle ADB + \angle BDC + \angle CDA = 360^\circ,$$

one of these angles is obtuse.



Now, suppose on the contrary that at most 2 triangles are non-acute. Note that each triangle belongs to exactly 2 quadrilaterals, and there are $\binom{5}{4} = 5$ quadrilaterals in total. Therefore, there must be a quadrilateral which does not consist of any non-acute triangles, contradicting the above observation. Therefore, there are at least 3 non-acute triangles.

- (b) There are $\binom{100}{5}$ groups of 5 points formed from the 100 points. By part (a), there are at least 3 non-acute triangles in each group. Since each triangle belongs to $\binom{97}{2}$ groups of 5 points, there are at least

$$3 \binom{100}{5} \div \binom{97}{2}$$

non-acute triangles. As there are $\binom{100}{3}$ triangles in total, at least

$$3 \binom{100}{5} \div \binom{97}{2} \div \binom{100}{3} = \frac{3}{10} = 30\%$$

of the triangles are non-acute.

5. Let $b_n = n^2 a_n + 2$ for all $n \geq 1$. The recurrence relation becomes

$$(n+1)(b_{n+1} - 2) = 2(2n+1)(b_n - 2) + 2(3n+1).$$

This is the same as

$$b_{n+1} = \frac{2(2n+1)}{n+1} b_n.$$

It follows that

$$b_n = \frac{2(2n-1)}{n} b_{n-1} = \frac{2^2(2n-1)(2n-3)}{n(n-1)} b_{n-2} = \dots = \frac{2^{n-1}(2n-1)!!}{n!} b_1,$$

where $(2n-1)!! = (2n-1)(2n-3)\dots(1)$. Since $b_1 = a_1 + 2 = 2$, we have

$$b_n = \frac{2^n(2n-1)!!}{n!} = \frac{(2n)!!(2n-1)!!}{(n!)^2} = \frac{(2n)!}{(n!)^2} = \binom{2n}{n}.$$

Therefore, we obtain

$$a_n = \frac{b_n - 2}{n^2} = \frac{1}{n^2} \left[\binom{2n}{n} - 2 \right].$$

By Vandermonde's identity, we can rewrite this as

$$a_n = \frac{1}{n^2} \left(\sum_{k=0}^n \binom{n}{k}^2 - 2 \right) = \frac{1}{n^2} \left(\sum_{k=1}^{n-1} \binom{n}{k}^2 \right).$$

When $n = p$ is a prime, we have $p \mid \binom{p}{k}$ for $1 \leq k \leq n-1$, and so $p^2 \mid \binom{p}{k}^2$. This implies $a_p \in \mathbb{Z}$. Clearly, $a_p > 0$. Since there are infinitely many prime numbers, there are infinitely many terms which are positive integers.

6. (IMO Shortlist 1998 N1/Problem 4) The solutions are $(x, y) = (11t^2, 11t), (7, 1), (121, 1), (10, 2)$ where $t \in \mathbb{Z}^+$.

The condition means $xy^2 + y + 11 \mid x^2y + x + y$. Since

$$y(x^2y + x + y) - x(xy^2 + y + 11) = y^2 - 11x,$$

we have

$$xy^2 + y + 11 \mid y^2 - 11x.$$

- If $y^2 - 11x > 0$, then we have $xy^2 + y + 11 \leq y^2 - 11x < y^2$. But then $xy^2 + y + 11 > xy^2 \geq y^2$. This is a contradiction.
- If $y^2 - 11x = 0$, then we can write $y = 11t$ for some $t \in \mathbb{Z}^+$. This yields $x = 11t^2$. Now, we check that

$$\frac{(11t^2)^2(11t) + (11t^2) + (11t)}{(11t^2)(11t)^2 + (11t) + 11} = \frac{11t(121t^4 + t + 1)}{11(121t^4 + t + 1)} = t \in \mathbb{Z},$$

and so all pairs $(x, y) = (11t^2, 11t)$ are solutions.

- If $y^2 - 11x < 0$, then we have $xy^2 + y + 11 \leq 11x - y^2 < 11x$. It follows that $11x > xy^2 + y + 11 > xy^2$, and hence $y^2 < 11$. This implies $y = 1, 2, 3$.
 - When $y = 1$, we need $x + 12 \mid x^2 + x + 1$. This is equivalent to

$$x + 12 \mid (-12)^2 + (-12) + 1 = 133 = 7 \times 19.$$

The solutions are $(x, y) = (7, 1), (121, 1)$.

- When $y = 2$, we need $4x + 13 \mid 2x^2 + x + 2$. Since $4x + 13$ is odd, this is the same as $4x + 13 \mid 8(2x^2 + x + 2)$. As

$$8(2x^2 + x + 2) = (4x + 13)(4x - 11) + 159,$$

this means $4x + 13 \mid 159 = 3 \times 53$. The only solution is $(x, y) = (10, 2)$.

- When $y = 3$, we need $9x + 14 \mid 3x^2 + x + 3$. Since $3 \nmid 9x + 14$, this is the same as $9x + 14 \mid 27(3x^2 + x + 3)$. As

$$27(3x^2 + x + 3) = (9x + 14)(9x - 11) + 235,$$

this means $9x + 14 \mid 235 = 5 \times 47$. There is no solution.

Therefore, we conclude that the solutions are those listed at the beginning.

Test 2

1. We have $AD : DP = 1 : 1$.

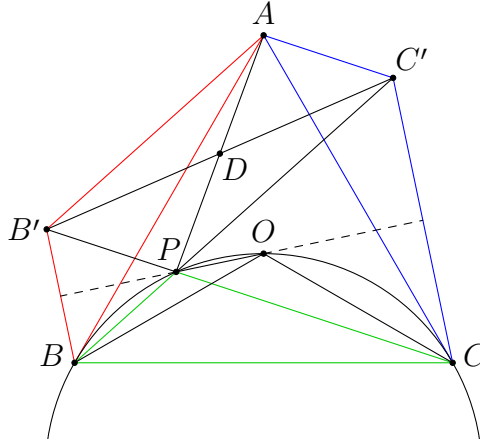
Since $\angle BPC = 120^\circ = \angle BOC$, the points B, C, O, P are concyclic. As

$$\angle BPO = 180^\circ - \angle OCB = 150^\circ,$$

we have $\angle B'PB = 2(180^\circ - 150^\circ) = 60^\circ$, and hence $\triangle PB'B$ is equilateral. Similarly, as

$$\angle CPO = \angle CBO = 30^\circ,$$

we have $\angle CPC' = 60^\circ$, and hence $\triangle PCC'$ is equilateral.



Now, since $\triangle BPB'$ and $\triangle BCA$ are directly similar, by spiral similarity, we have $\triangle BAB' \sim \triangle BCP$. In the same way, we have $\triangle ACC' \sim \triangle BCP$. Therefore, we obtain $\triangle BAB' \sim \triangle ACC'$. Indeed, they are congruent since $BA = AC$. This implies $AC' = BB' = B'P$ and $B'A = C'C = PC'$. Thus, $AB'PC'$ is a parallelogram. The diagonals AP and $B'C'$ bisect each other. In particular, we have $AD : DP = 1 : 1$.

2. Let P be the point on the side AB such that $AP = AD$. Then we have

$$BP = AB - AP = AD + BC - AD = BC.$$

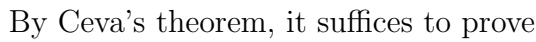
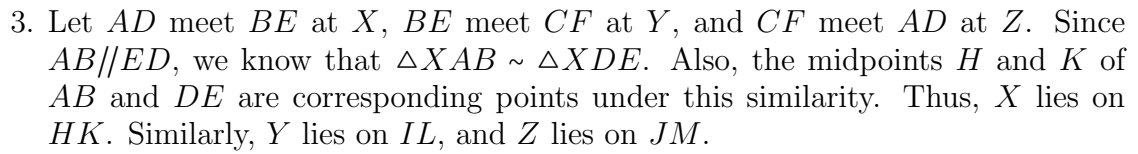
Let the internal angle bisector of $\angle ADC$ meet AB at Q . We find that

$$\angle QPC = \angle BPC = 90^\circ - \frac{1}{2}\angle CBP = \frac{1}{2}(180^\circ - \angle CBA) = \frac{1}{2}\angle ADC = \angle QDC.$$

This implies C, D, P, Q are concyclic. Similarly, we have

$$\angle DCQ = \angle DPA = 90^\circ - \frac{1}{2}\angle PAD = \frac{1}{2}(180^\circ - \angle BAD) = \frac{1}{2}\angle DCB.$$

This shows CQ bisects $\angle BCD$. Therefore, the internal angle bisectors of $\angle ADC$ and $\angle BCD$ intersect at the point Q on AB .



Firstly, since $AH = HB$ and $\triangle XAB \sim \triangle XDE$, we have

By symmetry, we have $\frac{\sin \angle EYL}{\sin \angle LYF} = \frac{CF}{BE}$ and $\frac{\sin \angle CZJ}{\sin \angle JZD} = \frac{AD}{CF}$. Therefore, the left-hand side of (1) is

as desired.

4. $\frac{AB}{BC}$ can be $\frac{\sqrt{2}}{2}$ or $\frac{\sqrt{10}}{2}$.

It is well-known that Q is the midpoint of AC . We first consider the case when P lies inside $\triangle ABC$. Let X be the projection of C on BQ , and let Y be the point on the line BQ such that $XY = XC$, and X lies between B and Y .

Note that $\triangle QAP \cong \triangle QCX$, and hence $PQ = XQ$. By the given condition, we have

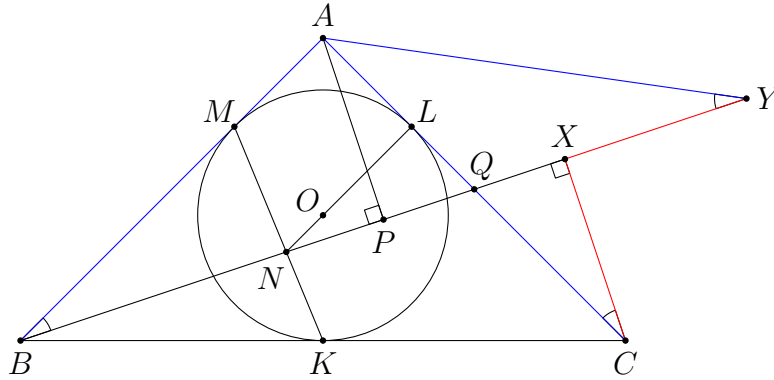
$$BP = AP + 2PQ = CX + PX = XY + PX = PY.$$

Together with $AP \perp BY$, we know that $\triangle ABY$ is an isosceles triangle, with $AY = AB = AC$. Thus, AX is the perpendicular bisector of CY . This implies

$$\angle ACX = \angle AYX = \angle YBA.$$

Therefore, A, B, C, X are concyclic, and $\angle BAC = \angle BXC = 90^\circ$. Conversely, when $\angle BAC = 90^\circ$, one can work backward and show that $BP = AP + 2PQ$. (In that proof, one should define Y as the point on the line BQ such that $AB = AY$, and then show that $XC = XY$.) Thus, this gives one possibility, where

$$\frac{AB}{BC} = \frac{1}{\sqrt{2}}.$$



Next we consider the case when P lies outside $\triangle ABC$. Similarly, let X be the projection of C on BQ . As above, we have $\triangle QAP \cong \triangle QCX$, and hence $QX = QP$. By the given condition, we have

$$BP = AP + 2PQ = CX + PX.$$

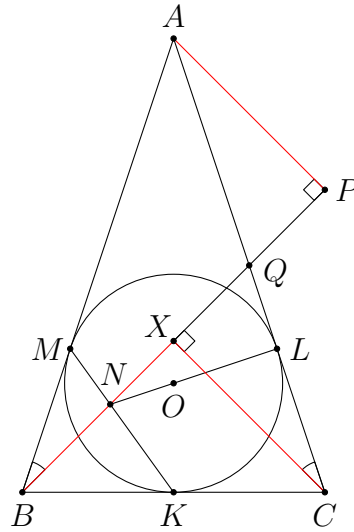
This implies $BX = CX$. Therefore, $BC = \sqrt{2}CX = \sqrt{2}AP$. Let $PQ = QX = x$ and $CX = AP = y$. Noting that $\angle PBA = \angle XCQ$, we have $\triangle PAB \sim \triangle XQC$.

It follows that $\frac{PA}{XQ} = \frac{PB}{XC}$. This implies

$$\frac{y}{x} = \frac{2x + y}{y}.$$

$$\frac{AB}{BC} = \frac{\sqrt{PA^2 + PB^2}}{BC} = \frac{\sqrt{y^2 + (2x + y)^2}}{\sqrt{2}y} = \frac{\sqrt{10}}{2}.$$

When $\frac{AB}{BC} = \frac{\sqrt{10}}{2}$, it is not hard to find that $\frac{BQ}{AB} = \frac{3}{2\sqrt{5}}$, and $\cos \angle CBQ = \frac{1}{\sqrt{2}}$. Then one can deduce $BX = CX$, and the condition holds.



To conclude, $\frac{AB}{BC}$ can be $\frac{\sqrt{2}}{2}$ or $\frac{\sqrt{10}}{2}$.

Year 2008

Test 1

1. The minimum value of xy is $\frac{1}{16120225}$.

Note that the left-hand side is positive. Therefore, by rewriting the right-hand side as

$$(x - y + 1) + \frac{1}{x - y + 1},$$

we know that $x - y + 1 > 0$. Thus, we can apply the AM-GM inequality to obtain

$$(x - y + 1) + \frac{1}{x - y + 1} \geq 2.$$

But then $1 + \cos^2(2007x + 2008y - 1) \leq 2$. This shows equality should hold, and hence

$$x - y + 1 = 1 \quad \text{and} \quad \cos(2007x + 2008y - 1) = \pm 1.$$

The first relation gives $x = y$. The second relation gives $4015x - 1 = k\pi$ for some $k \in \mathbb{Z}$, and so

$$x = y = \frac{1 + k\pi}{4015}.$$

Since k is an integer, it is obvious that $|1 + k\pi| \geq 1$. This yields

$$xy = \frac{(1 + k\pi)^2}{4015^2} \geq \frac{1}{16120225}.$$

Equality holds when $x = y = \frac{1}{4015}$.

2. Note that the product of three consecutive integers is divisible by 6. Let the product be $6k$ for some $k \in \mathbb{Z}$. Then we easily check that

$$(k + 1)^3 + (k - 1)^3 + (-k)^3 + (-k)^3 = 6k.$$

This completes the proof.

3. (a) Let $x = \frac{b + c - a}{2}$, $y = \frac{c + a - b}{2}$ and $z = \frac{a + b - c}{2}$. Then we have

$$\begin{aligned} & \frac{a}{b + c - a} + \frac{b}{c + a - b} + \frac{c}{a + b - c} \geq 3 \\ \Leftrightarrow & \frac{\frac{y + z}{2x}}{\frac{y + z}{2x} + \frac{z + x}{2y} + \frac{x + y}{2z}} + \frac{\frac{z + x}{2y}}{\frac{y + z}{2x} + \frac{z + x}{2y} + \frac{x + y}{2z}} + \frac{\frac{x + y}{2z}}{\frac{y + z}{2x} + \frac{z + x}{2y} + \frac{x + y}{2z}} \geq 3 \\ \Leftrightarrow & \frac{y}{x} + \frac{z}{x} + \frac{z}{y} + \frac{x}{y} + \frac{x}{z} + \frac{y}{z} \geq 6. \end{aligned}$$

This holds by the AM-GM inequality. Equality holds when $x = y = z$, i.e. $a = b = c$.

(b) By the power mean inequality and the triangle inequality, we have

$$a^k + b^k \geq 2 \left(\frac{a+b}{2} \right)^k > 2 \left(\frac{c}{2} \right)^k = \frac{c^k}{2^{k-1}}.$$

(c) By the power mean inequality, we have

$$a^k + b^k + c^k \geq 3 \left(\frac{a+b+c}{3} \right)^k = 3 \left(\frac{p}{3} \right)^k = \frac{p^k}{3^{k-1}}.$$

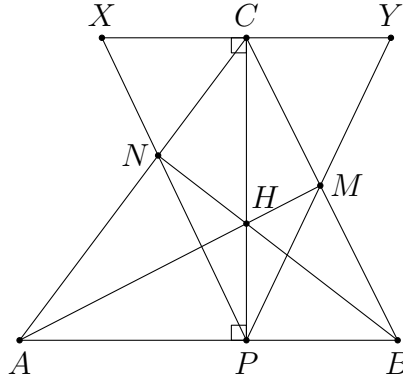
Equality holds when $a = b = c$.

4. (Blanchet's theorem) Suppose the line passing through C and parallel to AB meets PN and PM at X and Y respectively. Since $PC \perp XY$, the result is the same as $CX = CY$ (as this implies $\triangle PCX \cong \triangle PCY$).

By Ceva's theorem, we have

$$\frac{AP}{PB} \times \frac{BM}{MC} \times \frac{CN}{NA} = 1.$$

Note that $\frac{BM}{MC} = \frac{PB}{YC}$ and $\frac{CN}{NA} = \frac{XC}{PA}$ by similar triangles. It follows that $CX = CY$ as desired.



5. Note that for any positive integers x and y satisfying $x + y = 2^r$, we have

$$v_2(x) = v_2(2^r - y) = v_2(y),$$

where $v_2(n)$ is the largest m such that $2^m \mid n$. Therefore, WLOG we may assume $x_k < y_k$ in some step. Since

$$v_2(x_{k+1}) = v_2(2x_k) = v_2(x_k) + 1,$$

the sequence $\{v_2(x_k)\}_{k \geq 1}$ is strictly increasing. But then the sequence is bounded above by $r-1$ since $x_k < 2^r$. Therefore, it must be a finite sequence, which means the procedure will end in finitely many steps.

6. The answer is 28800.

Define

$$\begin{aligned} X &= \{2x_j : 1 \leq j \leq 10\} = \{2, 4, 6, \dots, 20\}, \\ Y &= \{3x_j : 1 \leq j \leq 10\} = \{3, 6, 9, \dots, 30\}. \end{aligned}$$

After removing the absolute value signs of the given expression, we get a sum of 10 terms from $X \cup Y$ minus the sum of the remaining 10 terms from $X \cup Y$. Therefore, we must have

$$\begin{aligned} S &\leq (30 + 27 + 24 + 21 + 18 + 15 + 20 + 18 + 16 + 14) \\ &\quad - (3 + 6 + 9 + 12 + 2 + 4 + 6 + 8 + 10 + 12) \\ &= 131. \end{aligned}$$

In the following, we will show that equality can be attained, and hence $S = 131$. Indeed, we shall count the number of permutations such that the sum is S .

In order that each of $2s$ for $s = 7, 8, 9, 10$ and $3t$ for $t = 5, 6, 7, 8, 9, 10$ is the larger term of the pair in the same absolute value sign, the *large numbers* 7, 8, 9, 10 cannot be adjacent terms (where x_{10} and x_1 are considered as adjacent). Also, 5, 6 cannot be the term immediately after the large numbers. Similarly, the *small numbers* 1, 2, 3, 4 cannot be adjacent, and 5, 6 cannot be the term immediately before the small numbers. Conversely, whenever all these conditions are satisfied, the given expression is equal to 131.

Note that for any pair of large numbers, there must be a small number in between (possibly together with 5 and/or 6), and vice versa. WLOG assume 1 is the first term. There are $3! \times 4! = 144$ ways to arrange the small numbers and the large numbers (for example, 1, 7, 2, 8, 3, 9, 4, 10). Afterwards, we can only place 5 and 6 in the 4 gaps between a small number and a large number (but not between a large number and a small number). By some basic counting, we know that there are $4 \times 5 = 20$ ways to do so (4 ways to place 5, and then 5 ways to place 6 since there is one more position in the same gap as 5). As we can shift all terms cyclically in 10 ways, the final answer is

$$144 \times 20 \times 10 = 28800.$$

Test 2

1. $f(x)$ can be any polynomial of the form

$$pa_p \binom{x}{p} + a_{p-1} \binom{x}{p-1} + a_{p-2} \binom{x}{p-2} + \dots + a_1 \binom{x}{1} + a_0,$$

where $a_0, a_1, \dots, a_p$ are arbitrary integers with $a_p \neq 0$.

Note that we can always write

$$f(x) = c_p \binom{x}{p} + c_{p-1} \binom{x}{p-1} + \cdots + c_1 \binom{x}{1} + c_0$$

for some rational coefficients $c_0, c_1, \dots, c_p$, since $\binom{x}{p}, \binom{x}{p-1}, \dots, \binom{x}{0}$ have different degrees. By putting $x = 0$, we get $c_0 = f(0) \in \mathbb{Z}$. By putting $x = 1$, we get $c_1 + c_0 = f(1) \in \mathbb{Z}$, and hence $c_1 \in \mathbb{Z}$. Similarly, by putting $x = 2, 3, \dots, p$, we find that all c_j 's are integers. Conversely, if all c_j 's are integers, it is obvious that $f(m) \in \mathbb{Z}$ for any $m \in \mathbb{Z}$.

Now, putting $x = p + 1$ and using $\binom{p+1}{k} = \binom{p}{k} + \binom{p}{k-1}$, we obtain

$$f(p+1) - f(1) = \sum_{k=0}^p c_k \binom{p+1}{k} - (c_1 + c_0) = \sum_{k=1}^p c_k \left[\binom{p}{k} + \binom{p}{k-1} \right] - c_1.$$

Since $p \mid \binom{p}{r}$ whenever $1 \leq r \leq p-1$, we have

$$f(p+1) - f(1) \equiv c_p \pmod{p}.$$

Therefore, $p \mid f(p+1) - f(1)$ if and only if $p \mid c_p$. This shows $f(x)$ can be any polynomial of the form mentioned at the beginning.

2. (a) We have

$$\begin{aligned} & \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - \left(\frac{1}{a} - \frac{1}{b} \right)^2 - \left(\frac{1}{b} - \frac{1}{c} \right)^2 - \left(\frac{1}{c} - \frac{1}{a} \right)^2 \\ &= 2 \left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \right) - \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) \\ &= \left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \right) - \frac{1}{2} \left[\left(\frac{1}{a} - \frac{1}{b} \right)^2 + \left(\frac{1}{b} - \frac{1}{c} \right)^2 + \left(\frac{1}{c} - \frac{1}{a} \right)^2 \right] \\ &\leq \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = \frac{a+b+c}{abc} \\ &= \frac{a+b+c}{4RS} = \frac{\sin A + \sin B + \sin C}{2S}, \end{aligned}$$

where we have used the fact $S = \frac{abc}{4R}$ and the extended sine law. Now, note that $f(x) = \sin x$ is a concave function on $(0, \pi)$ since $f''(x) = -\sin x < 0$. Therefore, by Jensen's inequality, we have

$$\frac{\sin A + \sin B + \sin C}{2S} \leq \frac{3}{2S} \sin \frac{A+B+C}{3} = \frac{3\sqrt{3}}{4S}.$$

The result follows readily. Equality holds when $a = b = c$.

(b) Let r and s be the inradius and the semiperimeter of $\triangle ABC$. Then we have $\tan \frac{A}{2} = \frac{r}{s-a}$, etc. Also, using $S = rs = \sqrt{s(s-a)(s-b)(s-c)}$, we have

$$\begin{aligned}
& \tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} \leq \frac{9R^2}{4S} \\
\Leftrightarrow & \frac{r}{s-a} + \frac{r}{s-b} + \frac{r}{s-c} \leq \frac{9R^2}{4S} \\
\Leftrightarrow & \frac{rs[(s-a)(s-b) + (s-b)(s-c) + (s-c)(s-a)]}{s(s-a)(s-b)(s-c)} \leq \frac{9R^2}{4S} \\
\Leftrightarrow & 3s^2 - 2(a+b+c)s + (ab+bc+ca) \leq \frac{9R^2}{4} \\
\Leftrightarrow & -(a+b+c)^2 + 4(ab+bc+ca) \leq 9R^2.
\end{aligned}$$

Recall that

$$9R^2 - (a^2 + b^2 + c^2) = OH^2 \geq 0,$$

where O and H are the circumcentre and orthocentre of $\triangle ABC$ respectively. Therefore, it suffices to prove

$$-(a+b+c)^2 + 4(ab+bc+ca) \leq a^2 + b^2 + c^2.$$

This is equivalent to

$$ab+bc+ca \leq a^2 + b^2 + c^2,$$

which is a well-known inequality. Thus, the inequality is proven. Equality holds when $\triangle ABC$ is equilateral.

3. (Bulgaria MO 2007 Problem 3 simplified) Let $\alpha = \cos \frac{\pi}{7}$. As $\cos \frac{4\pi}{7} + \cos \frac{3\pi}{7} = 0$, we have

$$2(2\alpha^2 - 1)^2 - 1 + (4\alpha^3 - 3\alpha) = 0.$$

This is the same as $(\alpha + 1)(8\alpha^3 - 4\alpha^2 - 4\alpha + 1) = 0$. Clearly, $\alpha \neq -1$. Therefore, α is a root of

$$P(x) = 8x^3 - 4x^2 - 4x + 1.$$

Suppose on the contrary that $\alpha = p + \sqrt{q} + \sqrt[3]{r}$. Then α is also a root of

$$Q(x) = (x - p - \sqrt{q})^3 - r = x^3 - 3(p + \sqrt{q})x^2 + 3(p + \sqrt{q})^2x - (p + \sqrt{q})^3 - r.$$

Consider the field $F = \{a + b\sqrt{q} : a, b \in \mathbb{Q}\}$ (which is just $\mathbb{Q}$ if $q = 0$). We claim that $P(x)$ has a root in F .

Firstly, note that $P(x) \neq 8Q(x)$. Indeed, if $P(x) = 8Q(x)$, we must have $q = 0$ in order that all coefficients of $Q(x)$ are rational. Then by comparing the coefficients of x^3 and x^2 , we need $p = \frac{1}{6}$. But then the coefficients of x do not match. Therefore, $R(x) = \gcd(P(x), Q(x))$ (over $F[x]$) has degree 1 or 2.

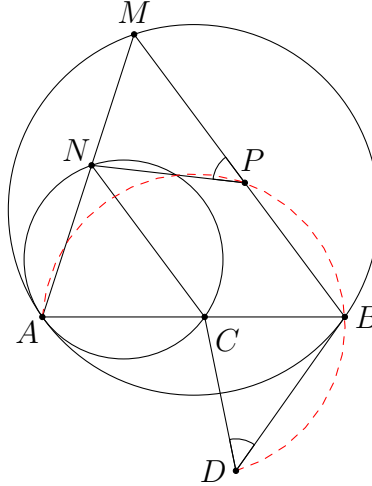
- If $\deg R = 1$, then since α is a root of $P(x)$ and $Q(x)$, it is a root of $R(x)$, and hence belongs to F .
- If $\deg R = 2$, we can write $P(x) = R(x)S(x)$ where $\deg S = 1$. Then $S(x)$ has a root in F , which is also a root of $P(x)$.

Now, let $\beta = a + b\sqrt{q}$ be a root of $P(x)$ in F . Then β is a root of $(x - a)^2 = b^2q$, and so the minimal polynomial of β over $\mathbb{Q}$ has degree at most 2. As this minimal polynomial divides $P(x)$, we find that $P(x)$ is reducible. As $\deg P = 3$, it must consist of a linear factor. Thus, $P(x)$ has a rational root. However, it is routine to check that none of

$$\pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}$$

is a root of $P(x)$. By the rational root theorem, $P(x)$ does not have any rational root. This is a contradiction. Therefore, α cannot be expressed in the form $p + \sqrt{q} + \sqrt[3]{r}$.

4. Let AB meet the inner circle at C . Let D be the point on the tangent at B to the outer circle such that $\angle BDC = \theta$, and D lies on opposite side of AB as M . We claim that the locus of P is $\widehat{ABD}$.



Firstly, since A is the homothetic centre of the two circles, we have $NC \parallel MB$. It follows that $\frac{AN}{NM} = \frac{AC}{CB}$. Secondly, since $\angle NMP = \angle CBD$ by the tangent and $\angle MPN = \angle BDC = \theta$, we have $\triangle MNP \sim \triangle BCD$. Note that the triangles are directly similar. Therefore, $MNAP$ is similar to $BCAD$. Thus, A is the centre of spiral similarity mapping MP to BD . It follows that $MB \cap PD = P$ lies on (ABD) , which is fixed.

It remains to notice that P approaches A when M approaches A , and P approaches D when M approaches B . Due to continuity, the locus of P is $\widehat{ABD}$ from A to D .

Year 2009

Test 1

1. The maximum value is 1004.

Note that $\sin \theta_j \cos \theta_{j+1} \leq \frac{1}{2}(\sin^2 \theta_j + \cos^2 \theta_{j+1})$ for any j , where the indices are taken modulo 2008. It follows that

$$\sum_{j=1}^{2008} \sin \theta_j \cos \theta_{j+1} \leq \frac{1}{2} \sum_{j=1}^{2008} (\sin^2 \theta_j + \cos^2 \theta_j) = \frac{1}{2} \sum_{j=1}^{2008} 1 = 1004.$$

Equality holds when $\theta_j = \frac{\pi}{4}$ for each j .

2. There is only 1 solution modulo 37.

The first four equations can be rewritten as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where the entries are taken modulo 37. The last equation means $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1$. Therefore, we need to count the number of 2×2 matrices A with entries in $\mathbb{F}_{37}$ such that $A^2 = A$ and $\det A = 1$.

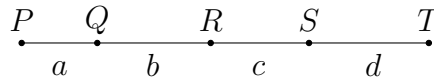
Since $\det A = 1 \neq 0$, A is invertible. Thus, we multiply both sides of $A^2 = A$ by A^{-1} to obtain $A = I$. Clearly, the identity matrix I satisfies the conditions. So $A = I$ is the only solution. Equivalently, this means $a \equiv d \equiv 1 \pmod{37}$ and $b \equiv c \equiv 0 \pmod{37}$.

3. We first states two results which will be used later.

Claim. Let P, Q, R, S, T be five points lying on a line in this order. Let $PQ = a$, $QR = b$, $RS = c$ and $ST = d$. If $b(a + b) = c(c + d)$, then

$$\frac{a+b}{c+d} \times \frac{b+c+d}{a+b+c} = 1, \tag{1}$$

$$\frac{b}{a+b} \times \frac{a+b+c+d}{b+c+d} \times \frac{a+b+c}{b+c} = 1. \tag{2}$$



Proof. For (1), we have

$$\begin{aligned} (a+b)(b+c+d) &= b(a+b) + (a+b)(c+d) = c(c+d) + (a+b)(c+d) \\ &= (c+d)(a+b+c). \end{aligned}$$

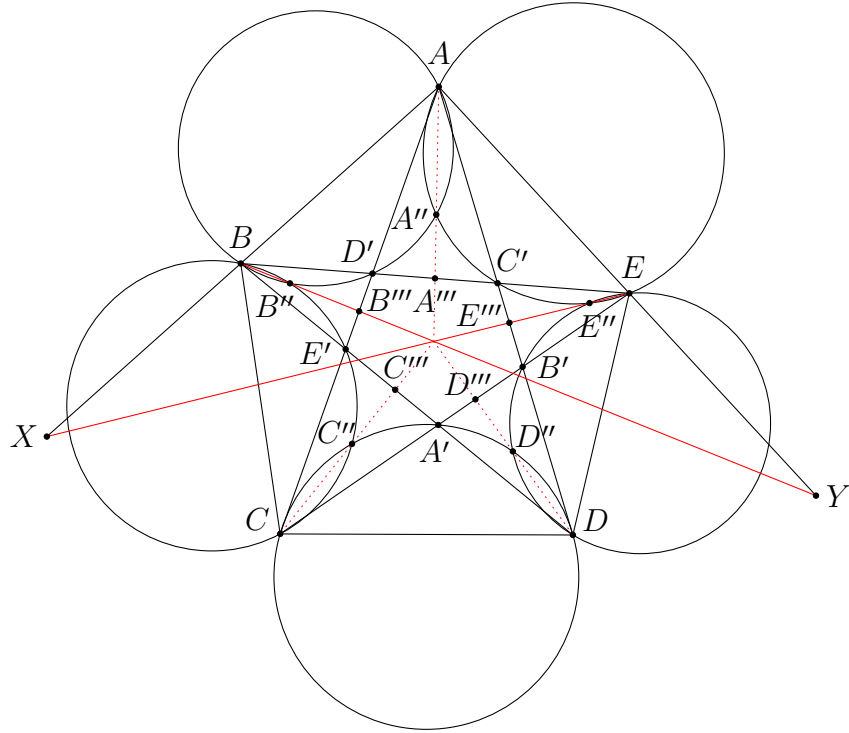
For (2), using (1), we only need to prove $b(a+b+c+d) = (c+d)(b+c)$. Indeed, we have

$$b(a+b+c+d) = b(a+b) + b(c+d) = c(c+d) + b(c+d) = (c+d)(b+c).$$

□

Let $AA'' \cap EB = A'''$, $BB'' \cap AC = B'''$, $CC'' \cap BD = C'''$, $DD'' \cap CE = D'''$ and $EE'' \cap DA = E'''$. By symmetry, it suffices to prove EE''' , AA''' , BB''' are concurrent. Let $X = AB \cap EE'''$ and $Y = AE \cap BB'''$. Applying Ceva's theorem to $\triangle ABE$, we need to prove

$$\frac{BA'''}{A'''E} \times \frac{EY}{YA} \times \frac{AX}{XB} = 1. \quad (3)$$



We apply Menelaus' theorem to obtain the following relations.

$$\begin{aligned} \frac{EY}{YA} \times \frac{AB'''}{B'''D'} \times \frac{D'B}{BE} &= -1 \quad (\triangle AD'E \text{ and line } BB'''Y), \\ \frac{AX}{XB} \times \frac{BE}{EC'} \times \frac{C'E'''}{E'''A} &= -1 \quad (\triangle ABC' \text{ and line } EE'''X), \\ \frac{C'E}{ED'} \times \frac{D'C}{CA} \times \frac{AB'}{B'C'} &= -1 \quad (\triangle AD'C' \text{ and line } CB'E), \\ \frac{C'B}{BD'} \times \frac{D'E'}{E'A} \times \frac{AD}{DC'} &= -1 \quad (\triangle AD'C' \text{ and line } BE'D). \end{aligned}$$

Multiplying these and rearranging the terms, we find that

$$\begin{aligned} & \frac{BA'''}{A'''E} \times \frac{EY}{YA} \times \frac{AX}{XB} \\ &= \left(\frac{BA'''}{A'''E} \times \frac{ED'}{BC'} \right) \times \left(\frac{B'''D'}{AB'''} \times \frac{AC}{CD'} \times \frac{AE'}{E'D'} \right) \times \left(\frac{E'''A}{C'E'''} \times \frac{C'B'}{B'A} \times \frac{C'D}{DA} \right). \end{aligned} \quad (4)$$

Since A''' lies on the radical axis AA'' of (ABD') and $(AC'E)$, we consider the powers and obtain $A'''D' \times A'''B = A'''C' \times A'''E$. This shows the condition in the claim is satisfied. By (1), we have

$$\frac{BA'''}{A'''E} \times \frac{ED'}{BC'} = 1.$$

Similarly, the points A, D', B''', E', C and A, C', E''', B', D satisfy the condition in the claim. By (2), we have

$$\frac{B'''D'}{AB'''} \times \frac{AC}{CD'} \times \frac{AE'}{E'D'} = \frac{E'''A}{C'E'''} \times \frac{C'B'}{B'A} \times \frac{C'D}{DA} = 1.$$

Putting all these into (4), we obtain (3). The proof is complete.

4. (a) The smallest number of committees in the school is 6.

If a student joins at most 2 committees, that student shares a common committee with at most $2(1004 - 1) = 2006 < 2007$ students, which contradicts the assumption. Therefore, each student joins at least 3 committees. Thus, there are at least

$$\frac{3 \times 2008}{1004} = 6$$

committees.

The minimum value can be attained. For example, partition all students into 8 groups $A, B, \dots, H$, each consisting of 251 students. We form the following 6 committees:

$$\begin{aligned} & \{A, B, C, D\}, \{A, E, F, G\}, \{A, B, E, H\}, \\ & \{B, F, G, H\}, \{C, D, G, H\}, \{C, D, E, F\}. \end{aligned}$$

Each committee contains $251 \times 4 = 1004$ students. One can check that every pair of groups belongs to at least one common committee, and so every pair of students joins a common committee.

- (b) Yes. One can check that every pair of committees in the example provided in part (a) consists of at most 7 groups only, and so the union consists of at most

$$251 \times 7 = 1757 < 1800$$

students.

5. The answer is any real number r such that $1 \leq r < 2$.

Firstly, we have $\frac{a^2 + b^2 + c^2}{ab + bc + ca} \geq 1$ since this is equivalent to

$$\frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2) \geq 0.$$

Secondly, by the triangle inequality, we have

$$\frac{a^2 + b^2 + c^2}{ab + bc + ca} < \frac{a(b+c) + b(c+a) + c(a+b)}{ab + bc + ca} = 2.$$

Thus, we have proved the proposed bounds. Now, for any $r \in [1, 2)$, we show that the triangle with side lengths $a = b = 1$ and $c = r - \sqrt{r^2 + r - 2}$ satisfies $\frac{a^2 + b^2 + c^2}{ab + bc + ca} = r$. Note that

$$f(r) = r - \sqrt{r^2 + r - 2} = \frac{2 - r}{r + \sqrt{r^2 + r - 2}}$$

is a decreasing function on $[1, 2)$. Since $f(1) = 1$ and $f(2) = 0$, we have $0 < c \leq 1$. Therefore, the triangle inequality is satisfied, and a, b, c can be the side lengths of a triangle. Now,

$$\begin{aligned} & \frac{a^2 + b^2 + c^2}{ab + bc + ca} = r \\ \Leftrightarrow & c^2 + 2 = r(2c + 1) \\ \Leftrightarrow & c^2 - 2rc + (2 - r) = 0 \\ \Leftrightarrow & c = r \pm \sqrt{r^2 + r - 2}. \end{aligned}$$

This proves the claim, and so we are done.

6. Since $p \leq 100$, we have $(p-1, 37) = 1$ (by checking $1, 38, 75$ are not primes). Thus, there exists $k \in \mathbb{Z}^+$ such that $37k \equiv 1 \pmod{p-1}$. Thus, by the Fermat little theorem, we have

$$(11^k)^{37} = 11^{37k} \equiv 11 \pmod{p}.$$

Therefore, $(x, y) = (0, 11^k)$ is a solution to the given system.

Test 2

1. (a) We have $f(x) = x^3 + 4x^2 - 4$.

Label the equation as follows.

$$f(x+y) = f(x) + f(y) + axy(x+y) + bxy + c(x+y) + 4 \quad (1)$$

Putting $x = y = 0$ in (1), we have

$$f(0) = f(0) + f(0) + 4,$$

which implies $f(0) = -4$. Next, putting $x = 1$ and $y = 0$ in (1), we have

$$f(1) = f(1) + f(0) + c + 4.$$

This yields $c = 0$. Now, putting $x = y = 1$ in (1), we have

$$f(2) = f(1) + f(1) + 2a + b + 4.$$

As it is given that $f(1) = 1$ and $f(2) = 20$, we obtain

$$2a + b = 14. \quad (2)$$

Also, we put $x = y = 2$ in (1). This gives

$$f(4) = f(2) + f(2) + 16a + 4b + 4.$$

This means $f(4) = 16a + 4b + 44$. Moreover, putting $x = 4$ and $y = -4$ in (1), we find that

$$f(0) = f(4) + f(-4) - 16b + 4.$$

This becomes

$$4a - 3b = -12. \quad (3)$$

Solving (2) and (3), we have

$$a = \frac{3(2a + b) + (4a - 3b)}{10} = 3$$

and $b = 14 - 2a = 8$. Equation (1) becomes

$$f(x + y) = f(x) + f(y) + 3xy(x + y) + 8xy + 4. \quad (4)$$

Putting $y = 1$ in (4), we get

$$f(x + 1) = f(x) + 3x(x + 1) + 8x + 5.$$

For $x \in \mathbb{Z}^+$, adding

$$\begin{aligned} f(x) &= f(x - 1) + 3(x - 1)x + 8(x - 1) + 5, \\ f(x - 1) &= f(x - 2) + 3(x - 2)(x - 1) + 8(x - 2) + 5, \\ &\dots, \\ f(2) &= f(1) + 3(1)(2) + 8(1) + 5, \end{aligned}$$

we obtain

$$\begin{aligned} f(x) &= f(1) + 3 \sum_{k=1}^{x-1} k(k + 1) + 8 \sum_{k=1}^{x-1} k + 5(x - 1) \\ &= (x - 1)x(x + 1) + 4(x - 1)x + 5x - 4 \\ &= x^3 + 4x^2 - 4. \end{aligned}$$

This also holds when $x = 0$ as $f(0) = -4$. Now, putting $y = -x$ in (4), we have

$$f(0) = f(x) + f(-x) - 8x^2 + 4.$$

For $x < 0$, this yields

$$f(x) = -f(-x) + 8x^2 - 8 = -((-x)^3 + 4(-x)^2 - 4) + 8x^2 - 8 = x^3 + 4x^2 - 4.$$

So the same formula holds for all $x \in \mathbb{Z}$.

Lastly, for $f(x) = x^3 + 4x^2 - 4$, we check that

$$\begin{aligned} f(x+y) &= (x+y)^3 + 4(x+y)^2 - 4 \\ &= x^3 + 3x^2y + 3xy^2 + y^3 + 4x^2 + 8xy + 4y^2 - 4 \end{aligned}$$

and

$$\begin{aligned} f(x) + f(y) + 3xy(x+y) + 8xy + 4 \\ = x^3 + 4x^2 - 4 + y^3 + 4y^2 - 4 + 3xy(x+y) + 8xy + 4. \end{aligned}$$

It is routine to check that the two sides are identical. Thus, this is the only solution.

(b) The greatest possible value of m is -1 .

Note that

$$\begin{aligned} x^3 + 4x^2 - 4 &\geq mx^2 + (5m+1)x + 4m \\ \Leftrightarrow x^3 + 4x^2 - x - 4 &\geq mx^2 + 5mx + 4m \\ \Leftrightarrow (x+1)(x+4)(x-1) &\geq m(x+1)(x+4) \\ \Leftrightarrow x-1 &\geq m. \end{aligned}$$

By putting $x = 0$, we need $m \leq -1$. When $m = -1$, this inequality holds for all $x \geq 0$. Thus, the greatest possible value of m is -1 .

2. (IMO Shortlist 2001 C3) We consider two cases.

Case 1. There exist two 3-cliques sharing two common people.

Suppose the two 3-cliques are $\{A, B, C\}$ and $\{A, B, D\}$. If all 3-cliques contain A or B , we are done as we can remove A and B . If there exists a 3-clique without A and B , it must be $\{C, D, E\}$ since every pair of 3-cliques has a common person. We claim that there is no 3-clique if C and D are removed.

Indeed, note that $\{A, C, D\}$, $\{B, C, D\}$ and $\{C, D, E\}$ are existing 3-cliques since all pairs of people in these 3-cliques know each other from above. Therefore, the only possible 3-clique without C and D is $\{A, B, E\}$. Then $\{A, B, C, D, E\}$ forms a 5-clique, which is a contradiction.

Case 2. Any pair of 3-cliques shares exactly one common person.

Suppose two 3-cliques are $\{A, B, C\}$ and $\{A, D, E\}$. If all 3-cliques contain A , we are done as we can remove A . If there exists a 3-clique without A , it must be $\{B, D, F\}$ up to renaming of the people. Then we have a 3-clique $\{A, B, D\}$ sharing two common people with $\{A, B, C\}$. This is a contradiction.

The proof is complete since all cases are covered.

3. By Dirichlet's theorem, there are infinitely many primes p of the form $3k + 2$. We claim that $N_p = p^2$ for all such primes p .

Indeed, for each of the p^2 pairs of (y, z) modulo p , we need to solve

$$x^3 \equiv -3^{-1}(4y^3 + 5z^3 - y^4z) \pmod{p}.$$

As $(p-1, 3) = 1$, this equation has a unique solution in x modulo p . This proves $N_p = p^2$.

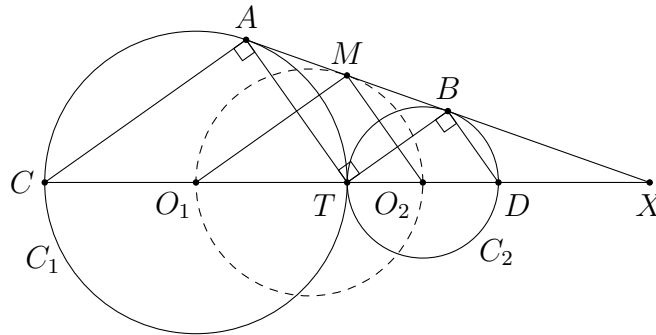
4. (a) Let O_1 and O_2 be the centres of C_1 and C_2 respectively. Let O_1O_2 meet C_1 and C_2 again at C and D respectively. Note that CT and DT are diameters of the two circles. Therefore, we have

$$\angle CAT = \angle BTA = \angle TBD = 90^\circ.$$

This implies $AC \parallel BT$ and $AT \parallel BD$, and so $\triangle ACT \sim \triangle BTD$. Since the radii are distinct, the triangles are not congruent. Thus, AB meets CD at a point X . Note that $\triangle XAC \sim \triangle XBT$. Using the similar triangles, we obtain

$$\frac{XT}{XC} = \frac{BT}{AC} = \frac{TD}{CT},$$

which is a fixed ratio. Therefore, X is a fixed point (as directed lengths are used). In other words, all such lines AB are concurrent at X .



(b) The locus is the circle with diameter O_1O_2 .

Let M be the midpoint of AB . Since M and O_1 are the midpoints of AB and CT , we have $AC \parallel MO_1 \parallel BT$. Similarly, we have $AT \parallel MO_2 \parallel BD$. This implies $\angle O_1MO_2 = 90^\circ$, and hence M lies on the circle Γ with diameter O_1O_2 .

When A approaches C from above, the point B approaches T from above. This shows M approaches O_1 from above. When A approaches C from below, the point B approaches T from below. This shows M approaches O_1 from below. Also, $M = O_1$ only when $A = C$ and $B = T$. By continuity, M runs through all points on Γ , and so Γ is the locus of M .

Year 2010

Test 1

1. n can only be 25.

We first provide an example for the case $n = 25$. Consider 10 participants $A_1, A_2, \dots, A_{10}$ in group $\mathcal{A}$, and 15 participants $B_1, B_2, \dots, B_{15}$ in group $\mathcal{B}$. Suppose the games played between 2 participants in the same group end in a draw. For the games played between 2 participants in different groups, the results are given below, where O means a draw, while an empty box means the group B participant wins.

	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}
B_1	O	O	O	O	O	O				
B_2	O	O	O	O	O	O				
B_3		O	O	O	O	O	O			
B_4			O	O	O	O	O	O		
B_5			O	O	O	O	O	O		
B_6				O	O	O	O	O	O	
B_7					O	O	O	O	O	O
B_8					O	O	O	O	O	O
B_9	O					O	O	O	O	O
B_{10}	O	O					O	O	O	O
B_{11}	O	O					O	O	O	O
B_{12}	O	O	O					O	O	O
B_{13}	O	O	O	O					O	O
B_{14}	O	O	O	O					O	O
B_{15}	O	O	O	O	O					O

The score of each participant in group $\mathcal{A}$ is $9 + 9 = 18$, where 9 points come from the games played between them. The score of each participant in group $\mathcal{B}$ is $14 + (6 + 2 \times 4) = 28$, where 14 points come from the games played between them. So the conditions are satisfied.

Now, note that the two participants in a game always gain 2 points in total. Therefore, the total score of all participants is

$$2 \binom{n}{2} = n(n-1).$$

For the 10 participants with the lowest scores, their total score gained by the games played between them is $2 \binom{10}{2} = 90$. Therefore, their total score is

$$2 \times 90 = 180.$$

For the other $n - 10$ participants, their total score gained by the games played between them is $2 \binom{n-10}{2} = (n-10)(n-11)$. Note that this is half of their total

score since the remaining half comes from the other games. Therefore, their total score is

$$2(n-10)(n-11).$$

Combining all these, we need $180 + 2(n-10)(n-11) = n(n-1)$. This is the same as $(n-16)(n-25) = 0$. Thus, $n = 16, 25$.

For $n = 16$, the 10 participants with the lowest scores obtain 180 points in total, while the other 6 participants obtain $2(16-10)(16-11) = 60$ points in total. Since $\frac{180}{10} = 18 > 10 = \frac{60}{6}$, it is impossible that all those 10 participants have the lowest score. Therefore, $n = 16$ is rejected, and $n = 25$ is the only possibility.

2. Let $x = bc$, $y = ca$ and $z = ab$. The condition becomes $xyz = 1$. Note that

$$\sum_{\text{cyc}} \frac{1}{a^2(b+c)} = \sum_{\text{cyc}} \frac{bc}{a(b+c)} = \sum_{\text{cyc}} \frac{x}{y+z}.$$

It remains to prove

$$\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \geq \frac{3}{2}.$$

This is exactly Nesbitt's inequality. Equality holds when $x = y = z = 1$, i.e. $a = b = c = 1$.

3. The answer is 10.

Let $a \leq b$ be two roots of $x^2 - px + q = 0$ such that $b > 20$. Note that

$$a \leq \frac{a+b}{2} = \frac{p}{2} \leq 15.$$

This shows it is impossible to have both roots greater than 20. Also, we have

$$b = \frac{p + \sqrt{p^2 - 4q}}{2} < \frac{p+p}{2} = p.$$

This implies $p \geq 21$, and so p can only be $21, 22, \dots, 30$. Thus, at most 10 equations may have a root greater than 20.

Consider the equations $x^2 - (20+k)x + k = 0$ for $k = 1, 2, \dots, 10$. Note that

$$\Delta = (20+k)^2 - 4k = k^2 + 36k + 400 > 0,$$

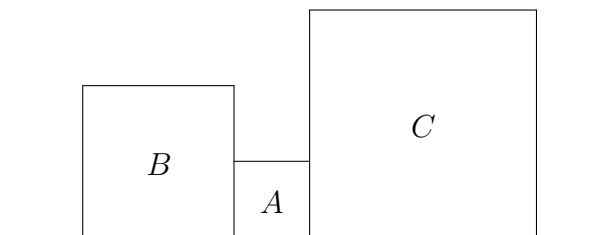
so the roots are real. The larger root is

$$\frac{(20+k) + \sqrt{k^2 + 36k + 400}}{2} > \frac{(20+k) + (20-k)}{2} = 20.$$

This shows it is possible to have 10 roots larger than 20.

4. The smallest possible area is 35.

Let a be the length of the smallest square A . If A touches a side but not a corner, then it is surrounded by two larger squares B and C . Then there is a gap between B and C with length a . It is not possible to place some larger squares in this gap, contradiction. Similarly, the smallest square cannot be placed at a corner.



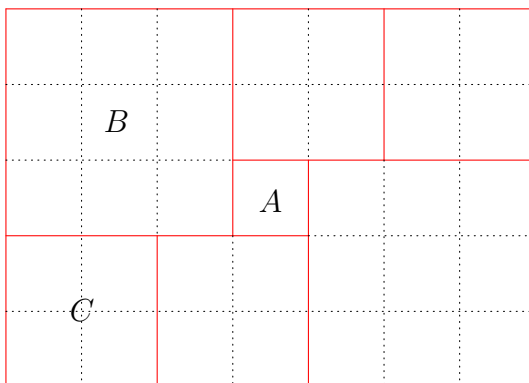
The distance from A to a side is at least $a + 1$. Therefore, each side has length at least

$$(a + 1) \times 2 + a = 3a + 2 \geq 5.$$

If both side lengths are at least 6, then the area is at least 36. Thus, we may assume there is a side with length 5, and $a = 1$.

Since there is no square of side length 1 touching this side, either there is a 5×5 square, or a 3×3 and a 2×2 squares touching this side. The former case can be ignored since we can remove such a 5×5 square. Now, suppose there is a 3×3 square B and a 2×2 square C as shown. Then A cannot be placed in the first 3 columns (or there is a gap of length 1). The same holds for the opposite side. Therefore, the other side length of the rectangle is at least $3 + 1 + 3 = 7$. This shows the area is at least $5 \times 7 = 35$.

An example of a 5×7 rectangle is given below.



5. The answer is 37.

We first prove the following result.

Claim. Consider a positive rational number $\frac{a}{b}$ where $(10a, b) = 1$. The smallest period of this number when expressed as a decimal number is the order of 10 modulo b .

Proof. Note that $\frac{a}{b}$ has period k if and only if the tail part of $10^k \cdot \frac{a}{b}$ is the same as that of $\frac{a}{b}$. This holds if and only if $(10^k - 1) \cdot \frac{a}{b}$ is a terminating decimal, i.e. $10^m(10^k - 1)\frac{a}{b} \in \mathbb{Z}$ for some nonnegative integer m . Equivalently, this means

$$b \mid 10^m(10^k - 1)a.$$

As $(10a, b) = 1$, this is the same as $b \mid 10^k - 1$. By definition, the smallest positive integer k for which this holds is the order of 10 modulo b . $\square$

Now, when n runs through all positive integers, the denominator of $\frac{n}{567}$ in the lowest term can be any positive divisor of $567 = 3^4 \times 7$. Let d_b be the order of 10 modulo b . By the claim, it suffices to find d_b for each positive divisor b of 567.

- Since $10 \equiv 1 \pmod{9}$, we have $d_1 = d_3 = d_9 = 1$.
- Since $10 \not\equiv 1 \pmod{27}$ and $10^3 = 1000 \equiv 1 \pmod{27}$, we have $d_{27} = 3$.
- Since $10^3 = 1000 \not\equiv 1 \pmod{81}$ and
$$10^9 = 1000^3 = (27 \times 37 + 1)^3 \equiv 1 \pmod{81}$$
by the binomial theorem, we have $d_{81} = 9$.
- Since $\varphi(7) = 6$ and $10^2, 10^3 \not\equiv 1 \pmod{7}$, we have $d_7 = 6$.
- For $b = 3^s 7$ where $s = 1, 2, 3, 4$, we have $d_b = [d_{3^s}, d_7]$. Therefore, we easily deduce $d_{21} = d_{63} = d_{189} = 6$, $d_{567} = 18$.

It follows that the answer is $1 + 3 + 6 + 9 + 18 = 37$.

6. Let AO meet BC at D . Since

$$\angle DAC = \angle BAD = \frac{1}{2} \angle BAC = \angle ACD,$$

we have $DA = DC$. Consider $\triangle OCA$ and $\triangle OCD$. By considering their areas, we find that

$$\frac{AO}{OD} = \frac{[OCA]}{[OCD]} = \frac{AC \times d(O, AC)}{CD \times d(O, CD)} = \frac{AC}{CD}.$$

Now, we show that this is the only solution. Let $g(x) = 3x - 1 - f(x)$ for any $x \in \mathbb{R}$. If $g(x) = 0$ and $x \neq 0$, then $f(x) = 3x - 1$ and so $\frac{f(x)}{x} = 3 - \frac{1}{x}$. By condition (i), there are finitely many x such that $g(x) = 0$.

For those $x \in \mathbb{R}$ with $g(x) \neq 0$, we rewrite condition (ii) as follows:

$$\frac{f(g(x))}{g(x)} = \frac{3(f(x) - 1 - 3x)}{3x - 1 - f(x)} = -3 - \frac{6}{3x - 1 - f(x)} = -3 - \frac{6}{g(x)}.$$

By condition (i), the left-hand side only takes finitely many values. This implies $g(x)$ only takes finitely many values (possibly 0).

Consider any $y \in \mathbb{R}$ such that $|f(y) - 3y| = |g(y) + 1|$ is maximized. For this y , we use (ii) to obtain

$$f(g(y)) - 3g(y) = 3(f(y) - 1 - 3y) - 3(3y - 1 - f(y)) = 6(f(y) - 3y).$$

This implies $|f(g(y)) - 3g(y)| = 6|f(y) - 3y|$. By the choice of y , we must have $f(y) - 3y = 0$. Since y is chosen to maximize $|f(y) - 3y|$, we can only have $f(x) = 3x$ for any $x \in \mathbb{R}$.

2. Let s be the semiperimeter of the triangle, and let $x = s - a$, $y = s - b$ and $z = s - c$. Note that $x, y, z > 0$ by the triangle inequality. Now,

$$\begin{aligned} & a^2 + b^2 + c^2 \geq 4\sqrt{3}T + (a - b)^2 + (b - c)^2 + (c - a)^2 \\ \Leftrightarrow & 2ab + 2bc + 2ca \geq 4\sqrt{3s(s - a)(s - b)(s - c)} + a^2 + b^2 + c^2 \\ \Leftrightarrow & 2 \sum_{\text{cyc}} (y + z)(x + z) \geq 4\sqrt{3(x + y + z)xyz} + (y + z)^2 + (z + x)^2 + (x + y)^2 \\ \Leftrightarrow & 4(xy + yz + zx) \geq 4\sqrt{3(x + y + z)xyz} \\ \Leftrightarrow & (xy + yz + zx)^2 \geq 3((xy)(yz) + (yz)(zx) + (zx)(xy)). \end{aligned}$$

This is a well-known inequality (for example, the difference of the left-hand side and the right-hand side is $\frac{1}{2} \sum_{\text{cyc}} (xy - yz)^2$). Equality holds when $xy = yz = zx$, i.e. $a = b = c$.

3. Fix one of the points A . We label the other points $B_1, B_2, \dots, B_{2008}$ such that the rays $AB_1, AB_2, \dots, AB_{2008}$ align in the clockwise direction. Then it is easy to see that among

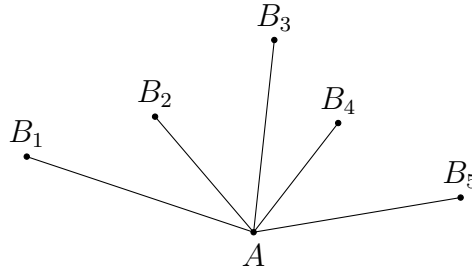
$$\triangle AB_1B_2, \triangle AB_2B_3, \dots, \triangle AB_{2007}B_{2008}, \triangle AB_{2008}B_1,$$

at most one triangle may contain some other points. Indeed, this only happens for $\triangle AB_jB_{j+1}$ when the clockwise angle from AB_j to AB_{j+1} is larger than 180° .

It follows that there are at least 2007 triangles with A as a vertex which do not contain any other points. By considering all choices of A , since each triangle is counted at most 3 times, there are at least

$$\frac{1}{3} \times 2009 \times 2007 = 1344021$$

such triangles.



4. (IMO Shortlist 1988 modified) There are 645 odd coefficients.

In the following, we consider polynomials in $\mathbb{F}_2[x]$. This means all coefficients are taken modulo 2. Let $f(n)$ be the number of odd coefficients in $(x^2 - x + 1)^n$.

Claim 1. We have $(x^2 - x + 1)^{2^k} = x^{2^{k+1}} - x^{2^k} + 1$ for any nonnegative integer k .

Proof. It suffices to note that

$$(x^2 - x + 1)^2 = x^4 - 2x^3 + 3x^2 - 2x + 1 = x^4 - x^2 + 1.$$

The result follows easily by induction. □

Claim 2. We have $f(2^k a + b) = f(a)f(b)$ for any positive integers a, b satisfying $b < 2^{k-1}$.

Proof. By claim 1, we have

$$(x^2 - x + 1)^{2^k a + b} = ((x^2 - x + 1)^{2^k})^a (x^2 - x + 1)^b = (x^{2^{k+1}} - x^{2^k} + 1)^a (x^2 - x + 1)^b.$$

Let

$$(x^{2^{k+1}} - x^{2^k} + 1)^a = \sum_{i=1}^s c_i x^{2^k \alpha_i},$$

$$(x^2 - x + 1)^b = \sum_{j=1}^t d_j x^{\beta_j}.$$

Then their product is

$$\sum_{i=1}^s \sum_{j=1}^t c_i d_j x^{2^k \alpha_i + \beta_j}.$$

Since we have $\beta_j \leq 2b < 2^k$, different pairs (i, j) correspond to different exponents $2^k \alpha_i + \beta_j$. Also, $c_i d_j$ is odd if and only if both c_i and d_j are odd. As there are $f(a)$ odd coefficients c_i and $f(b)$ odd coefficients d_j , the number of odd coefficients is $f(a)f(b)$. $\square$

Claim 3. We have $f(2^k - 1) = \frac{2^{k+2} - (-1)^k}{3}$ for any positive integer k .

Proof. It suffices to make the following observations. For odd k , the coefficients of $(x^2 - x + 1)^{2^k - 1}$ follow the pattern

$$\underbrace{110110 \cdots 110}_{\frac{2^k - 2}{3} \text{ triples } 110} \underbrace{111011011 \cdots 011}_{\frac{2^k - 2}{3} \text{ triples } 011},$$

while for even k , the coefficients of $(x^2 - x + 1)^{2^k - 1}$ follow the pattern

$$\underbrace{110110 \cdots 110}_{\frac{2^k - 1}{3} \text{ triples } 110} \underbrace{1011011 \cdots 011}_{\frac{2^k - 1}{3} \text{ triples } 011}.$$

It is not hard to prove these by induction and claim 1, since $2^{k+1} - 1 = (2^k - 1)2^k$. We omit the details. $\square$

Now, by the above claims, we obtain

$$\begin{aligned} f(2009) &= f(2^6 \times 31 + 25) = f(31)f(25) \\ &= f(31)f(2^3 \times 3 + 1) = f(31)f(3)f(1) \\ &= \frac{2^7 + 1}{3} \cdot \frac{2^4 - 1}{3} \cdot \frac{2^3 + 1}{3} = 645. \end{aligned}$$

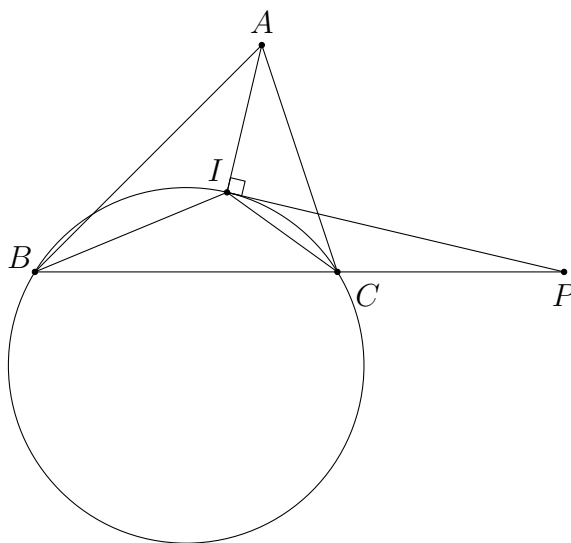
Year 2011

Test 1

1. We have

$$\angle CIP = \angle CIA - 90^\circ = \angle CBI.$$

This shows IP is tangent to (BCI) . Hence, $IP^2 = BP \cdot CP$ by considering the power of P with respect to (BCI) .



2. (IMO 1977 Problem 5 modified) The solutions are $(a, b) = (31, 53), (53, 31)$.

We are given that

$$a^2 + b^2 = Q(a + b) + R.$$

Since $R < a + b$, we have

$$(a + b)^2 \leq 2(a^2 + b^2) < 2(Q + 1)(a + b),$$

which implies $a + b < 2(Q + 1)$, and hence $R < 2(Q + 1)$. Thus, we have

$$Q^2 \leq Q^2 + R \leq Q^2 + 2Q + 1 = (Q + 1)^2.$$

Since $Q^2 + R = 2010$, we must have $Q = \lfloor \sqrt{2010} \rfloor = 44$, and so $R = 2010 - 44^2 = 74$. It remains to solve the equation

$$a^2 + b^2 = 44(a + b) + 74.$$

This can be rewritten as

$$(a - 22)^2 + (b - 22)^2 = 1042.$$

Let $x = (a - 22)^2$ and $y = (b - 22)^2$. WLOG assume $x \geq y$. Then we have $521 \leq x \leq 1042$. So it suffices to check $23^2 \leq x \leq 32^2$. Also, x and y must be odd since $x^2 + y^2 \equiv 2 \pmod{4}$. We compute the values as follows.

x	529	625	729	841	961
y	513	417	313	201	81

Among these values, only $y = 81$ is a perfect square. Therefore, we can only have $(a - 22, b - 22) = (\pm 31, \pm 9)$. Since $a, b \in \mathbb{Z}^+$, this implies $(a, b) = (53, 13), (53, 31)$. However, we need $a + b > R = 74$. Therefore, only $(a, b) = (53, 31)$ is possible. We check that when $(a, b) = (53, 31)$, we have

$$a^2 + b^2 = 3770 = 44(53 + 31) + 74.$$

Therefore, this is the only solution up to permutation.

3. (i) WLOG assume $a \geq b \geq c \geq d$. Then $\frac{a-1}{a} \geq \frac{b-1}{b} \geq \frac{c-1}{c} \geq \frac{d-1}{d}$. By Chebyshev's inequality, we have

$$\sum_{\text{cyc}} \left(a \cdot \frac{a-1}{a} \right) \geq \frac{1}{4} \left(\sum_{\text{cyc}} a \right) \left(\sum_{\text{cyc}} \frac{a-1}{a} \right).$$

Also, by the AM-GM inequality, we have

$$\sum_{\text{cyc}} \frac{a-1}{a} \geq 4 \sqrt[4]{\frac{(a-1)(b-1)(c-1)(d-1)}{abcd}}.$$

Combining these, we obtain

$$(a + b + c + d - 4) \geq (a + b + c + d) \sqrt[4]{\frac{(a-1)(b-1)(c-1)(d-1)}{abcd}}.$$

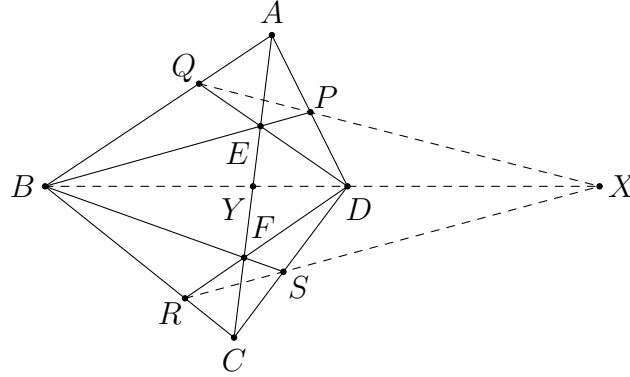
Raising both sides to the fourth power and rearranging the terms, we obtain the desired inequality. Equality holds when $a = b = c = d$.

- (ii) We have

$$\begin{aligned} & \frac{d}{a+b+c} + \frac{a}{b+c+d} + \frac{b}{c+d+a} + \frac{c}{d+a+b} \geq \frac{4}{3} \\ \Leftrightarrow & \frac{a+b+c+d}{a+b+c} + \frac{a+b+c+d}{b+c+d} + \frac{a+b+c+d}{c+d+a} + \frac{a+b+c+d}{d+a+b} \geq \frac{16}{3} \\ \Leftrightarrow & (a+b+c+d) \sum_{\text{cyc}} \frac{1}{a+b+c} \geq \frac{16}{3} \\ \Leftrightarrow & \left(\sum_{\text{cyc}} (a+b+c) \right) \left(\sum_{\text{cyc}} \frac{1}{a+b+c} \right) \geq 16. \end{aligned}$$

This is obviously true by the Cauchy-Schwarz inequality. Equality holds when $a = b = c = d$.

4. Let Y be the intersection point of AC and BD . Since AY, BP, DQ are concurrent, QP meets BD at the harmonic conjugate X of Y with respect to B, D (note that X could be a point at infinity). Similarly, since CY, BS, DR are concurrent, RS meets BD at X . This shows QP, BD, RS are concurrent in the projective sense. This means they are concurrent or parallel.



Remarks. Suppose B, Y, D, X lie on the same line (in this order). We say that X and Y are harmonic conjugate with respect to B and D if $\frac{BX}{XD} = -\frac{BY}{YD}$, where the lengths are directed lengths. Indeed, for this problem, if we let X be the intersection point of QP and BD , then by applying Ceva's theorem and Menelaus' theorem to $\triangle ABD$, we obtain

$$\begin{aligned} \frac{BX}{XD} \times \frac{DP}{PA} \times \frac{AQ}{QB} &= -1, \\ \frac{BY}{YD} \times \frac{DP}{PA} \times \frac{AQ}{QB} &= 1. \end{aligned}$$

Combining these, we get $\frac{BX}{XD} = -\frac{BY}{YD}$. Since directed lengths are used, there is a unique point X satisfying this relation. By doing the same thing for $\triangle CBD$, we see that RS also passes through X .

5. The answer is $1 + \sqrt{2}$.

Let a and b be the numerator and the denominator respectively. For each $n = 1, 2, \dots, 2024$, we have the identity

$$\sqrt{45 + \sqrt{2025 - n}} + \sqrt{45 - \sqrt{2025 - n}} = \sqrt{2} \cdot \sqrt{45 + \sqrt{n}}.$$

This can be easily proved by squaring both sides and simplifying. Now, summing over all n 's, we obtain

$$a + b = \sqrt{2}a.$$

This implies

$$\frac{a}{b} = \frac{a}{\sqrt{2}a - a} = \frac{1}{\sqrt{2} - 1} = 1 + \sqrt{2}.$$

6. At most 21523361 phone numbers can be selected, and there is only 1 way to select this amount of phone numbers.

Consider the situation when the phone numbers have n digits instead of 8. We shall prove the following by induction: at most $\frac{9^n + 1}{2}$ phone numbers can be selected, and there is a unique way to select the phone numbers, which is to choose all phone numbers whose sum has the same parity as n .

When $n = 1$, we can choose at most one phone number from each group

$$(1, 2), (3, 4), (5, 6), (7, 8), (9).$$

Thus, at most $\frac{9 + 1}{2} = 5$ phone numbers can be selected. Clearly, the only way to select 5 phone numbers is to select all odd numbers 1, 3, 5, 7, 9.

Assume the case $n - 1$ is true. Consider the case n . For those phone numbers not starting with the digit 9, we can pair them up such that $\overline{a_1 a_2 \cdots a_n}$ and $(a_1 + 1)a_2 \cdots a_n$ belong to the same pair if $a_1 = 1, 3, 5, 7$. Clearly, at most one phone number can be selected from each pair. Also, for the remaining phone numbers starting with 9, we can remove the digit 9 and regard this as the case $n - 1$. By the inductive hypothesis, we can select at most $\frac{9^{n-1} + 1}{2}$ phone numbers from these. In total, we can select at most

$$\frac{8 \cdot 9^{n-1}}{2} + \frac{9^{n-1} + 1}{2} = \frac{9^n + 1}{2}$$

phone numbers.

To choose this amount of phone numbers, since we need to select $\frac{9^{n-1} + 1}{2}$ phone numbers from those starting with 9, by the inductive hypothesis, the only way is to select all numbers $\overline{9a_2 \cdots a_n}$ with $a_2 + \cdots + a_n \equiv n - 1 \pmod{2}$. Equivalently, this means $9 + a_2 + \cdots + a_n \equiv n \pmod{2}$.

Instead of the above pairing, we provide another pairing. For those phone numbers not starting with 1, we can pair them up such that $\overline{a_1 a_2 \cdots a_n}$ and $(a_1 + 1)a_2 \cdots a_n$ belong to the same pair if $a_1 = 2, 4, 6, 8$. For those phone numbers starting with 1, we handle them as in the case $n - 1$. Arguing as above, we have to select one phone number from each pair. In particular, as those numbers $\overline{9a_2 \cdots a_n}$ with $9 + a_2 + \cdots + a_n \not\equiv n \pmod{2}$ are not selected, we have to select

all numbers $\overline{8a_2 \cdots a_n}$ with $8 + a_2 + \cdots + a_n \equiv n \pmod{2}$. Then by using the first pairing, we find that all numbers $\overline{7a_2 \cdots a_n}$ with $7 + a_2 + \cdots + a_n \equiv n \pmod{2}$ should be selected, etc. In the same way, we find that all numbers with sum congruent to n modulo 2 should be selected. Clearly, the condition is satisfied if we only select those numbers. Also, as exactly one number from each pair is selected, there are $\frac{9^n + 1}{2}$ such phone numbers. This proves the inductive step.

When $n = 8$, we can select at most $\frac{9^8 + 1}{2} = 21523361$ phone numbers.

Test 2

1. Firstly, consider the power of B with respect to $(ADME)$. This gives

$$BD \times BA = BM^2.$$

Similarly, we get

$$CE \times CA = CM^2$$

by considering the power of C with respect to the same circle. As M is the midpoint of BC , the two expressions are equal, so that

$$BD \times BA = CE \times CA. \quad (1)$$

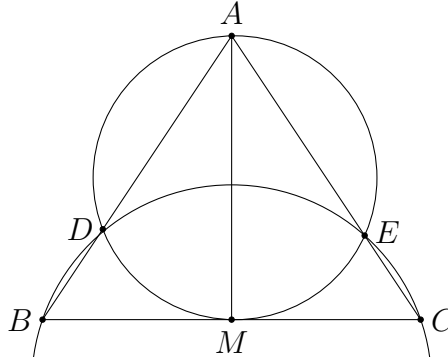
Next, since B, C, E, D are concyclic, we have

$$AD \times AB = AE \times AC \quad (2)$$

by considering the power of A . Adding (1) and (2), we get

$$AB^2 = AB \times BD + AB \times AD = AC \times CE + AC \times AE = AC^2.$$

This gives $AB = AC$.



2. Adding the inequalities

$$\begin{aligned} a_n &\geq a_{n-1} + n + 1, \\ a_{n-1} &\geq a_{n-2} + n, \\ &\dots, \\ a_2 &\geq a_1 + 3, \end{aligned}$$

we obtain $a_n \geq a_1 + (3 + 4 + \dots + (n + 1)) = \frac{n^2 + 3n + 4}{2} > \frac{(n + 1)(n + 2)}{2}$. It follows that

$$\sum_{n=1}^{2010} \frac{1}{a_n} < \sum_{n=1}^{\infty} \frac{2}{(n + 1)(n + 2)} = 2 \sum_{n=1}^{\infty} \left(\frac{1}{n + 1} - \frac{1}{n + 2} \right) = 2 \cdot \frac{1}{2} = 1 < \frac{3}{2}.$$

3. The only possible (and hence the maximum) n is 448.

Indeed, by the friendship theorem, there must be one person who knows everyone else. We give a proof as follows.

WLOG, assume A and B do not know each other. If X knows A , then X and B know a person Y in common. Note that the function mapping X to Y is well-defined, and it is a bijection between the set of persons known by A and the set of persons known by B . Hence any two persons who do not know each other know the same number of persons.

Let S_1 be the set of all persons who know exactly k persons, where k is chosen such that $|S_1| \geq 1$. Let S_2 be the complement of S_1 . Note that each person in S_2 must know every person in S_1 . Otherwise, this person must know exactly k persons from the above observation, and hence this person should belong to S_1 by definition.

If $|S_2| \geq 2$, then we choose two persons X and Y in S_2 arbitrarily. Since they only know one person in common, we must have $|S_1| = 1$. In this case the person in S_1 knows everyone.

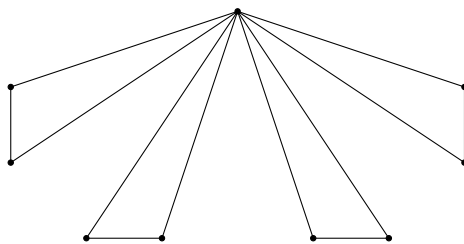
If $|S_2| = 1$, then the person in S_2 knows everyone.

If $|S_2| = 0$, then everyone knows exactly k persons. On the one hand, there are $\binom{451}{2}$ pairs of persons in this group. On the other hand, there are $\binom{k}{2}$ pairs of persons knowing a particular person in common. Therefore, we have

$$451 \binom{k}{2} = \binom{451}{2}.$$

But this implies $k(k-1) = 450$, which has no solution. (In this case, we make use of the fact that 451 is not of the form $k^2 - k + 1$. But the friendship theorem does apply to any number of persons.)

Therefore, we have proven there exists a person A who knows everyone. For every other person B , in order that A and B know exactly one person in common, B must know exactly one person apart from A . Thus, the other persons can be paired up such that only the two persons in the same pair know each other. Hence, the only possible n is $450 - 2 = 448$.



4. (IMO Shortlist 2003 N6/Problem 6 simplified)

- (a) Suppose on the contrary that all prime divisors of $\frac{p^p - 1}{p - 1}$ are congruent to 1 modulo p^2 . Then clearly we have $\frac{p^p - 1}{p - 1} \equiv 1 \pmod{p^2}$. But then

$$\frac{p^p - 1}{p - 1} = p^{p-1} + p^{p-2} + \dots + 1 \equiv p + 1 \not\equiv 1 \pmod{p^2}.$$

This is a contradiction, and so we are done.

- (b) Suppose on the contrary that $n^p - p \equiv 0 \pmod{q}$ for some $n \in \mathbb{Z}^+$. Since $q \mid \frac{p^p - 1}{p - 1}$, we have $p^p \equiv 1 \pmod{q}$. Therefore, we find that

$$n^{p^2} = (n^p)^p \equiv p^p \equiv 1 \pmod{q}.$$

By the Fermat little theorem, we have $n^{q-1} \equiv 1 \pmod{q}$. Let d be the order of n modulo q . Then we have $d \mid p^2$ and $d \mid q - 1$. Since $p^2 \nmid q - 1$ by part (a), this implies $d = 1, p$. Thus, we have

$$n^p \equiv 1 \pmod{q}.$$

This further implies $p \equiv 1 \pmod{q}$. But then we find that

$$\frac{p^p - 1}{p - 1} = p^{p-1} + p^{p-2} + \dots + 1 \equiv 1 + 1 + \dots + 1 = p \not\equiv 0 \pmod{q}.$$

This is a contradiction. Thus, $n^p - p \not\equiv 0 \pmod{q}$ for all $n \in \mathbb{Z}^+$.

Year 2012

Test 1

1. The solutions are $(x, y, z) = (2, -3, -3), \left(\frac{8 \mp 2\sqrt{10}}{3}, 1 \pm \sqrt{10}, 1 \pm \sqrt{10}\right)$.

Clearly, $x, y, z \neq 0$. Since

$$(y+1)zx = 12 = (z+1)xy,$$

we have $zx = xy$. This implies $y = z$. By the first equation $(x+1)y^2 = 27$, we obtain

$$x = \frac{27}{y^2} - 1.$$

Putting this into the second equation, we have

$$(y+1)y\left(\frac{27}{y^2} - 1\right) = 12.$$

After expansion, this becomes $y^3 + y^2 - 15y - 27 = 0$, i.e.

$$(y+3)(y^2 - 2y - 9) = 0.$$

Clearly, the solutions are $y = -3, 1 \pm \sqrt{10}$. Correspondingly, we find all solutions

$$(x, y, z) = (2, -3, -3), \left(\frac{8 \mp 2\sqrt{10}}{3}, 1 \pm \sqrt{10}, 1 \pm \sqrt{10}\right).$$

It is not hard to check that all these are indeed solutions.

2. A possible function is $f(x) = 0$ for any $x \in [0, 1)$ and $f(x) = 3(4^k - 1)$ for any $x \in [2^k - 1, 2^{k+1} - 1)$ where $k \in \mathbb{Z}^+$.

It suffices to check the above function satisfies the conditions. It is clear that $f(x) \geq 0$ for any $x \geq 0$. If $x \in [0, 1)$, then $2x+1 \in [1, 3)$. Therefore, we have

$$f(2x+1) = 3(4^1 - 1) = 9 = 4(0) + 9 = 4f(x) + 9.$$

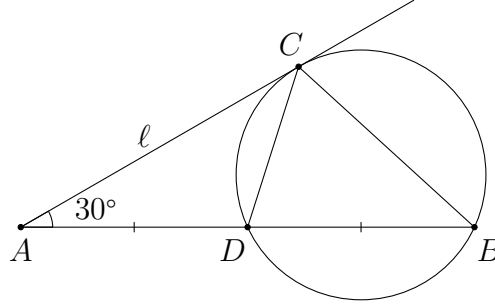
If $x \in [2^k - 1, 2^{k+1} - 1)$ where $k \in \mathbb{Z}^+$, then $2x+1 \in [2^{k+1} - 1, 2^{k+2} - 1)$. Therefore, we have

$$f(2x+1) = 3(4^{k+1} - 1) = 4 \cdot 3(4^k - 1) + 9 = 4f(x) + 9.$$

This completes the proof.

3. The maximum possible value of $\tan \angle DCB$ is $\frac{4\sqrt{2} + 3\sqrt{3}}{5}$.

To maximize $\tan \angle DCB$, we need to maximize $\angle DCB$ in the range $(0^\circ, 90^\circ)$. First we fix the positions of A and B (and hence D). Let ℓ be the ray AC , which is fixed. Note that the locus of points C with fixed $\angle DCB$ is a circular arc from B to D . When $\angle DCB$ decreases from 180° , the arc is getting closer and closer to ℓ . Therefore, $\angle DCB$ is maximized when ℓ is tangent to (BCD) , and C is the contact point of ℓ with (BCD) .



Consider the case when (BCD) is tangent to ℓ at C . By using powers, we have

$$AC^2 = AD \times AB = 2 \times 4 = 8,$$

and hence $AC = 2\sqrt{2}$. Using the cosine formula, we obtain

$$CD^2 = AC^2 + AD^2 - 2AC \times AD \cos 30^\circ = 12 - 4\sqrt{6}$$

and

$$BC^2 = AC^2 + AB^2 - 2AC \times AB \cos 30^\circ = 24 - 8\sqrt{6}.$$

Now, since $BC^2 + CD^2 - BD^2 = 32 - 12\sqrt{6} > 0$, we know that $\angle DCB < 90^\circ$. Therefore, this choice of C indeed maximizes $\tan \angle DCB$. Again by the cosine formula, we have

$$\cos \angle DCB = \frac{32 - 12\sqrt{6}}{2\sqrt{(12 - 4\sqrt{6})(24 - 8\sqrt{6})}} = \frac{32 - 12\sqrt{6}}{2\sqrt{2}(12 - 4\sqrt{6})} = \frac{3\sqrt{2} - \sqrt{3}}{6}.$$

Finally, we find that

$$\begin{aligned} \tan \angle DCB &= \sqrt{\sec^2 \angle DCB - 1} = \sqrt{\left(\frac{6\sqrt{2} + 2\sqrt{3}}{5}\right)^2 - 1} \\ &= \sqrt{\frac{59 + 24\sqrt{6}}{25}} = \frac{4\sqrt{2} + 3\sqrt{3}}{5}. \end{aligned}$$

4. The maximum number is 870.

For $k = 0, 1, \dots, 44$, let a_k be the number of persons having exactly k friends. Note that $k \leq 45 - a_k$, since each person counted in a_k can only be a friend of those $45 - a_k$ persons who are not counted in a_k . This yields $a_k \leq 45 - k$.

Now, the total number of pairs of friends is $\frac{1}{2} \sum_{k=0}^{44} k a_k$, since each pair of friends is counted twice when we sum over all k 's. Using $a_k \leq 45 - k$, we have

$$\begin{aligned} \frac{1}{2} \sum_{k=0}^{44} k a_k &= \frac{1}{2} \sum_{k=1}^{44} \sum_{j=1}^k a_k = \frac{1}{2} \sum_{j=1}^{44} \sum_{k=j}^{44} a_k \\ &= \frac{1}{2} \sum_{k=36}^{44} (k - 35) a_k + \frac{1}{2} \sum_{j=1}^{35} \sum_{k=j}^{44} a_k \\ &\leq \frac{1}{2} \sum_{k=36}^{44} (k - 35)(45 - k) + \frac{1}{2} \sum_{j=1}^{35} 45 \\ &= 870. \end{aligned}$$

This maximum value can be attained. For example, we partition the 45 persons into 9 groups of 1, 2, $\dots$, 9 persons respectively. Two persons are friends if and only if they belong to different groups. Then the persons in these groups have 44, 43, $\dots$, 36 friends respectively. Then the total number of pairs of friends is

$$\frac{1}{2} (1 \times 44 + 2 \times 43 + \dots + 9 \times 36) = 870.$$

5. We have $\frac{DB}{DC} = \frac{7}{2}$.

Firstly, since $\angle EDB = 90^\circ$ and $BF = EF$, we have $DF = BF$. This implies $\angle FDB = \angle FBD = \theta$, and hence $\angle DFE = 2\theta$. It follows that C lies on the segment BD , and

$$\angle CDF = 3\theta - 2\theta = \theta.$$

Let P be the intersection point of AD and FC , and let M be the midpoint of BD . Note that $FM \perp BD$ since $\triangle FBD$ is isosceles. This shows

$$\angle DMF = 90^\circ = \angle DPF.$$

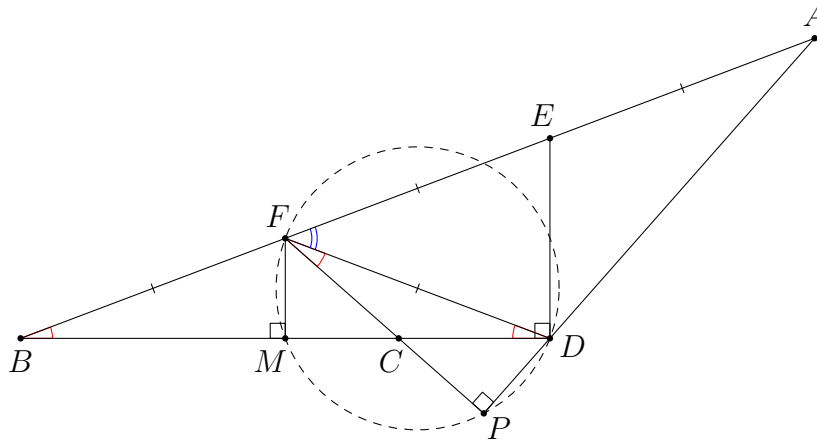
Therefore, F, M, P, D are concyclic. Since $\angle CDF = \angle CFD$, $FMPD$ is an isosceles trapezoid.

Now, applying Menelaus' theorem to $\triangle BCF$, we obtain

$$\frac{BD}{DC} \times \frac{CP}{PF} \times \frac{FA}{AB} = 1.$$

Note that $BD = 2MD = 2PF$ and $\frac{FA}{AB} = \frac{2}{3}$. Therefore, we get $\frac{CP}{DC} = \frac{3}{4}$. This implies $\frac{CM}{DC} = \frac{CP}{DC} = \frac{3}{4}$, and hence

$$\frac{DB}{DC} = \frac{2DM}{DC} = \frac{2(DC + CM)}{DC} = \frac{7}{2}.$$



6. There are 16 such positive integers m .

Let $a_k = m^k$. We would like to represent a_{x_0} as a sum of 2011 numbers of the form a_{x_j} . To get such a representation, we can do in the following way. Starting from the number a_{x_0} , each time we choose a number a_{k+1} and break it into m numbers a_k , so that the sum of all numbers remains a_{x_0} .

In view of the base m representation of the numbers, if we can write

$$a_{x_0} = a_{x_1} + a_{x_2} + \cdots + a_{x_{2011}},$$

then we are able to obtain the numbers on the right by using the above operation repeatedly. Conversely, if we can obtain these numbers after some operations, then clearly their sum is a_{x_0} by construction. Therefore, it remains to check for which m we can apply finitely many operations so that we can generate exactly 2011 numbers.

Note that there are $m-1$ numbers more after applying each operation. Initially, there is only one number. Therefore, it is the same as finding those $m \in \mathbb{Z}^+$ such that $m-1 \mid 2011-1 = 2010$. As $2010 = 2 \times 3 \times 5 \times 67$, there are $d(2010) = 16$ solutions.

Test 2

1. The answer is 518400.

We claim that $3^4 \cdot 4^4 \cdot 5^2 \mid xyzuvw$.

Firstly, observe that $3 \mid x$ or $3 \mid y$, since otherwise

$$u^2 = x^2 + y^2 \equiv 1 + 1 = 2 \pmod{3},$$

which is impossible. Similarly, 3 divides x or z , and 3 divides y or z . Thus, two of x, y, z are divisible by 3, say x and y . Then we have $3 \mid u$, and the first equation becomes $\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = \left(\frac{u}{3}\right)^2$. Again, 3 divides one of $\frac{x}{3}$ and $\frac{y}{3}$. Thus, we have $3^4 \mid xyu$. In general, we must have $3^4 \mid xyzuvw$.

Secondly, observe that $4 \mid x$ or $4 \mid y$, since otherwise

$$u^2 = x^2 + y^2 \equiv 1 + 1 = 2 \pmod{4},$$

which is impossible. As above, this implies $4^4 \mid xyzuvw$.

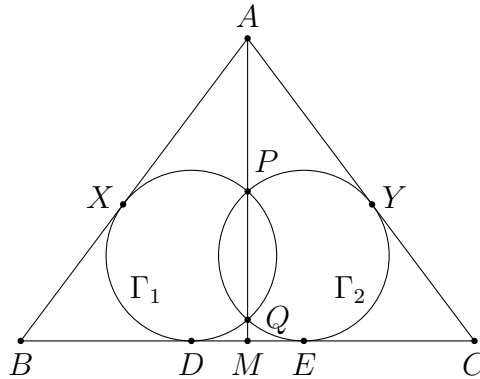
Thirdly, observe that 5 divides x, y or u , since otherwise

$$x^2 + y^2 \equiv \pm 1 \pm 1 = \pm 2 \not\equiv \pm 1 \equiv u^2 \pmod{5}.$$

In the same way, 5 divides one of x, z, v , 5 divides one of y, z, w . It is not hard to see 5 divides at least two of x, y, z, u, v, w , and hence $5^2 \mid xyzuvw$.

Combining these, we obtain $3^4 \cdot 4^4 \cdot 5^2 = 518400 \mid xyzuvw$.

2. Let Γ_1 be the circle touching AB and BC , and let Γ_2 be the circle touching AC and BC . Let Γ_1 touch AB and BC at X and D respectively. Let Γ_2 touch AC and BC at Y and E respectively.



Suppose on the contrary that $\Gamma_1 \neq \Gamma_2$. Then $D \neq E$ since otherwise both circles pass through three same points. By using powers, we have

$$MD^2 = MP \times MQ = ME^2.$$

This implies M is the midpoint of D and E . As M is also the midpoint of B and C , we have $BD = CE$.

Similarly, we have $AX^2 = AP \times AQ = AY^2$, and so $AX = AY$. It follows that

$$AB = AX + BX = AX + BD = AY + CE = AY + CY = AC,$$

contradicting the assumption that $\triangle ABC$ is scalene. Therefore, $\Gamma_1 = \Gamma_2$, which is the incircle of $\triangle ABC$.

3. The smallest possible value of r is $2n$ if n is even, and is $2n + 1$ if n is odd.

Case 1. $n = 2k$ is even

Let (i, j) be the cell in row i and column j . Consider the following $2n$ diagonals:

$$\begin{aligned} \{(i, j) : i + j = m\} & \text{ for } m = 2, 3, \dots, k + 1, \\ \{(i, j) : j - i = m\} & \text{ for } m = k, k + 1, \dots, 2k - 1, \\ \{(i, j) : i - j = m\} & \text{ for } m = k, k + 1, \dots, 2k - 1, \\ \{(i, j) : i + j = m\} & \text{ for } m = 3k + 1, 3k + 2, \dots, 4k. \end{aligned}$$

A	B	C	D	E	F
B	C			D	E
C					D
I					J
H	I			J	K
G	H	I	J	K	L

Obviously, these diagonals are disjoint. Therefore, one chess piece is needed for each diagonal, and so $r \geq 2n$.

Now, we put $2n$ chess pieces in the following cells:

$$\begin{aligned} (1, j) & \text{ for } j = 1, 2, \dots, k, \\ (i, n) & \text{ for } i = 1, 2, \dots, k, \\ (i, 1) & \text{ for } i = k + 1, k + 2, \dots, 2k, \\ (n, j) & \text{ for } j = k + 1, k + 2, \dots, 2k. \end{aligned}$$

X	X	X			X
					X
					X
X					
X					
X			X	X	X

Then we see that each row, each column and each diagonal consists of a chess piece. This proves $r = 2n$ when n is even.

Case 2. $n = 2k + 1$ is odd

Consider the following $2n + 2$ diagonals:

$$\begin{aligned} \{(i, j) : i + j = m\} & \text{ for } m = 2, 3, \dots, k + 2, \\ \{(i, j) : j - i = m\} & \text{ for } m = k, k + 1, \dots, 2k, \\ \{(i, j) : i - j = m\} & \text{ for } m = k, k + 1, \dots, 2k, \\ \{(i, j) : i + j = m\} & \text{ for } m = 3k + 2, 3k + 3, \dots, 4k + 2. \end{aligned}$$

A	B	C, D	E	F
B	C		D	E
C, I				D, J
H	I		J	K
G	H	I, J	K	L

Only 4 pairs of these diagonals are not disjoint. We call these C, D, I, J -diagonals respectively, as labelled in the above example. One chess piece is needed for each of those $4k = 2n - 2$ diagonals other than the C, D, I, J -diagonals.

If $r \leq 2n$, then we can put at most 2 chess pieces in the C, D, I, J -diagonals. By the pigeonhole principle, one chess piece must occupy at least two such diagonals. WLOG assume there is a chess piece in $(1, k + 1)$ occupying the C, D -diagonals. In order to occupy the I, J -diagonals simultaneously, there must be another chess piece in $(2k + 1, k + 1)$. But then there is no chess piece in row $k + 1$. This is a contradiction. Thus, we have $r \geq 2n + 1$.

Now, we put $2n + 1$ chess pieces in the following cells:

$$\begin{aligned} (1, j) & \text{ for } j = 1, 2, \dots, k + 1, \\ (i, n) & \text{ for } i = 1, 2, \dots, k, \\ (i, 1) & \text{ for } i = k + 2, k + 3, \dots, 2k + 1, \\ (n, j) & \text{ for } j = k + 1, k + 2, \dots, 2k + 1, \\ (k + 1, k + 1). \end{aligned}$$

X	X	X		X
				X
		X		
X				
X		X	X	X

Then we see that each row, each column and each diagonal consists of a chess piece. This proves $r = 2n + 1$ when n is odd.

4. The only solution is $f(x) = x$ for any $x \in \mathbb{R}$.

Using $f(x+1) = f(x) + 1$, it can be proved easily by induction that

$$f(x+n) = f(x) + n \quad (1)$$

for any integer n .

Consider any rational number $\frac{a}{b}$ where $a, b \in \mathbb{Z}$. For any positive integer k , we substitute $x = \frac{a}{b} + kb^2$ in (ii) to obtain

$$f\left(\frac{a^3}{b^3} + 3ka^2 + 3k^2ab^3 + k^3b^6\right) = f(x^3) = f(x)^3.$$

Since $k, a, b \in \mathbb{Z}$, by (1), we obtain

$$f\left(\frac{a^3}{b^3} + 3ka^2 + 3k^2ab^3 + k^3b^6\right) = f\left(\frac{a^3}{b^3}\right) + 3ka^2 + 3k^2ab^3 + k^3b^6.$$

Also, by letting $z = f\left(\frac{a}{b}\right)$, we have

$$f(x)^3 = (z + kb^2)^3 = z^3 + 3kb^2z^2 + 3k^2b^4z + k^3b^6.$$

Comparing the two sides, since $f\left(\frac{a^3}{b^3}\right) = z^3$, we find that

$$3ka^2 + 3k^2ab^3 = 3kb^2z^2 + 3k^2b^4z.$$

This implies $b^2z^2 + kb^4z - a^2 - kab^3 = 0$, and hence

$$(bz - a)(bz + a + kb^3) = 0.$$

Since $b \neq 0$, we cannot have $bz + a + kb^3 = 0$ for all $k \in \mathbb{Z}^+$. Therefore, we must have $bz - a = 0$, i.e. $z = \frac{a}{b}$. This proves $f(x) = x$ for any $x \in \mathbb{Q}$.

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, for any $x \in \mathbb{R}$, we can find a sequence $\{x_n\}$ of rational numbers converging to x . By the continuity, we have

$$f(x) = f\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} x_n = x.$$

It is easy to check that the identity function satisfies both conditions, so it is the only solution.

Year 2013

Test 1

1. The only possible value is 0.

It suffices to show that $10 \mid a^2 - ab + b^2$ implies $100 \mid a^2 - ab + b^2$. Note that $10 \mid a^2 - ab + b^2$ implies

$$10 \mid (a+b)(a^2 - ab + b^2) = a^3 + b^3.$$

Now,

$$a^3 \equiv -b^3 \pmod{10} \Rightarrow a^9 \equiv -b^9 \pmod{10} \Rightarrow a \equiv -b \pmod{10}$$

by the Euler-Fermat theorem since $\varphi(10) = 4$. Now we have

$$0 \equiv a^2 - ab + b^2 \equiv b^2 + b^2 + b^2 = 3b^2 \pmod{10}.$$

This implies $10 \mid b$. Therefore, both a and b are divisible by 10, and so obviously $100 \mid a^2 - ab + b^2$.

2. The greatest n is 145.

One can easily check that

$$145 \mapsto 42 \mapsto 20 \mapsto 4 \mapsto 16 \mapsto 37 \mapsto 58 \mapsto 89 \mapsto 145,$$

so that $f^8(145) = 145$. It remains to check $n \leq 145$.

We first prove that n has at most 3 digits. Suppose on the contrary that $n = \overline{a_1 a_2 \dots a_k}$ where $k \geq 4$. For any $m \leq n$, we have

$$f(m) \leq f(\underbrace{99 \dots 9}_{k \text{ times}}) = 81k < 10^{k-1} \leq n.$$

(To prove $81k < 10^{k-1}$ for $k \geq 4$, we check the base case $324 < 1000$ and the inductive step $81(k+1) < 10(81k) < 10^k$.) Using this, we can easily prove that $f^s(n) < n$ for any $s \in \mathbb{Z}^+$. Thus, it is impossible that $f^s(n) = n$.

Similarly, if $n > 243$, then for $m \leq n$, we have

$$f(m) \leq f(999) = 243 < n.$$

If $163 < n \leq 243$, then for $m \leq n$, we have

$$f(m) \leq f(199) = 163 < n.$$

Thus, we must have $n \leq 163$.

Note that n is also the image of some term in the cycle. Let $n = f(t)$ for some $t \leq 163$. If $n > 145 = 8^2 + 9^2$, then the only possibility is $t = 99$. But then we have

$$99 \mapsto 162 \mapsto 41 \mapsto 17 \mapsto 50 \mapsto 25 \mapsto 29 \mapsto 85 \mapsto 89 \mapsto 145,$$

which enters the cycle with largest term 145, and so it never goes back to 162. Therefore, we must have $n \leq 145$. This completes the proof.

3. (IMO 1973 Problem 3) The minimum value of $a^2 + b^2$ is $\frac{4}{5}$.

WLOG assume $a \geq 0$. (Indeed, by flipping the sign of a , we only need to flip the sign of the root x .) Clearly, $x \neq 0$. Rewrite the equation as

$$x^2 + ax + b + \frac{a}{x} + \frac{1}{x^2} = 0.$$

This is the same as

$$\left(x + \frac{1}{x}\right)^2 + a\left(x + \frac{1}{x}\right) + b - 2 = 0.$$

Let $y = x + \frac{1}{x}$. Note that $|y| = |x| + \frac{1}{|x|} \geq 2$ by the AM-GM inequality. The equation becomes

$$y^2 + ay + (b - 2) = 0.$$

If $a > 1$, then $a^2 + b^2 > 1$. Thus, we only need to consider the case $a \leq 1$. This implies the axis of symmetry $y = -\frac{a}{2}$ lies between -2 and 2 .

- If $y \geq 2$, then we have

$$0 = y^2 + ay + (b - 2) \geq (2)^2 + a(2) + (b - 2) = 2a + b + 2.$$

This yields $b \leq -2a - 2 \leq -2$, and hence $a^2 + b^2 \geq 4$.

- If $y \leq -2$, then we have

$$0 = y^2 + ay + (b - 2) \geq (-2)^2 + a(-2) + (b - 2) = -2a + b + 2.$$

This yields $b \leq 2a - 2 \leq 0$, and hence

$$a^2 + b^2 \geq a^2 + (2a - 2)^2 = 5\left(a - \frac{4}{5}\right)^2 + \frac{4}{5} \geq \frac{4}{5}.$$

Equality holds when $a = \frac{4}{5}$ and $b = -\frac{2}{5}$. Correspondingly, this means $y = -2$ and $x = -1$.

To conclude, the minimum value of $a^2 + b^2$ is $\frac{4}{5}$.

4. There are 6768 triangles.

Since there are 36 points in S , there are $\binom{36}{3} = 7140$ triples of points. We have to remove those triples which contain collinear points.

Firstly, there are 6 rows, and there are $\binom{6}{3} = 20$ ways to choose a triple from each row. Therefore, there are $6 \times 20 = 120$ triples consisting of three points in the same row. By symmetry, there are 120 triples consisting of three points in the same column.

Secondly, we consider those diagonal lines with slope 1. The number of points in these diagonal lines are 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 respectively. Thus, the number of triples we can choose from these diagonal lines is

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \binom{6}{3} + \binom{5}{3} + \binom{4}{3} + \binom{3}{3} = 50.$$

By symmetry, there are 50 triples coming from the diagonal lines with slope -1 .

Thirdly, we consider those lines with slope 2. We can list out all triples lying on these lines as follows:

$$\begin{aligned} &\{(0,0), (1,2), (2,4)\}, \{(0,1), (1,3), (2,5)\}, \\ &\{(1,0), (2,2), (3,4)\}, \{(1,1), (2,3), (3,5)\}, \\ &\{(2,0), (3,2), (4,4)\}, \{(2,1), (3,3), (4,5)\}, \\ &\{(3,0), (4,2), (5,4)\}, \{(3,1), (4,3), (5,5)\}. \end{aligned}$$

There are 8 such triples. By symmetry, there are 8 triples coming from the lines with slope -2 , 8 triples coming from the lines with slope $\frac{1}{2}$, and 8 triples coming from the lines with slope $-\frac{1}{2}$.

Lastly, since any other line contains at most 2 points in S , the answer is

$$7140 - 120 \times 2 - 50 \times 2 - 8 \times 4 = 6768.$$

5. The minimum number of participants is 17.

Let a_n be the number of participants obtaining score n , and let $m = a_0 + a_1 + \cdots + a_{10}$ be the total number of participants. Using the mean, we have $\sum_{n=0}^{10} na_n = 3m$. This means

$$\sum_{n=4}^{10} (n-3)a_n = 3a_0 + 2a_1 + a_2. \quad (1)$$

Case 1. m is odd

Since 3 is the median, we must have $a_0 + a_1 + a_2 \leq \frac{m-1}{2}$. This implies

$$a_0 + a_1 + a_2 + 1 \leq \sum_{n=3}^{10} a_n. \quad (2)$$

Using (1) and (2), we obtain

$$\begin{aligned} 3a_0 + 2a_1 + a_2 &= \sum_{n=4}^{10} (n-3)a_n = \sum_{n=3}^{10} a_n + \sum_{n=4}^{10} (n-4)a_n - a_3 \\ &\geq (a_0 + a_1 + a_2 + 1) + (3a_7) - a_3. \end{aligned}$$

This implies $2a_0 + a_1 + a_3 \geq 3a_7 + 1$. Since the mode is 7, we have $a_0, a_1, a_3 \leq a_7 - 1$. This yields

$$3a_7 + 1 \leq 2a_0 + a_1 + a_3 \leq 4(a_7 - 1),$$

and hence $a_7 \geq 5$. Now, we have

$$m \geq a_0 + a_1 + a_3 + a_7 = (2a_0 + a_1 + a_3) - a_0 + a_7 \geq (3a_7 + 1) - (a_7 - 1) + a_7 \geq 17.$$

Equality holds when $a_0 = a_1 = a_3 = 4$, $a_7 = 5$ and all other a_j 's are 0. One checks that all conditions are satisfied.

Case 2. m is even

If $a_3 > 0$, then the two scores in the middle are 3. The bound $a_0 + a_1 + a_2 \leq \frac{m-1}{2}$ used in case 1 still holds, and so we still have $m \geq 17$.

If $a_3 = 0$, then we only have $a_0 + a_1 + a_2 \leq \frac{m}{2}$. This implies

$$a_0 + a_1 + a_2 \leq \sum_{n=4}^{10} a_n. \quad (3)$$

Using (1) and (3), we obtain

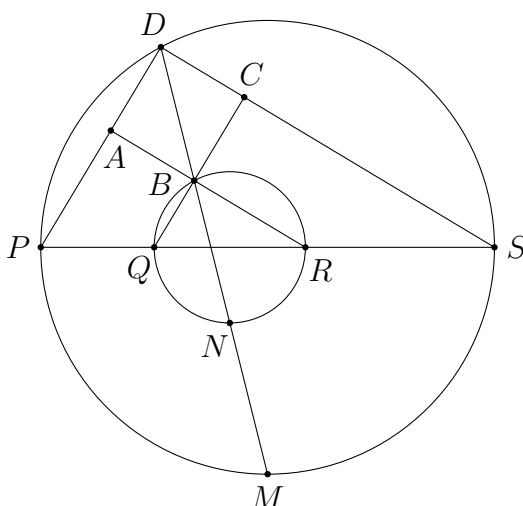
$$\begin{aligned} 3a_0 + 2a_1 + a_2 &= \sum_{n=4}^{10} (n-3)a_n = \sum_{n=4}^{10} a_n + \sum_{n=4}^{10} (n-4)a_n \\ &\geq (a_0 + a_1 + a_2) + (3a_7). \end{aligned}$$

This implies $2a_0 + a_1 \geq 3a_7$. But then one of a_0, a_1 is greater than or equal to a_7 , contradicting the fact that 7 is the mode.

In any case, we have $m \geq 17$, and so 17 is the minimum value.

6. Note that we can use compass and ruler to find the perpendicular bisector of two given points. Thus, we can use this to locate the midpoint of two points, and draw the circle having a given segment as a diameter.

Construct the circles with diameter PS and QR respectively. Construct the midpoints M and N of $\widehat{PS}$ and $\widehat{QR}$ (using the perpendicular bisectors) such that M and N lie on the same side of ℓ . Construct the intersection point B of the line MN and (QNR) , and construct the intersection point D of the line MN and (PMS) . Also, construct the intersection point A of PD and RB , and construct the intersection point C of SD and QB . We claim that $ABCD$ is the desired square.



By construction, we already know that P, Q, R, S lie on AD, BC, AB, CD respectively. Since PS and QR are diameters of the two circles, we easily obtain $\angle ADC = \angle ABC = 90^\circ$. Also, since M and N are the midpoints of $\widehat{PS}$ and $\widehat{QR}$, the line BD bisects $\angle ADC$ and $\angle ABC$. It follows that

$$\angle ADB = \angle CDB = \angle ABD = \angle CBD = 45^\circ,$$

which implies $\angle BAD = \angle BCD = 90^\circ$. Also, as $\angle ABD = \angle ADB$, we obtain $AB = AD$. This proves $ABCD$ is a square.

Test 2

1. A possible m is 5201836549.

Let $p_1, p_2, \dots, p_7$ be the primes 7, 13, 19, 31, 37, 43, 61 which are congruent to 1 modulo 3, and let $m = p_1 p_2 \cdots p_7$. Since $3 \mid p_j - 1$, the equation $x^3 \equiv 1 \pmod{p_j}$ has three distinct solutions modulo p_j by the cubing lemma (or primitive root).

Thus, there are $3^7 = 2187 > 2012$ tuples $(x_1, x_2, \dots, x_7)$ such that $x_j^3 \equiv 1 \pmod{p_j}$ for each j . By the Chinese remainder theorem, there exists x such that

$$x \equiv x_j \pmod{p_j}$$

for each j , and so

$$x^3 \equiv x_j^3 \equiv 1 \pmod{p_j}.$$

Since this holds for all j , we have $x^3 \equiv 1 \pmod{m}$. Note that different tuples $(x_1, x_2, \dots, x_7)$ and $(x'_1, x'_2, \dots, x'_7)$ correspond to different x 's modulo m , since $x_j \not\equiv x'_j \pmod{p_j}$ for at least one j . This completes the proof.

2. (i) The answer is $2^{2n-1} - \frac{1}{2} \binom{2n}{n}$.

A subset T of S is called a *bad* subset if in T the number of odd elements is greater than the number of even elements. A subset of S is neither good nor bad if it has exactly k odd elements and k even elements for some $k = 0, 1, \dots, n$. Since there are n odd elements and n even elements in S , the number of subsets which are neither good nor bad is

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$

by Vandermonde's identity.

Now, by symmetry, the number of good subsets is the same as the number of bad subsets. As there are 2^{2n} subsets of S in total, the total number of good subsets is

$$\frac{1}{2} \left[2^{2n} - \binom{2n}{n} \right] = 2^{2n-1} - \frac{1}{2} \binom{2n}{n}.$$

- (ii) The answer is $2^{2n-2}(2n^2 + n) - n^2 \binom{2n-1}{n}$.

We first count the number N_1 of good subsets containing a particular even number. Suppose such a good subset contains i more even numbers and j odd numbers. Then we need $i \geq j$. As there are $n-1$ even numbers remaining and n odd numbers in total, we have

$$N_1 = \sum_{i=0}^{n-1} \sum_{j=0}^i \binom{n-1}{i} \binom{n}{j} = \sum_{i=0}^{n-1} \sum_{k=n-i}^n \binom{n-1}{i} \binom{n}{k} = \sum_{i+k \geq n} \binom{n-1}{i} \binom{n}{k}$$

by using the change of variable $k = n - j$. Note that this is equal to the sum of coefficients of all x^m with $m \geq n$ in $(1+x)^{n-1}(1+x)^n = (1+x)^{2n-1}$. Thus, we obtain

$$N_1 = \sum_{m=n}^{2n-1} \binom{2n-1}{m} = \frac{1}{2} \sum_{m=0}^{2n-1} \binom{2n-1}{m} = 2^{2n-2}.$$

Similarly, we count the number N_2 of good subsets containing a particular odd number. Suppose such a good subset contains $i \geq 2$ even numbers and j more odd numbers. Then we need $i \geq j + 2$. This implies

$$\begin{aligned}
N_2 &= \sum_{i=2}^n \sum_{j=0}^{i-2} \binom{n}{i} \binom{n-1}{j} = \sum_{i=2}^n \sum_{k=n+1-i}^{n-1} \binom{n}{i} \binom{n-1}{k} \\
&= \sum_{i+k \geq n+1} \binom{n}{i} \binom{n-1}{k} = \sum_{m=n+1}^{2n-1} \binom{2n-1}{m} \\
&= \frac{1}{2} \left(\sum_{m=0}^{2n-1} \binom{2n-1}{m} - \binom{2n-1}{n-1} - \binom{2n-1}{n} \right) \\
&= 2^{2n-2} - \binom{2n-1}{n}.
\end{aligned}$$

Now, the sum of all the elements in all the good subsets of S is

$$\begin{aligned}
&N_1(2 + 4 + \cdots + 2n) + N_2(1 + 3 + \cdots + (2n-1)) \\
&= 2^{2n-2} \cdot n(n+1) + \left[2^{2n-2} - \binom{2n-1}{n} \right] \cdot n^2 \\
&= 2^{2n-2}(2n^2 + n) - n^2 \binom{2n-1}{n}.
\end{aligned}$$

3. Using the condition $a + b + c = 1$, we obtain

$$a^2 - 2bc + 2c = a^2 + 2c(1-b) = a^2 + 2c(a+c) = (c+a)^2 + c^2.$$

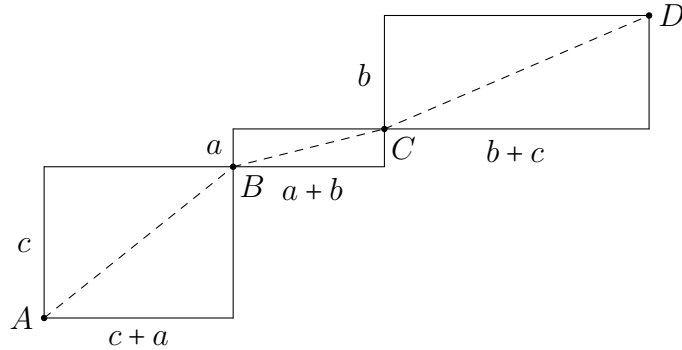
Similarly, the left-hand side of the inequality becomes

$$\sqrt{(c+a)^2 + c^2} + \sqrt{(a+b)^2 + a^2} + \sqrt{(b+c)^2 + b^2}.$$

By the triangle inequality, this is bounded below by

$$\sqrt{((c+a) + (a+b) + (b+c))^2 + (c+a+b)^2} = \sqrt{5}.$$

Equality holds when $a = b = c = \frac{1}{3}$.



4. Let ℓ_A meet II' at P . We claim that P is the intersection point of ℓ_A, ℓ_B, ℓ_C . It suffices to show that $\frac{II'}{I'P}$ is a constant.

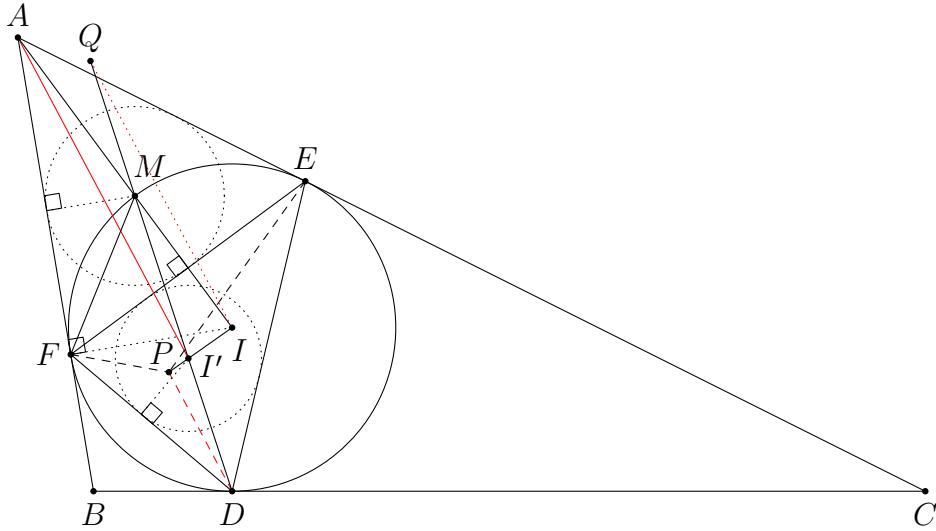
Let DI' meet the incircle of $\triangle ABC$ again at M , and meet the line passing through I and parallel to ℓ_A at Q . Let r and r' be the inradii of $\triangle ABC$ and $\triangle DEF$ respectively. By similar triangles, we obtain

$$\frac{II'}{I'P} = \frac{QI'}{I'D} = \frac{1}{I'D} \times \frac{MI' \times IA}{MA} = \frac{MI'}{I'D} \times \frac{IA}{MA}.$$

Since I' is the incentre of $\triangle DEF$, we have $MI' = MF$. Thus, we have

$$\frac{MI'}{I'D} = MF \times \frac{1}{I'D} = \frac{d(M, EF)}{\sin \angle EFM} \times \frac{\sin \angle I'DF}{r'} = \frac{d(M, EF)}{r'},$$

since $\angle EFM = \frac{1}{2} \angle EDF = \angle I'DF$.



Next, it is well-known that M is the incentre of $\triangle AEF$. Therefore, we have

$$\frac{IA}{MA} = \frac{d(I, AF)}{d(M, AF)} = \frac{r}{d(M, EF)}.$$

Combining these, we obtain

$$\frac{II'}{I'P} = \frac{MI'}{I'D} \times \frac{IA}{MA} = \frac{d(M, EF)}{r'} \times \frac{r}{d(M, EF)} = \frac{r}{r'}.$$

This is a constant, and so ℓ_A, ℓ_B, ℓ_C are concurrent at P .

Year 2014

Test 1

1. We have $f(8) = \frac{205}{8}$.

It is given that

$$\frac{2}{x}f(-x) + f\left(\frac{1}{x}\right) = x \quad (1)$$

for any $x \neq 0$. Replacing x by $-\frac{1}{x}$, we obtain

$$-2xf\left(\frac{1}{x}\right) + f(-x) = -\frac{1}{x}. \quad (2)$$

Using (1) and (2), we can solve for $f(-x)$. Indeed, we have

$$5f(-x) = 2x\left[\frac{2}{x}f(-x) + f\left(\frac{1}{x}\right)\right] + \left[-2xf\left(\frac{1}{x}\right) + f(-x)\right] = 2x^2 - \frac{1}{x}.$$

Replacing x by $-x$, we obtain

$$f(x) = \frac{2}{5}x^2 + \frac{1}{5x}.$$

For this function, we check that

$$\frac{2}{x}f(-x) + f\left(\frac{1}{x}\right) = \frac{2}{x}\left(\frac{2}{5}x^2 - \frac{1}{5x}\right) + \frac{2}{5x^2} + \frac{x}{5} = x.$$

Therefore, this is the only possible function, and so

$$f(8) = \frac{2}{5}(8)^2 + \frac{1}{5(8)} = \frac{205}{8}.$$

2. We have $\angle CAP = 42^\circ$.

Construct a point D such that $\angle DAC = \angle DCA = 72^\circ$, and B lies inside $\triangle DCA$. Since

$$\angle BCD = \angle ACD - \angle ACB = 72^\circ - 60^\circ = 12^\circ = \angle CBP,$$

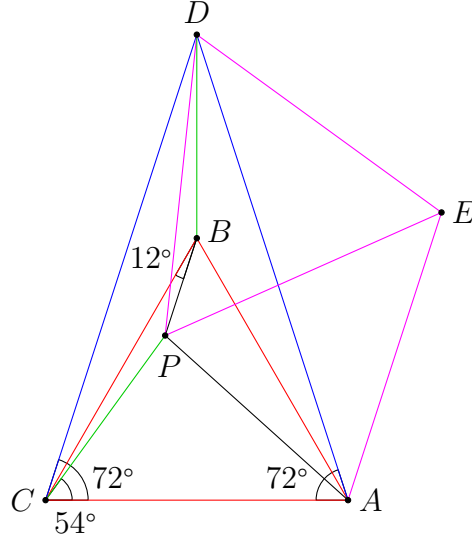
we have $DC \parallel BP$. Also, note that BD is the perpendicular bisector of CA . Therefore, we have

$$\angle CDB = \frac{1}{2}\angle CDA = 18^\circ$$

and

$$\angle PCD = \angle ACD - \angle ACP = 72^\circ - 54^\circ = 18^\circ = \angle CDB.$$

This implies $BDCP$ is an isosceles trapezoid.



Construct a point E such that $DCAE$ is an isosceles trapezoid and $DC \parallel EA$. Since $\triangle EDP$ is the image of reflection of $\triangle ACB$ in the perpendicular bisector of DC , it is equilateral. Now, as

$$\angle ADE = \angle CDE - \angle CDA = \angle ACD - \angle CDA = 72^\circ - 36^\circ = 36^\circ$$

and

$$\angle EAD = \angle CDA = 36^\circ = \angle ADE,$$

we have $EA = ED = EP$. It follows that

$$\begin{aligned} \angle PAC &= \angle EAC - \angle EAP \\ &= (180^\circ - \angle ACD) - \left(90^\circ - \frac{1}{2} \angle PEA\right) \\ &= 180^\circ - 72^\circ - 90^\circ + \frac{1}{2}(\angle DEA - \angle DEP) \\ &= 18^\circ + \frac{1}{2}(108^\circ - 60^\circ) \\ &= 42^\circ. \end{aligned}$$

3. We have $BP = \frac{1}{\sqrt{5}}$.

Note that $\angle CBA = 90^\circ$ and $AB = BC$. Since $\angle CBP = \angle BAP$, the line BC is tangent to (ABP) . As $\angle CBA = 90^\circ$, the centre of this circle must lie on AB . Thus, the centre is the midpoint M of AB . This implies $\angle APB = 90^\circ$.

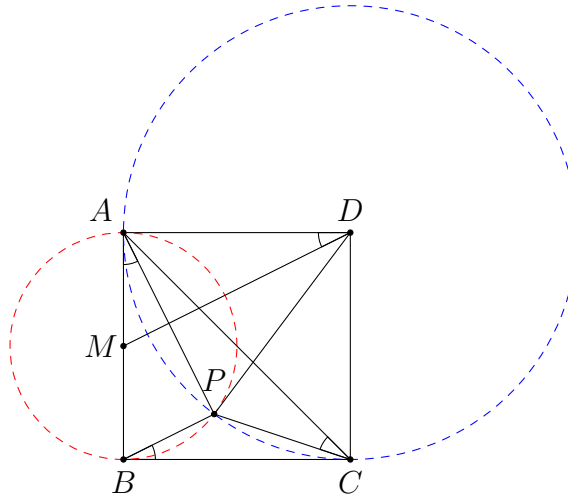
Since $\angle BAP = \angle ACP$, the line AB is tangent to (ACP) . The centre D of this circle must lie on the perpendicular line at A to AB , and lie on the perpendicular

bisector of AC . Clearly, D is the point such that $ABCD$ is a square. Note that MD is the perpendicular bisector of AP . Thus, we have

$$\angle ACP = \frac{1}{2} \angle ADP = \angle ADM.$$

Note that $\tan \angle ADM = \frac{AM}{AD} = \frac{1}{2}$. It follows that

$$BP = AB \sin \angle BAP = \frac{1}{\sqrt{5}}.$$



Alternatively, it is well-known that P is the Brocard point of $\triangle ABC$, and the Brocard angle $\omega = \angle PAB$ satisfies

$$\cot \omega = \cot A + \cot B + \cot C = 1 + 0 + 1 = 2.$$

4. The answer is 21087.

We use generating function. Since the first digit is nonzero, it corresponds to a factor $x + x^2 + \cdots + x^9$. All other digits can be one of $0, 1, \dots, 9$, and so each of them corresponds to a factor $1 + x + \cdots + x^9$. Thus, the answer is the coefficient of x^{18} in

$$f(x) = (x + x^2 + \cdots + x^9)(1 + x + \cdots + x^9)^5.$$

Now,

$$\begin{aligned} f(x) &= \frac{x - x^{10}}{1 - x} \cdot \left(\frac{1 - x^{10}}{1 - x} \right)^5 \\ &= (x - x^{10})(1 - x^{10})^5(1 - x)^{-6} \\ &= (x - x^{10}) \left(\sum_{r=0}^5 \binom{5}{r} (-1)^r x^{10r} \right) \left(\sum_{s=0}^{\infty} \binom{-6}{s} (-1)^s x^s \right) \end{aligned}$$

by the generalized binomial theorem. Note that

$$\binom{-6}{s} = \frac{(-6)(-7)\cdots(-5-s)}{s!} = (-1)^s \cdot \frac{(s+5)(s+4)\cdots(6)}{s!} = (-1)^s \binom{s+5}{5}.$$

To get the term x^{18} , we can only choose the terms $(x, 1, x^{17})$, (x, x^{10}, x^7) or $(x^{10}, 1, x^8)$ from the three brackets. So the coefficient is

$$-\binom{-6}{17} + \binom{5}{1}\binom{-6}{7} - \binom{-6}{8} = \binom{22}{5} - \binom{5}{1}\binom{12}{5} - \binom{13}{5} = 21087.$$

5. The answer is 181.

Let $n^2 = (m+1)^3 - m^3 = 3m^2 + 3m + 1$. This implies

$$(2n-1)(2n+1) = 4n^2 - 1 = 12m^2 + 12m + 3 = 3(2m+1)^2.$$

As $(2n-1, 2n+1) = (2n-1, 2) = 1$, one of $2n-1$ and $2n+1$ is a square and the other is 3 times a square.

- If $2n+1$ is a square, then $3 \mid 2n-1$, and hence $n \equiv 2 \pmod{3}$. But then $2n+1 \equiv 2 \pmod{3}$, so $2n+1$ cannot be a square. This is a contradiction.
- If $2n-1$ is a square, let $2n-1 = a^2$ and $2n+79 = b^2$. This implies

$$80 = b^2 - a^2 = (b-a)(b+a).$$

To maximize n , we need to maximize $b+a$. Since $b-a$ and $b+a$ have the same parity, the maximal case is $(b-a, b+a) = (2, 40)$. In that case, we have $(a, b, n) = (19, 21, 181)$. This means the largest n is 181.

6. The answer is 759.

For $m = 1, 2, 3$, the left-hand side becomes $\left\lceil \frac{3^n}{4} \right\rceil, \left\lceil \frac{5^n}{14} \right\rceil, \left\lceil \frac{7^n}{30} \right\rceil$. Since

$$\left\lceil \frac{3^8}{4} \right\rceil < 2013 < \left\lceil \frac{3^9}{4} \right\rceil, \quad \left\lceil \frac{5^6}{14} \right\rceil < 2013 < \left\lceil \frac{5^7}{14} \right\rceil, \quad \left\lceil \frac{7^5}{30} \right\rceil < 2013 < \left\lceil \frac{7^6}{30} \right\rceil,$$

there is no solution.

For $m \geq 4$, we have

$$2^{11} > 2014 > \frac{(2m+1)^n}{m(3m+1)} > \frac{(2m)^n}{m(4m)} = 2^{n-2}m^{n-2} \geq 2^{n-2}4^{n-2} = 2^{3n-6}.$$

This implies $n \leq 5$.

- For $n \leq 2$, we have $\frac{(2m+1)^n}{m(3m+1)} < \frac{(3m)^2}{m(3m)} = 3 < 2013$.

- For $n = 3$, we have

$$\frac{(2m+1)^3}{m(3m+1)} = \frac{8m^3 + 12m^2 + 6m + 1}{3m^2 + m} = \frac{8}{3}m + \frac{28}{9} + \frac{26m+9}{9(3m^2+m)}.$$

Clearly, the last fraction is less than 1. So we need $2012 < \frac{8}{3}m + \frac{28}{9} < 2014$.

The only integer solution is $m = 754$. We check that

$$\begin{aligned} & \left\lceil \frac{8}{3}(754) + \frac{28}{9} + \frac{26(754)+9}{9(3(754)^2+(754))} \right\rceil \\ &= \left\lceil 2013 + \frac{7}{9} + \frac{26(754)+9}{9(3(754)^2+(754))} \right\rceil \\ &= 2013 \end{aligned}$$

since $\frac{26(754)+9}{9(3(754)^2+(754))} < \frac{27(754)}{27(754)^2} < \frac{2}{9}$. So $m = 754$ is a solution.

- For $n \geq 4$, we claim that $f(m) = \frac{(2m+1)^n}{m(3m+1)}$ is increasing in m . Note that

$$\ln f(m) = n \ln(2m+1) - \ln m - \ln(3m+1),$$

and so

$$\frac{f'(m)}{f(m)} = \frac{2n}{2m+1} - \frac{1}{m} - \frac{3}{3m+1} \geq \frac{8}{2m+1} - \frac{3}{2m+1} - \frac{3}{2m+1} > 0.$$

Thus, $f'(m) > 0$, so that f is increasing.

Now, for $n = 4$, we check that

$$\frac{(2(18)+1)^4}{18(3(18)+1)} = \frac{37^4}{18 \cdot 55} = \frac{1369^2}{990} < 2013$$

and

$$\frac{(2(19)+1)^4}{19(3(19)+1)} = \frac{39^4}{19 \cdot 58} = \frac{1521^2}{1102} > 2014.$$

There is no solution.

For $n = 5$, we check that

$$\frac{(2(4)+1)^5}{4(3(4)+1)} = \frac{9^5}{4 \cdot 13} = \frac{59049}{52} < 2013,$$

$$\left\lceil \frac{(2(5)+1)^5}{5(3(5)+1)} \right\rceil = \left\lceil \frac{11^5}{5 \cdot 16} \right\rceil = \left\lceil \frac{161051}{80} \right\rceil = 2013$$

and

$$\frac{(2(6) + 1)^5}{6(3(6) + 1)} = \frac{13^5}{6 \cdot 19} = \frac{371293}{114} > 2014.$$

Thus, $m = 5$ is another solution.

To summarize, the answer is $754 + 5 = 759$.

Test 2

1. Let D be the intersection point of the internal angle bisector of $\angle BAC$ and BC .

By the angle bisector theorem, we have $\frac{BD}{CD} = \frac{c}{b}$, hence

$$BD = \frac{ac}{b+c} \quad \text{and} \quad CD = \frac{ab}{b+c}.$$

By Stewart's theorem, we have

$$i_a^2 = \frac{1}{a} \left(b^2 \cdot \frac{ac}{b+c} + c^2 \cdot \frac{ab}{b+c} \right) - \frac{ac}{b+c} \cdot \frac{ab}{b+c} = \frac{bc((b+c)^2 - a^2)}{(b+c)^2} = \frac{4bcs(s-a)}{(b+c)^2},$$

where s is the semiperimeter of $\triangle ABC$. Now, by the AM-GM inequality, since $s \neq s-a$, we have

$$i_a = \frac{2\sqrt{bc}}{b+c} \cdot \sqrt{s(s-a)} < \frac{2\sqrt{bc}}{b+c} \cdot \frac{s + (s-a)}{2} = \sqrt{bc} \leq \frac{b+c}{2}.$$

Similarly, we have $i_b < \frac{c+a}{2}$ and $i_c < \frac{a+b}{2}$. Therefore, we have

$$ai_a + bi_b + ci_c < ab + bc + ca \leq a^2 + b^2 + c^2.$$

This is bounded above by $9R^2$ since $9R^2 - (a^2 + b^2 + c^2) = OH^2$, where O and H are the circumcentre and orthocentre of $\triangle ABC$ respectively.

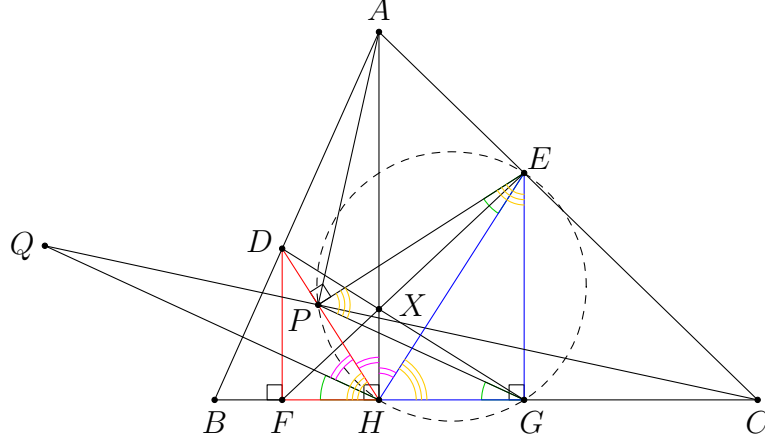
2. Suppose DG, EF, AH are concurrent at X . Let Q be the point on the line CP such that $QH \parallel PG$. We only work on the configuration as shown since the other cases are similar.

Firstly, by the parallel lines $AH \parallel EG$ and $QH \parallel PG$, we have $\triangle CAH \sim \triangle CEG$ and $\triangle CPG \sim \triangle CQH$. Thus, we have

$$\frac{AE}{EC} = \frac{HG}{GC} = \frac{QP}{PC}.$$

By the angle bisector theorem, it suffices to prove $\frac{AP}{PC} = \frac{AE}{EC}$. Equivalently, we need to prove $PA = PQ$.

By similar triangles, we have $\frac{DF}{EG} = \frac{DX}{GX} = \frac{FH}{GH}$. Together with the included angles $\angle DFH = \angle EGH = 90^\circ$, we have $\triangle DFH \sim \triangle EGH$. This implies $\angle DHF = \angle EHG$, and hence $\angle AHD = \angle AHE$.



Note that E, P, H, G are concyclic since $\angle EPH = \angle EGH = 90^\circ$. This yields

$$\angle PEH = \angle PGH = \angle QHB,$$

and so

$$\angle EHP = 90^\circ - \angle PEH = 90^\circ - \angle QHB = \angle AHQ.$$

It follows that $\angle DHQ = \angle EHA = \angle DHA$.

Next, note that $\angle PEG = \angle DHF = \angle GHE = \angle GPE$. This implies $PG = EG$. By similar triangles, we obtain

$$\frac{PG}{QH} = \frac{CG}{CH} = \frac{EG}{AH} = \frac{PG}{AH}.$$

Therefore, we have $QH = AH$. Thus, HP is the perpendicular bisector of AQ , and hence $PA = PQ$ as desired.

3. The greatest possible value of $d(m+n)$ is 166635.

Let $m = da$ and $n = db$ so that $(a, b) = 1$. The condition becomes $d \mid 3(a-b)$. WLOG assume $m > n$. Let $N = d(m+n) = d^2(a+b)$.

If $3 \nmid d$, then $d \mid a-b$. This implies $d \leq a-b$. Also, we have $a = \frac{m}{d} \leq \frac{2013}{d}$. This yields

$$N = d^2(a+b) \leq d^2(2a-d) \leq 4026d - d^3.$$

By differentiation, the maximum value of $4026d - d^3$ is attained at $d = \sqrt{\frac{4026}{3}}$. Thus,

$$N \leq \sqrt{\frac{4026}{3}} \left(4026 - \frac{4026}{3} \right) < 40 \cdot 3000 = 120000.$$

If $3 \mid d$, then $d \leq 3(a+b)$. Also, we have $a \leq \left\lfloor \frac{2013}{d} \right\rfloor$. This yields

$$N = d^2(a+b) \leq d^2 \left(2a - \frac{d}{3} \right) \leq d^2 \left(2 \left\lfloor \frac{2013}{d} \right\rfloor - \frac{d}{3} \right).$$

- For $d \leq 54$, we have $N \leq 4026d - \frac{d^3}{3} \leq 4026(54) - \frac{(54)^3}{3} = 164916$ since $f(d) = 4026d - \frac{d^3}{3}$ is increasing for $d \leq 54$ (again by differentiation).
- For $d = 57$, we have $N \leq (57)^2 \left(2 \left\lfloor \frac{2013}{57} \right\rfloor - \frac{57}{3} \right) = 165699$.
- For $d = 60$, we have $N \leq (60)^2 \left(2 \left\lfloor \frac{2013}{60} \right\rfloor - \frac{60}{3} \right) = 165600$.
- For $d = 63$, we have $N \leq (63)^2 \left(2 \left\lfloor \frac{2013}{63} \right\rfloor - \frac{63}{3} \right) = 162729$.
- For $d = 66$, we have $N \leq (66)^2 \left(2 \left\lfloor \frac{2013}{66} \right\rfloor - \frac{66}{3} \right) = 165528$.
- For $d = 69$, we have $N \leq (69)^2 \left(2 \left\lfloor \frac{2013}{69} \right\rfloor - \frac{69}{3} \right) = 166635$.
- For $d \geq 72$, we have $N \leq 4026d - \frac{d^3}{3} \leq 4026(72) - \frac{(72)^3}{3} = 165456$ since $f(d)$ is decreasing for $d \geq 54$.

To conclude, we have $N \leq 166635$. Equality holds when $d = 69$, $a = 29$ and $b = 6$. This means $(m, n) = (2001, 414)$.

4. The answer is 806.

By the pigeonhole principle, there is a club with $n \geq \frac{2013^2}{25}$ pairs. Suppose there are a boys and b girls in this club. Then the number of pairs is at most ab . By the AM-GM inequality, we have

$$\frac{a+b}{2} \geq \sqrt{ab} \geq \sqrt{n} \geq \frac{2013}{5}.$$

This implies the number of members of this club is $a + b \geq 806$.

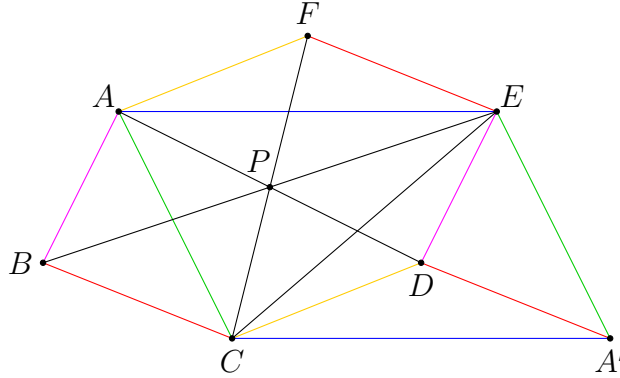
We now give a construction for which $k \leq 806$. We partition the boys into 5 groups $B_1, B_2, \dots, B_5$ such that each of B_1, B_2, B_3 has 403 boys, while each of B_4, B_5 has 402 boys. Similarly, we partition the girls into 5 groups $G_1, G_2, \dots, G_5$ in a similar way. Suppose each boy in group B_i and each girl in group G_j choose the club C_{ij} . Then each club consists of at most $403 + 403 = 806$ members. This proves $k \leq 806$.

It follows that the maximum k is 806.

Test 3

- Suppose AD, BE, CF meet at P . Since P is the midpoint of AD, BE, CF , all of $ABDE, BCEF$ and $C DFA$ are parallelograms. Let A' be the point such that $ACA'E$ is a parallelogram. Since $EA' = AC = FD$ and $EA' \parallel AC \parallel FD$, $FDA'E$ is also a parallelogram. Thus, we have $DA' = FE$. Together with $CD = AF$ and $CA' = AE$, we find that $\triangle DCA' \cong \triangle FAE$. By symmetry, we also have $\triangle DEA' \cong \triangle BAC$. It follows that

$$\begin{aligned} [ABCDEF] &= [ACE] + [ABC] + [CDE] + [EFA] \\ &= [ACE] + [EDA'] + [CDE] + [A'DC] \\ &= [ACA'E] = 2[ACE]. \end{aligned}$$



- The greatest k is 13.

Firstly, consider $a = b = \frac{2}{3}$ and $c = \frac{9}{4}$. They satisfy $abc = 1$. The inequality becomes

$$\frac{3}{2} + \frac{3}{2} + \frac{4}{9} + \frac{12k}{55} \geq \frac{k}{4} + 3,$$

i.e. $\frac{4}{9} \geq \frac{7k}{220}$. As k is a positive integer, this implies $k \leq 13$.

We now prove the inequality

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{13}{a+b+c+1} \geq \frac{13}{4} + 3 = \frac{25}{4}.$$

Let the left-hand side be $f(a, b, c)$. WLOG assume $c \geq b \geq a$. We first prove that

$$f(a, b, c) \geq f(\sqrt{ab}, \sqrt{ab}, c).$$

Note that the new triple still satisfies $(\sqrt{ab})(\sqrt{ab})(c) = 1$. Now,

$$\begin{aligned} & f(a, b, c) \geq f(\sqrt{ab}, \sqrt{ab}, c) \\ \Leftrightarrow & \frac{1}{a} + \frac{1}{b} - \frac{2}{\sqrt{ab}} \geq \frac{13}{2\sqrt{ab} + c + 1} - \frac{13}{a + b + c + 1} \\ \Leftrightarrow & \frac{(\sqrt{a} - \sqrt{b})^2}{ab} \geq \frac{13(\sqrt{a} - \sqrt{b})^2}{(2\sqrt{ab} + c + 1)(a + b + c + 1)}. \end{aligned}$$

It suffices to show $(2\sqrt{ab} + c + 1)(a + b + c + 1) \geq 13ab$. Indeed, since $a + b \geq 2\sqrt{ab}$, we only need to prove

$$2\sqrt{ab} + c + 1 \geq \sqrt{13ab}.$$

As $c \geq b \geq a$ and $abc = 1$, we have $c \geq 1$. Therefore, we have $\sqrt{c}(c + 1) \geq 2$. This implies

$$c + 1 \geq \frac{2}{\sqrt{c}} = 2\sqrt{ab} > (\sqrt{13} - 2)\sqrt{ab}.$$

This proves $f(a, b, c) \geq f(\sqrt{ab}, \sqrt{ab}, c)$.

After this mixing step, it remains to consider the case $f\left(t, t, \frac{1}{t^2}\right)$ where $t > 0$. Now,

$$\begin{aligned} & f\left(t, t, \frac{1}{t^2}\right) \geq \frac{25}{4} \\ \Leftrightarrow & \frac{2}{t} + t^2 + \frac{13t^2}{2t^3 + t^2 + 1} \geq \frac{25}{4} \\ \Leftrightarrow & 8t^6 + 4t^5 - 50t^4 + 47t^3 + 8t^2 - 25t + 8 \geq 0 \\ \Leftrightarrow & (t - 1)^2(8t^4 + 20t^3 - 18t^2 - 9t + 8) \geq 0. \end{aligned}$$

It remains to prove $8t^4 + 20t^3 - 18t^2 - 9t + 8 \geq 0$ for $t > 0$. Indeed, the left-hand side is equal to

$$(8t^2 + 30t)\left(t - \frac{5}{8}\right)^2 + \frac{1}{32}(524t^2 - 663t + 256).$$

The first term is nonnegative since $t > 0$. Since $663^2 - 4(524)(256) < 0$, the last term is nonnegative. This proves the inequality and we are done.

3. The maximum k is $2n - 1$.

Firstly, we show that $k \geq 2n - 1$. Suppose the first entry in a row is a . By adding $-a$ to this row, we make the first entry 0. Similarly, we do this for each row until the first entry of each row is 0. Next, suppose the first entry in a column is b . By adding $-b$ to this column, we make the first entry 0. Similarly, we do this for each column until the first entry of each column is 0. Note that we do nothing in the first column, and so all $2n - 1$ entries in the first row and the first column are 0.

Secondly, we show that $k \leq 2n - 1$. Let (i, j) be the cell in row i and column j , and let a_{ij} be the entry inside (i, j) . Suppose we can make $2n$ entries 0 after a finite number of steps. We call the cells containing these entries *special cells*. We prove the following result.

Claim. There exists a sequence of indices $i_1, i_2, \dots, i_s, j_1, j_2, \dots, j_s$ such that

$$(i_1, j_1), (i_1, j_2), (i_2, j_2), (i_2, j_3), \dots, (i_s, j_s), (i_s, j_1)$$

are distinct special cells.

Proof. Consider a graph as follows. Let the vertices be the $2n$ special cells. For each row and each column, we draw an edge between any pair of adjacent vertices. If there are $r_1, r_2, \dots, r_n$ vertices in the n rows, then the number of horizontal edges is at least

$$(r_1 - 1) + (r_2 - 1) + \dots + (r_n - 1) = 2n - n = n.$$

Similarly, there are at least n vertical edges. Thus, we have $2n$ edges in total. It is well-known that a graph with $2n$ vertices and at least $2n$ edges consists of a cycle.

This cycle in the graph corresponds to a ‘cycle’ of cells in the table. If this cycle consists of three consecutive cells $(i, j), (i, j'), (i, j'')$ in the same row, then we can remove (i, j') to shorten the cycle. We can do the same thing for consecutive cells in the same column. Eventually, we obtain a cycle such that the cells alternately lie in the same row and in the same column, which is exactly our claim. $\square$

Now, we choose a cycle of special cells as given by the claim. Note that the value

$$(a_{i_1 j_1} + a_{i_2 j_2} + \dots + a_{i_s j_s}) - (a_{i_1 j_2} + a_{i_2 j_3} + \dots + a_{i_s j_1}) \quad (1)$$

is unchanged after any operation. Indeed, if we add x to row i , and there are m indices among $i_1, i_2, \dots, i_s$ equal to i , then the value is changed by $+mx - mx = 0$. The same holds for column operations. Since all the entries involved eventually become 0, the initial value of (1) must be 0.

Consider the table consisting of the numbers $2, 2^2, \dots, 2^{n^2}$ in any order. Since each number is larger than the sum of all smaller numbers, the relation (1) cannot hold for any choice of the indices. This is a contradiction. Thus, it is impossible to have at least $2n$ special cells.

Combining the two parts, we know that the maximum k is $2n - 1$.

Test 4

1. A special set contains at most 4 elements.

To prove $n = |M| \leq 4$, it suffices to show $n = 5$ is impossible. Suppose on the contrary that there exists such a set M with 5 elements a, b, c, d, e .

If ab and ac are rational, it is impossible that $bc \in \mathbb{Q}$, since otherwise

$$a^2 = \frac{(ab)(ac)}{bc} \in \mathbb{Q}.$$

Also, it is impossible that $b + c \in \mathbb{Q}$. Indeed, if $b + c = r \in \mathbb{Q}$, then since

$$\frac{b}{c} = \frac{ab}{ac} \in \mathbb{Q},$$

we have $\frac{r - c}{c} \in \mathbb{Q}$. This implies $\frac{r}{c} \in \mathbb{Q}$, and so $c \in \mathbb{Q}$, contradicting $c^2 \notin \mathbb{Q}$.

Therefore, WLOG we may assume $a + b, a + c, a + d \in \mathbb{Q}$. It is impossible that $b + c \in \mathbb{Q}$, since otherwise

$$a = \frac{(a + b) + (a + c) - (b + c)}{2} \in \mathbb{Q},$$

contradicting $a^2 \notin \mathbb{Q}$. Thus, we have $bc \in \mathbb{Q}$. Similarly, $bd, cd \in \mathbb{Q}$. But this contradicts the above observation. This implies $n \leq 4$.

We now provide a set M with $n = 4$. Consider

$$M = \{a, b, c, d\} = \left\{ \frac{1 + \sqrt{13}}{2}, \frac{3 - \sqrt{13}}{2}, \frac{-1 + \sqrt{13}}{2}, \frac{-3 - \sqrt{13}}{2} \right\}.$$

We plainly check that

$$\begin{aligned} a + b &= 2 \in \mathbb{Q}, & ab &= \frac{-5 + \sqrt{13}}{2} \notin \mathbb{Q}, \\ a + c &= \sqrt{13} \notin \mathbb{Q}, & ac &= 3 \in \mathbb{Q}, \\ a + d &= -1 \in \mathbb{Q}, & ad &= -4 - \sqrt{13} \notin \mathbb{Q}, \\ b + c &= 1 \in \mathbb{Q}, & bc &= -4 + \sqrt{13} \notin \mathbb{Q}, \\ b + d &= -\sqrt{13} \notin \mathbb{Q}, & bd &= 1 \in \mathbb{Q}, \\ c + d &= -2 \in \mathbb{Q}, & cd &= \frac{-5 - \sqrt{13}}{2} \notin \mathbb{Q}, \end{aligned}$$

and

$$\begin{aligned}a^2 &= \frac{7 + \sqrt{13}}{2} \notin \mathbb{Q}, \\b^2 &= \frac{11 - 3\sqrt{13}}{2} \notin \mathbb{Q}, \\c^2 &= \frac{7 - \sqrt{13}}{2} \notin \mathbb{Q}, \\d^2 &= \frac{11 + 3\sqrt{13}}{2} \notin \mathbb{Q}.\end{aligned}$$

2. The only solution is $f(a) = 0$ for $a \notin S$ and $f(x) = x$ for $x \in S$.

Label the equations as follows.

$$f(a + x + y) + f(f(a)) + f(x) + f(y) = x + y \quad (1)$$

$$f(axy) + f(a) + f(x)f(y) = xy \quad (2)$$

Case 1. $0 \notin S$

Putting $a = 0$ in (2), we obtain

$$f(x)f(y) = xy - 2f(0) \quad (3)$$

for all $x, y \in S$. In particular, by putting $x = y$, we get

$$f(x)^2 = x^2 - 2f(0) \quad (4)$$

for all $x \in S$. Then we have

$$(x^2 - 2f(0))(y^2 - 2f(0)) = f(x)^2 f(y)^2 = (xy - 2f(0))^2,$$

which implies $2f(0)(x^2 + y^2) = 4f(0)xy$. This means

$$2f(0)(x - y)^2 = 0.$$

Since S has at least two elements, we can choose distinct $x, y \in S$ to conclude that $f(0) = 0$. Therefore, by (4),

$$f(x)^2 = x^2$$

for all $x \in S$. Since $f(x)f(y) = xy$ by (3), we see that the choice of the sign for $f(x)$ is independent of x . This means $f(x) = x$ for all $x \in S$ or $f(x) = -x$ for all $x \in S$.

Now, (2) is reduced to

$$f(axy) + f(a) = 0. \quad (5)$$

For fixed nonzero $a \notin S$, if $\frac{1}{a} \in S$, we may put $y = \frac{1}{a}$ in (5) to get $f(x) + f(a) = 0$. But this cannot be true as we can choose two different values for x (and hence $f(x)$). Therefore, we must have $\frac{1}{a} \notin S$. From this, we see that $x \in S$ implies $\frac{1}{x} \in S$, since otherwise $x = (x^{-1})^{-1} \notin S$.

Now, we can put $y = \frac{1}{x}$ in (5) to obtain $2f(a) = 0$, i.e. $f(a) = 0$ for any $a \notin S$. Equation (1) becomes

$$f(a + x + y) \pm (x + y) = x + y.$$

By putting $x = y$, we get

$$f(a + 2x) \pm 2x = 2x.$$

If the negative sign is chosen, then we have $f(a + 2x) = 4x$. Since $f(a + 2x)$ can only be $-(a + 2x)$ or 0 (depending on whether $a + 2x \in S$ or not), we need $a = -6x$ for any $a \notin S$ and $x \in S$ (note we cannot have $0 = 4x$). This is impossible as we can find two distinct elements in S . Therefore, $f(a) = 0$ for $a \notin S$ and $f(x) = x$ for $x \in S$.

Case 2. $0 \in S$

Putting $y = 0$ in (2), we obtain

$$f(0) + f(a) + f(x)f(0) = 0 \tag{6}$$

for all $a \notin S$, $x \in S$. In particular, by putting $x = 0$, we get

$$f(a) = -f(0) - f(0)^2. \tag{7}$$

Then equation (6) becomes

$$f(x)f(0) = f(0)^2.$$

If $f(0) \neq 0$, we need $f(x) = f(0)$ for all $x \in S$. Consider equation (1). It now becomes

$$f(a + x + y) + f(-f(0) - f(0)^2) + 2f(0) = x + y.$$

Note that the left-hand side can take at most two values (as $f(a + x + y)$ can be $-f(0) - f(0)^2$ or $f(0)$). However, as there are at least two elements in S , say 0 and $z \neq 0$, the right-hand side can take at least three different values, namely, 0, z , $2z$. This is a contradiction, and hence $f(0) = 0$. By (7), we obtain $f(a) = 0$ for $a \notin S$.

Next, we prove that $a \notin S$ implies $-a \notin S$. Indeed, suppose $-a \in S$. We put $y = -a$ in (1) to get

$$2f(x) + f(-a) = x - a. \tag{8}$$

By putting $x = 0$, we get

$$f(-a) = -a, \quad (9)$$

and hence $2f(x) = x$ for any $x \in S$ by (8). Now, by putting $x = -a$, we obtain $2f(-a) = -a$. But then we have $f(-a) = -a$ by (9). This forces $a = 0 \in S$, which is a contradiction. So the claim is true. From this, we see that $x \in S$ implies $-x \in S$. Thus, we can put $y = -x$ in (1). This gives

$$f(x) + f(-x) = 0 \quad (10)$$

for any $x \in S$.

Similarly, we shall prove that $z, w \in S$ implies $z + w \in S$. Suppose on the contrary that $z + w \notin S$. Note that $-w \in S$ from above. So we can put $a = z + w$ and $x = 0$, $y = -w$ in (1) to obtain

$$f(z) + f(-w) = -w.$$

Using (10), we get

$$f(z) - f(w) = -w.$$

By symmetry, we also have

$$f(w) - f(z) = -z.$$

Adding these, we obtain $z + w = 0 \in S$, which is a contradiction.

It is now clear that $a + x + y \notin S$ for any $a \notin S$ and $x, y \in S$. Otherwise, if $a + x + y \in S$, since $-x - y = (-x) + (-y) \in S$, we have

$$a = (a + x + y) + (-x - y) \in S.$$

It follows that (1) is reduced to $f(x) + f(y) = x + y$. Putting $x = y$, we get $f(x) = x$ for any $x \in S$.

In any case, $f(a) = 0$ for $a \notin S$ and $f(x) = x$ for $x \in S$ is the only possible function. One can check that this is indeed a solution if and only if $a + x + y, axy \notin S - \{0\}$ for any $a \notin S$ and $x, y \in S$.

Remarks. When S is a field or the multiplicative group of a field in $\mathbb{R}$, for example $S = \mathbb{Q}$, the condition mentioned at the end of the solution holds. Thus, there exists such a function f in these cases.

3. The smallest and the largest possible values of R are 0 and $\frac{n(n+1)(2n+1)}{6}$ respectively.

Let the points be $A_1, A_2, \dots, A_{2n+1}$. If we draw the vector $\overrightarrow{A_i A_j}$ whenever $i < j$, then for any $\triangle A_i A_j A_k$ with $i < j < k$, we have

$$\overrightarrow{A_i A_j} + \overrightarrow{A_i A_k} + \overrightarrow{A_j A_k} = 2\overrightarrow{A_i A_k} \neq \mathbf{0}.$$

Therefore, it is possible that $R = 0$.

For the maximum value, we call $\triangle A_i A_j A_k$ *special* if it does not have sum 0. Note that the vectors of a special triangle must be $\overrightarrow{A_i A_j}$, $\overrightarrow{A_i A_k}$ and $\overrightarrow{A_j A_k}$ up to symmetry. We say that A_i is the initial vertex and A_k is the terminal vertex of this triangle.

For each vertex A_i , suppose there are b_i vectors pointing away from A_i and c_i vectors pointing towards A_i . Note that $b_i + c_i = 2n$. Then there are $\binom{b_j}{2}$ pairs of vectors $\overrightarrow{A_i A_j}$ and $\overrightarrow{A_i A_k}$, and $\binom{c_j}{2}$ pairs of vectors $\overrightarrow{A_j A_i}$ and $\overrightarrow{A_k A_i}$. Therefore, there are $\binom{b_j}{2}$ special triangles with A_i as initial vertex (note that the vector between A_j and A_k is irrelevant), and $\binom{c_j}{2}$ special triangles with A_i as terminal vertex. As each special triangle is counted twice, it follows that the number of special triangles is

$$\frac{1}{2} \sum_{j=1}^{2n+1} \left[\binom{b_j}{2} + \binom{c_j}{2} \right] \geq \frac{1}{2} \sum_{j=1}^{2n+1} \left[\binom{n}{2} + \binom{n}{2} \right] = \frac{n(n-1)(2n+1)}{2}$$

by Jensen's inequality. Since there are $\binom{2n+1}{3}$ triangles in total, this implies

$$R \leq \binom{2n+1}{3} - \frac{n(n-1)(2n+1)}{2} = \frac{n(n+1)(2n+1)}{6}.$$

This maximum value $R = \frac{n(n+1)(2n+1)}{6}$ can be attained. For example, suppose we draw the vector $\overrightarrow{A_j A_{j+m}}$ for $1 \leq j \leq 2n+1$ and $m = 1, 2, \dots, n$, where the indices are taken modulo $2n+1$. Then $b_j = c_j = n$ for any j , and so equality holds.

Year 2015

Test 1

1. The answer is $\sqrt{2} - 1$.

Let $P(n)$ be the probability that there is eventually no alien on the Earth given that there are currently n aliens, where n is a positive integer. Let $P(1) = c$. Since different aliens act independently, we have $P(n) = c^n$.

By the given actions, we know that

$$P(1) = \frac{1}{4} + \frac{1}{4}P(2) + \frac{1}{4}P(3) + \frac{1}{4}P(1).$$

This becomes $4c = 1 + c^2 + c^3 + c$, which can be factorized as

$$(c - 1)(c^2 + 2c - 1) = 0.$$

Thus, we have $c = 1$ or $-1 \pm \sqrt{2}$. As c is a probability, we have $0 \leq c \leq 1$, so that $c = -1 - \sqrt{2}$ is rejected. We shall also reject $c = 1$ in the following.

Indeed, note that the expected number of aliens arisen from one of the aliens is $\frac{0 + 2 + 3 + 1}{4} = \frac{3}{2} > 1$. Therefore, the expected number of aliens increases indefinitely. So it is impossible that all aliens must eventually be destroyed. This shows $c < 1$.

Therefore, the probability is $c = \sqrt{2} - 1$.

2. The maximum number of sets is 7.

We first assume there are at least 8 sets. WLOG assume 1 and 2 are common elements of at least two sets.

- If there are 4 sets containing both 1 and 2, say $\{1, 2, 3, 4\}$, $\{1, 2, 5, 6\}$, $\{1, 2, 7, 8\}$, $\{1, 2, 9, 10\}$, then all other sets must contain both 1 and 2. Otherwise a set has to contain 1, 3, 5, 7, 9 (up to renaming of the elements) or 3, 4, ..., 10, which contradicts condition (i). But now condition (iii) is violated. So this case is not possible.
- If exactly 3 sets contain both a and b , say $\{1, 2, 3, 4\}$, $\{1, 2, 5, 6\}$, $\{1, 2, 7, 8\}$, then all other sets must contain one of 1, 2, one of 3, 4, one of 5, 6 and one of 7, 8. Consider the following triples of numbers:

$$\begin{aligned} &\{3, 5, 7\}, \{4, 6, 8\}, \\ &\{3, 5, 8\}, \{4, 6, 7\}, \\ &\{3, 6, 7\}, \{4, 5, 8\}, \\ &\{3, 6, 8\}, \{4, 5, 7\}. \end{aligned}$$

At most one triple in each row may be a subset of one of the sets, since otherwise the two sets containing the triples in the same row share at most one element. Therefore, the number of sets is at most $3+4 = 7$, contradiction.

- If exactly 2 sets contain both a and b , say $\{1, 2, 3, 4\}$ and $\{1, 2, 5, 6\}$, then except $\{3, 4, 5, 6\}$, all other sets must contain one of 1, 2, one of 3, 4 and one of 5, 6. Consider the following triples of numbers:

$$\begin{aligned} &\{1, 3, 5\}, \{2, 4, 6\}, \\ &\{1, 3, 6\}, \{2, 4, 5\}, \\ &\{1, 4, 5\}, \{2, 3, 6\}, \\ &\{1, 4, 6\}, \{2, 3, 5\}. \end{aligned}$$

Again, at most one triple in each row may be a subset of one of the sets. Therefore, the number of sets is at most $2 + 1 + 4 = 7$, contradiction.

Finally, we can find 7 sets satisfying all the given conditions, for example,

$$\{1, 2, 3, 4\}, \{1, 2, 5, 6\}, \{1, 2, 7, 8\}, \{1, 3, 5, 7\}, \{1, 3, 6, 8\}, \{1, 4, 5, 8\}, \{1, 4, 6, 7\}.$$

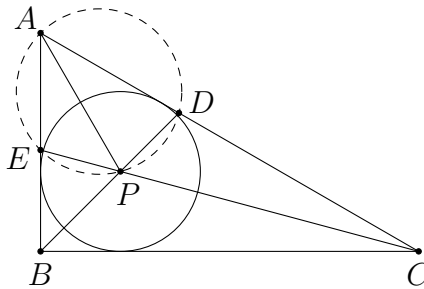
Therefore, the answer is 7.

3. The largest possible value of BC is $3 + \sqrt{3}$.

Since $PD = PE$ and AP bisects $\angle EAD$, the quadrilateral $AEPD$ is cyclic or it is a kite. The latter case is impossible since $\angle B > \angle C$. Thus, we have

$$180^\circ = \angle EAD + \angle DPE = A + \left(90^\circ + \frac{A}{2}\right) = 90^\circ + \frac{3A}{2}.$$

This implies $A = 60^\circ$.



As usual, let a, b, c and s be the lengths of BC, CA, AB and the semiperimeter respectively. Firstly, since the inradius is 1, we have

$$s - a = \cot \frac{A}{2} = \sqrt{3}.$$

By the sine law, we have $\frac{a}{\sin 60^\circ} = \frac{b}{\sin B} = \frac{c}{\sin C}$, and so

$$\frac{2a}{\sqrt{3}} = \frac{b + c - a}{\sin B + \sin C - \sin 60^\circ}.$$

Note that

$$\sin B + \sin C = 2 \sin \frac{B+C}{2} \cos \frac{B-C}{2} = \sqrt{3} \cos (60^\circ - C).$$

Therefore, we have

$$a = \frac{\sqrt{3}(s-a)}{\sin B + \sin C - \sin 60^\circ} = \frac{2\sqrt{3}}{2 \cos (60^\circ - C) - 1} \leq \frac{2\sqrt{3}}{2 \cos (60^\circ - 30^\circ) - 1} = 3 + \sqrt{3}.$$

Equality holds when $C = 30^\circ$ and $B = 90^\circ$.

4. Yes. By the pigeonhole principle, two of the numbers of the form $11 \dots 1$ leave the same remainder when divided by a_{2014} . Their difference, which is of the form $11 \dots 100 \dots 0$, is a multiple of a_{2014} . Since $(a_{2014}, 10) = 1$, we know that a_{2014} divides $\underbrace{11 \dots 1}_{m \text{ times}}$ for some $m \in \mathbb{Z}^+$. It is then obvious that $a_{2014} \mid a_{2014+km}$ for any $k \in \mathbb{Z}^+$, since

$$a_{2014+km} = a_{2014} \times 10^{km} + \underbrace{77 \dots 7}_{km \text{ times}}.$$

5. The maximum number of colours is 41.

An example of 41 colours is given below, where the numbers represent different colours.

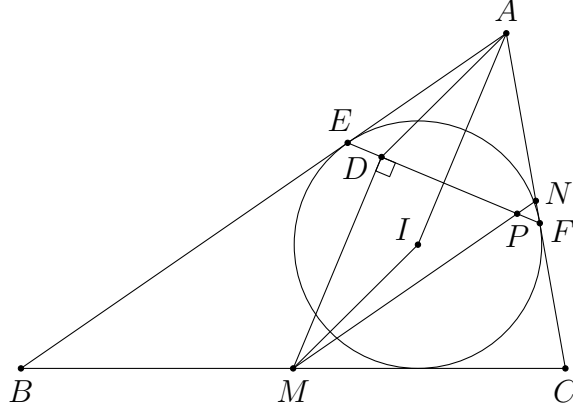
1	1	1	1	1	1	2	3	4	5
1	1	1	1	1	6	7	8	9	1
1	1	1	1	10	11	12	13	1	1
1	1	1	14	15	16	17	1	1	1
1	1	18	19	20	21	1	1	1	1
1	22	23	24	25	1	1	1	1	1
26	27	28	29	1	1	1	1	1	1
31	32	33	1	1	1	1	1	1	30
36	37	1	1	1	1	1	1	34	35
41	1	1	1	1	1	1	38	39	40

Now, assume at least 42 colours are used. By the pigeonhole principle, WLOG we may assume the first row contains $\left\lceil \frac{42}{10} \right\rceil = 5$ different colours. We need to use at least 37 more colours. Again by the pigeonhole principle, we may assume

the second row contains $\left\lceil \frac{37}{9} \right\rceil = 5$ different colours. Now, each column already has two different colours, and hence there can be at most 3 more new colours in each column. This gives a total of at most $5 + 5 + 3 \times 10 = 40$ colours, which is a contradiction. Therefore, at most 41 colours can be used.

6. The product is 9360.

Let a, b, c, r, s be the lengths of BC, CA, AB , the inradius and the semiperimeter of $\triangle ABC$ respectively. Let N be the midpoint of AC , and let P be the intersection point of EF and MN . It is well-known that $\angle BPC = 90^\circ$. Therefore, we have $MP = \frac{a}{2}$. (Alternatively, one can prove this by noting $MN = \frac{c}{2}$ and $PN = NF = \frac{c-a}{2}$.)



Note that $NM \parallel AB$ by the midpoint theorem. Since $ADMI$ is a parallelogram, we have

$$AI = DM = PM \sin \angle DPM = \frac{a}{2} \sin \angle FEA = \frac{a}{2} \cos \frac{A}{2}.$$

On the other hand, we have

$$AI = \frac{r}{\sin \frac{A}{2}} = \frac{bc \sin A}{2s \sin \frac{A}{2}} = \frac{bc}{s} \cos \frac{A}{2}.$$

Equating these, we obtain $2bc = as$. As it is given that $2s = 65$, this yields $4bc = 65a$. Then the given condition becomes $\frac{4bc}{65} + b + c = 65$. This can be factorized as

$$(4b + 65)(4c + 65) = 5 \cdot 65^2 = 5^3 \cdot 13^2.$$

As $4b + 65$ and $4c + 65$ are larger than 65, we must have $4b + 65 = 125$ and $4c + 65 = 169$ up to permutation. This gives $(a, b, c) = (24, 15, 26)$ or $(24, 26, 15)$. Therefore, $abc = 24 \times 15 \times 26 = 9360$.

Test 2

1. Suppose there are n ink stains, having areas $S_1, S_2, \dots, S_n$ (in mm^2) respectively. Suppose the lengths of the projections of the ink stains on the top edge of the answer sheet are $x_1, x_2, \dots, x_n$ (in mm) and the lengths of the projections of the ink stains on the left edge of the answer sheet are $y_1, y_2, \dots, y_n$ (in mm) respectively. Since any line parallel to an edge of the answer sheet intersects at most one ink stain, we have $\sum_{j=1}^n x_j \leq 210$ and $\sum_{j=1}^n y_j \leq 297$. By the condition $S_j \leq 1$ and the AM-GM inequality, we have

$$\sum_{j=1}^n S_j \leq \sum_{j=1}^n \sqrt{S_j} \leq \sum_{j=1}^n \sqrt{x_j y_j} \leq \sum_{j=1}^n \frac{x_j + y_j}{2} \leq \frac{1}{2}(210 + 297) = 253.5.$$

2. We have $a_{2015} = 4531$.

We claim that

$$a_{4k+1} = 9k + 1, a_{4k+2} = 9k + 3, a_{4k+3} = 9k + 4, a_{4k+4} = 9k + 7$$

for any integer $k \geq 0$. The base cases $k = 0, 1$ can be verified directly. Indeed, the first 8 terms are

$$1, 3, 4, 7, 10, 12, 13, 16.$$

Now, assume the claim holds for $k = 0, 1, \dots, n-1$. Observe that none of the previous terms is congruent to 2 modulo 3, and all numbers congruent to 1 modulo 3 less than $9n$ has appeared. Consider the case $k = n$.

- Since $(9n-1)+4 = 3(3n+1)$ and $9n+3 = 3(3n+1)$, we have $a_{4n+1} \neq 9n-1, 9n$. If $(9n+1)+a_i = 3a_j$, then $a_i \equiv 2 \pmod{3}$, contradiction. Thus, $a_{4n+1} = 9n+1$.
- Since $(9n+2)+1 = 3(3n+1)$, we have $a_{4n+2} \neq 9n+2$. If $(9n+3)+a_i = 3a_j$, then $3 \mid a_i$. By the inductive hypothesis, we must have $a_i = 9t+3$ for some integer t . Then we have $a_j = 3n+3t+2 \equiv 2 \pmod{3}$, contradiction. Thus, $a_{4n+2} = 9n+3$.
- If $(9n+4)+a_i = 3a_j$, then $a_i \equiv 2 \pmod{3}$, contradiction. Thus, $a_{4n+3} = 9n+4$.
- Since $(9n+5)+7 = 3(3n+4)$ and $(9n+6)+(9n+6) = 3(6n+4)$, we have $a_{4n+1} \neq 9n+5, 9n+6$. If $(9n+7)+a_i = 3a_j$, then $a_i \equiv 2 \pmod{3}$, contradiction. Thus, $a_{4n+4} = 9n+7$.

This proves our claim by induction. Therefore,

$$a_{2015} = a_{4 \times 503 + 3} = 9 \times 503 + 4 = 4531.$$

Therefore, $y = 4(x + y) - (4x + 3y) \geq 4(36) - 121 = 23$. The figure on the right gives one example using 23 L -shapes.

Test 3

1. Partition the square as shown. There are two congruent rectangles of size 1.3×1.5 above, and three congruent rectangles of size 1.7×1 below. By the pigeonhole principle, 2 of the 6 points must lie inside the same rectangle.

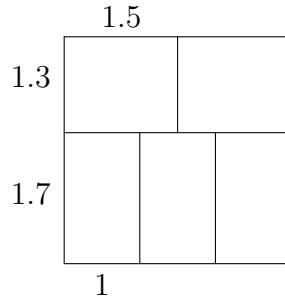
If there are 2 points belonging to the same 1.3×1.5 rectangle, their distance is at most

$$\sqrt{1.3^2 + 1.5^2} = \sqrt{3.94} < 2.$$

If there are 2 points belonging to the same 1.7×1 rectangle, their distance is at most

$$\sqrt{1.7^2 + 1^2} = \sqrt{3.89} < 2.$$

So we are done.



2. The length of PQ is $2R$.

Let O be the centre of ω_1 . Let S and T be the contact points of KP and KQ with ω_1 . WLOG assume $ACTS$ is convex. Let KB meet ST at X . Let M be the midpoint of ST , and let Y be the projection of B on KO .

Claim 1. We have $KB^2 = AB \times AC$ and $AB = KS$.

Proof. First, note that $\triangle ABK \sim \triangle KBD$ since $\angle AKD = \angle KBA = 90^\circ$. Therefore, we have

$$KB^2 = AB \times BD = AB \times AC.$$

Next, consider the power of B with respect to ω_1 . By the power of a point theorem, we have

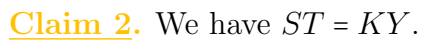
$$AB \times BC = R^2 - OB^2.$$

Using the above relation, we find that

$$AB \times BC = AB \times AC - AB^2 = KB^2 - AB^2.$$

$$AB^2 = (KB^2 + OB^2) - R^2 = OK^2 - OS^2 = KS^2.$$

This proves $AB = KS$.


$$KB^2 = AB \times AC = KS \times 2R = 2KS \times OS = 4[KS O] = 2[KS OT] = ST \times KO.$$
$$KB^2 = KY \times KO.$$

Claim 3. We have $XB \times KB = AB \times CB$.

$$\angle KSO = \angle KBO = \angle KTO = 90^\circ.$$

Also, K is the midpoint of $\widehat{ST}$. Thus, it is well-known that $\triangle KXS \sim \triangle KSB$, and hence

$$KX \times KB = KS^2 = AB^2$$

by claim 1. Also, claim 1 suggests $KB^2 = AB \times AC$. It follows that

$$XB \times KB = KB^2 - KX \times KB = AB \times AC - AB^2 = AB \times CB.$$

□

Now we prove that $\triangle KPQ \sim \triangle BTS$. Note that $\angle QKP = \angle SBT$ by the concyclic points. Also, since $PQ \parallel KB$, we have

$$\angle KPQ = \angle BKS = \angle BTS.$$

This proves $\triangle KPQ \sim \triangle BTS$. The ratio of similarity is the distance from K to PQ divided by the distance from B to TS , which is $\frac{BC}{YM}$. Therefore, we have

$$\frac{PQ}{TS} = \frac{BC}{YM}.$$

To show that $PQ = 2R = AC$, it suffices to prove

$$\frac{AC}{TS} = \frac{BC}{YM}.$$

By the claims, we have

$$\frac{TS}{YM} \stackrel{\text{claim 2}}{=} \frac{YK}{YM} = \frac{BK}{BX} \stackrel{\text{claim 3}}{=} \frac{KB^2}{AB \times BC} \stackrel{\text{claim 1}}{=} \frac{AC}{BC}.$$

This completes the proof.

3. (Japan MO 2014 Problem 5) The maximum possible value of k is 4.

Firstly, consider $a = 0$ and $b = c = \frac{1}{2}$. The inequality becomes

$$0 + \frac{2}{4+k} + \frac{2}{4+k} \geq \frac{1}{2}.$$

Hence, we need $k \leq 4$.

It remains to prove

$$\frac{a}{1+9bc+4(b-c)^2} + \frac{b}{1+9ca+4(c-a)^2} + \frac{c}{1+9ab+4(a-b)^2} \geq \frac{1}{2}$$

By the Cauchy-Schwarz inequality, we have

$$\left(\sum_{\text{cyc}} a(1+9bc+4(b-c)^2) \right) \left(\sum_{\text{cyc}} \frac{a}{1+9bc+4(b-c)^2} \right) \geq (a+b+c)^2 = 1.$$

Therefore, it suffices to prove

$$2 \geq \sum_{\text{cyc}} a(1 + 9bc + 4(b - c)^2) = \sum_{\text{cyc}} a + 3abc + 4 \sum_{\text{sym}} a^2b.$$

Using $a + b + c = 1$ and homogenizing the expression, we need to prove

$$(a + b + c)^3 \geq 3abc + 4(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b).$$

This is equivalent to

$$a^3 + b^3 + c^3 + 3abc \geq a^2b + a^2c + b^2a + b^2c + c^2a + c^2b.$$

This is exactly Schur's inequality. So the inequality is proved.

Test 4

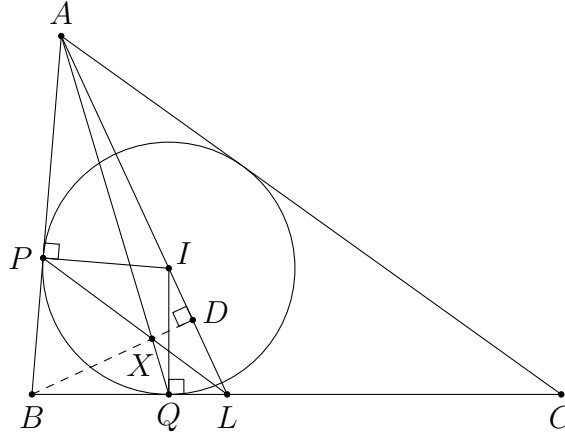
1. Note that $\ell(n) = d$ if and only if $n = 2^k d$ for some nonnegative integer k and odd integer d . Let a_d be the number of positive integers n with $\ell(n) = d$ and $1 \leq n \leq 2^{2015}$. Therefore, for any odd $d \leq 2^{2015}$, a_d is 1 plus the largest integer k such that $2^k d \leq 2^{2015}$. This inequality is equivalent to $2^{2014-k} < d \leq 2^{2015-k}$. Thus, for each $m \leq 2016$, only those odd d with $2^{2015-m} < d \leq 2^{2016-m}$ satisfy $a_d = m$. It follows that

$$\begin{aligned} \sum_{n=1}^{2^{2015}} \ell(n) &= \sum_{\substack{d=1 \\ 2 \nmid d}}^{2^{2015}} a_d d = \sum_{m=1}^{2016} m \sum_{\substack{2^{2015-m} < d \leq 2^{2016-m} \\ 2 \nmid d}} d \\ &= \sum_{m=1}^{2014} m \cdot \frac{(2^{2015-m} + 2^{2016-m})2^{2014-m}}{2} + 2016 \\ &= 3 \sum_{m=1}^{2014} m \cdot 2^{4028-2m} + 2016 \\ &= 3 \sum_{m=1}^{2014} \sum_{j=1}^m 2^{4028-2m} + 2016 \\ &= 3 \sum_{j=1}^{2014} \sum_{m=j}^{2014} 2^{4028-2m} + 2016 \\ &= \sum_{j=1}^{2014} (2^{4030-2j} - 1) + 2016 \\ &= \frac{2^{4030} - 4}{3} - 2014 + 2016 \\ &= \frac{2^{4030} + 2}{3}. \end{aligned}$$

2. (Belarus TST 2014 Test 4 Problem 1) Let I be the incentre of $\triangle ABC$. Let D be the projection of B on AL . Noting that $\triangle API \sim \triangle ADB$ and $\triangle LQI \sim \triangle LDB$, we have $\frac{AP}{AD} = \frac{PI}{DB}$ and $\frac{LQ}{LD} = \frac{QI}{DB}$. Therefore, we have

$$\frac{AP}{PB} \times \frac{BQ}{QL} \times \frac{LD}{DA} = \frac{AP}{AD} \times \frac{LD}{LQ} = \frac{PI}{DB} \times \frac{DB}{QI} = 1.$$

By Ceva's theorem, AQ, BD, PL are concurrent at X . Therefore, $BX \perp AL$.



3. (Iran TST 2014 Test 1 Problem 3) There are only two such tables as given below.

1	0	3	3	4
1	3	2	1	1
0	1	0	1	0
2	1	0	0	0
1	0	0	0	0

1	0	4	4	3
1	4	1	1	1
0	0	0	0	1
1	0	0	0	0
2	1	0	0	0

In the following, let a_{ij} be the entry in the cell (i, j) . Let r_i and c_j be the sum of all entries in row i and column j of the table respectively, and let S be the sum of all entries in the table.

Claim 1. We have $c_j = n$ for any j .

Proof. Firstly, we have

$$c_j = a_{0j} + a_{1j} + \cdots + a_{(n-1)j} \leq n \quad (1)$$

since the sum counts the number of $0, 1, \dots, n-1$ in row j . It suffices to show that none of the entries in row j is equal to n , since in that case equality holds (note that it is impossible for an entry to be larger than n).

Suppose on the contrary that $a_{ij} = n$ for some i, j . Then all entries in row j are equal to i . In particular, we have $a_{jj} = i$. If $i = j$, then $n = a_{jj} = i$. This

implies all numbers in row n are n , and so all numbers in the table are n . This is impossible. Now, since $i \neq j$, by (1), we have

$$n \geq c_j \geq a_{ij} + a_{jj} = n + i.$$

This yields $i = 0$.

Now, this means $a_{0j} = n$. As $c_j \leq n$, we must have $a_{ij} = 0$ for $1 \leq i \leq n-1$. This implies row i (for $1 \leq i \leq n-1$) contains at least one 0, and hence $a_{0i} \geq 1$. If $a_{00} = 0$, then it suggests there is no 0 in row 0, which contradicts its own value. If $a_{00} \geq 1$, then it suggests there is some 0 in row 0, which is again a contradiction. Therefore, none of the terms in the table is n . Thus, equality of (1) holds. $\square$

By claim 1, we have $S = c_0 + c_1 + \dots + c_{n-1} = n^2$. Also, we know that $a_{ij} \leq n-1$ for any i, j from the proof. This shows there cannot be a row of 0. Indeed, if all entries in row i are 0, then $a_{0i} = n$, contradiction. Thus, we have $r_i \geq 1$ for any i .

Note that each $r_i = a_{i0} + a_{i1} + \dots + a_{i(n-1)}$ counts the number of i 's in the table. As all entries are at most $n-1$, every entry is counted in some r_i . It follows that

$$r_1 + 2r_2 + \dots + (n-1)r_{n-1} = S = n^2. \quad (2)$$

Claim 2. We have $a_{00} = 1$, $a_{01} = 0$ and $a_{0j} \geq 1$ for $j \geq 2$ (whenever these terms exist).

Proof. As in the proof of claim 1, we must have $a_{00} \geq 1$. If $a_{00} \geq 2$, there are at least two 0's in row 0, say $a_{0j_1} = a_{0j_2} = 0$. This shows all entries in rows j_1 and j_2 are at least 1. In particular, $a_{j_1j}, a_{j_2j} \geq 1$ for $1 \leq j \leq n-1$, which implies row j contains the numbers j_1 and j_2 . Thus, $r_j \geq j_1 + j_2$. By (2), we have

$$n^2 = \sum_{j=1}^{n-1} jr_j \geq \sum_{j=1}^{n-1} j(j_1 + j_2) \geq \sum_{j=1}^{n-1} j(1+2) = \frac{3n(n-1)}{2}.$$

This implies $n \leq 3$. Since $0, j_1, j_2$ are distinct, we must have $n = 3$. Then row 0 must be $2, 0, 0$. Also, all remaining entries are at least 1. But then $c_1 \geq 2 + 1 + 1 = 4 > 3$, which contradicts claim 1. So we have proven $a_{00} = 1$.

2	0	0

Next, suppose $a_{0j} = 0$ for some $j \geq 3$. As above, we can show that $r_k \geq j \geq 3$ for $1 \leq k \leq n-1$, and then deduce $n \leq 3$. But then the term a_{0j} should not exist. So we must have $a_{0j} \geq 1$ for $j \geq 3$.

Now, suppose $a_{02} = 0$. Then all entries in row 2 are at least 1. So each row contains the number 2. This further implies $r_i \geq 2$ for all i , and hence each number from 0 to $n-1$ has appeared in the table at least twice. Moreover, as row 2 contains one 2 and all other entries are at least 1, we have $r_2 \geq n+1$. This implies there are at least $n+1$ entries equal to 2 in the table. Thus,

$$n^2 = S \geq 2 \sum_{\substack{k=0 \\ k \neq 2}}^{n-1} k + (n+1) \cdot 2 = n^2 + n - 2.$$

It follows that $n \leq 2$. But then the term a_{02} should not exist. So we must have $a_{02} \geq 1$.

Lastly, as $a_{00} = 1$, there is one 0 in row 0. The only possibility is $a_{01} = 0$. $\square$

Claim 3. We have

$$r_1 \geq 2n - 2, \tag{3}$$

$$r_0 + r_2 + r_3 + \cdots + r_{n-1} \geq n^2 - 2n + 1. \tag{4}$$

Proof. Since $a_{01} = 0$ by claim 2, all entries in row 1 are at least 1. Let $a_{11} = t$. If $t = 1$, then there is only one 1 in row 1, and so all other entries are at least 2. This gives

$$r_1 \geq 1 + 2(n-1) = 2n - 1 > 2n - 2.$$

If $t \geq 2$, then in row 1 there is at least one t , exactly t entries equal to 1, and all other entries are at least 2. This yields

$$r_1 \geq t + t + 2(n-t-1) = 2n - 2.$$

This proves (3).

For the other inequality, note that r_0 is the number of 0 in the table. Since there is one 0 in row 0 by claim 2 and no 0 in row 1, all the remaining $r_0 - 1$ entries 0 appear in the other rows. And then all other $n(n-2) - (r_0 - 1)$ entries in those rows are at least 1, which yields

$$r_2 + r_3 + \cdots + r_{n-1} \geq n(n-2) - (r_0 - 1).$$

This is exactly (4). $\square$

By (3) and (4), we find that

$$r_0 + r_1 + r_2 + \cdots + r_{n-1} \geq (2n - 2) + (n^2 - 2n + 1) = n^2 - 1.$$

Note that the left-hand side is exactly n^2 . This shows the inequality is almost an equality. Indeed, we must have $r_1 = 2n - 1$ or $2n - 2$. In view of the proof of (4), rows 2, 3, $\dots$, $n-1$ contain at most one entry larger than 1, which must be 2 if it

exists. As a result, we have $a_{ij} = 0$ for $i \geq 3$, $j \geq 2$, and $a_{22} + a_{23} + \cdots + a_{2(n-1)} \leq 1$. As there are at least $n - 2$ entries 0 in row $i \geq 3$, we have $a_{0j} \geq n - 2$ for $j \geq 3$. We can now use the information we have obtained to find all tables. We first list out the facts that we have found so far.

- (i) $a_{ij} \leq n - 1$ for any i, j
 - (ii) $c_j = n$ for $0 \leq j \leq n - 1$
 - (iii) $r_i \geq 1$ for any i
 - (iv) $S = r_0 + r_1 + \cdots + r_{n-1} = r_1 + 2r_2 + \cdots + (n - 1)r_{n-1} = n^2$
 - (v) $a_{00} = 1$ and $a_{01} = 0$
 - (vi) $r_1 = 2n - 2$ or $2n - 1$
 - (vii) $r_0 + r_2 + r_3 + \cdots + r_{n-1} = n^2 - 2n + 1$ or $n^2 - 2n + 2$
 - (viii) rows $2, 3, \dots, n - 1$ contain at most one entry larger than 1, which must be 2 if it exists
 - (ix) $a_{ij} = 0$ for $i \geq 3$ and $j \geq 2$
 - (x) $a_{22} + a_{23} + \cdots + a_{2(n-1)} \leq 1$
 - (xi) $a_{0j} = n - 2$ or $n - 1$ for $j \geq 3$
- If $n \geq 6$, since there are at least $n - 3$ terms in row 0 equal to $n - 2$ or $n - 1$ by (xi), we have

$$a_{(n-2)0} + a_{(n-1)0} \geq n - 3.$$

But the left-hand side is at most $1 + 2 = 3$ by (viii). Thus, we must have $n = 6$. Now, by (x), we know that at most one of $a_{22}, a_{23}, a_{24}, a_{25}$ can be nonzero. Thus, $a_{02} \geq 3$. Together with (v) and (xi), we know that there is no 2 in row 0. This yields $a_{20} = 0$, which further implies $a_{02} \geq 4$. This shows there are indeed 4 terms in row 0 equal to 4 or 5, and so

$$a_{40} + a_{50} \geq 4.$$

Again, this is a contradiction since the left-hand side is at most $1 + 2 = 3$ by (viii).

- If $n = 2$, since $a_{01} = 0$ and $c_1 = 2$ by (v) and (ii), we have $a_{11} = 2$, contradicting (i).

1	0
	2

1	0	
2	1	2
0	2	

- If $n = 3$, then $r_1 = 4, 5$ by (vi). By (iv), we have $r_0 + r_1 + r_2 = r_1 + 2r_2 = 9$. Since $r_i \geq 1$ for any i by (iii), we must have $(r_0, r_1, r_2) = (2, 5, 2)$.

Now, as $r_1 = 5$ and $a_{11} \leq 2$ by (i), in view of the proof of (vi), row 1 must be 2, 1, 2. Using the column sum $c_j = 3$ by (ii), we find that row 2 is 0, 2, ?. But since $a_{12} = 2$, there should be two 1's in row 2, contradiction.

- If $n = 4$, then $r_1 = 6, 7$ by (vi).

- If $r_1 = 6$, then $r_0 + r_2 + r_3 = 2r_2 + 3r_3 = 10$ by (iv). The only positive solution is $(r_0, r_2, r_3) = (6, 2, 2)$. By the proof of (vii), there is exactly one 2 in rows 2 and 3. Thus, another number 2 comes from rows 0 and 1. Now, by the proof of (vi), as row 1 cannot contain two 2's, row 1 must be 1, 3, 1, 1.

1	0	2	3
1	3	1	1
1			
1			

1	0	3	2
1	3	1	1
1			
1			

Recall that $a_{02}, a_{03} \geq 2$ by (xi). Also, there is only one 2 in row 0 from above. Thus, row 0 must be 1, 0, 2, 3 or 1, 0, 3, 2. In either case, this implies $a_{20} = a_{30} = 1$. But then $a_{21} + a_{31}$ cannot be 1 as all 6 entries 1 are used, contradicting $c_1 = 4$.

- If $r_1 = 7$, then $r_0 + r_2 + r_3 = 2r_2 + 3r_3 = 9$ by (iv). The only positive solution is $(r_0, r_2, r_3) = (5, 3, 1)$. Now, since $r_1 = 7$, by the proof of (vi), row 1 can only be 2, 1, 2, 2, or $a_{11} = 2$, $\{a_{10}, a_{12}, a_{13}\} = \{1, 1, 3\}$. In the former case, this implies $a_{21} = 3$, contradicting (viii). In the latter case, this implies $a_{31} = 1$. As $r_3 = 1$, we have $a_{30} = a_{32} = a_{33} = 0$. This yields $a_{03} = 3$, contradicting $a_{30} = 0$.

1	0		3
3	2	1	1
0	1	0	0

1	0		3
1	2	3	1
0	1	0	0

1	0		3
1	2	1	3
0	1	0	0

- If $n = 5$, then $r_1 = 8, 9$ by (vi).

- If $r_1 = 9$, then $r_0 + r_2 + r_3 + r_4 = 2r_2 + 3r_3 + 4r_4 = 16$ by (iv). The solutions are $(r_0, r_2, r_3, r_4) = (10, 3, 2, 1), (11, 1, 2, 2)$. Also, we know that all numbers in rows 2, 3, 4 are 0 or 1 by (viii). This implies $a_{ij} = 0$ for $2 \leq i, j \leq 4$. It follows that $r_2 \leq 2$, and so $(r_0, r_2, r_3, r_4) = (11, 1, 2, 2)$ is the only possibility.

Now, as $r_3 = r_4 = 2$, rows 3 and 4 must be 1, 1, 0, 0, 0. This implies $a_{03} = a_{04} = 3$, but this contradicts $a_{30} = 1$.

1	0		3	3
		0	0	0
1	1	0	0	0
1	1	0	0	0

– If $r_1 = 8$, then $r_0 + r_2 + r_3 + r_4 = 2r_2 + 3r_3 + 4r_4 = 17$ by (iv). The solutions are $(r_0, r_2, r_3, r_4) = (12, 1, 1, 3), (11, 3, 1, 2), (10, 5, 1, 1), (11, 2, 3, 1)$. There are several possibilities for a_{11} in view of the proof of (vi).

* If $a_{11} = 2$, then there are two 1's and three 2's in row 1. This implies $a_{21} = 3$, which contradicts (viii).

* If $a_{11} = 3$, then there are three 1's, one 2 and one 3 in row 1. This implies $a_{21} = a_{31} = 1$ and $a_{41} = 0$. We can fill in some known entries by (v) and (ix). By (iii), row 4 cannot be a zero row. Thus, $a_{04} = 4$. Then $a_{14} = 1$ and $a_{24} = 0$ by using $c_4 = 5$. Now, as $a_{14} = 0$, we have $a_{40} = 1$.

1	0	3	3	4
1	3	2	1	1
0	1	0	1	0
2	1	0	0	0
1	0	0	0	0

Next, note that $a_{40} = 1$ implies there is only one 4 in row 0. This shows $a_{03} \leq 3$. By (xi), we must have $a_{03} = 3$, and hence $a_{30} \neq 0$. In view of the above solutions, as $r_4 = 1$ and $r_3 \geq 2$, we must have $(r_0, r_2, r_3, r_4) = (11, 2, 3, 1)$. This gives $a_{30} = 2$. So we need two 3's in row 0, implying $a_{02} = 3$.

As $r_2 = 2$, there are two 1's in row 2. Also, there is one 1 in row 3. Thus, we have $a_{12} = 2$ and $a_{13} = 1$, and hence $a_{10} = 1$. Finally, using $c_j = 5$, we easily obtain $a_{20} = a_{22} = 0$ and $a_{23} = 1$. It is routine to check that this table satisfies all requirements.

* If $a_{11} = 4$, then $a_{1j} = 1$ for $j \neq 1$. This implies $a_{21} = a_{31} = 0$ and $a_{41} = 1$. As $a_{13} = 1$, there is one 1 in row 3, which implies $a_{30} = 1$. Thus, there are four 0's in row 3, yielding $a_{03} = 4$. Then by $c_3 = 5$, we obtain $a_{23} = 0$.

1	0	4	4	3
1	4	1	1	1
0	0	0	0	1
1	0	0	0	0
2	1	0	0	0

Note that there are already two 4's in the table. This shows $r_4 \geq 2$. In view of the solutions, either $r_4 = 2$ or 3. But then if $r_4 = 2$, we must have $a_{40} = 1$. This contradicts $a_{14} = 1$, as there are now two 1's in row 4. Therefore, we must have $r_4 = 3$. This implies $a_{40} = 2$. It follows that $a_{04} = 3$ and $a_{24} = 1$ by counting the numbers of 0's and 2's in row 4. Since we have shown there are three 4's in the table (because $r_4 = 3$), it must be the case $(r_0, r_2, r_3, r_4) = (12, 1, 1, 3)$. Now

it is easy to work out $a_{02} = 4$ and $a_{20} = a_{22} = 0$. It is routine to check that this table satisfies all requirements.

To summarize, there are only two 5×5 tables satisfying all requirements, which are given at the beginning.

Year 2016

Test 1

1. The only solution is $n = 7$.

Observe that

$$\begin{aligned}
 n &\equiv n && (\text{mod } 7), \\
 n^2 + 10 &\equiv (n + 2)(n - 2) && (\text{mod } 7), \\
 n^2 - 2 &\equiv (n + 3)(n - 3) && (\text{mod } 7), \\
 n^3 + 6 &\equiv (n - 1)(n^2 + n + 1) && (\text{mod } 7), \\
 n^5 + 36 &\equiv (n + 1)(n^4 - n^3 + n^2 - n + 1) && (\text{mod } 7).
 \end{aligned}$$

Since $n - 3$, $n - 2$, $n - 1$, n , $n + 1$, $n + 2$ and $n + 3$ are 7 consecutive integers, one of them must be a multiple of 7. It follows that one of the numbers is equal to 7, or otherwise it cannot be a prime. Clearly, $n^2 + 10 > 7$ and $n^5 + 36 > 7$.

If $n = 7$, the 5 numbers are 7, 59, 47, 349, 16843, all of which are primes.

If $n^2 - 2 = 7$, then $n = 3$. But then $n^3 + 6 = 33$ is not a prime.

If $n^3 + 6 = 7$, then $n = 1$, which is not a prime.

2. The largest possible n is 3.

Note that

$$\begin{aligned}
 &(3x_i - x_j)(x_i - 3x_j) \geq (1 - x_i x_j)^2 \\
 \Leftrightarrow &3x_i^2 - 10x_i x_j + 3x_j^2 \geq 1 - 2x_i x_j + x_i^2 x_j^2 \\
 \Leftrightarrow &3x_i^2 - 6x_i x_j + 3x_j^2 \geq 1 + 2x_i x_j + x_i^2 x_j^2 \\
 \Leftrightarrow &3(x_i - x_j)^2 \geq (1 + x_i x_j)^2 \\
 \Leftrightarrow &\left| \frac{x_i - x_j}{1 + x_i x_j} \right| \geq \frac{1}{\sqrt{3}}. \tag{1}
 \end{aligned}$$

WLOG, we may assume $x_1 < x_2 < \dots < x_n$. For $1 \leq j \leq n$, we can find $\theta_j \in \mathbb{R}$ such that $0 < \theta_j < \frac{\pi}{2}$ and $x_j = \tan \theta_j$. Now for $1 \leq i < j \leq n$, the inequality (1) is equivalent to

$$\tan(\theta_j - \theta_i) \geq \frac{1}{\sqrt{3}},$$

which means $\theta_j - \theta_i \geq \frac{\pi}{6}$.

If $n \geq 4$, by the pigeonhole principle, there exists j with $1 \leq j \leq 3$ such that $\theta_{j+1} - \theta_j < \frac{\pi}{6}$. This contradicts (1). Therefore, we must have $n \leq 3$.

It is possible to have $n = 3$. For example, we can take $x_j = \tan \frac{(2j-1)\pi}{12}$ for $j = 1, 2, 3$. Following the above arguments, we see that all inequalities are satisfied.

3. We have $\angle API = 90^\circ$.

Let N be the point on Γ such that MN is a diameter. Note that

$$\angle MBD = \angle MAC = \angle BAM = \angle BPM.$$

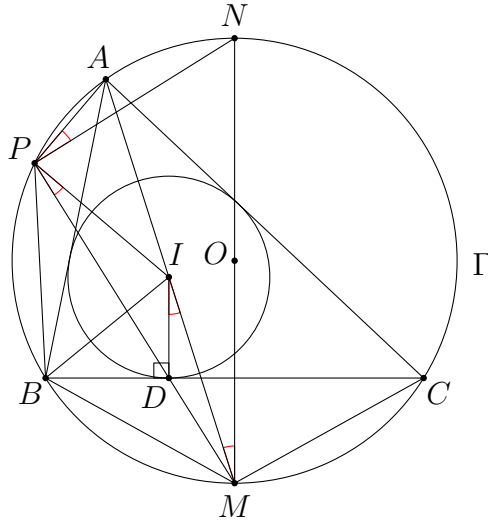
This implies $\triangle MBD \sim \triangle MPB$, and hence $MD \times MP = MB^2$. It is well-known that $MI^2 = MB^2$. Thus, we have $MD \times MP = MI^2$. This implies $\triangle MDI \sim \triangle MIP$.

Now, we have $ID \parallel NM$ since both lines are perpendicular to BC . It follows that

$$\angle MPI = \angle DIM = \angle NMI = \angle NMA = \angle NPA.$$

Thus, we have

$$\angle API = \angle IPN + \angle NPA = \angle IPN + \angle MPI = \angle MPN = 90^\circ.$$



4. There is no such triple.

First note that

$$2^m p^2 = q^7 - 1 = (q - 1)(q^6 + q^5 + q^4 + q^3 + q^2 + q + 1).$$

Since $q^6 + q^5 + q^4 + q^3 + q^2 + q + 1$ is odd and is larger than 1, we have $q - 1 = 2^m$ or $q - 1 = 2^m p$. The latter is impossible by Zsigmondy's theorem, since $q^7 - 1$ has

a prime divisor not dividing $q - 1$ (alternatively, one can obtain a contradiction by using some simple bounds). Thus, we have $q = 2^m + 1$.

If m is odd, then $q = 2^m + 1 \equiv 2 + 1 \equiv 0 \pmod{3}$. As q is a prime, we must have $q = 3$, and so $m = 1$. But then $2p^2 + 1 = 2187$ has no solution.

If m is even, let $m = 2k$. The equation becomes $q^7 - (2^k p)^2 = 1$. By Catalan's conjecture, there is no solution.

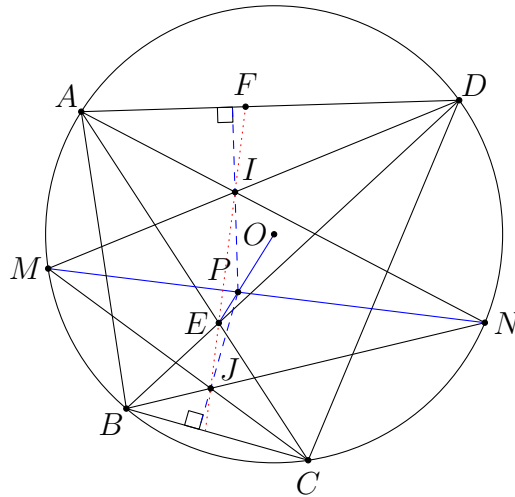
5. We have $\frac{MN}{NP} = \frac{23}{13}$.

Let I and J be the incentres of $\triangle ADE$ and $\triangle BCE$ respectively. Note that I is the intersection point of AN and DM , while J is the intersection point of BN and CM . Applying Pascal's theorem to $ANBDMC$, we find that I, J, E are collinear.

Since $\angle IAD = \angle JBC$ and $\angle IDA = \angle JCB$, we have $\triangle AID \sim \triangle BJC$. Thus, we have

$$\triangle MIN \sim \triangle AID \sim \triangle BJC \sim \triangle MJN,$$

with $\triangle MIN \cong \triangle MJN$. This implies $IJ \perp MN$. Also, the isogonal line of IJ with respect to $\angle MIN$, the isogonal line of IJ with respect to $\angle MJN$ and the line MN are concurrent at some point P' by symmetry. Observe that $IP' \perp AD$ since $\triangle MIN$ and $\triangle AID$ are oppositely similar. Similarly, $JP' \perp BC$. We now claim that $P' = P$.



Let E_1, P_1, O_1 be the projection of E, P', O on AD , and let E_2, P_2, O_2 be the projection of E, P', O on BC . Note that E_1, P_1, O_1 are the foot of altitude from E , the contact point of the incircle with the side AD , and the midpoint of AD

in $\triangle EAD$. Similar results hold for E_2, P_2, O_2 in $\triangle EBC$. As $\triangle EAD \sim \triangle EBC$, we have $\frac{E_1P_1}{P_1O_1} = \frac{E_2P_2}{P_2O_2}$. Therefore, E, P', O are collinear. This implies P' lies on both EO and MN , and hence $P' = P$.

Now, let F be the intersection of AD and IJ . Then $\frac{MP}{NP} = \frac{AF}{DF}$ due to the isogonal lines. By the angle bisector theorem, we have $\frac{AF}{DF} = \frac{AE}{DE}$.

Note that $\triangle EAD \sim \triangle EBC$, and the ratio is $\frac{AD}{BC} = 2$. Let $BE = x$ and $CE = y$ so that $AE = 2x$ and $DE = 2y$. It is given that $2x + y = 11$ and $x + 2y = 12$. We easily find that $x = \frac{10}{3}$ and $y = \frac{13}{3}$. Therefore,

$$\frac{MN}{NP} = \frac{MP + NP}{NP} = \frac{AE}{DE} + 1 = \frac{x}{y} + 1 = \frac{23}{13}.$$

6. The maximum number of acute-angled triangles is 2729148240.

Let $n = 2015$, so that there are $2n + 1$ lines. We fix one of the lines ℓ and place it as the x -axis of the coordinate plane. Then the other $2n$ lines can be partitioned into two groups, one consisting of those lines with positive slopes and one consisting of those lines with negative slopes. Let a and b be the sizes of these two groups respectively.

Observe that every pair of lines belonging to the same group form an obtuse triangle with ℓ . Therefore, the number of obtuse triangles having ℓ as a sideline that is adjacent to the obtuse angle is

$$\binom{a}{2} + \binom{b}{2} \geq 2\binom{n}{2} = n(n-1)$$

by Jensen's inequality, since the binomial function $\binom{x}{2}$ is convex.

By considering all choices of ℓ , each obtuse triangle is counted twice. Therefore, the number of acute-angled triangles is at most

$$\binom{2n+1}{3} - \frac{(2n+1)n(n-1)}{2} = \frac{n(n+1)(2n+1)}{6}.$$

This bound is attainable. Indeed, consider the sidelines of a regular polygon with $2n + 1$ sides. Then for every line ℓ being the x -axis, the number of lines with positive slopes is equal to the number of lines with negative slopes by symmetry. Therefore, equality of the above deduction holds.

When $n = 2015$, the answer is $\frac{2015 \times 2016 \times 4031}{6} = 2729148240$.

Test 2

1. Yes. Consider any competition C_1 . Since every other competition shares a common winner with C_1 , there must be a winner of C_1 who won at least $\left\lceil \frac{43}{7} \right\rceil = 7$ more competitions by the pigeonhole principle. WLOG assume A won the competitions $C_1, C_2, \dots, C_8$.

Consider any other competition C_k . By the pigeonhole principle, one of the winners of C_k won at least $\left\lceil \frac{8}{7} \right\rceil = 2$ competitions among $C_1, C_2, \dots, C_8$. But since the common winner of these competitions is A , we know that A won C_k as well. Thus, A won all the competitions.

2. Let AX meet Γ again at P , and let AY meet Γ again at Q . If $PQ \parallel \ell$, the figure is symmetric with respect to AB , and so (AXY) and $(AX'Y')$ are tangent at A . In the following, we only consider the configuration as shown.

Firstly, since

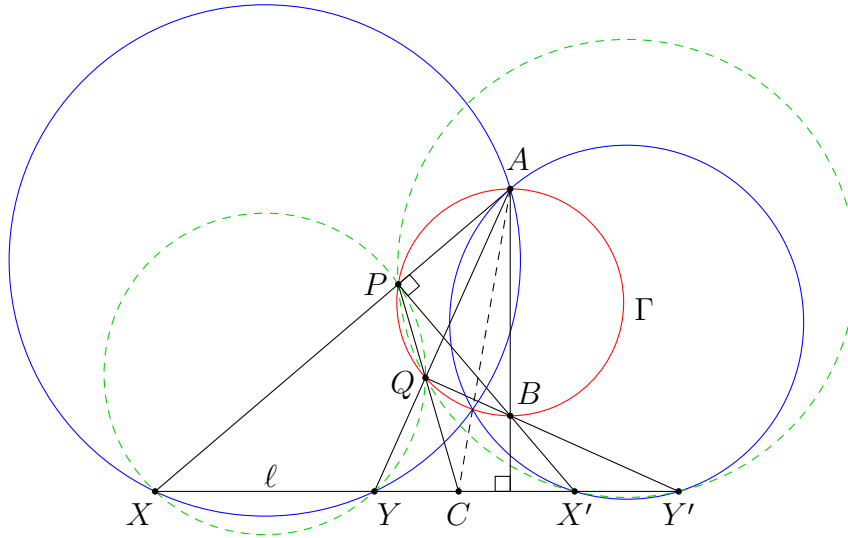
$$\angle AQP = \angle ABP = 90^\circ - \angle PAB = \angle YXP,$$

the points Q, P, X, Y are concyclic. Similarly, P, Q, X', Y' are concyclic.

Now, let PQ meet ℓ at C . Then we have

$$CX \times CY = CP \times CQ = CX' \times CY'.$$

Hence, C has the same power with respect to (AXY) , $(AX'Y')$ and Γ . As all three circles pass through A , the line AC is the common radical axis of these circles. Thus, the circles are coaxial as desired.



3. The minimum value is $\frac{27}{2}$.

Let $x = \frac{1}{a}$, $y = \frac{1}{b}$ and $z = \frac{1}{c}$. Then we have $xyz = 1$. By the AM-GM inequality, we have

$$a^3 + 2 = a^3 + 1 + 1 \geq 3\sqrt[3]{(a^3)(1)(1)} = 3a.$$

Therefore,

$$\frac{a^3 + 8}{a^3(b+c)} \geq \frac{3a+6}{a^3(b+c)} = \frac{3a^2bc + 6abc}{a^3(b+c)} = \frac{3x + 6x^2}{y+z}.$$

Similarly, we have $\frac{b^3 + 8}{b^3(c+a)} \geq \frac{3y + 6y^2}{z+x}$ and $\frac{c^3 + 8}{c^3(a+b)} \geq \frac{3z + 6z^2}{x+y}$. Thus, it remains to minimize

$$3\left(\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}\right) + 6\left(\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y}\right).$$

By Nesbitt's inequality, we have

$$\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \geq \frac{3}{2}.$$

Also, by the Cauchy-Schwarz inequality, we have

$$\left((y+z) + (z+x) + (x+y)\right) \left(\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y}\right) \geq (x+y+z)^2.$$

This implies

$$\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y} \geq \frac{x+y+z}{2} \geq \frac{3\sqrt[3]{xyz}}{2} = \frac{3}{2}$$

by the AM-GM inequality. Therefore, the given expression is at least

$$3 \cdot \frac{3}{2} + 6 \cdot \frac{3}{2} = \frac{27}{2}.$$

Equality holds when $a = b = c = 1$.

4. Nora has a winning strategy.

Let Mable write down $p_1 < p_2 < \dots < p_{2015}$ in step (1). Then $P = p_1 p_2 \dots p_{2015}$ is Mable's score.

Let $q_k = p_1 p_2 \dots p_k$ for each k . Nora needs to write down a positive integer Q in step (2) such that $(P, Q + q_j) = 1$ for all j . Indeed, for each $1 \leq k \leq 2015$, we choose $r_k \in \mathbb{Z}$ such that

$$r_k \not\equiv -q_j \pmod{p_k}$$

for all $1 \leq j \leq 2015$. Such an r_k exists because there are at most $k < p_k$ residues that r_k cannot take (as $p_k \mid q_j$ if $j \geq k$).

Now, there exists $Q \in \mathbb{Z}^+$ such that $Q \equiv r_k \pmod{p_k}$ for all k by the Chinese remainder theorem. For this choice of Q , we have

$$Q + q_j \equiv r_k + q_j \not\equiv 0 \pmod{p_k}$$

for any j and k . Therefore, we have $(P, Q + q_j) = 1$ for all j as desired.

Test 3

1. (Bulgaria MO 2014 Problem 4) The only solutions are $(p, q) = (2, 3), (3, 2)$.

When $p = 2$, we need $q^2 \mid 63$. The only possibility is $q = 3$. This is a solution since $4 \mid 28$. When $p = 3$, we need $q^2 \mid 728$. The only possibility is $q = 2$, which is another solution since $9 \mid 9$.

Now, we assume $p \geq 5$. Note that

$$p^2 \mid q^3 + 1 = (q + 1)(q^2 - q + 1).$$

As $(q + 1, q^2 - q + 1) = (q + 1, 3) \leq 3$, we must have $p^2 \mid q + 1$ or $p^2 \mid q^2 - q + 1$. Thus, $p^2 \leq q^2 - q + 1 < q^2$, and so $p < q$.

Next, observe that

$$q^2 \mid p^6 - 1 = (p - 1)(p + 1)(p^2 - p + 1)(p^2 + p + 1).$$

Since $p < q$ and clearly $p \neq q - 1$, we have $q \nmid p - 1$ and $q \nmid p + 1$. Also, since

$$(p^2 - p + 1, p^2 + p + 1) = (p^2 - p + 1, 2p) = 1,$$

we have $q^2 \mid p^2 - p + 1$ or $q^2 \mid p^2 + p + 1$. Thus,

$$q^2 \leq p^2 + p + 1 \leq (q - 2)^2 + (q - 2) + 1 = q^2 - 3q + 3.$$

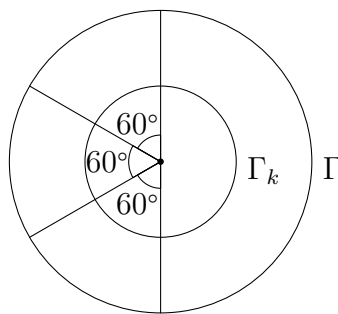
This is impossible. Therefore, there is no solution when $p \geq 5$.

2. (Iran TST 2014 Test 1 Problem 6) Let O_1 and O_2 be the circumcentres of $\triangle AC_1B'$ and $\triangle AB_1C'$ respectively. Let M and N be the midpoints of AC' and AB' respectively. Also, let MO_2 meet NO_1 at P , and let O be the circumcentre of $\triangle AIT$.

Since $BA = BB_1$, BI is the perpendicular bisector of AB_1 . Thus, O_2 lies on BI . Similarly, O_1 lies on CI . Next, since MO_2 and NO_1 are the perpendicular bisectors of AC' and AB' respectively, P is the circumcentre of $\triangle AC'B'$. Note that $AC' = AB'$ since $AI \perp C'B'$ and AI bisects $\angle C'AB'$. Therefore, P lies on AI .

Now, since BM, IP, CN are concurrent at A , we can apply Desargues' theorem to $\triangle BIC$ and $\triangle MPN$. This shows $BI \cap MP = O_2$, $IC \cap PN = O_1$ and $BC \cap MN$

- Consider a coordinate plane such that the line joining any pair of centres of the circles is not parallel to the coordinate axes. We label the circles as $\Gamma_1, \Gamma_2, \dots, \Gamma_{2016}$ such that the x -coordinate of the centre of Γ_i is less than that of Γ_j if $i < j$.



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two of these centres is at most 2, which shows the two circles intersect. Thus, we may assume there are at most 25 centres in each sector. This implies at most 75 centres among $\Gamma_1, \Gamma_2, \dots, \Gamma_{k-1}$ lie in Γ , or equivalently at most 75 of these circles intersect Γ_k .

Now, we colour the circles $\Gamma_1, \Gamma_2, \dots, \Gamma_{2016}$ one by one in 76 colours. Suppose we have coloured $\Gamma_1, \Gamma_2, \dots, \Gamma_k$. Since Γ_{k+1} intersects at most 75 of the previous circles, we can colour it in a way such that its colour is different from the colours of all circles among $\Gamma_1, \Gamma_2, \dots, \Gamma_k$ which intersect Γ_{k+1} . By the pigeonhole principle, we can find

$$\left\lceil \frac{2016}{76} \right\rceil = 27$$

circles having the same colour. This means these 27 circles are pairwise disjoint. So we are done.

Test 4

1. (Japan MO 2014 Problem 1) We have $\angle BAC = 120^\circ$.

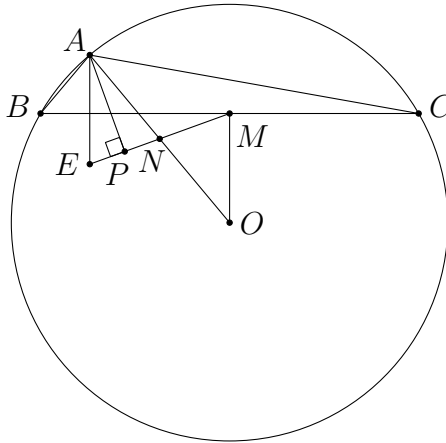
Let M be the midpoint of BC . Suppose ℓ meets the altitude from A , the angle bisector of $\angle BAC$, and the line AO at E, P, N respectively.

Note that $AE \parallel MO$. In order that EM bisects AO , the quadrilateral $AEOM$ must be convex. Therefore, $\angle BAC$ must be obtuse.

Now, since $AE \parallel MO$ and $AN = ON$, we have $\triangle AEN \cong \triangle OMN$. It is well-known that AE and AO are isogonal lines with respect to $\angle BAC$. Therefore, AP bisects $\angle EAN$. As we also have $AP \perp EN$, this shows $\triangle AEN$ is isosceles, with $AE = AN$. Therefore,

$$OM = AE = AN = \frac{1}{2}OA = \frac{1}{2}OB.$$

As $\angle BMO = 90^\circ$, this yields $\angle OBM = 30^\circ$. Thus, $\angle BAC = 120^\circ$.



2. (Turkey MO 2014 Problem 5) n can be any odd integer greater than 1.

For even n , suppose $a_j + \frac{(-1)^j}{a_j} = b_j$ for $1 \leq j \leq n$, where $\{b_j\}$ is a permutation of $\{a_j\}$. The relation can be rewritten as

$$a_j^2 - a_j b_j = (-1)^{j+1}.$$

Summing over all j 's, we obtain

$$\sum_{j=1}^n (a_j^2 - a_j b_j) = 0$$

since n is even. As $\{b_j\}$ is a permutation of $\{a_j\}$, this implies

$$\frac{1}{2} \sum_{j=1}^n (a_j - b_j)^2 = 0.$$

Hence, we must have $a_j = b_j$, which is impossible since $a_j + \frac{(-1)^j}{a_j} = b_j$. Therefore, there is no such sequence.

For $n = 1$, we need $a_1 - \frac{1}{a_1} = a_1$, which is impossible.

So it remains to show that such a sequence exists when $n > 1$ is odd. First we construct a sequence $a_1, a_2, \dots, a_{n+1}$ such that $a_1 = 1 + x$ with $x > 0$, and

$$a_{j+1} = a_j + \frac{(-1)^j}{a_j}$$

for $1 \leq j \leq n$. It suffices to show $a_1, a_2, \dots, a_n$ are pairwise distinct, and choose x such that $a_{n+1} = a_1$.

Note that

$$a_{j+2} = a_{j+1} + \frac{(-1)^{j+1}}{a_{j+1}} = a_j + \frac{(-1)^j}{a_j} + (-1)^{j+1} \left(a_j + \frac{(-1)^j}{a_j} \right)^{-1} = a_j + \frac{1}{a_j^3 + (-1)^j a_j}.$$

For odd j , since $a_1 > 1$, we can prove inductively that $a_{j+2} > a_j > 1$. Also, for even j , since $a_2 = a_1 - \frac{1}{a_1} > 0$, we can prove inductively that $a_{j+2} > a_j > 0$. Therefore, if we can prove that $a_{n+1} = a_1$, then we have

$$0 < a_2 < a_4 < \dots < a_{n+1} = a_1 < a_3 < \dots < a_n,$$

and so all terms are pairwise distinct positive real numbers.

By using the recurrence relation, it is obvious that a_{n+1} is a continuous function of a_1 . When a_1 goes to 1, we have

$$a_{n+1} - a_1 \geq a_4 - a_1 = a_2 + \frac{1}{a_2^3 + a_2} - a_1 \rightarrow \infty$$

since $a_2 \rightarrow 0$. This shows $a_{n+1} - a_1 > 0$ for some x .

On the other hand, for even j , as $a_j \geq a_2 = 1 + x - \frac{1}{1+x} = \frac{2x+x^2}{1+x} > x$, we have

$$a_{j+2} = a_j + \frac{1}{a_j^3 + a_j} < a_j + \frac{1}{x^3},$$

and hence $a_{n+1} < a_2 + \frac{n-1}{2x^3}$ by summing over all j 's. It follows that

$$a_{n+1} - a_1 < a_2 + \frac{n-1}{2x^3} - a_1 = \frac{n-1}{2x^3} - \frac{1}{1+x} = \frac{(n-1) + (n-1)x - 2x^3}{2x^3(1+x)} < 0$$

for sufficiently large x .

Now, by the intermediate value theorem, there exists $x > 0$ such that $a_{n+1} = a_1$. So we are done.

3. (Korea MO 2014 Problem 5) Consider all numbers of the form $a + bk$ where a, b are integers with $0 \leq a, b < \sqrt{p}$. Since there are more than $(\sqrt{p})^2 = p$ numbers, by the pigeonhole principle, two of them must be congruent modulo p . Their difference, which is of the form $c + dk$ with $|c|, |d| < \sqrt{p}$ and $(c, d) \neq (0, 0)$, is divisible by p . Now, note that

$$(c + dk)(c - dk) = c^2 - d^2k^2 \equiv c^2 + 5d^2 \pmod{p}.$$

Therefore, $p \mid c^2 + 5d^2$. Since

$$0 < c^2 + 5d^2 < p + 5p = 6p,$$

we must have $c^2 + 5d^2 = rp$ with $r = 1, 2, 3, 4, 5$.

- If $c^2 + 5d^2 = p$, then

$$(c^2 - 5d^2)^2 + 5(2cd)^2 = (c^2 + 5d^2)^2 = p^2.$$

Clearly, $c, d \neq 0$. So we can take $(m, n) = (|c^2 - 5d^2|, 2|cd|)$.

- If $c^2 + 5d^2 = 2p$, then c, d are odd as $2p \equiv 2 \pmod{4}$. Therefore, we have

$$\left(\frac{c^2 - 5d^2}{2}\right)^2 + 5(cd)^2 = \left(\frac{c^2 + 5d^2}{2}\right)^2 = p^2.$$

So we can take $(m, n) = \left(\left|\frac{c^2 - 5d^2}{2}\right|, |cd|\right)$.

- If $c^2 + 5d^2 = 3p$, then $3 \nmid c, d$ as $9 \nmid 3p$. Therefore, $3 \mid c \pm d$ for some choice of the sign. Then we have

$$\left(\frac{c \mp 5d}{3}\right)^2 + 5\left(\frac{c \pm d}{3}\right)^2 = \frac{2(c^2 + 5d^2)}{3} = 2p$$

with $\frac{c \mp 5d}{3}$ and $\frac{c \pm d}{3}$ nonzero. Thus, this is reduced to the second case.

- If $c^2 + 5d^2 = 4p$, then c, d are even. Therefore, we have

$$\left(\frac{c}{2}\right)^2 + 5\left(\frac{d}{2}\right)^2 = p,$$

which is reduced to the first case.

- If $c^2 + 5d^2 = 5p$, then $5 \mid c$. Therefore, we have

$$d^2 + 5\left(\frac{c}{5}\right)^2 = p,$$

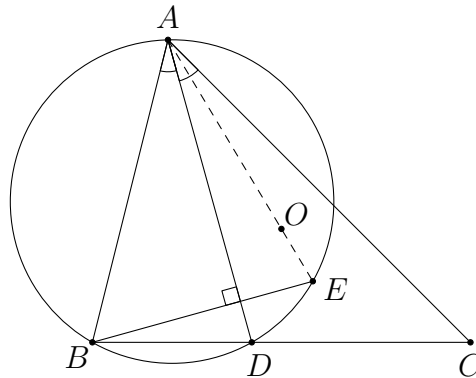
which is reduced to the first case.

Since all cases are exhausted, we are done.

Test 1

- By simple angle chasing, we find that

As O and E lie on the same side of AB as C in this case, this implies A, O, E are collinear.



- Suppose A and B are enemies. By the given condition, every other member cannot be a friend of both A and B . Since each of A and B has 2 more enemies other than themselves, we must have $n \leq 2 + 2 \times 2 = 6$. Also, as each member has 3 enemies, we have $n \geq 4$, and there are $\frac{3n}{2}$ pairs of enemies. This shows n is even, and hence n can only be 4 or 6.

To give examples for $n = 4$ and $n = 6$, we use terminologies in graph theory. Suppose vertices denote the members and edges join pairs of enemies. Then K_4 and $K_{3,3}$ are examples for $n = 4$ and $n = 6$ respectively.



3. The largest possible value of $f(-1)$ is -125 .

Let $f(x) = (x-a)(x-b)(x-c)$ where $a, b, c \geq 0$ and $abc = -f(0) = 64$. Then we have

$$\begin{aligned} f(-1) &= -(1+a)(1+b)(1+c) \\ &= -1 - (a+b+c) - (ab+bc+ca) - abc \\ &\leq -1 - 3\sqrt[3]{abc} - 3\sqrt[3]{a^2b^2c^2} - abc \\ &= -125. \end{aligned}$$

Equality holds when $a = b = c = 4$, i.e. $f(x) = (x-4)^3$.

4. There are $2^{2015} + 2^{4031}$ such sequences.

Let a_n be the number of such sequences with n terms instead of 2016. Since there are 4^n sequences formed by 1, 2, 3, 4, there are $4^n - a_n$ sequences with an odd number of 1's among these.

Consider a sequence with n terms formed by 1, 2, 3, 4 having an even number of 1's. If the last digit is 1, then there is an odd number of 1's in the first $n-1$ digits. There are $4^{n-1} - a_{n-1}$ choices for these digits. If the last digit is 2, 3 or 4, then there is an even number of 1's in the first $n-1$ digits. There are a_{n-1} choices for these digits. This gives the recurrence relation

$$a_n = 4^{n-1} - a_{n-1} + 3a_{n-1} = 2a_{n-1} + 4^{n-1}.$$

Adding up

$$\begin{aligned} a_n &= 2a_{n-1} + 4^{n-1}, \\ 2a_{n-1} &= 2^2a_{n-2} + 2 \cdot 4^{n-2}, \\ &\dots \\ 2^{n-2}a_2 &= 2^{n-1}a_1 + 2^{n-2} \cdot 4, \end{aligned}$$

we obtain

$$\begin{aligned} a_n &= 2^{n-1}a_1 + \sum_{k=0}^{n-2} (2^k \cdot 4^{n-1-k}) = 2^{n-1}(3) + \sum_{k=0}^{n-2} 2^{2n-2-k} \\ &= 3 \cdot 2^{n-1} + 2^{2n-1} - 2^n = 2^{n-1} + 2^{2n-1}. \end{aligned}$$

In particular, $a_{2016} = 2^{2015} + 2^{4031}$.

5. The answer is 6.

Let $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$. Firstly, we find that

$$a_{n+1} - a_n = \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1} = \frac{1}{(2n+1)(2n+2)} > 0.$$

This shows the sequence is strictly increasing. Thus, we easily find that

$$a_{1008} > a_3 = \frac{37}{60} > 0.6.$$

Secondly, we can prove by induction that $a_n \leq 0.7 - \frac{1}{4n}$ for any $n \geq 3$. The base case $a_3 = \frac{37}{60} \leq 0.7 - \frac{1}{12}$ holds. Assuming this holds for some $n = k$, we find that

$$a_{k+1} = a_k + \frac{1}{(2k+1)(2k+2)} \leq 0.7 - \frac{1}{4k} + \frac{1}{(2k+1)(2k+2)}.$$

Now,

$$\begin{aligned} & 0.7 - \frac{1}{4k} + \frac{1}{(2k+1)(2k+2)} \leq 0.7 - \frac{1}{4(k+1)} \\ \Leftrightarrow & \frac{1}{(2k+1)(2k+2)} \leq \frac{1}{4k(k+1)} \\ \Leftrightarrow & 4k(k+1) \leq (2k+1)(2k+2) \\ \Leftrightarrow & 0 \leq 2k+2. \end{aligned}$$

This clearly holds. By induction, we have $a_n \leq 0.7 - \frac{1}{4n} < 0.7$ for any $n \geq 3$. In particular, $a_{1008} < 0.7$. Therefore, the first digit of a_{1008} after the decimal point is 6.

Remarks. For the upper bound, alternatively, we can use

$$a_n < \int_n^{2n} \frac{1}{x} dx = \left[\ln x \right]_{x=n}^{x=2n} = \ln 2n - \ln n = \ln 2 < 0.7,$$

since $\frac{1}{x}$ is a decreasing function.

6. a_1 can only be 2^{2015} .

Let $s_n = a_n + b_n$ and $p_n = a_n b_n$ for all $n \geq 1$. The relations become $s_{n+1} = \frac{s_n}{2}$ and $p_{n+1} = \sqrt{p_n}$. Inductively, we find that $s_n = \frac{s_1}{2^{n-1}}$ and $p_n = {}^{2^{n-1}}\sqrt{p_1}$.

Since a_n and b_n are real roots of $x^2 - s_n x + p_n = 0$, we have $s_n^2 - 4p_n \geq 0$ for any n . This implies

$$\frac{s_1^2}{2^{2n-2}} - 4 {}^{2^{n-1}}\sqrt{p_1} \geq 0.$$

If $p_1 > 0$, then the left-hand side approaches $0 - 4 = -4 < 0$ when n goes to infinity. This is a contradiction. Thus, we must have $p_1 = 0$. This yields $p_n = 0$ for any $n \geq 1$, and hence $a_n = 0$ or $b_n = 0$.

As $b_{2016} = 1$, we need $a_1 = 0$. Using $s_{2015} = 2s_{2016} = 2$ and $p_{2015} = 0$, we easily deduce $\{a_{2015}, b_{2015}\} = \{0, 2\}$. Similarly, by backward induction, we find that $\{a_n, b_n\} = \{0, 2^{2016-n}\}$ for $1 \leq n \leq 2016$. In particular, $a_1 = 2^{2015}$ since $a_1 > 0$.

It is possible that $a_1 = 2^{2015}$. An example of the sequences is $a_1 = 2^{2015}$, $b_1 = 0$, and $a_n = 0$, $b_n = 2^{2016-n}$ for $n \geq 2$.

Test 2

1. (a_1, a_2) can be $(1, -2015), (15, -69), (70, -14), (2016, 0)$ or their permutations.

Firstly, taking the difference of the two relations $a_n + a_{n+1} = 2a_{n+2}a_{n+3} + 2016$ and $a_{n+1} + a_{n+2} = 2a_{n+3}a_{n+4} + 2016$, we obtain

$$a_{n+2} - a_n = 2a_{n+3}(a_{n+4} - a_{n+2}).$$

Inductively, this easily implies

$$a_{n+2} - a_n = 2^k a_{n+3} a_{n+5} \cdots a_{n+2k+1} (a_{n+2k+2} - a_{n+2k})$$

for any $k \in \mathbb{Z}^+$. This shows $2^k \mid a_{n+2} - a_n$ for any $k \in \mathbb{Z}^+$, and hence $a_{n+2} = a_n$. Thus, we may assume all odd terms are equal to $a_1 = b$ and all even terms are equal to $a_2 = c$. Now the only condition is

$$b + c = 2bc + 2016.$$

This means

$$(2b - 1)(2c - 1) = 2(2bc - b - c) + 1 = -4031 = -29 \times 139.$$

Thus, we obtain the solutions $(b, c) = (1, -2015), (15, -69), (70, -14), (2016, 0)$ up to permutation.

2. Firstly, since

$$\angle EPF_1 = \angle ED_1F_1 = \angle C_1QF_1,$$

we have $PE \parallel QC_1$. By symmetry, we also have $BP \parallel QC_1$. Therefore, B, P, E are collinear, with $BE \parallel QC_1$.

Secondly, since $\angle D_1DB_1 = \angle D_1CB_1$ and $\angle EDB = \angle ECB$, we know that B_1, C, D, D_1 and B, C, D, E are two groups of concyclic points. By Reim's theorem, we have $BE \parallel B_1D_1$ (alternatively, since $\angle C_1B_1D_1 = \angle DCE = \angle C_1BE$). Thus, $B_1D_1 \parallel BE \parallel QC_1$.



4. (IMO Shortlist 1998 N5 modified) n can be any nonnegative integral power of 2.

If n has an odd divisor $d \geq 3$, then $2^d - 1 \mid 2^n - 1 \mid m^2 + 81$. Since $2^d - 1 \equiv 3 \pmod{4}$ and $3 \nmid 2^d - 1$, there exists an odd prime $p > 3$ such that $p \equiv 3 \pmod{4}$ and $p \mid 2^d - 1$. This implies $p \mid m^2 + 81$. However, this means $m^2 \equiv -9^2 \pmod{p}$, and hence $(9^{-1}m)^2 \equiv -1 \pmod{p}$. This is impossible as $p \equiv 3 \pmod{4}$. Therefore, n has no odd divisor greater than 1. Thus, $n = 2^k$ for some nonnegative integer k .

It remains to find an m such that $2^{2^k} - 1 \mid m^2 + 81$. Firstly, note that

$$2^{2^k} - 1 = (2 + 1)(2^2 + 1) \cdots (2^{2^{k-1}} + 1).$$

The factors on the right are pairwise relatively prime. Indeed, if $r < s$, then $2^{2^r} + 1 \mid 2^{2^s} - 1$, and $(2^{2^s} - 1, 2^{2^s} + 1) = (2^{2^s} - 1, 2) = 1$, so that $(2^{2^r} + 1, 2^{2^s} + 1) = 1$. Now, by the Chinese remainder theorem, there exists $m \in \mathbb{Z}^+$ such that $3 \mid m$ and

$$m \equiv 9 \cdot 2^{2^{j-1}} \pmod{2^{2^j} + 1}$$

for $j = 1, 2, \dots, k-1$. For this m , we have $3 \mid m^2 + 81$ and

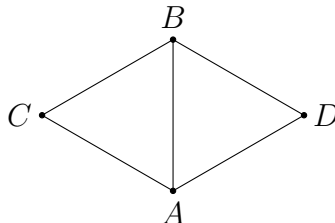
$$m^2 + 81 \equiv 81(2^{2^j} + 1) \equiv 0 \pmod{2^{2^j} + 1}.$$

Hence, $2^{2^k} - 1 \mid m^2 + 81$ as desired.

Test 3

1. (a) No. Let the centres be A, B, C, D, E . WLOG assume A is the centre of (BCD) . One of the following must occur.

- If B is the centre of (ACD) , then $AC = AD = AB = BC = BD$. So both $\triangle ABC$ and $\triangle ABD$ are equilateral triangles. Note that at least one of A, B must lie on the circle with centre C . So E lies on the circle with centre C that passes through A and B . Similarly, E lies on the circle with centre D that passes through A and B . This is a contradiction since the two circles only meet at A and B .

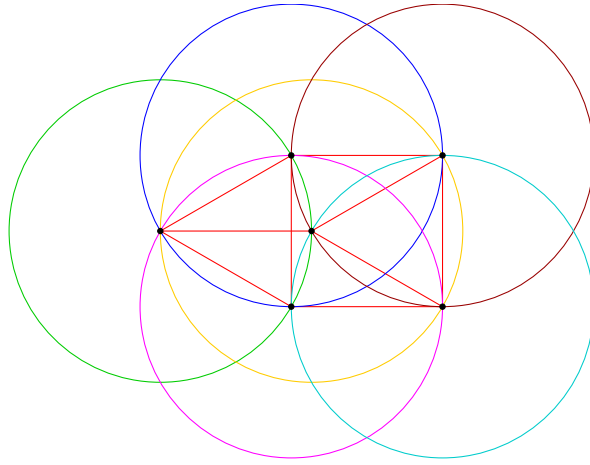


- If B is the centre of (ACE) , then $AC = AD = AB = BC = BE$. So $\triangle ABC$ is an equilateral triangle. Thus, there must be 3 points lying on the circle with centre C that passes through A and B . In either case $CA = CB = CD$ or $CA = CB = CE$, we get the same configuration as above and hence it is a contradiction.

- If B is the centre of (CDE) , we may also assume C is the centre of (BDE) since otherwise it is included in the above two cases by symmetry. Then this is exactly the same as the first case. Again, this yields a contradiction.

Therefore, there do not exist such circles.

- (b) Yes. Consider two equilateral triangles of side lengths 1 having parallel sides and the same orientation such that each pair of corresponding vertices is at a distance 1 apart. It is clear that each point has a distance 1 to exactly three other points. Therefore, we can draw a circle with each point as centre and radius 1 that passes through exactly 3 other points.



2. The average score is 670000.

Let $n = 100$, and let S be any subset of $\{1, 2, \dots, 2n\}$ such that $|S| = n$. There are $\binom{2n}{n}$ such sets S . The score of S is

$$\begin{aligned}
 \left(\sum_{a \in S} a - \sum_{b \notin S} b \right)^2 &= \left(\sum_{k=1}^{2n} k - 2 \sum_{a \in S} a \right)^2 \\
 &= \left(\frac{2n(2n+1)}{2} \right)^2 - 4 \left(\frac{2n(2n+1)}{2} \right) \sum_{a \in S} a + 4 \left(\sum_{a \in S} a \right)^2 \\
 &= n^2(2n+1)^2 - 4n(2n+1) \sum_{a \in S} a + 4 \sum_{a \in S} a^2 + 4 \sum_{\substack{a, b \in S \\ a \neq b}} ab.
 \end{aligned}$$

In the last sum, the pairs (a, b) and (b, a) are considered as different pairs.

Firstly, we have

$$\sum_S \sum_{a \in S} a = \sum_{a=1}^{2n} \sum_{a \in S} a = \sum_{a=1}^{2n} a \binom{2n-1}{n-1} = n(2n+1) \binom{2n-1}{n-1}$$

since there are $\binom{2n-1}{n-1}$ sets S containing a (as we need to choose $n-1$ more elements from the remaining $2n-1$ elements other than a).

Secondly, we have

$$\begin{aligned}
\sum_S \left(\sum_{a \in S} a^2 + \sum_{\substack{a, b \in S \\ a \neq b}} ab \right) &= \sum_{a=1}^{2n} \sum_{a \in S} a^2 + \sum_{\substack{a, b=1 \\ a \neq b}}^{2n} \sum_{a, b \in S} ab \\
&= \sum_{a=1}^{2n} a^2 \binom{2n-1}{n-1} + \sum_{\substack{a, b=1 \\ a \neq b}}^{2n} ab \binom{2n-2}{n-2} \\
&= \sum_{a=1}^{2n} a^2 \binom{2n-1}{n-1} + \sum_{a=1}^{2n} a \left(\sum_{b=1}^{2n} b - a \right) \binom{2n-2}{n-2} \\
&= \sum_{a=1}^{2n} a^2 \left[\binom{2n-1}{n-1} - \binom{2n-2}{n-2} \right] + n(2n+1) \binom{2n-2}{n-2} \sum_{a=1}^{2n} a \\
&= \frac{n(2n+1)(4n+1)}{3} \binom{2n-2}{n-1} + n^2(2n+1)^2 \binom{2n-2}{n-2}.
\end{aligned}$$

Therefore, the average score is

$$\begin{aligned}
&n^2(2n+1)^2 + \binom{2n}{n}^{-1} \left[-4n^2(2n+1)^2 \binom{2n-1}{n-1} + \frac{4n(2n+1)(4n+1)}{3} \binom{2n-2}{n-1} \right. \\
&\quad \left. + 4n^2(2n+1)^2 \binom{2n-2}{n-2} \right] \\
&= n^2(2n+1)^2 + \left(-2n^2(2n+1)^2 + \frac{2n^2(2n+1)(4n+1)}{3(2n-1)} + \frac{2(n-1)n^2(2n+1)^2}{2n-1} \right) \\
&= \frac{n^2(2n+1)}{3} \\
&= 670000.
\end{aligned}$$

3. (Bulgaria MO 2016 Problem 2) The maximum possible value of n is 18.

We first show that $n \geq 19$ is impossible. Let A, B, C be the answers in each problem. For $n \geq 19$, by the pigeonhole principle, we may assume $\left\lceil \frac{19}{3} \right\rceil = 7$ students answer A in problem 1. Then by the pigeonhole principle, we may assume $\left\lceil \frac{7}{3} \right\rceil = 3$ students answer A in problem 2. These 3 students must have different answers to the remaining problems. Thus, we may assume their answers are

$$AAAAAA, AABBBB, AACCCC.$$

Consider another student choosing answer A in problem 1. By the pigeonhole principle, this student chooses the same choice in $\left\lceil \frac{4}{3} \right\rceil = 2$ of the problems among

problems 3 to 6. WLOG assume the answers are $A?AA??$. This shares 3 same answers with the first student with answers $AAAAAA$, which is a contradiction. Therefore, we must have $n \leq 18$.

We now provide an example for $n = 18$. The following lists out the answers given by 18 students.

<i>AAAAAA</i>	<i>AABBCC</i>	<i>ABCACB</i>	<i>ACBCAB</i>	<i>ABACBC</i>	<i>ACCBBA</i>
<i>BBBBBB</i>	<i>BBCCAA</i>	<i>BCABAC</i>	<i>BACABC</i>	<i>BCBACA</i>	<i>BAACCB</i>
<i>CCCCCC</i>	<i>CCAABB</i>	<i>CABCBA</i>	<i>CBABCA</i>	<i>CACBAB</i>	<i>CBBAAC</i>

It is routine to check that any two students in the same column share 0 same answer, while any two students not in the same column share 2 same answers. (Note that there is a cyclic symmetry in each column.)

Test 4

1. Yes. Consider the permutation

$$3, 2, 1, 12, 11, \dots, 4, 23, 22, \dots, 13, 57, 56, \dots, 24, 6666, 6665, \dots, 58.$$

More precisely, we define

$$a_k = \begin{cases} 2^2 - k & \text{if } 1 \leq k \leq 3, \\ 4^2 - k & \text{if } 4 \leq k \leq 12, \\ 6^2 - k & \text{if } 13 \leq k \leq 23, \\ 9^2 - k & \text{if } 24 \leq k \leq 57, \\ 82^2 - k & \text{if } 58 \leq k \leq 6666. \end{cases}$$

The result follows readily.

2. (Romania TST 2016 Day 1 Problem 1) Let $\Gamma = (ACD)$. Since

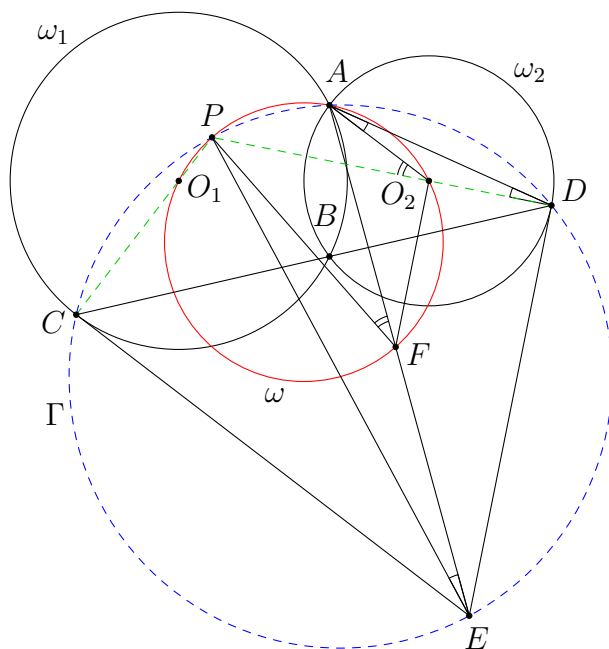
$$\angle ECA + \angle EDA = (180^\circ - \angle ABC) + (180^\circ - \angle ABD) = 180^\circ,$$

the point E lies on Γ . It is well-known that there is a spiral similarity with centre A mapping O_1O_2 to CD (equivalently, $\triangle AO_1O_2 \sim \triangle ACD$). Therefore, the intersection of CO_1 and DO_2 , called P , lies on both ω and Γ .

By Reim's theorem, $O_2F \parallel DE$. Hence, $\angle PO_2F = \angle PDE = 90^\circ$. This shows PF is a diameter of ω . It remains to prove $FE = FP$. Indeed, we have

$$\angle AFP = \angle AO_2P = 2\angle ADP = 2\angle AEP,$$

and hence $\angle FPE = \angle FEP$. This yields $FE = FP$ as desired.



3. (Turkey TST 2016 Problem 4) For any function $f(x)$ and any integer $m \geq 0$, define

$$\Delta_m f(x) = \sum_{n=0}^m f(x+n)(-1)^n \binom{m}{n}.$$

Claim. Let $Q(x)$ be a polynomial of degree d . Then $\Delta_m Q(x) = 0$ for all $m \geq d$.

Proof. This is a standard result about finite differences of polynomials. Indeed, let $Q_0(x) = Q(x)$ and $Q_{k+1}(x) = Q_k(x) - Q_k(x+1)$ for $k \geq 0$. It is easy to show that $Q_k(x) = \Delta_k Q(x)$ for any k . Also, it can be proved by induction that $Q_k(x)$ is a polynomial of degree $d-k$ for $k = 0, 1, \dots, d$ (essentially because $x^j - (x+1)^j$ has degree $j-1$), and hence $Q_k(x) = 0$ for all $k \geq d$. This proves the claim. $\square$

Consider any function $f(x)$ such that $f(n) = a_n$ for all integer $n \geq 0$. We are given that $\Delta_m f(0) = 0$ for all $m \geq N$ where $N \in \mathbb{Z}^+$. By the Lagrange interpolation formula, construct a polynomial P of degree at most $N-1$ such that $P(n) = f(n)$ for $n = 0, 1, \dots, N-1$. By the claim, we have $\Delta_N P(0) = 0$. Also, we know that $\Delta_N f(0) = 0$. Equating these, we find that

$$\sum_{n=0}^N P(n)(-1)^n \binom{N}{n} = \sum_{n=0}^N f(n)(-1)^n \binom{N}{n}.$$

Since $P(n) = f(n)$ for $n = 0, 1, \dots, N-1$, this yields $P(N) = f(N)$. Inductively, we obtain $P(n) = f(n) = a_n$ for all nonnegative integer n . This completes the proof.

Year 2018

Test 1

1. n can be 47 or 17.

Suppose $n^2 + 32n + 8 = m^2$ where m is a positive integer. Note that this can be rewritten as $(n + 16)^2 - 248 = m^2$, which implies

$$(n + m + 16)(n - m + 16) = 248 = 2^3 \times 31.$$

Hence, $(n + m + 16, n - m + 16) = (248, 1), (124, 2), (62, 4), (31, 8)$. For each pair, we can solve the system of two linear equations in two unknowns. The only positive integer solutions are $(n, m) = (47, 61), (17, 29)$.

2. The only polynomial is $x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + 1$.

Suppose one of the coefficients of $f_0(x) = f(x)$, say the coefficient of x^k , is at least 2. Define $f_1(x) = f_0(x) - 2x^k + x^{k+1}$. Then $f_1(2) = f_0(2) = 2017$ and $f_1(1) < f_0(1) = 7$. Also, the coefficients of f_1 are nonnegative integers. Repeating the same process, since the sequence $f_0(1), f_1(1), \dots$ of nonnegative integers is strictly decreasing, the process must end at some step. Then we obtain a polynomial $g(x)$ with coefficients 0 or 1 such that $g(2) = 2017$ and $g(1) \leq 7$.

Since the binary representation of 2017 is $11111100001_{(2)}$, we know that g is uniquely determined, and is given by

$$g(x) = x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + 1.$$

As $g(1) = 7$, the equality of $g(1) \leq 7$ should hold. This means we do not need to carry out any process, and f must be the same as g .

3. The answer is $k = 23$.

We first show $k \leq 23$. We list the students as $S_1, S_2, \dots, S_{1200}$ and the clubs as $C_1, C_2, \dots, C_{24}$. Consider the following construction. For $1 \leq j \leq 24$, student S_j joins clubs $C_1, \dots, C_{j-1}, C_{j+1}, \dots, C_{24}$. For $25 \leq j \leq 1200$, student S_j joins the same clubs as S_1 . Then we can see every 23 students have a common club, but all 1200 students do not have a common club.

Now we show $k \geq 23$. Suppose a student S joins clubs $C_1, C_2, \dots, C_k$. Because all 1200 students do not join a common club, for each $1 \leq j \leq k$, there is a student S_j not joining the club C_j . Hence the students $S, S_1, S_2, \dots, S_k$ do not have a common club, forcing $k \geq 23$. The proof is complete.

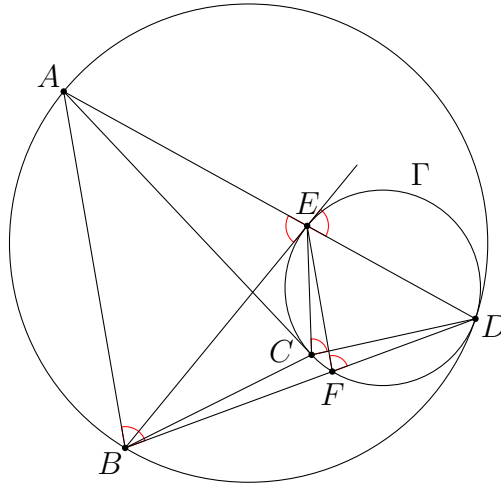
The circle HBC is the reflection of the circumcircle of $\triangle ABC$ in BC . It follows that M is the reflection of O in BC . Let D be the midpoint of BC . Recall that $OD = \frac{1}{2}AH$. Therefore, we have $OM = AH$. As both of them are perpendicular to BC , $AHMO$ is a parallelogram. Hence, AM passes through the midpoint of OH . The same holds for BN and CP . Thus, AM, BN, CP and OH are concurrent.

Test 2

1. Note that $AB^2 = AC^2 = AE \times AD$. This implies $\triangle ABE \sim \triangle ADB$. It follows that $\angle AEB = \angle ABD$. Let BD meet Γ again at F . Consider the homothety with centre D taking Γ to the circumcircle of $\triangle ABD$. Since EF is mapped to AB , the lines EF and AB are parallel. Hence, we have

$$\angle AEB = \angle ABD = \angle EFD = \angle ECD.$$

This implies BE is tangent to Γ .



2. (a, b, c) can be any integer triple with $a, b, c \geq 2015$ except $(2015, 2015, 2015)$.

It is obvious that no operation can be carried out in the exceptional case. For the other cases, we claim that it is always possible to increase the total number of coins in some operations. If this is true, then the total number of coins will exceed 3×2017^{2017} after some operations, and hence there is a pile with at least 2017^{2017} coins.

Consider any integer triple (a, b, c) with $S = a + b + c \geq 6046$ (which is $3 \times 2015 + 1$). Note that operation (1) does not change the sum S , while operation (2) always increases S by 1. Since we always have $\left\lfloor \frac{S}{3} \right\rfloor > 2015$, there must be a pile with

at least 2016 coins after each operation, and hence one of the operations can be carried out.

Suppose on the contrary that there exists a case with $S \geq 6046$ such that S can no longer be increased. Thus, only operation (1) is possible, which means the number of coins in each pile is always even or at most 2015. Applying operation (1) if necessary, we may assume there is an empty pile. Afterwards, we apply operation (1) repeatedly to the nonempty pile with the smallest number of coins until this is not possible. The number of coins of the nonempty pile with the smallest number of coins is halved in each operation. Therefore, it must become an odd number after some steps.

We are now in a situation where the number of coins are $0, x, y$ respectively, where x is odd and $0 < x \leq 2015 < y$. As $x + y \geq 6046$, we deduce $y \geq 4031$. Note that y must be even since otherwise no more operation (1) can be carried out. Hence, we have $y \geq 4032$. Next, we apply operation (1) on the pile with y coins. The number of coins becomes $0, \frac{y}{2}, x + \frac{y}{2}$. Each of the nonempty piles contains at least $\frac{y}{2} \geq 2016$ coins. As $S = x + y$ is odd, one of these piles has an odd number of coins. This contradicts our assumption that there cannot be an odd pile with more than 2016 coins.

Therefore, the total number of coins can always be increased. The goal can be attained in a finite number of operations whenever the total number of coins is at least 6046.

3. The solutions are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) = x$ for all $x \in \mathbb{R}$.

We label the equation as follows.

$$f(f(xy - x)) + f(x + y) = yf(x) + f(y) \quad (1)$$

Putting $x = 0$ in (1), we have $f(f(0)) = yf(0)$. When $y = 0$, we have $f(f(0)) = 0$. Then $yf(0) = 0$ for all $y \in \mathbb{R}$ so that $f(0) = 0$. Putting $y = 1$ in (1), we have

$$f(x + 1) = f(x) + f(1). \quad (2)$$

Next, we replace y by $y + 1$ in (1). Using (2), we find that

$$f(f(xy)) + f(x + y) = yf(x) + f(x) + f(y).$$

Swapping x and y , and comparing to this equation, we get $yf(x) = xf(y)$ for all $x, y \in \mathbb{R}$. It follows that $\frac{f(x)}{x}$ is a constant for nonzero x . This means $f(x) = cx$ for some constant c , which also holds for $x = 0$. Substituting this into (1), we have

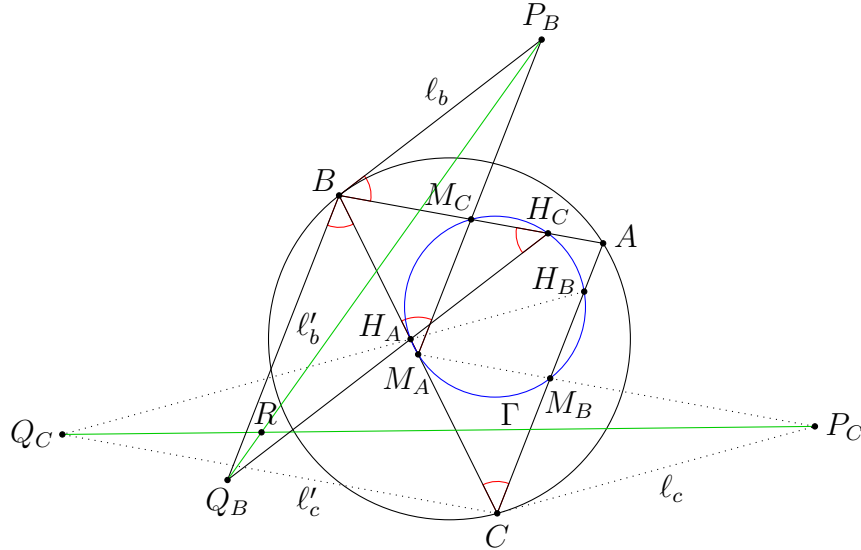
$$c^2(xy - x) + c(x + y) = cxy + cy.$$

This holds for all $x, y \in \mathbb{R}$ if and only if all coefficients of like terms are equal. This holds if and only if $c^2 = c$, i.e. $c = 0$ or $c = 1$. Therefore, the only functions are the zero function and the identity function.

4. Firstly, since $M_C M_A \parallel AC$, we have $\angle M_C B P_B = \angle ACB = \angle M_C M_A B$ so that ℓ_b is tangent to $(BM_A M_C)$. This implies $P_B B^2 = P_B M_C \times P_B M_A$, and hence P_B lies on the radical axis of B and the nine-point circle Γ of $\triangle ABC$.

Secondly, we have $\angle B H_C H_A = \angle ACB = \angle A B P_B = \angle Q_B B H_A$ so that ℓ'_b is tangent to $(B H_A H_C)$. This implies $Q_B B^2 = Q_B H_A \times Q_B H_C$, and hence Q_B lies on the radical axis of B and Γ .

Therefore, $P_B Q_B$ is the radical axis of B and Γ . Similarly, $P_C Q_C$ is the radical axis of C and Γ . So R is the radical centre of B, C and Γ . Thus, we have $RB^2 = RC^2$, which implies $RB = RC$ as desired.



Test 3

1. (International Zhautykov Olympiad 2014 Problem 4 modified/Vietnam MO 2017 Problem 2) No. Suppose such a polynomial P exists.

Let $Q(x) = P(x+1) - 1$. Then $Q(\sqrt[3]{2}) = \sqrt[3]{2}$ and $Q(\sqrt{5}) = 1 + 3\sqrt{5}$. Note that $\sqrt[3]{2}$ is a root of $Q(x) - x = 0$. Since the minimal polynomial of $\sqrt[3]{2}$ is $x^3 - 2$, we have

$$Q(x) - x = (x^3 - 2)R(x)$$

for some polynomial $R(x)$ with integer coefficients. Let $R(\sqrt{5}) = a + b\sqrt{5}$ for some $a, b \in \mathbb{Z}$. Putting $x = \sqrt{5}$, we obtain

$$1 + 2\sqrt{5} = (5\sqrt{5} - 2)(a + b\sqrt{5}) = (-2a + 25b) + (5a - 2b)\sqrt{5}.$$

Thus, we have $-2a + 25b = 1$ and $5a - 2b = 2$. This gives

$$a = \frac{2(-2a + 25b) + 25(5a - 2b)}{121} = \frac{52}{121} \notin \mathbb{Z}.$$

This is a contradiction. So such a polynomial P does not exist.

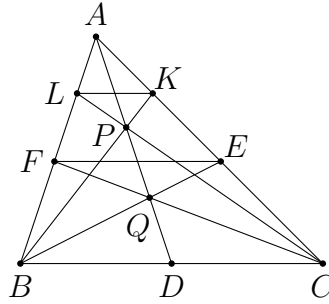
2. (Belarus TST 2017 Test 2 Problem 4) The only possible value is 1.

We use projective geometry. Consider the following projection.

$$AB(A, L, F, B) \xrightarrow{C} AD(A, P, Q, D) \xrightarrow{B} AC(A, K, E, C).$$

This shows (A, L, F, B) and (A, K, E, C) are perspective, and so LK, FE, BC are concurrent. Since $KL \parallel EF$, this shows $KL \parallel EF \parallel CB$.

Now, since AD, BK, CL are concurrent, $KL \cap CB$ is the harmonic conjugate of D with respect to BC . As $KL \cap CB$ is a point at infinity, D must be the midpoint of BC .



3. The biggest k is $n - m + 1$.

To show that $k \leq n - m + 1$, consider the initial board such that only the cells (i, j) (lying in row i and column j) with $i = j$ are black. If we colour at least $n - m + 2$ columns red, then at least 2 of the first m columns are coloured. WLOG assume columns 1 and 2 become red. Then the first two rows are alike no matter which other columns are coloured.

It remains to prove we can always colour $n - m + 1$ columns red. Indeed, it suffices to show that whenever $n \geq m$, we can always colour one column red so that no two rows are still alike, as we can repeat the same process for the remaining columns until the number of columns not in red is less than m . At this stage, exactly $n - m + 1$ columns are red.

Suppose on the contrary that it is not possible to colour a column red that fulfill the condition. We construct a simple graph as follows. Let the m vertices represent the rows. For each of the n columns, we can find two rows such that

these rows only differ in that column, or otherwise we can colour that column red. Then we draw an edge joining the two corresponding vertices. Note that there may be more than one way to draw this edge for each column, but we only draw one such edge.

Now, the resultant graph is simple since each pair of rows may differ in at most one column. Also, the simple graph has exactly $n \geq m$ edges. It is well-known that such a simple graph must consist of a cycle. WLOG assume the vertices corresponding to rows $1, 2, \dots, d$ form a cycle, and the edges joining them correspond to columns $1, 2, \dots, d$. Also, we may assume the cells in row 1 are $WWW \dots W$ (where W stands for an uncoloured white cell, while B stands for a black cell). Since row 2 differs from row 1 in column 1, row 2 must be $BWW \dots W$. Similarly, row 3 must be $BBW \dots W$, etc., and row d must be $\underbrace{BB \dots B}_d W \dots W$. But then rows 1 and d differ in $d \geq 3$ columns, which contradicts

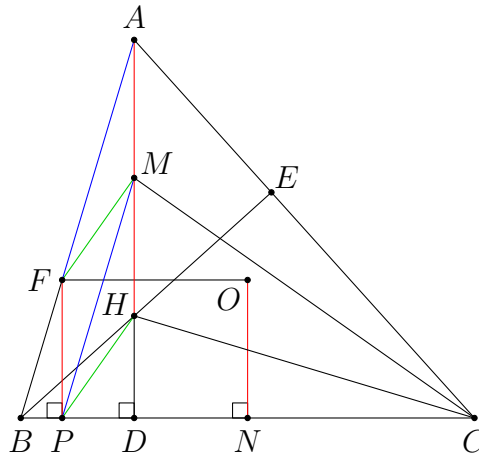
the assumption that they only differ in one column. Therefore, we can always colour a column red. This completes the proof.

Test 4

1. Let P and N be the projection of F and O on BC respectively. Recall that $ON = \frac{1}{2}AH$. Therefore, $FP = AM = MH$.

Since FP and AM are perpendicular to BC , they are parallel. Thus, $AFPM$ is a parallelogram. This implies $MP \parallel AF$, and hence $MP \perp CH$. Also, we have $MH \perp PC$. Therefore, H is the orthocentre of $\triangle MPC$. It follows that $PH \perp CM$.

Now, as $FP \parallel MH$ and $FP = MH$, we know that $FPHM$ is another parallelogram. This shows $FM \parallel PH$, and hence $FM \perp CM$. In other words, $\angle FMC = 90^\circ$.



2. The only possible n is 4.

We call those edges which are the sides of a base the *base edges*, and call the remaining edges the *lateral edges*. Let s be the sum of all numbers surrounding each face, and let x be the sum of all numbers on the lateral edges.

Firstly, we consider the sum of all numbers on the base edges. Since they are the sides of the two bases, this is equal to $2s$. Also, they are the numbers not used on the lateral edges, and hence this is equal to $1 + 2 + \dots + 3n - x$. This yields

$$x + 2s = \frac{3n(3n + 1)}{2}. \quad (1)$$

Secondly, we consider the sides of the lateral faces. Each base edge is a side of 1 lateral face, while each lateral edge is a side of 2 lateral faces. The total sum of numbers on all these edges is $1 + 2 + \dots + 3n + x$. These edges together surround n faces, and so the total sum is ns . It follows that

$$ns - x = \frac{3n(3n + 1)}{2}. \quad (2)$$

By (1) and (2), we obtain

$$\begin{aligned} s &= \frac{(x + 2s) + (ns - x)}{n + 2} = \frac{3n(3n + 1)}{n + 2}, \\ x &= \frac{3n(3n + 1)}{2} - 2s = \frac{3n(3n + 1)(n - 2)}{2(n + 2)}. \end{aligned}$$

As $s \in \mathbb{Z}$, we have $n + 2 \mid 3n(3n + 1)$. This implies $n + 2 \mid 30$.

Next, we have the bound $x \leq 3n + (3n - 1) + \dots + (2n + 1)$ since there are only n lateral edges. Now,

$$\begin{aligned} \frac{3n(3n + 1)(n - 2)}{2(n + 2)} &\leq \frac{(5n + 1)n}{2} \\ \Leftrightarrow 3(3n + 1)(n - 2) &\leq (5n + 1)(n + 2) \\ \Leftrightarrow 2(2n^2 - 13n - 4) &\leq 0. \end{aligned}$$

Thus, we have $n \leq 6$. Together with $n + 2 \mid 30$ and the trivial bound $n \geq 3$, we easily obtain the only possibilities $n = 3, 4$.

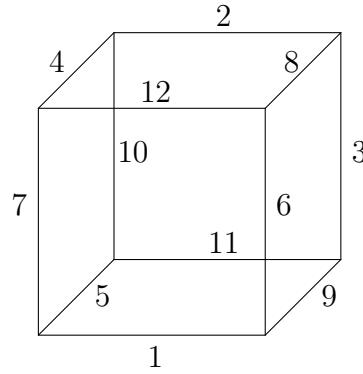
For $n = 3$, we have $s = 18$ and $x = 9$.

- If the numbers on the lateral edges are 1, 2, 6, then the numbers surrounding the three lateral surfaces are 1, 2, a, b ; 1, 6, c, d and 2, 6, e, f respectively. Since $a + b = s - 1 - 2 = 15$, we must have $a = 7$ and $b = 8$. But then $c + d = 18 - 1 - 6 = 11$ has no solution with $c, d \in \{3, 4, 5, 9\}$.

- If the numbers on the lateral edges are 1, 3, 5, then the numbers surrounding the three lateral surfaces are 1, 3, a, b ; 1, 5, c, d and 3, 5, e, f respectively. Since $a + b = 14$, we must have $a = 6$ and $b = 8$. But then $c + d = 12$ has no solution with $c, d \in \{2, 4, 7, 9\}$.
- If the numbers on the lateral edges are 2, 3, 4, then the numbers surrounding the three lateral surfaces are 2, 3, a, b ; 2, 4, c, d and 3, 4, e, f respectively. Since $a + b = 13$, $c + d = 12$ and $e + f = 11$, none of a, b, c, d, e, f is equal to 1, which is impossible.

Therefore, it is not possible to have $n = 3$.

For $n = 4$, we have $s = 26$ and $x = 26$. An example is given below.



3. (Bulgaria MO 2017 Problem 4) The solutions are $(p, a, m) = (2, 1, 1), (2, 3, 2), (2, 7, 3), (2, 15, 4), (3, 1, 1), (3, 7, 2), (3, 25, 3), (5, 1, 2), (5, 101, 3)$.

When $p = 2$, we need $1 + a = 2^m$ with $a \leq 20$. This yields $(a, m) = (1, 1), (3, 2), (7, 3), (15, 4)$.

When $p = 3$, we need $2 + a = 3^m$ with $a \leq 45$. This yields $(a, m) = (1, 1), (7, 2), (25, 3)$.

When $p = 5$, we need $24 + a = 5^m$ with $a \leq 125$. This yields $(a, m) = (1, 2), (101, 3)$.

Now, suppose $p \geq 7$. By Wilson's theorem, we have $-1 + a \equiv 0 \pmod{p}$. Also, it is clear that $a \equiv 1 \pmod{p-1}$. Thus, we can write $a = kp(p-1) + 1$ for some integer $k \geq 0$. As it is given that $5p^2 \geq a$, we easily deduce $k \leq 5$ since $p \geq 7$.

The equation now becomes $(p-1)! + kp(p-1) = p^m - 1$. This is the same as

$$(p-2)! + kp = p^{m-1} + p^{m-2} + \dots + 1.$$

Note that the terms 2 and $\frac{p-1}{2}$ both appear in $(p-2)!$, and they are distinct since $p \geq 7$. Thus, we have $p-1 \mid (p-2)!$. It follows that $k \equiv m \pmod{p-1}$. If

$m \geq p-1$, then

$$(p-2)! + kp = p^{m-1} + p^{m-2} + \dots + 1 > p^{p-2}.$$

On the other hand, we have

$$(p-2)! + kp \leq (p-2)! + 5p = (2)(3)(4)(5)\dots(p-2) + 5p < p^{p-3} + p^{p-3} < p^{p-2}$$

since $p \geq 7$. This yields a contradiction. Thus, we must have $m < p-1$. Since we also have $k \leq 5 < p-1$, the only possibility for $k \equiv m \pmod{p-1}$ is $k = m$. It remains to handle a few cases.

- If $k = m = 1$, the equation becomes $(p-2)! + p = 1$. This is impossible.
- If $k = m = 2$, the equation becomes $(p-2)! + 2p = p + 1$. Again, this is impossible.
- If $k = m = 3$, the equation becomes $(p-2)! + 3p = p^2 + p + 1$. We need $p-2 \mid p^2 - 2p + 1$, which means $p-2 \mid 1$. This cannot hold for $p \geq 7$.
- If $k = m = 4$, the equation becomes $(p-2)! + 4p = p^3 + p^2 + p + 1$. We need $p-2 \mid p^3 + p^2 - 3p + 1$, which means $p-2 \mid 7$. This cannot hold for prime $p \geq 7$.

Therefore, there is no solution for $p \geq 7$.

Year 2019

Test 1

1. The answer is $-\frac{9}{2}$.

The condition means $\frac{a}{x-1} + \frac{1}{x-2} + \frac{1}{x-6} \leq \frac{a}{3}$ for $3 < x < 5$. Now, noting that $x-1 > 0$, $x-2 > 0$ and $x-6 < 0$, we have

$$\begin{aligned} & \frac{a}{x-1} + \frac{1}{x-2} + \frac{1}{x-6} \leq \frac{a}{3} \\ \Leftrightarrow & 3a(x-2)(x-6) + 3(x-1)(x-6) + 3(x-1)(x-2) \geq a(x-1)(x-2)(x-6) \\ \Leftrightarrow & ax^3 - (12a+6)x^2 + (44a+30)x - (48a+24) \leq 0 \\ \Leftrightarrow & (x-4)(ax^2 - (8a+6)x + (12a+6)) \leq 0. \end{aligned}$$

As $x=4$ is the maximum, there must be another linear factor $x-4$ coming from the quadratic factor. It follows from the factor theorem that $-4a-18=0$, i.e. $a = -\frac{9}{2}$.

When $a = -\frac{9}{2}$, the above inequality becomes $(x-4)^2 \left(-\frac{9}{2}x + 12\right) \leq 0$, which is true for $3 < x < 5$. Therefore, $x=4$ is indeed the maximum.

2. The answer is 120.

The probability that the chord does not intersect the triangle is $\frac{11}{25}$. The only way this can happen is when the two points are chosen on the same arc between two of the triangle vertices. The probability that a point is chosen on one of the arcs opposite to one of the base angles is $\frac{x}{180}$ and the probability that a point is chosen on the arc between the two base angles is $\frac{180-2x}{180}$. Therefore we have

$$2\left(\frac{x}{180}\right)^2 + \left(\frac{180-2x}{180}\right)^2 = \frac{11}{25}.$$

This simplifies to $x^2 - 120x + 3024 = 0$, and can be factorized as $(x-84)(x-36) = 0$. Hence the sum of roots is $84 + 36 = 120$.

3. The answer is 2019.

Clearly, the answer should not be an integer greater than 2018. Hence we may remove the last 4 terms on the left-hand side of the equation.

Next, we may assume that $x = a(6!) + b(5!) + c(4!) + d(3!) + e(2!) + f(1!)$, where a, b, c, d, e, f are nonnegative integers and $a \leq 6, b \leq 5, c \leq 4, d \leq 3, e \leq 2, f \leq 1$.

Substituting into the original equation gives

$$1237a + 206b + 41c + 10d + 3e + f = 2019.$$

Since $206b + 41c + 10d + 3e + f \leq 1231$, we have $788 \leq 1237a \leq 2019$. This gives $a = 1$. Then we have

$$206b + 41c + 10d + 3e + f = 782.$$

Similarly, we find that $b = 3, c = 4, d = e = f = 0$. Hence,

$$x = 1(6!) + 3(5!) + 4(4!) = 1176$$

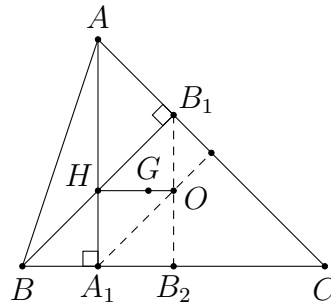
is the only solution.

4. The answer is 12.

Let H be the orthocentre of $\triangle ABC$. By the property of the Euler line, we have $OH = 3OG = 3$ and $HO \parallel BC$.

Let AA_1 and BB_1 be the altitudes of $\triangle ABC$ and let B_2 be the midpoint of BC . Since $\angle ACB = 45^\circ$, we have $BB_1 = B_1C$. Thus, all the points B_1, O, B_2 lie on the perpendicular bisector of BC , and hence $B_1B_2 \perp BC$. This implies $B_1B_2 \parallel HA_1$.

Similarly, we can prove $A_1O \parallel BB_1$. Since $HO \parallel BC$, the sides of $\triangle HA_1O$ are parallel to the sides of $\triangle B_2B_1B$. Thus, $\triangle HA_1O$ is the medial triangle of $\triangle B_2B_1B$. It follows that $BC = 2BB_2 = 4OH = 12$.



5. No. Suppose such a choice exists. Let $n_1, n_2, \dots, n_{2019}$ be the number of points on the planes respectively. Note that $n_j \geq 3$ for $1 \leq j \leq 2019$. It is clear that

$$2024 = \binom{24}{3} = \binom{n_1}{3} + \binom{n_2}{3} + \dots + \binom{n_{2019}}{3}.$$

We first show $n_j \leq 4$ for all $1 \leq j \leq 2019$. Indeed, if at least one is greater than 4, then

$$2024 = \binom{n_1}{3} + \binom{n_2}{3} + \cdots + \binom{n_{2019}}{3} \geq \underbrace{1 + 1 + \cdots + 1}_{2018 \text{ times}} + \binom{5}{3} = 2028,$$

which is impossible.

Now suppose among all the n_j 's, a of them are equal to 3 and b of them are equal to 4. Then we have $a + b = 2019$ and

$$a + 4b = a \binom{3}{3} + b \binom{4}{3} = 2024.$$

This gives $b = \frac{(a + 4b) - (a + b)}{3} = \frac{5}{3}$, which is again impossible. Therefore, such a choice does not exist.

6. The answer is 466.

Note that

$$\begin{aligned} a^3 + 5a^2 - 57a + 99 &= (a + 11)(a - 3)^2 \geq 0, \\ b^3 + 5b^2 - 88b + 208 &= (b + 13)(b - 4)^2 \geq 0, \\ c^3 + 5c^2 - 125c + 375 &= (c + 15)(c - 5)^2 \geq 0. \end{aligned}$$

Adding these inequalities, we find that

$$a^3 + b^3 + c^3 + 5a^2 + 5b^2 + 5c^2 \geq 57a + 88b + 125c - (99 + 208 + 375) \geq 466.$$

Equality holds when $a = 3$, $b = 4$ and $c = 5$.

Test 2

1. The sequence can be any constant sequence $p, p, p, \dots$ where p is a prime

The recurrence relation can be rewritten as

$$p_{n+2} + k = (p_{n+1} + k) + (p_n + k).$$

Since the characteristic equation $\lambda^2 - \lambda - 1 = 0$ has roots $\frac{1 \pm \sqrt{5}}{2}$, we have

$$p_n = A \left(\frac{1 + \sqrt{5}}{2} \right)^n + B \left(\frac{1 - \sqrt{5}}{2} \right)^n - k,$$

for some constants A, B .

Furthermore, as there are at most p_1^2 possibilities for the remainders of the pair (p_{n+1}, p_n) modulo p_1 , and the remainders of the subsequent terms are uniquely determined by these, the sequence must be periodic modulo p_1 . Thus, there are infinitely many terms which are congruent to p_1 modulo p_1 . Since they are primes, they must be p_1 .

If $A \neq 0$, the term $\left(\frac{1+\sqrt{5}}{2}\right)^n$ dominates the expression for p_n , and so the sequence is eventually strictly increasing (if $A > 0$) or decreasing (if $A < 0$). This contradicts our observation. Thus, we must have $A = 0$. Similarly, if $B \neq 0$, the term $\left(\frac{1-\sqrt{5}}{2}\right)^n$ now dominates the expression for p_n , which again implies $p_n \neq p_1$ for all sufficiently large n . Thus, we must have $B = 0$. It follows that $p_n = -k$ is a constant.

Conversely, if the sequence is $p, p, p, \dots$, where p is a prime, we can take $k = -p$ so that the recurrence relation holds.

2. If 5 is a primitive root modulo p , then there exists m such that $1 \leq m \leq p-1$ and $5^m \equiv 7^{-1} \pmod{p}$. Note that $m \neq p-1$, since otherwise $1 \equiv 7^{-1} \pmod{p}$, which contradicts $p > 10$. Therefore, we can choose this m and $n = 1$ so that $m+n \leq p-1 < p$ and $p \mid 5^m 7 - 1$.

Similarly, we are done if 7 is a primitive root modulo p .

It remains to consider the case when the orders d_1 and d_2 of 5 and 7 modulo p are both less than $p-1$. Note that d_1 and d_2 divide $p-1$. Thus, $d_1, d_2 \leq \frac{p-1}{2}$. We can choose $m = d_1$ and $n = d_2$ so that $m+n \leq p-1 < p$ and

$$5^{d_1} 7^{d_2} - 1 \equiv (1)(1) - 1 \equiv 0 \pmod{p}.$$

This completes the proof.

3. (a) Note that we have $\angle CAE = \angle CBE = \angle DBF = \angle DAF$. As $AP \perp CD$, we see that AP bisects $\angle EAF$. Since P also lies on the perpendicular bisector of EF , the points A, E, P, F are concyclic. Thus,

$$\angle EPF = 180^\circ - \angle EAF = \angle CAE + \angle DAF = 2\angle CAE.$$

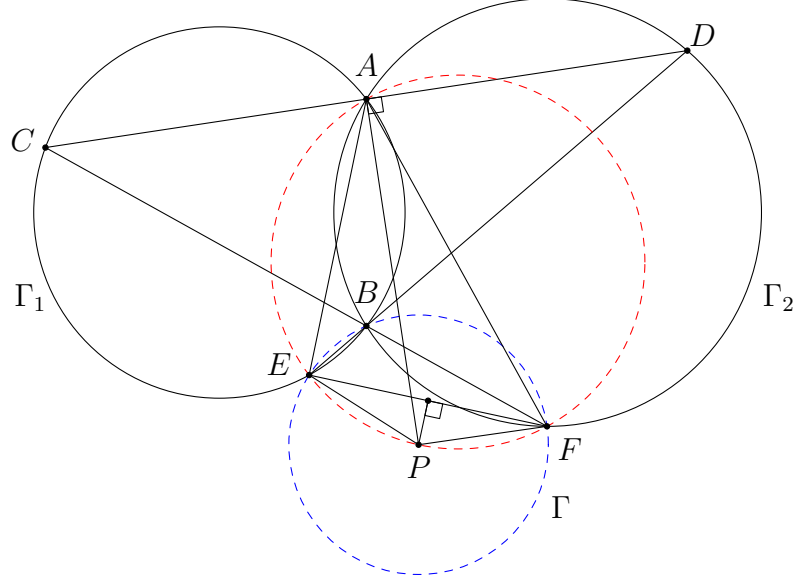
- (b) (CMO 2013 Problem 1 modified) By part (a), since $\angle EPF = 2\angle CBE$, the point B lies on the circle Γ with centre P and radius PE . By the power of a point theorem, we have

$$2CA^2 = CA \cdot CD = CB \cdot CF = CP^2 - PE^2.$$

Hence,

$$AP^2 = CP^2 - CA^2 = (2CA^2 + PE^2) - CA^2 = CA^2 + PE^2,$$

as desired.



4. The answer is 8100.

Suppose we choose all points (x, y) with $1 \leq x, y \leq 10$. For each chosen point B , there are 9 choices of chosen points A which have the same y -coordinate as B , and there are 9 choices of chosen points C which have the same x -coordinate as B . Thus, we have

$$N = 100 \times 9 \times 9 = 8100$$

in this case.

Consider any way of choosing 100 points. Suppose the points lie in k different horizontal lines. Let there be $a_1, a_2, \dots, a_k$ chosen points on these horizontal lines respectively. We count the number of triples (A, B, C) with A and B lying on the first horizontal line ℓ , and B and C having the same x -coordinate. For any point C not on ℓ , there is at most one point B on ℓ having the same x -coordinate as C , and there are exactly $a_1 - 1$ such triples if B exists. So there are at most $(a_1 - 1)(100 - a_1)$ triples corresponding to ℓ . It follows that

$$N \leq \sum_{j=1}^k (a_j - 1)(100 - a_j).$$

Using the fact that $a_1 + a_2 + \dots + a_k = 100$, we get

$$N \leq 101 \sum_{j=1}^k a_j - \sum_{j=1}^k a_j^2 - 100k = 10100 - \sum_{j=1}^k a_j^2 - 100k.$$

By the Cauchy-Schwarz inequality, we have

$$\sum_{j=1}^k a_j^2 \geq \frac{1}{k} \left(\sum_{j=1}^k a_j \right)^2 = \frac{10000}{k}.$$

This yields

$$N \leq 10100 - \left(\frac{10000}{k} + 100k \right) \leq 10100 - 2000 = 8100$$

by the AM-GM inequality. The result follows readily.

Test 3

1. (IMO Shortlist 2018 A1) See the official solution.
2. (IMO Shortlist 2018 G2) See the official solution.
3. (IMO Shortlist 2018 C3) See the official solution.

Test 4

1. (IMO Shortlist 2018 N1) See the official solution.
2. (IMO Shortlist 2018 C1) See the official solution.
3. (IMO Shortlist 2018 A7) See the official solution.

Year 2020

Test 1

1. The only solution is the function

$$f(n) = \begin{cases} p & \text{if } n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 1 & \text{otherwise.} \end{cases}$$

For $n = 1$, the only positive divisor is 1, and hence $f(1) = 1$.

For any prime p , we prove by induction on k that $f(p^k) = p$. For the base case $k = 1$, we have $f(p) = f(1)f(p) = p$. Assuming $f(p^j) = p$ for $j = 1, 2, \dots, k$, we have

$$p^k f(p^{k+1}) = f(1)f(p) \cdots f(p^{k+1}) = p^{k+1}.$$

This implies $f(p^{k+1}) = p$. So $f(p^k) = p$ for all $k \in \mathbb{Z}^+$ by induction.

Now, consider $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ where $s \geq 2$ and $\alpha_j \in \mathbb{Z}^+$ for each j . The positive divisors of n include the prime powers p_j^k . Since

$$\prod_{j=1}^s \prod_{k=1}^{\alpha_j} f(p_j^k) = \prod_{j=1}^s \prod_{k=1}^{\alpha_j} p_j = \prod_{j=1}^s p_j^{\alpha_j} = n = \prod_{d|n} f(d),$$

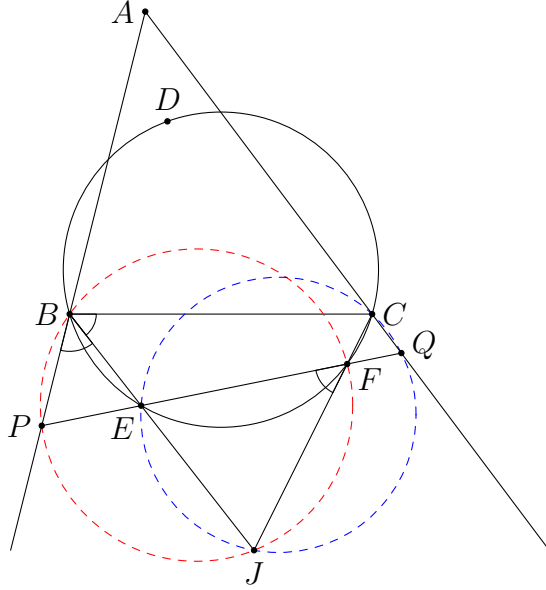
we have $\prod_{\substack{d|n \\ d \neq p_j^k}} f(d) = 1$. Thus, the images of those positive divisors of n different from a power of prime are 1. In particular, $f(n) = 1$.

Lastly, we check that the proposed function satisfies the condition. Indeed, for any $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s} > 1$, we have

$$\prod_{d|n} f(d) = \prod_{j=1}^s \prod_{k=1}^{\alpha_j} f(p_j^k) = n.$$

2. Let BE and CF intersect at the A -excentre J of $\triangle ABC$. We claim that J is the desired fixed point.

We only consider the configuration as shown since the other cases are similar. Since BE is the external angle bisector, we have $\angle PBE = \angle EBC$. This is equal to $\angle EFJ$ as B, E, F, C are concyclic. This implies P, B, F, J are concyclic. Similarly, Q, C, E, J are concyclic. This shows the two circles always pass through J , which is independent of D .



3. The answer is $2^{674} + 672$.

We claim that both $(2^r + 1, 2^s + 1)$ and $(2^r + 1, 2^s - 1)$ are of the form $2^a + 1$ or 1 for any $r, s \in \mathbb{Z}^+$. We prove this by induction on $\min\{r, s\}$. The base case $\min\{r, s\} = 1$ is trivial since the greatest common divisor can only be 1 or 3. Assume this holds for all smaller cases and consider the inductive step.

For $(2^r + 1, 2^s + 1)$, WLOG assume $r \leq s$. Let $s = qr + b$ where $0 \leq b < r$. Then

$$2^s + 1 = 2^b(2^r)^q + 1 \equiv 2^b(-1)^q + 1 = \pm 2^b + 1 \pmod{2^r + 1}.$$

This shows $(2^r + 1, 2^s + 1) = (2^r + 1, 2^b \pm 1)$. As $b < r$, we are done by the inductive hypothesis.

Similarly, for $(2^r + 1, 2^s - 1)$, if $r \leq s$, then we write $s = qr + b$ for some $0 \leq b < r$ so that $(2^r + 1, 2^s - 1) = (2^r + 1, 2^b \pm 1)$. If $s < r$, then we write $r = qs + b$ for some $0 \leq b < s$ so that $(2^r + 1, 2^s - 1) = (2^b + 1, 2^s - 1)$. In any case, we are done by the inductive hypothesis.

So we have proven the claim by induction. Next, note that $2^a + 1 \mid 2^r + 1$ if and only if $r = da$ for some odd integer d . (Again, by writing $r = qa + b$ with $0 \leq b < a$, we can show that $2^a + 1 \mid 2^b(-1)^q + 1$ and hence $2^a + 1 \leq |2^b(-1)^q + 1|$ unless $2^b(-1)^q + 1 = 0$. As $b < a$, the former case is impossible.) Therefore, the greatest common divisor of two terms is $2^a + 1$ only if both are of the form $2^{(2k-1)a} + 1$. In particular, we cannot have $a > 673$ since otherwise one term is at least $2^{3(674)} + 1 > 2^{2019} + 1$.

On the other hand, a can be one of $1, 2, \dots, 673$ as $(2^a + 1, 2^{3a} + 1) = 2^a + 1$. Obviously, the greatest common divisor can also be 1. So the answer is

$$1 + (2^1 + 1) + (2^2 + 1) + \dots + (2^{673} + 1) = 2^{674} + 672.$$

4. The answer is 24.

Let p be an odd prime. Using the binomial theorem, we have

$$(2 + \sqrt{5})^p + (2 - \sqrt{5})^p = \sum_{k=0}^{\frac{p-1}{2}} \binom{p}{2k} 2^{p+1-2k} \cdot 5^k = 2^{p+1} + \sum_{k=1}^{\frac{p-1}{2}} \binom{p}{2k} 2^{p+1-2k} \cdot 5^k,$$

which is an integer. Since p is an odd prime, we have $(2 - \sqrt{5})^p < 0$. From this it follows that $\lfloor (2 + \sqrt{5})^p \rfloor = (2 + \sqrt{5})^p + (2 - \sqrt{5})^p$. Thus,

$$\lfloor (2 + \sqrt{5})^p \rfloor - 2^{p+1} = \sum_{k=1}^{\frac{p-1}{2}} \binom{p}{2k} 2^{p+1-2k} \cdot 5^k.$$

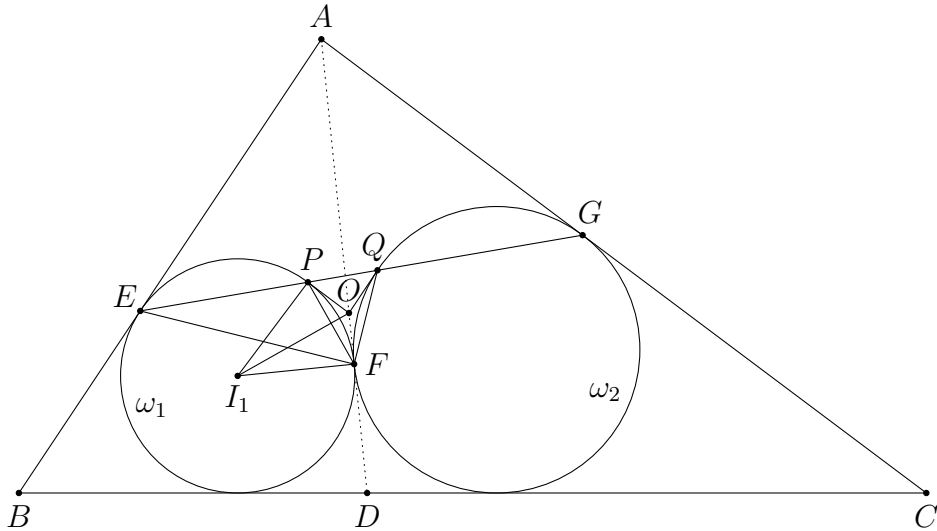
Since $\binom{p}{r}$ is divisible by p for $1 \leq r \leq p-1$, $\lfloor (2 + \sqrt{5})^p \rfloor - 2^{p+1}$ is divisible by p .

It is easy to check that the expression is not divisible by p when $p = 2$. Since there are 25 primes less than 100, the answer is 24.

5. Let O be the circumcentre of $\triangle FPQ$. First of all,

$$\angle OFP = 90^\circ - \angle PQF = \angle FQG - 90^\circ = \frac{\angle FAG}{2} = \angle GEF = \angle AFP$$

where the second last equality follows from the fact that $AE = AF = AG$. This shows that O lies on AD .



Next, Let I_1 be the centre of ω_1 . Note that $\triangle OFI_1 \cong \triangle OPI_1$ as their corresponding sides are equal. Thus $\angle I_1PO = 90^\circ$, which means OP is tangent to ω_1 . Similarly OQ is tangent to ω_2 .

Hence, the line AD , the tangent to ω_1 at P and the tangent to ω_2 at Q are concurrent at O , as desired.

6. (Czech and Slovak Olympiad 2017 Round III Category A Problem 4 modified)

The answer is $2020 \binom{4038}{2019}$.

We show that for all sequences with n zeros and n ones, the sum of all continuous runs is given by $(n+1) \binom{2n}{n}$.

We first count the number of changes, which means a pair of consecutive digits which is 01 or 10. There are $2n-1$ positions for which changes may occur. For each fixed position, there are 2 ways to choose the pair of digits for this change, and $\binom{2n-2}{n-1}$ ways to put the remaining $n-1$ zeros and $n-1$ ones. Hence, there are

$$2(2n-1) \binom{2n-2}{n-1}$$

changes in total.

Now, for each sequence, the number of runs is 1 more than the number of changes. Since there are $\binom{2n}{n}$ sequences in total, the total number of runs is

$$\begin{aligned} & 2(2n-1) \binom{2n-2}{n-1} + \binom{2n}{n} \\ &= 2n \binom{2n-1}{n} + \binom{2n}{n} = 2n \binom{2n-1}{n-1} + \binom{2n}{n} \\ &= n \binom{2n}{n} + \binom{2n}{n} = (n+1) \binom{2n}{n}, \end{aligned}$$

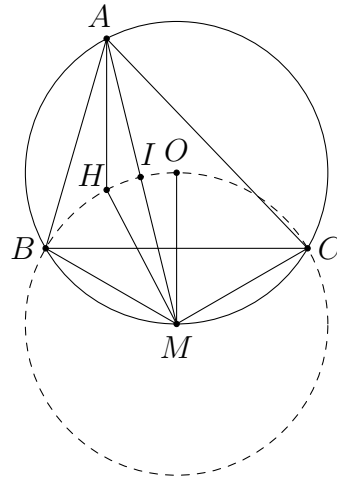
using the fact $k \binom{m}{k} = m \binom{m-1}{k-1}$.

Test 2

- Let O be the circumcentre of $\triangle ABC$. Since $BM = CM = IM = OM$ (where OM is the circumradius of $\triangle ABC$), B, C, I, O are concyclic. Therefore, we have $\angle BIC = \angle BOC$. This yields $90^\circ + \frac{A}{2} = 2A$, and hence $A = 60^\circ$. Now

$$\angle BHC = 180^\circ - A = 120^\circ = 2A = \angle BOC,$$

which implies H lies on $(BCOI)$. So $HM = IM$. Noting the triangle inequality $AH + HM \geq AM$, we conclude $AH \geq AI$ as desired.

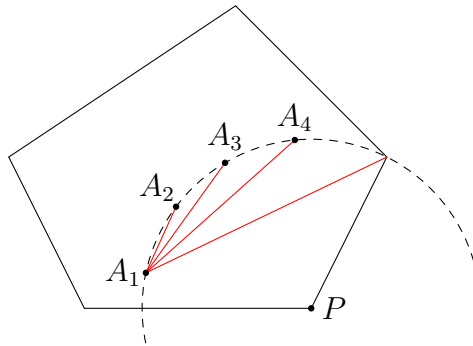


2. Suppose k is less than 44. Choose a point P on the boundary of the convex hull of this set of 2019 points.

By the pigeonhole principle, since $\left\lceil \frac{2018}{43} \right\rceil > 45$, there is a circle centered at P on which at least 45 of the other points lie. Moreover, these 45 points all lie on the same semicircle since P is on the boundary of the convex hull. Let these points be $A_1, A_2, \dots, A_{45}$ in that order. Then the distances

$$A_1A_2 < A_1A_3 < \dots < A_1A_{45}$$

are distinct, which contradicts the assumption that there are less than 44 distances. Thus, $k \geq 44$.



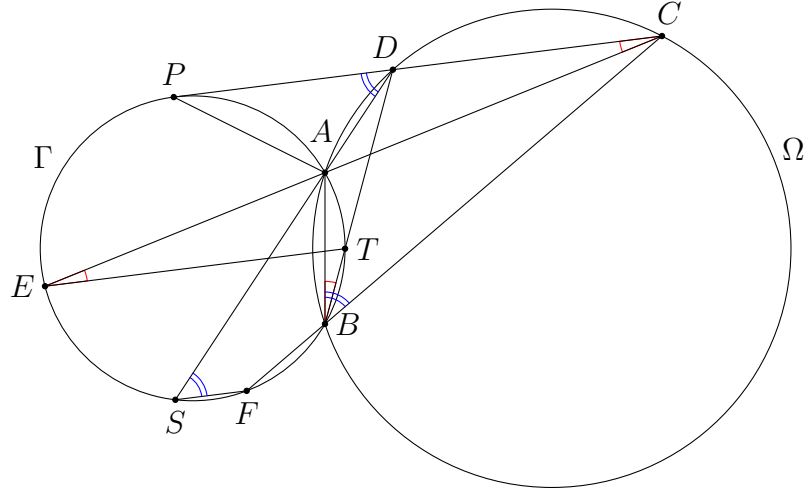
3. Since

$$\angle TEC = \angle TEA = \angle TBA = \angle DBA = \angle DCA = \angle DCE,$$

we get $PC \parallel ET$. Similarly, since

$$\angle FSD = \angle FSA = \angle CBA = \angle PDA = \angle PDS,$$

we get $PC \parallel SF$.



4. The answer is any constant function f .

Clearly, constant functions are solutions. In the following, we show that there is no solution if f is not a constant function.

Label the equation

$$f(f(x) + y) + f(x + f(y)) = 2f(xf(y)). \quad (1)$$

Swapping x and y , we get $f(f(y) + x) + f(y + f(x)) = 2f(yf(x))$. Comparing with (1), we get

$$f(xf(y)) = f(yf(x)). \quad (2)$$

Putting $x = 0$ in (2), we find that $f(0) = f(yf(0))$. If $f(0) \neq 0$, then $yf(0)$ runs through all real values. This implies f is a constant function, which is a contradiction. So we have $f(0) = 0$. Putting $x = 1$ in (2), we obtain

$$f(f(y)) = f(yf(1)). \quad (3)$$

Next, we put $x = 0$ in (1) to get

$$f(y) + f(f(y)) = 0. \quad (4)$$

Replacing x by $f(x)$ in (1), and using (4), we find that

$$f(-f(x) + y) + f(f(x) + f(y)) = 2f(f(x)f(y)).$$

Swapping x and y , and comparing with this equation again, we get

$$f(-f(x) + y) = f(-f(y) + x). \quad (5)$$

Putting $y = f(x)$ in (5) and using (4), we have $f(f(x) + x) = 0$. Then we put $x = y = 1$ in (1) to obtain $f(f(1)) = 0$. It follows from (4) by putting $y = 1$ that $f(1) = 0$. Thus, (3) becomes $f(f(y)) = 0$. Together with (4), we find that $f(y) = 0$ for any $y \in \mathbb{R}$. This is a contradiction as we have assumed f is non-constant.

Therefore, the only solutions are the constant functions.

Test 3

1. (IMO Shortlist 2019 G1) See the official solution.
2. (IMO Shortlist 2019 N2) See the official solution.
3. (IMO Shortlist 2019 A2) See the official solution.

Test 4

1. (IMO Shortlist 2019 C2) See the official solution.
2. (IMO Shortlist 2019 A5) See the official solution.
3. (IMO Shortlist 2019 G5) See the official solution.

Year 2021

Test 1

1. The only solution is $(a, b, c) = \left(0, \frac{1}{2}, \frac{3}{2}\right)$.

Note that $(2^a - 1)^2 \geq 0$, $(2^b - \sqrt{2})^2 \geq 0$ and $(2^c - \sqrt{8})^2 \geq 0$. These yield

$$2^{2a} + 1 \geq 2^{a+1},$$

$$2^{2b} + 2 \geq 2^{b+\frac{3}{2}},$$

$$2^{2c} + 8 \geq 2^{c+\frac{5}{2}}.$$

Multiplying these inequalities, we obtain

$$(2^{2a} + 1)(2^{2b} + 2)(2^{2c} + 8) \geq 2^{a+b+c+1+\frac{3}{2}+\frac{5}{2}} = 2^{a+b+c+5}.$$

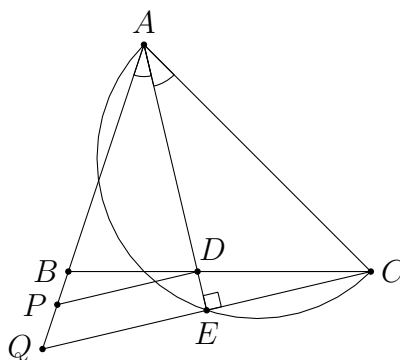
This shows all equalities should hold. This means $(a, b, c) = \left(0, \frac{1}{2}, \frac{3}{2}\right)$.

2. The answer is $\frac{2}{k+1}$.

Let CE meet AB at Q . Let P be the point on AQ such that $PD \parallel QC$. Note that E is the midpoint of QC since $\triangle AEQ \cong \triangle AEC$. Also, we have $\triangle APD \sim \triangle AQE$ and $\triangle BPD \sim \triangle BQC$. Thus, we have

$$\frac{AD}{AE} = \frac{PD}{QE} = 2 \cdot \frac{PD}{QC} = 2 \cdot \frac{BD}{BC}.$$

By the angle bisector theorem, $\frac{DB}{DC} = \frac{AB}{AC} = \frac{1}{k}$. It follows that $\frac{AD}{AE} = \frac{2}{k+1}$.



3. Miss Adams needs to perform at least two operations.

A strategy is as follows. In the first operation, she puts the coins of weights 1 to 15 on one side, and the three coins of weights 39, 40 and 41 on the other side.

Then the two groups will be of the same weight (120). Since the total weight of 15 coins is at least $1 + 2 + \dots + 15 = 120$ and the total weight of 3 coins is at most $39 + 40 + 41 = 120$, this is the only situation for which a group of 15 coins has the same weight as a group of 3 coins. So Mr. Bean knows the two unused coins are of weight 37 and 38. Miss Adams then puts the 1-gram coin and the 37-gram coin on one side, and the 38-gram coin on the other side. Hence Mr. Bean will know which coin has weight 1.

Now we show that one operation is not enough. If one operation is enough, then the 1-gram coin should be the only coin in its group (a group on the balance or the unused group).

- If the 1-gram coin is put on the balance and the other group on the balance contains more than one coin, the situation is the same if the 1-gram coin and the 2-gram coin are swapped.
- If the 1-gram coin is put on the balance and the other group on the balance contains exactly one coin, the situation is the same if the 1-gram coin is replaced by any unused coin.
- If the 1-gram coin is the only unused coin and one side of the balance is heavier, then that side has weight at least 158 as the total weight of all coins is 315, while the other side has weight at most 156. So the situation is the same if the 1-gram coin and the 2-gram coin are swapped.
- If the 1-gram coin is the only unused coin and the two sides of the balance have the same weight, we call the group containing the 3-gram coin group A , and the other group B . If group A contains a coin of weight $k \neq 3$ and group B contains a coin of weight $k + 1$, then we can swap these two coins, and swap the 1-gram coin and the 3-gram coin. The two groups still have the same weight. In the remaining case, group A should contain the heaviest coins among those with weight $1, 2, \dots, 15$ (possibly none), and the heaviest coins among those with weight $37, 38, 39, 40, 41$. But then the total weight of group A cannot be 157. The computations are illustrated below.

$$- 2 + 3 + \dots + 15 = 119 < 157$$

$$- 4 + 5 + \dots + 15 + 41 = 155 < 157 < 158 = 3 + 4 + \dots + 15 + 41$$

$$- 10 + 11 + \dots + 15 + 40 + 41 = 156 < 157 < 165 = 9 + 10 + \dots + 15 + 40 + 41$$

$$- 14 + 15 + 39 + 40 + 41 = 149 < 157 < 162 = 13 + 14 + 15 + 39 + 40 + 41$$

$$- 38 + 39 + 40 + 41 = 158 > 157$$

Therefore, in any case Mr. Bean cannot tell whether the single coin has weight 1 or not. So at least two operations are needed.

4. (Dutch TST 2019 Test 3 Problem 2 modified) No, it is not possible. Indeed, if $n^2 + 3n + 3 = ab$, then

$$(a + b)^2 = (a - b)^2 + 4ab \geq 4ab = 4n^2 + 12n + 12 > 4n^2 + 12n + 9 = (2n + 3)^2.$$

Since both sides are squares, this yields $(a + b)^2 \geq (2n + 4)^2$. It follows that

$$(a - b)^2 = (a + b)^2 - 4ab \geq (2n + 4)^2 - (4n^2 + 12n + 12) = 4n + 4 = 4(n + 1).$$

Therefore, $|a - b| \geq 2\sqrt{n + 1}$.

5. Let E be the intersection of the lines AB and CD . Let O be the circumcentre of $\triangle BDE$. Since

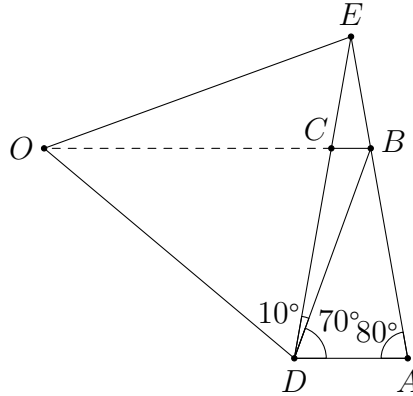
$$\angle EBD = \angle BAD + \angle BDA = 80^\circ + 70^\circ = 150^\circ,$$

we have $\angle DOE = 360^\circ - 2 \times 150^\circ = 60^\circ$. So $\triangle DEO$ is an equilateral triangle. Also, since

$$\angle EBO = \frac{180^\circ - \angle BOE}{2} = 90^\circ - \angle BDE = 80^\circ = \angle EBC,$$

the points B, C, O are collinear. Note that $\triangle EDA \cong \triangle OBE$ because they are isosceles triangles with the same base angles and $ED = EO$. This implies $DA = BE$. Also, from $\triangle BCE \sim \triangle ADE$, we have $\frac{BC}{AD} = \frac{BE}{AE}$, which implies $AD \cdot BE = BC \cdot AE$. Hence we obtain

$$AD^2 = AD \cdot BE = BC \cdot AE = BC(BE + AB) = BC(AD + AB).$$

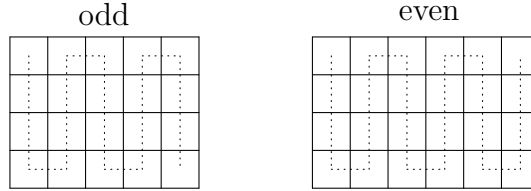


6. For convenience, suppose all cells are unit squares. We first prove two claims.

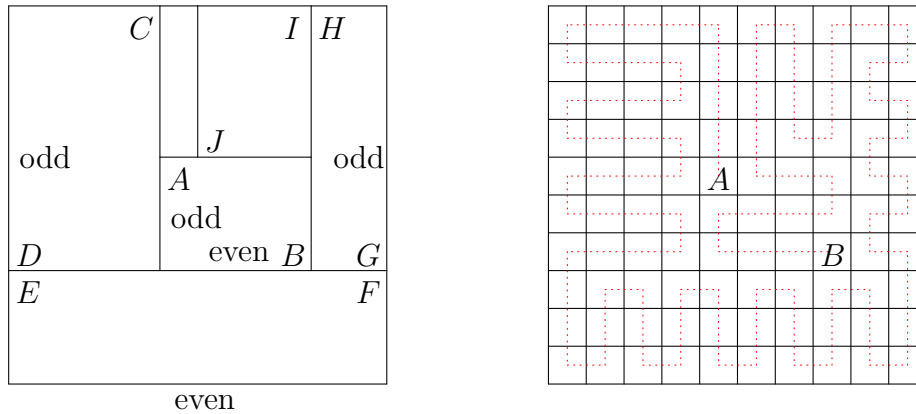
Claim 1. For any rectangular chessboard, a chess piece can move from one corner to the opposite corner by visiting every cell exactly once if at least one side has an odd length.

Claim 2. For any rectangular chessboard, a chess piece can move from one corner to an adjacent corner by visiting every cell exactly once if the side containing the two corners has an even length.

These can be easily proved by considering a snake-shaped path as shown. The assumptions in the claims are sufficient by checking the parity.



Now, we provide a construction of a path from A to B . Since A and B have distinct colours, the side lengths of the rectangular chessboard with A and B as opposite corners must have different parities. WLOG, assume the vertical side has an odd length and the horizontal side has an even length. Since the whole chessboard has an even length, we may assume there is an even number of rows above A . Partition the chessboard as shown below (the left figure).



- Firstly, the chess piece moves from A upwards until it reaches the top edge.
- Then it enters the top-left rectangle and moves from corner C to corner D (by claim 1, since the vertical side has an odd length).
- Next, it enters the bottom rectangle and moves from corner E to corner F (by claim 2, since the horizontal side has an even length).
- Afterwards, it enters the top right rectangle and moves from corner G to corner H (by claim 1, since the vertical side has an odd length).

- Then it enters the top rectangle and moves from corner I to corner J (by claim 1, since the horizontal side has an odd length).
- Lastly, it goes back to the rectangle with A and B as opposite corners and moves to B (by claim 1, since the vertical size has an odd length).

An example is given above (the right figure).

Test 2

1. There are at least 4037 points in M .

Since there are finitely many points in S , we can find a coordinate system such that all points in S have distinct x -coordinates. Indeed, we just need to find a line which is not perpendicular to any line joining two of the points in S .

Now, let the $n = 2020$ points be $P_j = (x_j, y_j)$ where $j = 1, 2, \dots, n$. WLOG assume $x_1 < x_2 < \dots < x_n$. Since

$$\frac{x_1 + x_2}{2} < \frac{x_1 + x_3}{2} < \dots < \frac{x_1 + x_n}{2} < \frac{x_2 + x_n}{2} < \dots < \frac{x_{n-1} + x_n}{2},$$

the midpoints of $P_1P_2, P_1P_3, \dots, P_1P_n, P_2P_n, \dots, P_{n-1}P_n$ have pairwise different x -coordinates, and so they must be distinct. This shows $|M| \geq 2n - 3 = 4037$.

The case 4037 is attainable. For example, we choose the points $0, 2, 4, 6, \dots, 2n-2$ on the real number line. Then the midpoints can only be $1, 2, 3, \dots, 2n-3$. So there are exactly $2n - 3 = 4037$ midpoints.

2. (IMO Shortlist 1992 modified) We claim that $g(x) = x^3 - 2019x - 2021$ is irreducible over $\mathbb{Q}$. Since $\deg g = 3$, if g were reducible, then it must have a linear factor. In other words, g would have a rational root c . Since g is monic, c must be an integer by the rational root theorem. However, $c^3 - 2019c$ must be even, so it is impossible that $c^3 - 2019c - 2021 = 0$. This proves our claim.

Note that α is a root of $g(x)$. It follows from the above claim that g is the minimal polynomial of α over $\mathbb{Q}$ (see remark). From the given condition $g(f(\alpha)) = 0$, we know that $g(x) \mid g(f(x))$. Replacing x by $f(x)$, we obtain $g(f(x)) \mid g(f^2(x))$. Since $g(x)$ divides $g(f(x))$, and $g(f(x))$ divides $g(f^2(x))$, we have $g(x) \mid g(f^2(x))$. Similarly, for any positive integer n , we have $g(f^{n-1}(x)) \mid g(f^n(x))$. By induction, this yields $g(x) \mid g(f^n(x))$. Therefore, $g(f^n(\alpha)) = 0$. This is exactly the same as $(f^n(\alpha))^3 - 2019f^n(\alpha) = 2021$ as desired.

Remarks. We say that g is the minimal polynomial of α over $\mathbb{Q}$ if g is the monic polynomial with rational coefficients of smallest degree such that $g(\alpha) = 0$. An

important property of the minimal polynomial is that for any polynomial h , $h(\alpha) = 0$ if and only if g divides h . Note that this also implies the minimal polynomial is the only irreducible monic polynomial having α as a root.

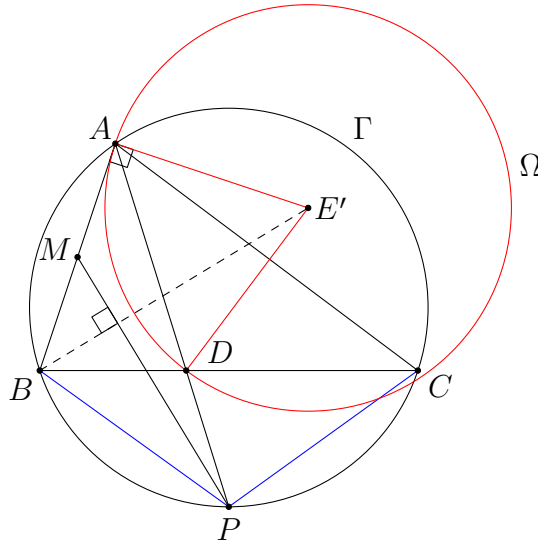
Here is a proof. For any polynomial h such that $h(\alpha) = 0$, let

$$h(x) = g(x)q(x) + r(x)$$

where $\deg r < \deg g$. Then $h(\alpha) = g(\alpha)q(\alpha) + r(\alpha)$, which implies $r(\alpha) = 0$. Due to the minimal choice of the degree of g , r must be the zero polynomial. This means $g(x) \mid h(x)$. Conversely, if $g(x) \mid h(x)$, then it is obvious that $h(\alpha) = 0$.

3. Since P is the midpoint of the minor arc BC , AD is the internal angle bisector of $\angle BAC$.

Let E' be the intersection of the perpendicular bisector of AD and the line through A perpendicular to AB . Using the method of false position, it suffices to show $BE' \perp MP$, and then conclude $E' = E$. Let Ω be the circle with centre E' that passes through A (and hence D).



Since $AB \perp AE'$, AB is tangent to Ω . As $MA = MB$, the powers of M with respect to Ω and B are equal. Since $\angle PBD = \angle PAC = \angle PAB$, we have $\triangle PBD \sim \triangle PAB$, so $PB^2 = PD \times PA$. Therefore, the powers of P with respect to Ω and B are equal.

From above, both M and P lie on the radical axis of Ω and B . Hence, MP is perpendicular to the line joining the centres of the two circles, i.e. $E'B$. The result follows from above.

4. Yes. We shall prove that $P(x) = (x^2 + 1)(x^2 - 2)(x^2 + 2)(x^2 + 7)$ satisfies all properties. It is not hard to see that P has no rational root. It remains to show it satisfies the second condition.

Consider any odd prime p . Using facts about quadratic residues, we have the following.

- If $p \equiv 1 \pmod{4}$, then $x^2 + 1 \equiv 0 \pmod{p}$ is solvable.
- If $p \equiv 7 \pmod{8}$, then $x^2 - 2 \equiv 0 \pmod{p}$ is solvable.
- If $p \equiv 3 \pmod{8}$, then $x^2 + 2 \equiv 0 \pmod{p}$ is solvable since

$$\left(\frac{-2}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{2}{p}\right) = (-1)(-1) = 1.$$

This shows there exists x and $c \in \{1, -2, 2\}$ such that $x^2 + c \equiv 0 \pmod{p}$. Note that $p \nmid x$. Therefore, we have

$$(x^2 + c)' = 2x \not\equiv 0 \pmod{p}.$$

By Hensel's lifting lemma, $x^2 + c \equiv 0 \pmod{p^k}$ is solvable for any positive integer k .

Next, we want to show for any $k \geq 1$, there exists an integer b_k such that $2^k \mid (b_k^2 + 7)$. Indeed, the statement is true for $k = 1, 2, 3$, by simply taking $b_k = 1$. Assume now the statement is valid for some $k \geq 3$. For the case $k + 1$, take $b_{k+1} = b_k + t2^{k-1}$, with t to be determined. Now, since b_k is odd,

$$b_{k+1}^2 + 7 = (b_k + t2^{k-1})^2 + 7 \equiv (b_k^2 + 7) + t2^k \pmod{2^{k+1}}.$$

By the inductive hypothesis, $2^k \mid (b_k^2 + 7)$. So we just take $t \equiv \frac{b_k^2 + 7}{2^k} \pmod{2}$ to get $2^{k+1} \mid (b_{k+1}^2 + 7)$.

Now, let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be the prime factorization of n . From above we know that for every $1 \leq j \leq s$, there exists an integer m_j such that $p_j^{\alpha_j} \mid P(m_j)$. By the Chinese remainder theorem, we can find $m \equiv m_j \pmod{p_j^{\alpha_j}}$ for $1 \leq j \leq s$. Then $P(m) \equiv P(m_j) \equiv 0 \pmod{p_j^{\alpha_j}}$. So $n \mid P(m)$. Therefore, the polynomial $P(x)$ satisfies the second condition, and we are done.

Test 3

1. (IMO Shortlist 2020 A3) See the official solution.
2. (IMO Shortlist 2020 G1) See the official solution.

3. (IMO Shortlist 2020 N6) See the official solution.

Test 4

1. (IMO Shortlist 2020 C1) See the official solution.
2. (IMO Shortlist 2020 A2) See the official solution.
3. (IMO Shortlist 2020 G6) See the official solution.

Year 2022

Test 1

1. Note that $a_0 \equiv 1 \pmod{5}$. We claim that $a_n \equiv 3 \pmod{5}$ if $n \geq 1$ is odd, and $a_n \equiv 2 \pmod{5}$ if $n \geq 1$ is even. Indeed, we check that

$$a_1 = a_0 + \frac{2}{a_0} \equiv 1 + 2 \cdot 1^{-1} = 3 \pmod{5}.$$

Whenever $a_n \equiv 3 \pmod{5}$, then

$$a_{n+1} = a_n + \frac{2}{a_n} \equiv 3 + 2 \cdot 3^{-1} \equiv 3 + 2 \cdot 2 \equiv 2 \pmod{5}.$$

Whenever $a_n \equiv 2 \pmod{5}$, then

$$a_{n+1} = a_n + \frac{2}{a_n} \equiv 2 + 2 \cdot 2^{-1} \equiv 3 \pmod{5}.$$

Therefore, the claim follows by induction. Since 2 and 3 are not squares modulo 5, none of the a_n 's with $n \geq 1$ is a square of a rational number.

2. The answer is 30.

Firstly, since $a^4 + a + 1 = 0$, we have

$$a^5 + 2a + 1 = a(a^4 + a + 1) - a^2 + a + 1 = -a^2 + a + 1.$$

Let $y = -x^2 + x + 1$. Then $x = \frac{1 \pm \sqrt{-4y+5}}{2}$. Now,

$$\begin{aligned} 0 &= x^4 + x + 1 \\ &= \left(\frac{3 - 2y \pm \sqrt{-4y+5}}{2} \right)^2 + \frac{1 \pm \sqrt{-4y+5}}{2} + 1 \\ &= \frac{2y^2 - 8y + 7 \pm (3 - 2y)\sqrt{-4y+5}}{2} + \frac{1 \pm \sqrt{-4y+5}}{2} + 1 \\ &= y^2 - 4y + 5 \pm (2 - y)\sqrt{-4y+5}. \end{aligned}$$

This implies $(y^2 - 4y + 5)^2 = (y - 2)^2(-4y + 5)$, which is a degree 4 polynomial equation. Therefore, $a^5 + 2a + 1$, $b^5 + 2b + 1$, $c^5 + 2c + 1$ and $d^5 + 2d + 1$ are roots of the equation

$$f(y) = (y^2 - 4y + 5)^2 - (y - 2)^2(-4y + 5) = 0.$$

Note that f is monic. So it is exactly $x^4 + px^3 + qx^2 + rx + s$. Thus,

$$p + 2q + 4r + 8s = 8f\left(\frac{1}{2}\right) - \frac{1}{2} = 30.$$

3. There are 278400 ways to assign the people.

Note that each family can be partitioned in either a $(4, 1)$ fashion (where exactly one parent sits with three children and the other is alone) or a $(3, 2)$ fashion (where one parent sits with two children and the other sits with one).

Case 1. All families are partitioned in $(4, 1)$ fashion.

The 4-group of each family occupies a table, and the other four parents take the remaining table. There are 2 ways to choose a parent of each family. So the number of ways is $2^4 = 16$ if we regard the tables as the same.

Case 2. Three families are partitioned in $(4, 1)$ fashion and one is partitioned in $(3, 2)$ fashion.

There are 4 ways to choose which family to be in $(3, 2)$ fashion. The 3-group must sit with one of the three parents from another family, while the 2-group must sit with the remaining two parents. There are 2 ways to choose a parent of each family, and 3 ways to choose a child of the family in $(3, 2)$ fashion to be put in the 2-group. So the number of ways is $4 \times 3 \times 2^4 \times 3 = 576$.

Case 3. Two families are partitioned in $(4, 1)$ fashion and two are partitioned in $(3, 2)$ fashion.

There are $\binom{4}{2} = 6$ ways to choose which two families to be in $(3, 2)$ fashion. The two 2-groups must sit together, while the 3-groups must sit with one of the two parents from the other families. There are 2 ways to choose a parent of each family, and 3 ways to choose a child of each family in $(3, 2)$ fashion to be put in the 2-groups. So the number of ways is $6 \times 2 \times 2^4 \times 3^2 = 1728$.

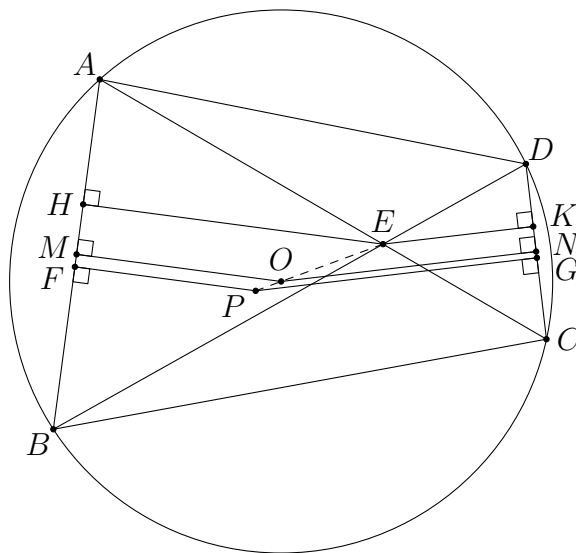
It is easy to check that no other cases are possible. Since the tables are different, there are $5! = 120$ ways to arrange the tables. Therefore the answer is

$$120 \times (16 + 576 + 1728) = 278400.$$

4. The answer is $\frac{PE}{PO} = 5$.

Note that $\frac{AF}{FB} = \frac{DG}{GC}$. Now if we let M and N be the perpendicular feet from O to AB and CD respectively, we have $\frac{AM}{MB} = \frac{DN}{NC}$. Also if we let H and K be the perpendicular feet from E to AB and CD respectively, then by the fact

$\triangle EAB \sim \triangle EDC$, we have $\frac{AH}{HB} = \frac{DK}{KC}$. Therefore, P, O, E are collinear, and $\frac{PE}{PO} = \frac{FH}{FM} = \frac{GK}{GN}$.



Now $AF = 15$ and $AM = \frac{AF + FB}{2} = 14$. Since $\triangle EAB \sim \triangle EDC$ with $\frac{AB}{DC} = 2$, we have $EA = 2ED = 26$. By the cosine formula, we have

$$AH = AE \cos A = 26 \times \frac{26^2 + 28^2 - 30^2}{2 \times 26 \times 28} = 10.$$

Hence we conclude $\frac{PE}{PO} = \frac{FH}{FM} = \frac{AF - AH}{AF - AM} = \frac{15 - 10}{15 - 14} = 5$.

5. The answer is 898.

Indeed, let $\omega = e^{\frac{2\pi i}{3}}$ be a cube root of unity. By the binomial theorem, we have

$$\begin{aligned}(1+1)^n &= \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}, \\ (1+\omega)^n &= \binom{n}{0} + \binom{n}{1}\omega + \cdots + \binom{n}{n}\omega^n, \\ (1+\omega^2)^n &= \binom{n}{0} + \binom{n}{1}\omega^2 + \cdots + \binom{n}{n}\omega^{2n}.\end{aligned}$$

It is known that $1 + \omega^k + \omega^{2k} = 0$ for any $3 \nmid k$, and $1 + \omega^k + \omega^{2k} = 3$ for any $3 \mid k$.

Thus, adding up the above 3 equations, we obtain

$$\begin{aligned}
& 3 \left(\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots + \binom{n}{3 \lfloor \frac{n}{3} \rfloor} \right) \\
&= (1+1)^n + (1+\omega)^n + (1+\omega^2)^n \\
&= 2^n + (-\omega^2)^n + (-\omega)^n \\
&= \begin{cases} 2^n + 2 & \text{if } n \equiv 0 \pmod{6}, \\ 2^n + 1 & \text{if } n \equiv 1, 5 \pmod{6}, \\ 2^n - 1 & \text{if } n \equiv 2, 4 \pmod{6}, \\ 2^n - 2 & \text{if } n \equiv 3 \pmod{6}. \end{cases}
\end{aligned}$$

Therefore, we have $f(n) = \frac{1}{3}$ if $3 \nmid n$, and $f(n) = \frac{2}{3}$ if $3 \mid n$. As $\left\lfloor \frac{2021}{3} \right\rfloor = 673$, we have

$$f(1) + f(2) + \cdots + f(2021) = 673 \left(\frac{2}{3} \right) + (2021 - 673) \left(\frac{1}{3} \right) = 898.$$

6. No, it is not possible. Indeed, let n_i be the number of 01-sequences in the set with 1 at the i th entry, where $1 \leq i \leq 200$. We also let S be the total number of different positions between all pairs of 01-sequences in the set. Since every pair differs at least at 101 positions, we have $S \geq 101 \binom{n}{2}$. On the other hand, there are $n_i(n - n_i)$ pairs of different digits in the i th entry. Thus, by considering the i th positions of all pairs of 01-sequences, we have $S = \sum_{i=1}^{200} n_i(n - n_i)$. Using the AM-GM inequality, we have

$$n_i(n - n_i) \leq \left(\frac{n_i + (n - n_i)}{2} \right)^2 = \frac{n^2}{4}.$$

Hence we have

$$50n^2 = \sum_{i=1}^{200} \frac{n^2}{4} \geq S \geq 101 \binom{n}{2}.$$

This implies $n \leq 101$. However, when $n = 101$, the inequality $\frac{n^2}{4} \geq n_i(n - n_i)$ is strict, as one side is an integer and the other side is not. Thus $n < 101$, or $n \leq 100$.

Test 2

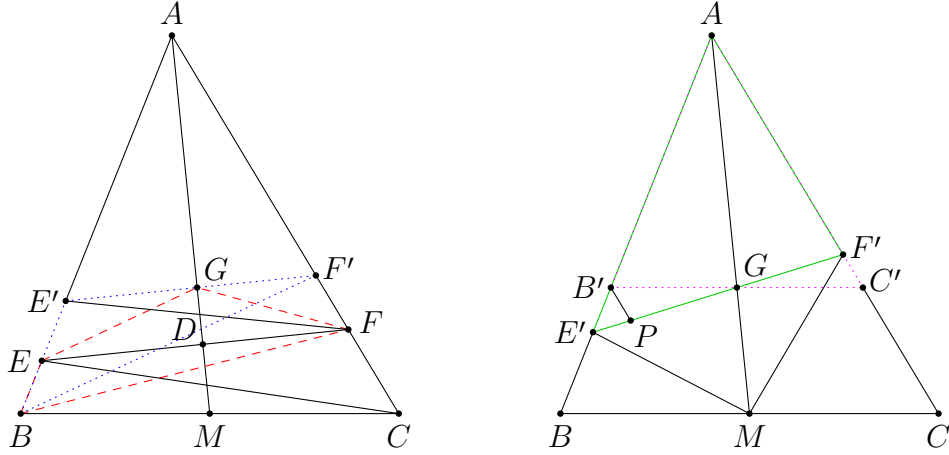
- Let E' and F' be the points on AB and AC such that E', G, F' are collinear and $E'F' \parallel EF$. Since $\triangle EFG$ and $\triangle EFE'$ have the same base EF and the same height, they have the same area. This yields

$$[BEGF] = [BEF] + [EFG] = [BEF] + [EFE'] = [BFE'] \geq [BF'E'].$$

The last inequality holds since $\triangle BFE'$ and $\triangle BF'E'$ have the same base BE , but F' is closer to the sideline BE' than F . Equality holds when $F = F'$, i.e. $D = G$. Similarly, we have $[CFGE] \geq [CE'F']$, with equality $D = G$. Thus, we have

$$[BEGF] + [CFGE] \geq [BF'E'] + [CE'F'] = 2[MF'E'] = [AE'F']$$

since M is the midpoint of BC and $AG : GM = 2 : 1$. It suffices to prove $[AE'F'] \geq \frac{4}{9}[ABC]$.



Let B' and C' be the points on AB and AC such that B', G, C' are collinear and $B'C' \parallel BC$. As $\triangle AB'C' \sim \triangle ABC$ with ratio $2 : 3$, we have $[AB'C'] = \frac{4}{9}[ABC]$. Therefore, it remains to show $[AE'F'] \geq [AB'C']$. WLOG assume E' lies on the segment BB' and F' lies on the segment AC' . Then we only need to show $[GB'E'] \geq [GC'F']$. Indeed, let P be the symmetric point of F' in G . Then $\triangle B'GP \cong \triangle C'GF'$, so that P lies on the line $F'G$. Also, P must lie on the segment GE' since

$$\angle GPB' = \angle GF'C' = \angle F'E'A + \angle E'AF' > \angle PE'B'.$$

It follows that $[GB'E'] \geq [GB'P] = [GC'F']$ as desired. Equality holds when $E = B'$ and $F = C'$, i.e. $D = G$ and $EF \parallel BC$.

2. Since $a_0 = 0 = a_m$, we can prove by induction that $a_k = a_{m+k}$ for all k . If there are two consecutive terms of $\{a_n\}$ which are the same, then all subsequent terms are the same. By the periodicity, all terms are the same, and they are equal to $a_0 = 0$. This gives $a_1 = 0$ and we are done.

Now, assume $d_n = a_{n+1} - a_n \neq 0$ for all n . Using the fact $x - y \mid P(x) - P(y)$ for all $x, y \in \mathbb{Z}$, we know that d_n divides $P(a_{n+1}) - P(a_n) = a_{n+2} - a_{n+1} = d_{n+1}$ for all n . This gives

$$|a_1 - a_0| = |d_0| \leq |d_1| \leq |d_2| \leq \dots \leq |d_m| = |a_{m+1} - a_m| = |a_1 - a_0|.$$

Therefore, all equalities hold, and we have $|d_n| = |d_0|$ for all n , which means $d_n = \pm d_0$ for all n . Now, note that

$$d_0 + d_1 + \cdots + d_{m-1} = a_m - a_0 = 0.$$

Since $d_n \neq 0$, there are both positive and negative terms among $d_0, d_1, \dots, d_{m-1}$. WLOG assume d_k and d_{k+1} have different signs for some $k < m$. Then

$$a_{k+2} = a_k + d_k + d_{k+1} = a_k \pm d_0 \mp d_0 = a_k.$$

This implies $a_{n+2} = a_n$ for all $n \geq k$, and so $a_2 = a_{m+2} = a_m = 0$ as desired.

3. (a) The roots of $z^2 + 3z + 8 = 0$ are $\frac{-3 \pm \sqrt{23}i}{2}$. Let $\alpha = \frac{-3 + \sqrt{23}i}{2}$. Then

$$\bar{\alpha} = \frac{-3 - \sqrt{23}i}{2}, \text{ and hence } x^2 + 3xy + 8y^2 = (x - \alpha y)(x - \bar{\alpha} y). \text{ Note that}$$

$$\begin{aligned} (x_1 - \alpha y_1)(x_2 - \alpha y_2) &= (x_1 x_2 + \alpha^2 y_1 y_2) - \alpha(x_1 y_2 + x_2 y_1) \\ &= (x_1 x_2 + (-3\alpha - 8)y_1 y_2) - \alpha(x_1 y_2 + x_2 y_1) \\ &= (x_1 x_2 - 8y_1 y_2) - \alpha(x_1 y_2 + x_2 y_1 + 3y_1 y_2). \end{aligned}$$

Thus, defining $s = x_1 x_2 - 8y_1 y_2$ and $t = x_1 y_2 + x_2 y_1 + 3y_1 y_2$, we have

$$\begin{aligned} &(x_1^2 + 3x_1 y_1 + 8y_1^2)(x_2^2 + 3x_2 y_2 + 8y_2^2) \\ &= (x_1 - \alpha y_1)(x_1 - \bar{\alpha} y_1)(x_2 - \alpha y_2)(x_2 - \bar{\alpha} y_2) \\ &= (x_1 - \alpha y_1)(x_2 - \alpha y_2)\overline{(x_1 - \alpha y_1)(x_2 - \alpha y_2)} \\ &= (s - \alpha t)\overline{(s - \alpha t)} \\ &= (s - \alpha t)(s - \bar{\alpha} t) \\ &= s^2 + 3st + 8t^2. \end{aligned}$$

This clearly proves the result.

- (b) No. Suppose on the contrary that $x^2 + 3xy + 8y^2 \equiv 7 \pmod{23}$ for some integers x and y . This implies $4x^2 + 12xy + 32y^2 \equiv 28 \pmod{23}$, and hence $(2x + 3y)^2 \equiv 5 \pmod{23}$. However, we can check that 5 is not a square modulo 23 by computing the Legendre symbol $\left(\frac{5}{23}\right) = \left(\frac{23}{5}\right) = \left(\frac{3}{5}\right) = -1$ (or by testing $0^2, (\pm 1)^2, \dots, (\pm 11)^2 \pmod{23}$). This is a contradiction, and so there is no such integer in S .

4. (Graham-Pollak theorem) The answer is 99.

Clearly the tournament can be run in 99 days if we assign, on day k (where $1 \leq k \leq 99$),

- player k to be ‘white’;
- players $k + 1, k + 2, \dots, 100$ to be ‘black’; and
- all other players to be ‘idle’.

In this way player i and player j (where $1 \leq i < j \leq 100$) would have met exactly once on day i .

Now suppose the number of days is less than 99, and we would derive a contradiction. Note that we can assign a real number x_i to each player i , such that

- (1) at least one x_i is nonzero;
- (2) the sum of all x_i is zero; and
- (3) the sum of the x_i ’s of all ‘white’ players on each day is zero.

This is because (2) and (3) represent a homogeneous system of fewer than 100 linear equations in 100 unknowns, and hence must have a nonzero solution (i.e. condition (1)). It follows that

$$0 = (x_1 + x_2 + \dots + x_{100})^2 = (x_1^2 + x_2^2 + \dots + x_{100}^2) + 2 \sum_{1 \leq i < j \leq 100} x_i x_j. \quad (4)$$

Since any two players have met exactly once, we have

$$\sum_{1 \leq i < j \leq 100} x_i x_j = \sum_k \left(\sum_{w \in W_k} x_w \sum_{b \in B_k} x_b \right)$$

where W_k and B_k denote the set of ‘white’ and ‘black’ players on day k respectively. Since the sum of the x_i ’s of all ‘white’ players on each day is zero, the above summation is equal to 0. On the other hand, $x_1^2 + x_2^2 + \dots + x_{100}^2 > 0$ as at least one x_i is nonzero. This gives a contradiction to (4).

Test 3

1. (IMO Shortlist 2021 A1) See the official solution.
2. (IMO Shortlist 2021 N3) See the official solution.
3. (IMO Shortlist 2021 G4) See the official solution.
4. (IMO Shortlist 2021 C5) See the official solution.

Year 2023

Test 1

1. If each ring-tone lasts for exactly 3 minutes, the alarm will be ringing at 5:30–5:33, 5:39–5:42, 5:45–5:51 and 5:54–6:30. It will ring for $3 + 3 + 6 + 36 = 48$ minutes in total. Therefore, we must have $x > 3$.

Let $x = 3 + y$ where $0 < y < 3$. In addition to the above periods, the alarm will ring for y more minutes after each of 5:33, 5:42 and 5:51. Thus, we need $48 + 3y = 50$. The only solution is $y = \frac{2}{3}$, and hence $x = \frac{11}{3}$.

2. Clearly, $p \neq 2, 5$. Let $m = 5^{4n} - 5^{3n} + 5^{2n} - 5^n + 1$. Then

$$(2 \cdot 5^{2n} - 5^n + 2)^2 - 5 \cdot 5^{2n} = 4m \equiv 0 \pmod{p}.$$

This gives $5 \equiv (5^{-n}(2 \cdot 5^{2n} - 5^n + 2))^2 \pmod{p}$. Using the Legendre symbol, we have $\left(\frac{5}{p}\right) = 1$. On the other hand, we have

$$(5^{2n} - 5^n + 1)^2 + 5^n(5^n - 1)^2 = m \equiv 0 \pmod{p}.$$

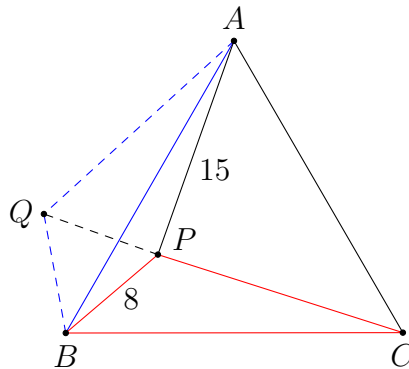
As above, this implies $\left(\frac{-5^n}{p}\right) = 1$. It follows that

$$\left(\frac{-1}{p}\right) = \left(\frac{-5^n}{p}\right) \left(\frac{5^n}{p}\right) = 1 \cdot 1^n = 1.$$

Therefore, $p \equiv 1 \pmod{4}$.

3. Rotate $\triangle PBC$ about point B by 60° in the anticlockwise direction, so that BA is the image of BC after rotation. Let Q be the image of P after rotation. Then

$$[ABP] + [BCP] = [PAQB] = [PQB] + [QPA].$$



Now, $\triangle PQB$ is an equilateral triangle with side length 8, and so its area is $\frac{\sqrt{3}}{4} \times 8^2 = 16\sqrt{3}$. Also, we have

$$[QPA] = \frac{1}{2} \times 8 \times 15 \times \sin \angle QPA \leq \frac{1}{2} \times 8 \times 15 = 60.$$

Thus, the sum of areas is at most $60 + 16\sqrt{3}$. Equality holds when $\angle QPA = 90^\circ$, i.e. $\angle BPA = 150^\circ$ (which fixes $\triangle ABC$ and point P).

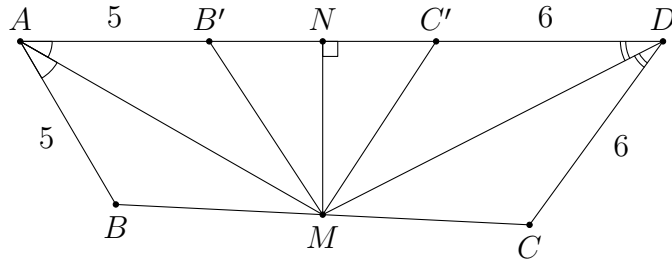
4. By the triangle inequality, we have

$$\begin{aligned} \frac{|x| + |x + 4y| + |y + 7z| + 2|z|}{|x + y + z|} &\geq \frac{|x| + \frac{4}{11}|x + 4y| + \frac{1}{11}|-y - 7z| + 2|z|}{|x + y + z|} \\ &\geq \frac{\left| x + \frac{4}{11}(x + 4y) + \frac{1}{11}(-y - 7z) + 2z \right|}{|x + y + z|} \\ &= \frac{15}{11}. \end{aligned}$$

Equality holds when $(x, y, z) = (28k, -7k, k)$ for some $k \neq 0$. So the minimum value is $\frac{15}{11}$.

5. Let M be the midpoint of BC . Let B' and C' be points on AD such that $AB' = AB = 5$ and $DC' = DC = 6$. Then $B'C' = 17 - 5 - 6 = 6$. Note that $\triangle ABM \cong \triangle AB'M$ and $\triangle DCM \cong \triangle DC'M$. Note also that $\triangle MB'C'$ is isosceles as $MB' = MB = MC = MC'$. Let N be the midpoint of $B'C'$. Then we have $B'N = C'N = 3$ and $\angle MB'C' = \angle MC'B'$. Observe that $MCDB'$ is cyclic since MD bisects $\angle B'DC$ and $MB' = MC$ (while $DB' \neq DC$). Thus,

$$2\angle B'MA = \angle B'MB = \angle C'DC = 2\angle MDC'.$$



It follows that $\triangle AB'M \sim \triangle MC'D$ and $\frac{AB'}{MB'} = \frac{MC'}{DC'}$. Therefore,

$$MB' = MC' = \sqrt{AB' \cdot DC'} = \sqrt{5 \times 6} = \sqrt{30}.$$

Finally we have $MN = \sqrt{MC'^2 - NC'^2} = \sqrt{30 - 9} = \sqrt{21}$. Therefore,

$$\begin{aligned}[ABCD] &= 2([AB'M] + [DC'M] + [MB'N]) \\ &= (AB' + C'D + B'N) \times MN \\ &= (5 + 6 + 3)\sqrt{21} = 14\sqrt{21}.\end{aligned}$$

6. (a) Answer: 65

Note that

$$\binom{64}{2} = 2016 < 2022 < 2080 = \binom{65}{2}.$$

As each pair of straight lines produces at most one intersection point, 64 lines produce at most 2016 intersection points, which is insufficient. Therefore, at least 65 lines are needed.

It is possible to have 65 lines. For example, suppose there are 65 pairwise non-parallel lines $\ell_1, \ell_2, \dots, \ell_{65}$ such that for each $k = 1, 2, \dots, 29$, the lines ℓ_{2k-1} and ℓ_{2k} intersect at a distinct point on ℓ_{65} , and there are no concurrent lines other than $\ell_{2k-1}, \ell_{2k}, \ell_{65}$ for $k = 1, 2, \dots, 29$. Then the number of intersection points is

$$\binom{65}{2} - 2 \cdot 29 = 2022.$$

(b) Answer: 65

As in part (a), at least 65 lines are needed. An example of 65 lines is given below. Suppose there are 65 lines $\ell_1, \ell_2, \dots, \ell_{65}$ where $\ell_{64} \parallel \ell_{65}$ and there is no other pair of parallel lines. Also, for each $k = 1, 2, \dots, 28$, the lines ℓ_{2k-1} and ℓ_{2k} intersect at a distinct point on ℓ_{65} , and there are no concurrent lines other than $\ell_{2k-1}, \ell_{2k}, \ell_{65}$ for $k = 1, 2, \dots, 28$. Then the number of intersection points is

$$\binom{65}{2} - 1 - 2 \cdot 28 = 2023.$$

Test 2

1. No, it is not possible.

Suppose it is possible that after finitely many moves, the square grid contains exactly 574 black unit squares. Then there are exactly $24 \times 24 - 574 = 2$ white unit squares. No matter where the white unit squares are, we can always find a 2×2 square that contains exactly 1 white unit square and 3 black unit squares. One checks that the parity of the number of white cells in this 2×2 square is fixed no matter how we choose the rows or columns in a move. Therefore, it is

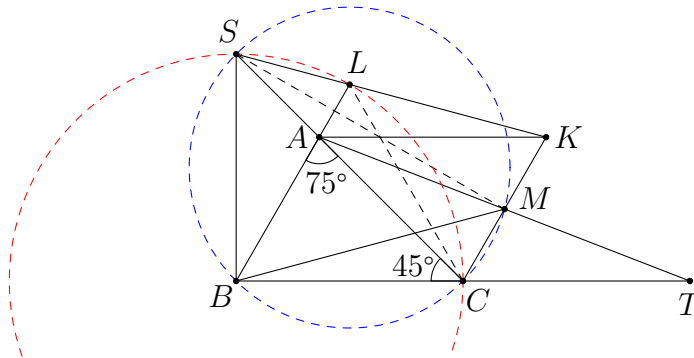
impossible that all the four cells are white initially. This is a contradiction, and so it is impossible to have exactly 574 black unit squares.

2. As M and C are midpoints of AT and BT respectively, we have $MC \parallel AB$ and $MC = \frac{1}{2}AB$ by the midpoint theorem. Extend CM to K so that $CM = MK$. Then $AB = CK$, and hence $ABCK$ is a parallelogram. Also, extend CA to S so that $\angle SKC = 75^\circ$. Then $\triangle SKC$ is an isosceles triangle with $SC = SK$. Therefore, $\angle SMC = \angle SMK = 90^\circ$.

Let BA meet SK at L . Then $LKCA$ is an isosceles trapezoid. It follows that $LC = AK = BC$. Together with $\angle B = 60^\circ$, we know that $\triangle CBL$ is equilateral, and hence $BC = BL$. Furthermore, $\angle LSC = 30^\circ = \frac{1}{2} \angle LBC$. Thus, S lies on the circle with centre B and radius BC . Therefore, $BS = BL = BC$, and

$$\angle SBC = 180^\circ - 2\angle BCA = 90^\circ.$$

Finally, S, B, C, M are concyclic since $\angle SBC + \angle SMC = 180^\circ$. It follows that $\angle BMC = \angle BSC = 45^\circ$.



3. The greatest M is 4 if $n = 4$, and is 3 if $n \geq 5$.

Let S be the sum $\sum_{a \in A} |a|$. For $n = 4$, by considering $A = \{-1, 0, 1, 2\}$, we need $M \leq 4$. Let $a < b < c < d$ be the elements in A . From the condition, we must have $b - a, c - b, d - c \geq 1$. This implies

$$S = (|a| + |d|) + (|b| + |c|) \geq |d - a| + |c - b| \geq 3 + 1 = 4.$$

This shows the largest M is 4 when $n = 4$.

For $n \geq 5$, by considering $A = \{-1, 0, \varepsilon, 2\varepsilon, \dots, (n-4)\varepsilon, 1 + (n-4)\varepsilon, 1 + (n-3)\varepsilon\}$ where $\varepsilon > 0$, we need $M \leq 3 + f(n)\varepsilon$ for some function f . When ε tends to 0, this yields $M \leq 3$.

Firstly, for $n = 5$, let $a < b < c < d < e$ be the elements in A . Since $c - a > c - b$ and $e - c > d - c$, we must have $c - a \geq 1$ and $e - c \geq 1$ by considering the element c . Also, at least one of $c - b \geq 1$ and $d - c \geq 1$ must hold. WLOG assume $c - b \geq 1$. Then we have

$$S \geq (|a| + |e|) + (|b| + |c|) \geq |e - a| + |c - b| \geq 2 + 1 = 3.$$

This shows the largest M is 3 when $n = 5$.

Secondly, for $n \geq 6$, let a be the element with the smallest absolute value. WLOG assume $a \geq 0$.

- If $0 \leq a \leq \frac{1}{2}$, then whenever $|x - a| \geq 1$, we must have $|x| \geq 1 - a$. Since there are three such x 's, we have $S \geq 3(1 - a) + (n - 3)a \geq 3(1 - a) + 3a = 3$.
- If $a > \frac{1}{2}$, then all elements have absolute value larger than $\frac{1}{2}$, and hence $S > \frac{n}{2} \geq 3$.

This shows the largest M is 3 when $n \geq 6$.

4. The condition on s means there exists a positive integer n and nonnegative integers a and b such that

$$10^a s \leq 5^n < 10^a(s + 1) \quad \text{and} \quad 10^b s \leq 4^n < 10^b(s + 1).$$

Note that $10^a s = 5^n$ and $10^b s = 4^n$ cannot hold simultaneously. Multiplying the second inequality and the square of the first inequality, we obtain

$$10^{2a+b} s^3 < 5^{2n} 4^n = 10^{2n} < 10^{2a+b} (s + 1)^3.$$

This implies $s^3 < 10^{2n-2a-b} < (s + 1)^3$. Since s is a two-digit number, we have

$$10^3 \leq s^3 < 10^{2n-2a-b} < (s + 1)^3 \leq 10^6.$$

Thus, $2n - 2a - b$ is equal to 4 or 5. This shows $s = \left\lfloor 10^{\frac{4}{3}} \right\rfloor = 21$ or $\left\lfloor 10^{\frac{5}{3}} \right\rfloor = 46$. The two special numbers are 21 and 46.

Test 3

1. (IMO Shortlist 2022 A1) See the official solution.
2. (IMO Shortlist 2022 N1) See the official solution.
3. (IMO Shortlist 2022 G3) See the official solution.